# JOB STRAIN AND ELECTROCARDIOGRAPHICALLY ASSESSED PATHOPHYSIOLOGIC MECHANISMS

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SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF DOCTOR OF SCIENCE IN ERGONOMICS
UNIVERSITY OF MASSACHUSETTS LOWELL

Date: Apr 23, 2003

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UMI Number: 3083322

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# JOB STRAIN AND ELECTROCARDIOGRAPH ASSESSED PATHOPHYSIOLOGIC MECHANISMS

BY

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ABSTRACT OF A DISSERTATION SUBMITTED TO THE FACULTY OF THE
DEPARTMENT OF WORK ENVIRONMENT
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF
DOCTOR OF SCIENCE IN ERGONOMICS
UNIVERSITY OF MASSACHUSETTS LOWELL
2003

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# ABSTRACT

## JOB STRAIN AND ELECTROCARDIOGRAPH ASSESSED PATHOPHYSIOLOGIC MECHANISMS

The goal of this dissertation was to assess the effect of stress in the work environment on cardiovascular health through assessment of electrocardiograph data. The hypothesis was that psychosocial stress from work affects cardiac regulation in a pathogenic manner. The demand/control model was used to quantify work stress. Cardiac regulation was measured by components of electrocardiogram output: high frequency power (HFP) which represents the parasympathetic autonomic nervous system (ANS); the ratio of low to high frequency power (lo/hi), residual heart rate as well as QT interval variability (QTVI) as measures of sympathetic activity. Thirty-Six males between the ages of 35 and 59 years and free of known heart disease were selected from a larger longitudinal study of healthy subjects and cardiovascular risk based on responses to a subset of questions from the Job Content Questionnaire (JCQ). Subjects were monitored with an ambulatory Holter electrocardiograph. Monitoring began on the morning of a workday followed for 48 continuous hours through the following rest day. Subjects were classified as high and low job strain based on occupational code, responses to the Job Content Questionnaire (JCQ) and a workday diary. High strain subjects exhibited lower parasympathetic activation during the entire 48-hour monitoring period,

while measures of sympathetic activation were elevated in the high strain subjects on the workday. Multilevel modeling demonstrated that strain reported on the ambulatory diary was significantly associated with reductions in parasympathetic activity within-subjects. JCQ assessed decision latitude had a significant effect on this relationship with greater within-subject reductions in parasympathetic activity with low decision latitude. One implication of this study is that demands seem associated with short-term alterations in cardiac regulatory control with long-term changes in parasympathetic activity being associated with job control. Preliminary results support the use of ECG data for further analysis of Karasek's Stress Disequilibrium Model. However, longitudinal studies on larger populations in a wide variety of occupations are needed to fully explore the methods and theories presented. Implications of findings for physiological linkages between strain and disease, testing new job strain based cardiac regulatory hypotheses, as well as for future studies of occupational monitoring are discussed.

# ACKNOWLEDGEMENTS

Only through the support, care and guidance of so many individuals has it been possible to complete this dissertation. I am indebted to my advisor Robert Karasek; he has been an invaluable mentor providing professional and personal guidance and insight. Kevin Costas has played a critical role in the overall project; I would not have been successful without his support and friendship. It was Professor Karasek's foresight, which led to the development of the HRV Project, the platform which made this dissertation possible. Additionally, it has been Professor Karasek's theoretical developments of the Stress Disequilibrium Theory that led to our collaboration on the empirical investigations presented in Chapter Four, and my biological reinterpretation proposed in Chapter One. As the author of this dissertation I plan on submitting Chapters Two and Three for publication as the primary author with the co-author assistance of Professor Karasek. Due to my respect for his work in the development of the theoretical basis for Chapter Four, where I provide initial empirical support for this theory, I would like Professor Karasek to be the primary author of this paper as we prepare it together for publication.

I am very grateful for the work and time of my committee members, Bryan Buccholz and David Kriebel, for their constant support and advice on so many diverse aspects of the of this project. I would like to gratefully acknowledge the members of the University of Massachusetts Medical Center and the SEASONS study, Dr. Ira Ockene, Philip Merriam, and Kathy Ruffino for their technical assistance during the data collection process. I am deeply appreciative to the Cardiovascular Electrophysiology Division at Columbia University Medical Center, specifically Dr. Thomas Bigger and Richard Steinman, for their timely and professional processing of the electrocardiogram data, and to the Cardiovascular Electrophysiology Division at Johns Hopkins Hospital, both Dr. Ronald Berger and Barry Fetics for their incredible degree of collaboration.

collegiality and hospitality in providing algorithms and training in signal processing of QT intervals.

Throughout this process I have had the friendship, mentoring and support of so many individuals from the College of Health Professions, Joseph Dorsey, Janice Stecchi, Susan O'Sullivan, Claire Chamberlain deserve individual mention to the special and critical roles they have each played in my development throughout my years at the university. No one could ask for a more professionally oriented, teaching focused, caring group of colleagues and friends than those that I have had the pleasure of knowing within the College of Health Professions.

From a very early age my parents, Nancy and Joseph, have instilled in me a deep sense of respect for education. Many times throughout my life that they saw much more in me than I could see. Their love and support, care and guidance, work ethic and admiration for the human spirit have always been and continue to provide a safe beacon of light in my life.

Through the many years of my academic, professional and personal life my family has provided me with the most love and support than it would seem humanly possible. I have been deeply touched by the love of my wife, Traci and my two sons, Timothy and Andrew. This process has placed them under considerable strain and I have, on too many times to count, asked for them to remain understanding of my work and time, which they have always done with smiles. They have provided not only joy and happiness with their caring, understanding and unconditional love, but they have also given me the desire to be a better man.

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# INTRODUCTION

This dissertation is organized in a multi-monograph format. Chapter One explains the goals, specific aims and background from which the dissertation is based. The dissertation is a result of the Heart Rate Variability Project at the Department of Work Environment at the University of Massachusetts Lowell and that project's research platform is discussed [1]. The chapter provides a general introduction to the Job Strain literature and raises questions regarding Job Strain and heart disease etiological mechanisms. A detailed description of ambulatory electrocardiograph output and the physiological systems analyzed is included. The chapter introduces basic concepts of a new theoretical model, Karasek's Stress Disequilibrium Model [2] and a new extension of this, focused on the physiological implications of environmental control that requires a new set of monitoring skills, which ambulatory electrocardiograph assessment may provide. Chapters two through four are stand-alone monographs each representing unique portions of the increasingly complex, integrated, and interdependent analytical approach used. Each chapter discusses the implications of its findings in furthering our understanding of job strain cardiac disease mechanisms and the potential role of ambulatory electrocardiograph in elucidating these mechanisms. Chapter Two reports on what has now become a "traditional analysis" of physiologically interpretable dependent variables between high and low job strain subjects. It tests the hypothesis that high job strain influences cardiac regulation in a manner suggestive of exhaustion, pathogenesis,

or deregulation depending on the dependent variable. Chapter Three reports on a within-subject analysis of job strain during a 48-hour period on cardiac vagal regulation using multilevel linear modeling. It expands on Chapter Two by analyzing associations connecting within subject cardiac vagal regulation with diary-based reports of job strain and a review of associations within selected sociological periods during the work and rest days. Chapter Four validates the use of sympathetic nervous system indices prior to their use, in the following chapter, for conducting analyses of autonomic nervous system coordination and variation. These analyses provide preliminary support for Karasek's Stress Disequilibrium theory showing coordination and adaptability differences in subjects with job strain, as well as subjects with self-reported exhaustion. Chapter four builds on the previous two chapters by extending the use of ambulatory electrocardiograph monitoring in the workplace from that which has been previously attempted by incorporating a new theoretical foundation based on previously disconnected literatures in psychophysiology and systems principles. Finally, Chapter Five summarizes conclusions and explicates recommendations for future work.

# CHAPTER ONE

# FOUNDATIONS FOR THE USE OF ELECTROCARDIOGRAPH MONITORING TO ELUCIDATE PATHOPHYSIOLOGIC MECHANISMS OF JOB STRAIN

## Heart Rate Variability Project

The overall goal of the Heart Rate Variability project is to assess the effect of stress in the work environment on cardiovascular health via ambulatory electrocardiograph (ECG) monitoring [1]. The project utilized well-understood measures of psychosocial work stress and measures of cardiovascular function to analyze the predictive combinations of measures that illustrate the association between work related stress and cardiovascular function.

Professor Robert Karasek in the Department of Work Environment at the University of Massachusetts Lowell initiated the Heart Rate Variability Project in 1996. The project protocol was designed to collect occupation and ambulatory electrocardiograph data on a population-based sample to allow contrasts for testing numerous physiological hypotheses related to the Demand Control Model [1]. Subjects were recruited from a sample participating in an ongoing study of seasonal variations in cholesterol (SEASONS) under the direction of Dr. Ira Ockene at the University of Massachusetts Worchester Medical Center [3]. The SEASONS study included a short

version of the Job Content Questionnaire (JCQ) to allow a pre-monitoring screening of subjects fitting age and gender criterion to identify those most likely in high and low strain jobs. The initial wave of data collection, performed by doctoral student Kevin Costas, was performed during 1996 – 1997. At that time 23 subjects were successfully monitored, however, only 3 of these subjects were categorized as high strain based on a full JCQ assessment (completed at the time of monitoring) using national sample cut points. Initial analyses illustrated that psychosocial response questions taken from the JCQ, and the similar diary questionnaires, could show a fairly predictable association with physiological indicators likely to be representative of general conditions of stress [4]. The investigation identified that a 48-hour Holter monitor device could be used successfully to gather real time ECG data in a work environment. However, after the initial wave of data collection the study had a major limitation in its subject recruiting method for high job stress subjects. These subjects did not participate in sufficient numbers to allow analysis of the intended hypotheses. Selective recruitment bias seemed to occur when high strain subjects were asked to participate in a complex study protocol [5].

I joined the Heart Rate Variability project team in September 1997 shortly after it had been concluded that the project, despite having a successful monitoring protocol, required modification to the recruitment strategies. Changes to the recruitment methods were implemented to allow a greater participation yield of high strain subjects without compromising monitoring rigor and are discussed elsewhere [5].

## Overall Hypothesis: Heart Rate Variability Project

The overall hypothesis is that psychosocial stress from work activity affects cardiac regulation in a pathogenic manner. The demand/control model [6-8] is used to quantify job stress. The pathogenic process is hypothesized to begin with stress-induced acute deregulation of cardiac control. Over the long-term, the deregulatory effects of chronic stress exposures must be compensated for by long-term regulatory shifts. These shifts can reduce adaptive capacity, and thereby increase the risk of morbid events. In this project cardiac regulatory control is measured by major components of the ECG: high frequency power which represents parasympathetic autonomic nervous system (ANS)-based cardiac regulation; residual heart rate, QT interval variability index and the ratio of low to high frequency power (lo/hi) which represent sympathetic ANS activity.

## Specific Aims: Dissertation

1. Investigate possible electrocardiograph determined etiologic factors linking work related stress with cardiovascular disease
2. Identify the effect of daily fluctuations of stress on cardiac regulatory control
3. Identify patterns of autonomic nervous system response indicative of altered cardiac regulatory coordination or reduced adaptive capacity that might indicate exhaustion
4. Investigate the possibility of utilizing ambulatory electrocardiograph monitoring to develop stress exposure-response relationships, leading to easier detection of high risk job situations and more sophisticated modeling for testing intervention effectiveness

## Background

### Job Strain Literature

Evidence is accumulating linking psychosocial stress at work to an increasing proportion of occupationally related illness with the majority of the problem stemming from poor work organization [9, 10]. More specifically, substantial research exists relating adverse psychosocial work environments with cardiovascular disease [11, 12].

Perhaps the most commonly utilized hypothesis associating risk of stress-related chronic disease with working conditions is the Demand / Control model (or demand/control/ support model) described by Karasek [6-8] and Johnson and Hall [13]. Stratifying jobs based on high and low psychological demands (such as work deadline pressure, conflict with colleagues) and decision latitude (control over the work situation), four groups are created. The primary hypothesis involves high psychological demands coupled with low decision latitude (defined as "job strain") increasing the risk of stress-related chronic disease (with low social support increasing the risk). A second set of hypotheses, related to what could be called "good stress", involve "active jobs" under conditions of high demands and high decision latitude predicting motivation and new learning behaviors and coping pattern development (with the reverse prediction for low demands coupled with low decision latitude, and of course the active behavior hypotheses are contingent on the demands not being too high). Other groups created with the demand control model include the passive quadrant (low psychological demands and

low decision latititude) and the low strain quadrant (low psychological demands and high decision latitude) [8].

Schnall et al. have recently published a comprehensive review of occupational stress and cardiovascular illness research[14], and specific to the demand/control model by Schnall, Landsbergis, and Baker[12],Kristensen [15, 16] and by Theorell and Karasek [17]. Schnall et al. [12] have noted that a large percentage of studies have found a significant association with job strain and cardiovascular mortality of morbidity, in studies utilizing a wide range of methodologies and designs. Augmenting their findings in their original review with several additional studies (Change based on Schnall et al. 2000) we have now tabulated 72 published studies of cardiovascular disease (CVD) or CVD risk factors associated with job strain. Of these, 36 studies investigated cardiovascular disease or mortality (including two of all cause mortality) and 25 show "positive" (i.e. either all significant, or mixed significant results) associations with job strain. Eleven of 19 of these are positive cohort studies. Low decision latitude has a positive association in 35 of 46 studies, and high psychological demands have a positive association in 14 of 40 studies with an inverse association in five.

The updated review above has also catalogued 41 studies of the major CVD risk factors associated with job strain. In general the picture of associations relating job strain to the major CVD risk factors is significantly weaker. Conventionally acknowledged major risk factors for cardiovascular disease are high blood pressure, high serum cholesterol (low LDL/HDL ratio), smoking, obesity, diabetes, and family history of CHD [18]. It is, of course, possible that the association between job strain and CHD are due to increases in levels of the risk factors that occur under conditions of job strain: for

example increased blood pressure due to job strain. Job strain is positively associated with high blood pressure in 4 of the 17 studies. Smoking is associated positively in three studies, with mixed positive associations in 6 more studies from a total of eleven studies. Serum cholesterol or its components are consistently not positively associated, while two studies out of seven have mixed positive associations. However, the picture improves when more sophisticated physiological monitoring techniques are employed. Using ambulatory blood pressure monitoring, of thirteen studies the association of job strain with high blood pressure is consistently positive and significant in six studies, while in the remaining seven mixed positive associations are found (while more laborious, this monitoring improves the ability to predict risk of CHD risk in general).

## Alternative Etiological Pathways of Stress-Related Risk

The most obvious conclusion, in light of the rather consistent associations between job strain and cardiovascular disease morbidity and less convincing picture of job strain and the conventional risk factors is that the association between job strain and coronary heart disease is mediated through other physiological pathways besides the conventional cardiovascular risk factors. The possibility increases potential interest in sudden cardiac causes of coronary disease mortality – which accounts for approximately 50% of total cardiac mortality [19]. In sudden cardiac death (SCD) defective control of cardiac rhythms is one of the etiological pathways [20-22]. While it must be clearly stated that the common opinion about cardiac death is that it occurs in individuals who often have high levels of the conventional risk factors it remains true that these factors are not

always considered the precipitating cause in the case of sudden cardiac death [23]. SCD that is not related to coronary artery disease risk factors is related to failures of the body to control cardiac output and to support cardiac muscle action. Moreover, the most common control mechanisms that achieve these cardiac controls are the parasympathetic control mechanisms – which coordinate heart activity by slowing it down based on impulses from the central nervous system (the brain stem / autonomic nervous system). Thus, failures in basic central nervous system control of the heart are implicated in the most substantial cause of cardiovascular disease mortality – and yield a conceivable basis for hypotheses related to levels of job strain, exhaustion and cardiac risk. These are precisely the pathways to cardiovascular disease that would be implicated by job strain research in the electrocardiograph area – which are indeed already documented, albeit in non-epidemiological research, in the human factors literature.

*Overview of Argument Integrating Three Research Traditions*

Research literature on human performance and on work overload risks have long sought to combine their understanding of physical load risks with an understanding of the implications of mental loads and social demands, particularly in the many jobs which are becoming such a significant portion of the work burden and potentially the workplace risks [9, 10]. For example, the field of human factor research has found ineffective performance and fatigue in response to an overload of mental demands, but often in laboratory studies [24]. While human factors literature has provided a detailed understanding of physiological response to stress and sophisticated monitoring methods

[25], it has rarely examined chronic disease consequences of stressors. Simultaneously, separate research traditions in the mental strain risks and chronic disease risks have been growing rapidly in the last several decades. However, the research traditions involving chronic disease and how mental stressors affect chronic disease risk have two separate shortcomings from our perspective. First, the now growing body of research on psychosocial stress exposure at work and chronic disease has confirmed that these workplace hazards are major factors, but has not provided a clear understanding of the physiological pathways that would be involved for monitoring the conditions of job risk. A second body of research implicates the alterations in the autonomic nervous system related to mental stressors as an etiology of sudden cardiac death. However, this literature is only just beginning to be connected with that of work stress literature [26-29], and has not been connected to the human factors literature.

The Heart Rate Variability Project attempts to integrate the strengths of understanding provided by these research traditions to illuminate how workplace psychosocial stressors could contribute to the development of chronic disease – through defined physiological pathways related to cardiac output regulation by the autonomic nervous system [1]. This understanding could in turn expand existing insights about the true nature of workplace risks and potential solutions. This combination of research traditions is made feasible by the fact that both the human factors literature and the cardiovascular disease literature have defined easily measured electrocardiograph based physiological parameters including heart rate variability (HRV) and QT interval variability that are regularly utilized to identify compromised physiological states. What remains is to link these traditions to further our understanding of the effects of workplace related stressors –

assisted by methods already developed in the existing literatures. Figure 1-1 illustrates the above three research traditions and how their integration allows an understanding into the etiology of cardiovascular disease secondary to psychosocial job stress.

Several important questions remain to be answered. A stressor is anything that causes stress. Stress, in its simplest form, is defined as "...bodily or mental tension resulting from factors that tend to alter an existent equilibrium" [30]. Stress results from overload experienced by the organism's "control systems" when they attempt to maintain integrated function in the face of environmental challenges [2]. A stress response refers to the compensatory reaction the organism's control systems to the stressor. A stress response is not "disease" - but merely a normal adaptive response to maintain homeostasis while the system responds to the stressor. What makes this process throughout the day dangerous? The concept of overload has been discussed [30, 31], but what constitutes overload? Finally, the concept of low control has the strongest associations in the work stress epidemiology/chronic disease literature [14]. How does low control impact the physiological response to stressors?

## Pathological Mechanisms

The nervous system comprises the brain and various types of nerves that carry sensory impulses from all parts of the body to the brain and through which messages are conducted from the brain to the entire body. The autonomic nervous system (ANS) conveys sensory impulses from the blood vessels, the heart and all of the organs in the chest, abdomen and pelvis through nerves to parts of the brain including the medulla, pons and hypothalamus. These impulses often do not reach our consciousness, but elicit largely automatic or reflex responses through the autonomic nerves, thereby allowing appropriate reactions of all organs in the body to variations in environmental temperature, posture, food intake, stressful experiences and other changes to which individuals are exposed.

There are two major components of the ANS, the sympathetic (SNS) and the parasympathetic systems (also referred to as the vagal system). The sensory nerves from both systems convey impulses from the body to the ANS controlling centers. From these centers impulses are conveyed to all parts of the body by the parasympathetic and sympathetic nerves. Through these nervous pathways, the autonomic nerves convey stimuli resulting in largely unconscious, reflex, bodily adjustments such as in the size of the pupil, the digestive functions of the stomach and intestines, the rate and depth of respiration, the rate and force of the heart's contraction and dilatation or constriction of the blood vessels.

Parasympathetic nervous system activity conveys stimuli resulting in system relaxation and biological restoration. Typical parasympathetic mediated responses

include a reduction in heart and breathing rate, as well as increased gastrointestinal activity. Sympathetic activity conveys stimuli for energy mobilization and system activation as described by Cannon's fight/flight response [32].

Traditional stress-related etiological models of disease development are "non-specific" in their predictions of relationships between exposure and disease. The best known theory, Selye's General Adaptation Syndrome [31, 33] was inspired by Selye's frustration as a student with medical school neglect of non-specific symptom information from patients. Selye's physiological research was based on broadly active hormone systems with cortico-steroids (cortisol) and pitutary-hypothalmic-andreno-cortical response (HPA axis). The 20th century's first stress response model, Cannon's "fight-flight" response (i.e., when a rabbit sees a wolf) [32], implicated the full range of sympathetic autonomic response and adreno-medullary hormone (adrenaline) responses (SAM axis) in the stress process. An implication of Cannon's work is that co-ordination of physiological sub-system response is crucial to species survival. McEwan's recent "Allostatic Load" extension of Selye's formulation [34] looks at a summation of chronic stressor loads and response exhaustion. Again, it provides no specificity of health outcome prediction. All of the physiological processes directly implicated in these research traditions are fundamental to human physiological function (cardiovascular, neurological, hormonal, etc.) and could hypothetically potentiate a very broad range of chronic diseases.

Extending this tradition, Karasek has formulated a stress theory -"the stress dis-equilibrium model"- to further account for the fact that "lack of control," while not a stressor by itself, appears to be the most consistent contributor to some important stress-

related chronic disease [2, 8, 35]. The primary hypothesis is that long term stress, through its effects on the central nervous system, causes deregulation of highly coordinated physiological processes, which in turn leads to chronic disease. The hypothesis implies that deregulation is increased by exposure to larger and longer-term stress risks, as well as lack of individual control over important work activities (limited external control translates into limited internal physiological possibilities for adjustment). Stressor exposure and limited control elevate risk of chronic disease, possibly synergistically. The process of "demands" leading to successively more serious levels of exhaustion and dis-function is further exacerbated by "lack of control" (the demand-control model interface).

## The Method: Electrocardiograph Analysis

The electrocardiograph (ECG) provides a graphic analysis of the heart's electrical activity. Cardiologists commonly use the ECG to detect arrhythmias, heart blocks, and myocardial perfusion. Holter monitoring is the recording of 24 to 48 hour ECG. Electrodes are connected to the subjects' chest, and wires connect these electrodes to a recording device. The waveforms of the ECG depict electrical activity of the heart – and these electrical activities correspond to a point in time in the cardiac cycle. Figure 1-1 illustrates ECG waveforms from three cardiac cycles. The P wave corresponds to contraction of the atria, the Q and R wave correspond to contraction of the ventricles (depolarization), and the T wave represents relaxation of the ventricles. The PR interval is the time period between contraction of the atrial and the ventricular chambers of the heart. The QT interval represents the time period between contraction of the ventricles and relaxation of the ventricles, termed repolarization. The RR interval represents the time period between each beat of the heart, and is the reciprocal of heart rate (i.e. RR interval of 1000 ms yields one beat per second or 60 beats per minute). The use of data derived from a Holter monitor can be used to assess the physiological response to stress. The monitor is non-invasive, is about the size and weight of a walk-man radio, and can be worn on the subject's belt. Wearing the electrodes can cause mild discomfort, and subjects cannot shower while being monitored. The electrocardiogram (ECG) obtained from a Holter monitor can be used to obtain beat-to-beat changes in heart rate (RR Interval), as well as the time interval of repolarization (the QT Interval).

Analysis of the time intervals between the ECG waveforms allows us to differentiate components of autonomic nervous system activity [36-38]. ECG intervals are influenced by autonomic nervous system (ANS) activity through its influence on the Sino-Atrial Node (the hearts intrinsic pacemaker located in the right atrium) and the heart muscle itself. The intervals are commonly proportional to each other, meaning as the RR interval shortens, (HR is higher) the PR interval and QT interval also become shortened.

## The Autonomic Nervous System and the ECG

The central nervous system coordinates all adaptive and regulatory functions of the body. Many systems are controlled without conscious decision through the autonomic nervous system (largely based in the brainstem). The electrical activity of the heart that causes mechanical contraction is self-initiated, however the autonomic nervous system influences the heart's intrinsic electrical activity through a variety of coordinated cardiac reflexes. The heart rate and the beat-to-beat rhythm of the heart are achieved through coordination of the autonomic nervous system. The chambers of the heart are richly innervated with fibers from both the sympathetic (SNS) and parasympathetic (PNS) divisions of the autonomic nervous system. These dual systems have opposite roles in cardiac control; parasympathetic activity decreases contractility, increases the PR interval, RR interval, the QT interval, and decreases HR. Sympathetic stimulus increases contractility, decreases the PR interval, RR interval, QT interval, and increases HR.

As the system charged with providing the body's transport of nutrient and oxygen carrying blood the cardiovascular system is highly regulated. Since blood flow is

maintained by a pressure gradient in the system, and since blood pressure is essential to the maintenance of this gradient, a closed loop feedback system works to maintain blood pressure at all times. As blood pressure is affected by environmental and physiological demands – the system must accordingly adjust to allow homeostasis. Blood pressure control is a vital regulatory system in human physiological response.

Short term regulation of blood pressure by the autonomic nervous system can be depicted in a simplified model, involving the coordination of three interrelated variables: heart rate, stroke volume and total peripheral resistance. The baroreflex represents the major source of feedback control for the coordination of these variables. When blood pressure increases, arterial walls are distended resulting in increased activity from the baroreceptors to reduce cardiac activity through increased parasympathetic activity (signal from the brainstem to slow down the heart.) When blood pressure drops sympathetic activity is increased. Changes in autonomic control over the heart manifests as changes in the aforementioned ECG intervals. Blood pressure is the product of cardiac output and total peripheral resistance. Total peripheral resistance is affected by vascular tone, which can be increased by increases in sympathetic activity, and decreased with decreased sympathetic activity. Cardiac output is the product of heart rate and stroke volume, both influenced by the autonomic nervous system. The increase in heart rate due to sympathetic activation is perhaps best understood. This occurs through a decrease in the refractory period of the SA node with a concomitant decrease in the RR interval, and proportional decreases in the PR and QT intervals. The effect of autonomic nervous system activity on the ECG represents the neural mediation of cardiac control. The autonomic nervous system also utilizes neuro-endocrine mechanisms (catacholamines

such as epinephrine and norepinephrine) to assist in cardiac regulation. Epinephrine and norepinephrine are released from the adrenal medulla in the brainstem during times of sympathetic nervous system activity for longer lasting activation.

*Heart Rate (Chronotropy) as a Limited Measure of Stress*

Heart rate has probably been used more often as a measure of mental workload than any other measure [25]. While it is true heart rate changes with the changing activity of the autonomic nervous system, these changes reflect the global state of the autonomic nervous system. Bernston [39] proposed in the "doctrine of autonomic space" that autonomic nervous system activity does not always follow a reciprocal pattern (i.e. that sympathetic autonomic nervous system activity increase is not necessarily coupled with decreased parasympathetic nervous system activity). In healthy subjects Bernston identified eight different modes of autonomic control [39]. Based on the possibility of different modes of autonomic nervous system control it becomes apparent that heart rate changes alone do not provide information about the specific activity of the sympathetic and parasympathetic branches. Bernston et al. [40] hypothesized that stress may impact the modes of autonomic nervous system control in its pathway towards disease outcomes. Therefore, HR, does not provide sensitive information regarding the activity of the autonomic nervous system for observing autonomic response patterns to occupational stress. In particular, there is a lack of information in HR alone for investigating potential pre-cursors to pathology due to occupational stress. However, more sophisticated analysis can provide such information.

## Heart Rate Variability: RR Interval Variability

Heart Rate Variability (HRV) refers to an assessment of the variance of the beat-to-beat interval, usually analyzed with a post hoc time series analysis. HRV represents oscillations within the autonomic nervous system as it influences the depolarization of the heart's sino-atrial node. This analysis of variability allows identification and quantification of autonomic nervous system components.

Heart rate variability can be analyzed in the time domain, or spectral (frequency) domain. Time domain analysis uses simple statistical analyses of a beat-to-beat interval time series [36, 41, 42]. The most frequently used parameter from the time domain analysis is the standard deviation of all normal RR intervals (SDNN). However, while time domain analysis provides information on heart rate variability, it does not distinguish the parasympathetic and sympathetic influences nearly as well as frequency domain analysis [36, 38]. $SDNN^2$ represents the total variance of the RR interval time series. According to Parseval's theorem, the total variance of the RR interval time series is equivalent to the total power of the time series. [42]. The units for power are as milliseconds squared.

Frequency domain analysis (spectral analysis) of HRV decomposes the time series into sinusoidal components [43]. Power (variance) is plotted as a function of frequency and the analysis computes power in defined frequency regions. The spectral analysis reveals the frequency of variations in the time series, and also the amount of power (variance) within certain frequency ranges. The most commonly used methods of performing spectral analysis utilize the Fast Fourier Transformation (FFT), or autoregression [36, 44]. In humans there are two frequency bands of oscillation that have

known physiological meaning and are related to the autonomic nervous system in its role controlling heart rate [37, 45, 46]. The high frequency power (HFP) spectrum (.15 to .4 Hz; equivalent to a frequency of once every 2 ½ to 6 seconds) is thought to be synchronous with respiratory sinus arrhythmia (RSA) [40]. RSA refers to the increase and decrease of RR intervals in phase relationship with inspiration and expiration [40]. RSA is present when the heart is controlled by the parasympathetic branch of the autonomic nervous system. HFP (as measured from a time series of RR intervals) increases in proportion to parasympathetic activity. The low frequency power (LFP) spectrum (.04 to .15 Hz) is altered with interventions that either raise or block the sympathetic drive to the heart [46-48]. Blocking sympathetic drive decreases LFP (reduction in power (variance) within the frequency range of LFP) and raising the sympathetic drive to the heart raises the LFP. However, LFP is also influenced by the parasympathetic nervous system activity, therefore it reflects sympathetic activity with parasympathetic [36-38]. Since LFP reflects sympathetic activity with parasympathetic modulation, normalized units (which reflect the relative amount of LFP as compared to HFP) have been proposed as a better indicator of sympathetic activity. Measures include the ratio of LFP/HFP [48], or the percent of total power in the high or low frequency range. These ranges of frequency oscillations thus provide physiologically interpretable information about relaxation.

Residuals from a linear regression between high frequency power (PNS activity) and heart rate are referred to as residual heart rate. Residual heart rate has been proposed as a relative index of sympathetic activation [49, 50]. Positive residuals indicate that heart rate is higher than estimated by parasympathetic activity and sympathetic activity is proposed as the major source of this increase. Conversely, when residuals are negative

heart rate is lower than estimated by vagal activity and therefore sympathetic activity may be reduced.

Lenneman and Backs have provided evidence that a factor analytic derived sympathetic factor (determined by the residual heart rate) is highly correlated within subjects with pre-ejection period [25]. Pre-ejection period (PEP) is determined by impedance cardiography and is considered the gold standard of non-invasive cardiac sympathetic activity based on a series of blockade studies [39]. Pearson product-moment correlations were computed for the residual heart rate and PEP. Correlations were computed within subjects. The within subject correlations were computed across ten tasks and five baselines for all subjects. The residual heart rate was correlated with PEP in 14 of 20 subjects ($.68 < r < .86$; $p < .05$).

*Stability of Heart Rate Variability Measures*

Time and frequency domain components of HRV have been shown to be stable indicators of an individual's autonomic nervous system function. Sinnreich et al [51] demonstrated that HRV measures taken in 294 healthy subjects (age 35-65) at two instances at least one month apart were highly correlated. They conclude that recordings of HRV in non-laboratory settings are stable over months and characteristic of an individual [51]. Using spectral analysis of HRV, Maliani [52] has been able to distinguish an individual's postural state (supine or standing) based on HRV responses, demonstrating stable responses indicative of an individual's physiological state.

*Clinical Uses of Heart Rate Variability*

HRV has been utilized in clinical studies to test a variety of health outcomes [38, 42, 53, 54]. In healthy populations, low HRV has been shown to be a risk factor for all cause and cardiac mortality [55-58], as well as new onset of hypertension [59]. Low HRV or Low HFP is also a risk for mortality in patients who have had a myocardial infarction [45, 60-63], have coronary artery disease [64] or congestive heart failure [65-67]. Reductions in HRV have also been identified in subjects that are depressed, anxious, paranoid, angry, or hostile in an attempt to address pathways of increased cardiovascular disease [68-72]. Exhaustion and similar conditions such as chronic fatigue, and over training are associated with alterations in both neuroendocrine indices of autonomic nervous system function as well as alterations in autonomic nervous components of heart rate variability [73-81].

*The Ergonomics Mental Workload Literature*

Sympathetic activity increases with mental loads and psychological stress including work [47, 48, 82-87]. Kalsbeek [88] studied the effect of mental loads on heart rate variability and found HRV decreased with increasing mental loads. Callister [84] found an increase in sympathetic activity through the assessment of nerve activity to skeletal muscle, as well as cardiac parameters such as heart rate and blood pressure. This study determined that skeletal muscle sympathetic activity decreased initially with mental tasks and became elevated only at cognitive challenges that were perceived as stressful. Heart rate and blood pressure were elevated throughout the task.

Based on the known changes in autonomic nervous system activity during mental stress, and the known HRV and HRV frequency changes with autonomic nervous system activity it has been proposed that mental stress will decrease the total variance of heart rate (HRV), increase LFP, decrease HFP and increase the LF/HF ratio. Studies have been performed to validate this hypothesis. In 1975, Hyndman induced mental loads on subjects with a choice task or a tracking task. The major finding was a decrease in total power with mental challenges that also involved physical responses. Hyndman was able to demonstrate through control studies that it was not the physical task that induced changes in total power, but rather the mental demands [85].

Moriguchi analyzed the impact of mental stress on power spectral frequencies with and without the impact of a beta adrenergic blocker (blocks sympathetic activity to the heart) [89]. Mental stress increased heart rate, blood pressure, LFP, and LF/HF Ratio. No association was identified with HFP, where we would have expected a decline. With

the beta blocker, which blocks sympathetic arousal of the heart, the basal (seated and relaxed) and pre stress (seated and waiting for the stressor – anticipation) heart rate, blood pressure and spectral density were not altered, but with mental stress there was a decrease in LFP, LF/HF Ratio and an increase in HFP, as well as blunted heart rate and blood pressure response. They concluded that the increase in blood pressure and heart rate that follow mental stress are the result of increased sympathetic activity [89].

## QT Interval Variability

The QT interval (Figure 1-1) represents the time for repolarization of the ventricular myocardium. Prolongation of the QT interval has been an established predictor of ventricular arrhythmias for quite some time [90]. Berger and coworkers [91] introduced a method to analyze temporal repolarization lability through the assessment of the QT interval variability index. The QT variability index (QTVI) is the log ratio between the QT interval and RR interval variabilities [91]. The QT interval is known to vary, in part, as a function of the RR interval (or heart rate). The QTVI measures the variability of the QT interval after controlling for the known effect of the RR interval. The sympathetic nervous system stimulus at the ventricular myocardium influences the time for electrical repolarization, which is then interpretable on the ECG as the QT interval. Both branches of the ANS have connections at the sino-atrial node and the ventricular myocardium. However, sympathetic branch connections predominate the ventricular myocardium receptor sites [92]. Studies have demonstrated that sympathetic nervous system activation increases the temporal variability of the QT interval in a HR

independent manner [93]. Increased sympathetic nervous system activation leads to increased QT interval variations that are independent of HRV [93]. Based on the calculation of the QTVI as the ratio of normalized variance for the QT interval and HR, the QTVI measures the amount of QT interval variability that is independent of HRV. HRV is mediated by variations due to depolarization of the sino-atrial node and as such is influenced by both branches of the ANS. However, an increase in QTVI has been proposed as only reflecting sympathetic activation [94, 95]. Yeragani and colleagues sought to test the hypothesis that QTVI is mediated by sympathetic activity in normal subjects [94]. In 19 subjects QTVI was significantly elevated after a change from supine to standing postures (sympathetic activation) as well as with pharmacological stimulation of $Beta_1$ and $Beta_2$ (primarily cardiac) sympathetic receptors with isoproterenol [94]. These findings occurred independent of changes in QT duration and therefore thought to represent a direct sympathetic effect on the ventricular myocardium influencing repolarization. Pohl et al. measured QTVI in a group of 6 panic disorder patients and 11 controls during isoproterenol infusions [96]. All subjects demonstrated significant elevations in QTVI during pharmacological sympathetic activation, with greater increases occurring in the panic disorder patients [96]. QTVI has been shown to predict risk of sudden cardiac death [91, 97]. Elevated QTVI has also been linked to certain emotional and mental health disorders such as that are predominated by elevated sympathetic nervous system activation including panic disorder and depression [95].

## Occupational Stress and Electrocardiograph Monitoring

Utilizing ambulatory electrocardiograph monitoring occupational stress research could integrate the strengths of understanding provided by the aforementioned research traditions (Figure 1-1) to illuminate how workplace psychosocial stressors contribute to the development of chronic disease – through defined physiological pathways related to cardiac regulation by the autonomic nervous system. Studies utilizing HRV measures for investigating the physiological impact of occupational stress in the work place are beginning to appear [27-29, 98-104]. A variety of methods have been utilized to categorize occupational stress as well as to measure HRV. A brief review of these studies follows.

Researchers from Japan have assessed the impact of work characteristics such as working hours, commute time and overtime on autonomic nervous system activity through HRV parameters. Sasaki [98] examined the impact of work hours on HRV parameters. Significant differences were identified between workers categorized as having short (Mean = 54.5 h/week) or long (Mean = 63.7 h/week) work hours. Workers in the younger age groups working longer hours had significantly lower Lo/Hi ratios and reduced noradrenaline excretion than those of the same age working shorter hours, both findings are related to reduced sympathetic nervous system activity. Sasaki discusses the role of sleep deprivation and exhaustion as possible etiologies for these differences. No such differences were identified in the older group of workers; however, this may be related to the relatively small sample size, the long work hours of even the short work hour group, and the reduction in total HRV in the older sample.

Kageyama [99] considered commuting time and overtime in an examination of HRV in white-collar workers. Workers working more than 60 hours per week had higher Lo/Hi Ratios and lower HFP when standing than workers working less than 60 hours per week. This is the reverse of the effect identified by Sasaki et al [98]. It is possible that factors left uncontrolled in the study by Kageyama et al. account for this disparity, such as age, and sleep hours. Yet other factors that differ between the groups studied such as the unmeasured job characteristics, years employed, and physical activity might also contribute to the discrepancy.

Iwasaki [100] measured HRV in salesmen from an engineering firm. The group was divided based on working hours (total hours at office and commuting) into long (mean = 64.5 hours) and short (mean = 57.2 hours) workweeks. Iwasaki also assessed symptoms of fatigue and drowsiness and categorized subjects as having no symptoms, or one or more symptoms. There were significantly more individuals with complaints of fatigue both before and after work in the long work time group. HFP was significantly lower in the group that complained of fatigue in the 30 – 60 year old group. Like Sasaki [98] the short working hours group in this study still had a long workweek.

It is important to note that these studies did not measure HRV with a Holter monitor for an extended duration, but rather measured HRV during a 3 minute [98, 100] or 5 minute [99] period in the morning of one day. This may make it difficult to accurately measure the impact of work characteristics on HRV parameters, and at the very least this makes the measurement of HRV less robust for identifying differences in the contrasted groups and therefore for testing the proposed hypotheses. These studies also have assessed groups of workers, that based on the theoretical foundations of the demand

control model, would be considered active jobs (Kageyama assessed male white collar workers, and Sasaki assessed male engineers), as opposed to high strain jobs. Characteristics of the workplace (in particular the job) were not monitored and therefore their impact on HRV parameters could not be assessed. To test hypotheses regarding work stress it is necessary to have valid measures of workplace characteristics, variation in workplace characteristics and some degree of contrast in work stress levels between subjects. Therefore, while these studies have taken an initial step in the use of HRV measures to evaluate workers, they have not adequately tested the impact of work stress.

Kobayashi [101] studied Holter recordings of 12 healthy nurses during normal day shifts and then during a third shift (Midnight until 8:30 AM). On the days with a half-day rest between shifts, the Lo/Hi Ratio remained elevated during 21 hours of the 24-hour period. Kobayashi concluded that this shift fluctuation contributed to alterations in the daily rhythm of the sympathetic nervous system. Kobayashi [102] also studied HRV in taxi drivers working 24 hour shifts and the 48 hours of rest following these shifts. The sample had both normotensive and hypertensive drivers that were not being treated for their hypertension. The hypertensive group, as compared to the reference group of normotensives, had lower HFP (parasympathetic) during the day, and the expected increase in Lo/Hi Ratio during the day lasted longer after the work-day (delayed recovery of sympathetic tone). Matsuzuki [103] studied male factory workers and reported significantly reduced HFP (parasympathetic) during day sleep after night work, as opposed to night sleep after day work. These studies were able to analyze 24-hour patterns of HRV and responses to work, rest and sleep because Holter monitors were used. The alterations observed in the circadian patterns are interesting and studies of this

type may lead to important observations and possible causal linkages between job characteristics and heart disease. The alterations in circadian patterns described in these studies are similar to those identified in patients with various forms of heart disease [54]. However, despite the advantage of 24 hour monitoring in the preceding studies [101-103] they still lacked the necessary measurements of and variations in job characteristics to test hypotheses relating occupational stress to altered autonomic control and higher risk of heart disease.

Measurements of HRV as a marker of autonomic function are an area of future growth in occupational stress research. More recent studies have incorporated the appropriate methodologies to test hypotheses linking occupational stress to alterations in autonomic control of the heart as a possible etiologic link to chronic illness. These studies have investigated occupational characteristics as independent variables and Holter monitoring to assess HRV [27-29, 104]. Utilizing Siegrist's [105] model of effort reward imbalance, Vrijkotte [29] characterized job stress in 109 male white-collar workers as having a high or low imbalance (ratio of demands on the job and rewards from the job,) as well as over-commitment (a coping style associated with the inability to withdraw from work obligations.) Workers were monitored for three days during the same work-week, two work days (Monday and Thursday) and one day off (Saturday or Sunday.) Subjects reporting high imbalance had significantly higher heart rates and lower parasympathetic tone (measured the RMSSD,) as compared with the low imbalance group. These effects were noted for both work and home. While heart rate differences disappeared during sleep, RMSSD remained lower in the high imbalance group during all periods. This may indicates a condition of lower parasympathetic tone and this may

predispose these subjects to greater heart rate reactivity [106]. The authors were surprised to find no significant imbalance by over-commitment interaction in this study, considering their hypothesis that over-commitment leads to the inability to unwind and possibly an exhaustive state.

Van Amelsvoort [28] combined three work place exposure measures; work place noise, shift work, and Karasek's Demand Control Model, to study their impact on HRV in a group of 135 subjects (113 male, 22 female.) Subjects wore a Holter monitor for a 24-hour period, at the beginning of a morning shift for shift workers, and a day shift for the day workers. Shift workers displayed reduced SDNN during sleep compared to day workers. Utilizing the Demand Control model subjects from the low strain group were compared to the active, high strain, and passive subjects. Subjects in the active, high strain and passive groups had elevated LFP as percent of total power (%LFP)(increased sympathetic tone) during work. Workers reporting greater noise levels had greater %LFP when compared to the low noise level workers. Significant findings for the high strain group is expected given the hypotheses of the Demand Control model, however difficult to find in this study. Only 14% of the population was categorized as having a high strain job (high demands, low control), and these subjects were (as were all subjects) young (mean age = 30.8 (7.5)) and had recently been hired at the job in which they were working when monitored (therefore difficult to assess chronic exposure). No previous exposure history is mentioned. Despite these reasons making contrasts difficult to identify altered autonomic regulation in the high strain subjects in this study, it is interesting that there was a very mild trend, where the subjects in the high strain category had lower HFP during sleep. This study applied sleep adjustments to the data since sleep

values are the least impacted by position and activity. However, studies should be cautious when applying sleep adjustments to the HRV data. If high strain workers have low HFP (reduced parasympathetic tone) throughout the 24 hours of a day, adjustments based on individual sleep values may reduce the between subjects variance, making significant, and physiological meaningful findings difficult to identify.

Riese [104] has studied HRV in 159 Dutch female nurses. Nurses wore a Holter monitor on a work-day and a leisure day. In this homogeneous population of young females (mean age = 35.9 (8.5)) no significant effect of job strain was found on HRV. Using multivariate analysis HRV was predicted during lying and sitting activities on the work-day by a job strain X social support interaction, however this finding was opposite of the Demand Control hypothesis. The study is in agreement with the general finding that job strain is a concept that does not apply to women as well as men [107] and, in particular to pre-menopausal women when describing cardiovascular risk [8]. Although older nurses have shown occupational strain related effects on health related quality of life [108]. A significant effect of day by child-care load, in all women HRV was lower at work than on the leisure day, except for single mothers. Single mothers had lower HRV values at home as compared to the work-day. Frankenhauser [109] has compared the psychoendocrinological responses to demands in women versus men and has discussed the more significant impact of home responsibilities on women.

Hanson [27] tested the effect of work stress based on the effort reward imbalance theory on parasympathetic control of the heart with the high frequency power of HRV. Seventy subjects (39 males, 31 females) from two different professions (health professionals and office workers) mean age (36.5). Subjects were monitored during the

work day and completed diary response questions aimed to test mental state, as well as activity. Multi-level analysis was performed with level one effects being HRV measures with in a person, and level two representing stable between subjects differences. After controlling for time of day, "need for control" had a negative effect on high frequency power of HRV (parasympathetic activity.) While this study choose occupations that are reportedly similar with regard to position and activity, the lack of control for these two parameters can impact the interpretation of the overall results. There is no way to determine whether reports of "need for control" were associated with greater activity, or standing positions, which are both associated with reductions in high frequency power of HRV. This investigation also lacks the necessary contrasts, between job characteristics, work versus rest day, and awake versus sleep, to test several of our hypotheses regarding the impact of job stress on the autonomic nervous system.

## Theoretical Extension for the Expanded use of Electrocardiograph Monitoring in Job Strain Research

Despite the growing number of studies using ECG monitoring in job stress research few clear conclusions can be drawn because the exact methods for either monitoring or analyses of data remain unclear [11, 27-29, 98-103, 110]. A benefit of ambulatory ECG monitoring is the ease of obtaining indicators of cardiac regulation over extended periods of time, leading to complicated datasets. The rationale for using HRV to measure exposure in job stress research has been twofold: first, it draws from the cardiovascular epidemiology evidence that links reduced HRV to an increased risk of cardiovascular disease [26-29, 98, 100, 102, 110]; and second, evidence that components of HRV are related to autonomic nervous system based cardiac regulation [26, 111]. These are not mutually exclusive since stress based alterations in autonomic nervous system regulation have long been proposed as an etiologic mechanism of chronic disease [31, 33]. Part of the problem in determining the exact methods for analysis of large ECG monitoring datasets in job stress studies may be the lack of a complete theoretical platform for stress induced autonomic nervous system deregulation. The Stress Disequilibrium Model provides propositions that assist in formulating hypotheses for the development of analytical methodology for using ECG data in quantifying job stress exposure and the pathogenic process.

In the Stress Disequilibrium Theory the pathogenic process is hypothesized to begin with stress-induced acute deregulation of physiologic control [2]. External demands (as stressors) are proposed to require the ordering of chaos to maintain a stable, homeostatic environment as the organism meets the energy requirements associated with the external demands [2]. Internal homeostasis is critical for higher mammals and

represents the platform for complex functioning and interaction of physiological systems. Based on this requirement, external demands call for internal coordination loads (allostatic). External lack of control is proposed to increase coordination difficulty of internal physiological coordination by disrupting parameters. Control over sub-systems creates and organizes the high-level organism response to turn the sub-systems disorganized random activity into ordered work. This is the physiological work at the systems level and it is the essence of the adaptive action in the internal environment that allows the organism to maintain stability. Over the long-term, the deregulatory effects of chronic stress exposures must be compensated for by long-term regulatory shifts. These shifts can reduce adaptive capacity, and thereby increase the risk of morbid events [2].

The action of physiological control systems is cyclical. Cycles of organization and ordered work deplete the organism's control capacity, and must be followed by periods of rest, cut off and protected from environmental challenges during which equilibrium (of disorder with maximum entropy under the 2nd law of thermodynamics) is restricted. This equilibrium must be achieved before the next cycle of ordered work is possible. It is the ordered energy / disordered energy cycle (a "thermodynamic model," consistent with such a proposition by Recordati [112]) that Karasek claims supports understanding of system level control capacity reduction that accompany low external control as restricting degrees of freedom of the physiological system's adjustment in long term context's leading to chronic disease.

This theory represents a systems-based understanding of disease development and prevention [2]. It implies understanding of the human physiological system's natural abilities to function as an integrated system, of understanding system limitations under

complex societal demands and of the attendant physiological warning signals. This approach differs from the currently popular medical research thrust toward molecular-level, reductionist understandings of the disease process.

Figure 1-3 illustrates the potential use of electrocardiographically determined indices of cardiac control, including heart rate variability. Clearly, there is interest in identifying more traditional associations (i.e. direct indications of HRV determined disease risk or differences in autonomic nervous system activity) in this project (Chapters Two and Three). These HRV indications, such as reduced HRV and reduced HFP, may indicate the presence of long term, irreversible regulatory shifts. However, we also hope to introduce new analytical methods combining knowledge from previously disparate research traditions upon the framework of the Stress Disequilibrium Theory. Figure 1-4 illustrates the multiple steps toward disease proposed as part of Karasek's theory [2]. Restricting analytical methodology in the area of job stress research using ECG indices to traditional approaches for identifying changes that may indicate long term regulatory shifts potentially waits until chronic disease is imminent or irreversible change has occurred. This issue has been identified in the area of job strain related high blood pressure. By the time clinical indications of high blood pressure have been identified (clinic based measurement) significant cardiac disease related structural changes have occurred. Yet, ambulatory measurements during work can identify high blood pressure prior to structural clinical disease [14, 113].

## Toward a Biological Reinterpretation of Job Strain Impact on Long Term Health Status

A plausible explanation of how environmental control can influence physiological control is required to investigate possible job strain disease mechanisms and approaches to monitor exposure via ECG measures. I will attempt to provide a biological interpretation. This interpretation provides a macro-level context of insight and is based on an integration of Karasek's Stress Disequilibrium Theory [2] with evolutionary biology and physiology [114, 115], adaptability theory [116], complex adaptive systems and emergence [117-120]. Karasek describes the adaptive capacity of a system as related to the nature of coordination of its integrated capacities, as well as the nature of these capacities [2]. With this formulation of adaptive capacity Karasek has essentially combined Ashby's Law of Requisite Variety with that of complex adaptive systems and emergence. Karasek's description of the nature of integrated capacities, referring to the range of responses available to the system, proposes that the system requires a certain number and range of responses, Ashby proposed that the number of possible responses of a system needs to be greater than the number of possible disturbances to minimize the deviation of the system from a state of stability (homeostasis) [117]. A fundamental difference is that Karasek makes specific micro level propositions based not only on the number and range of system responses but that the integration of system sub-components in a coordinated manner is what allows for, not only optimal physiological response, but the number and range of responses. Complex adaptive systems (CAS) and emergence theories are based on integration of subsystems at higher and higher organizational levels or complexity.

However CAS does not imply specific coordination responses and are geared to a particular ability or functional response.

Biological adaptability is the process whereby an organism fits itself to its environment and it is essential to life at all levels of organization [116, 119]. Adaptability involves both the functional and structural organization of systems as well as the physicochemical constitution of the organism. The study of a variety of CAS has led to the initial formulation of general principles common to all CAS [118, 119]. These principles include the concept that structural and functional organization occurs within and between systems. Each system consists of a variety of interacting sub-components that ultimately aggregate to create some measure of functional output [118, 119]. The functional output of each system interacts with other systems at a variety of functional levels. Emergence is a property that arises from the structural and functional organization, and is the principle that the whole system is more complex than the sum of its parts [120]. CAS adapt by changes at multiple levels of the total system including changes in the manner that sub-systems interact. Life emerges from the complex interactions of the hierarchically organized internally interacting structures [120, 121]. Applied to the area of job strain research an individual represents a system with numerous internal interacting sub-systems.

Another related theoretical perspective is that of biological phenotype. Phenotype refers to the biological and physiological internally interacting sub-systems that emerge to create an organism's traits. The phenotype represents the complex combinations of physiological and biological integration within the system and together they provide a range of capabilities for response to the environment. The ability of the phenotype to

adapt to its environment can be referred to as its adaptive capacity or its fitness homeostasis. In response to work environments we will refer to this as "work related adaptive capacity." While phenotypes are stable between individuals within a species based on genotype, some variation does exist between individuals due to variation in genotype. During the life of an individual a genotype is fixed, however within individual variation in phenotype is a function of the environment in which the individual exists [122]. The capacity of an individual to change their phenotype is referred to as phenotype plasticity. The individual, as a system, then interacts with the environment ultimately creating ever-increasing levels of emerging, self-organized, complexity that are interpreted as higher order systems. The individual can also change, ultimately adapting, the environment, which is the basis of all forms of human work.

*The importance of variability for adaptability*

Adaptability is one of the most striking properties of living organisms and it is heavily dependent on the presence of variation at all levels of organization [116]. Variation has significant functional implications when considering adaptability at a variety of time scales. However, this is not to say that more absolute variance is better. Variation must occur within boundaries with too little variation the organisms cannot adapt, too much variation and variation becomes the creative agent of change and dominates the state of the organism [115]. Biological variation for the process of adaptation is also isotropic, having no preferred pathway. Gould describes this as a dynamic sphere with the current state of the organism at the center and all radii equally accessible [115]. Extending this

geometric analogy, the movement of the center of the sphere represents changes in the state of the organism, with the availability of radii and the size of the sphere representing adaptive options to the organism. On short time scales this state could refer to activity of some physiological measure or process, such as heart rate, while on longer time scales, yet still within the lifetime of the organism, the state could relate to the organisms phenotype.

Traditional stress models utilize the perspective that some physiological processes vary in response to environmental changes to maintain stability of other processes, however they focus exclusively on environmental demands. McEwan's "allostatic," load comes from the Greek root "allo", meaning variable. Allostatic systems help keep the body stable by themselves being able to change. Within this context the individual maintains homeostasis when presented with a challenge by the interaction of the allostatic sub-systems. The variance must occur in the correct system, at the right time, in the right amount, and this requires coordination by higher organizational structures, such as the central nervous system.

*Phenotype plasticity*

Adaptability in the shortest time scales includes physiological responses to challenges imposed by environments, and in significantly longer time scales genotypic variation and selection. Between these vastly different aspects of adaptability is phenotypic plasticity. Phenotypic plasticity requires more time than physiological adaptations, yet occurs within the lifespan of the individual. It represents the extent to

which the individual can change its physiology, behavior or structure in response to the environment to allow more effective short-term physiological adaptations [122, 123]. To clearly illuminate, we are interested in two levels of adaptation. One exists in response to short term environmental demands to maintain homeostasis through allostatic mechanisms. The other, phenotype plasticity, refers to the overall physiological, behavioral, and structural capacity of the individual to adapt over longer time scales thereby adjusting the individual's phenotype. For example: the elevation in heart rate to increase cardiac output during increased energy demands is a short term physiological adaptation, while, the individuals overall fitness level and ability to increase energy production refers to his/her phenotype. The individual's phenotype can vary during his/her lifetime depending on the environment (i.e. how much they exercise), or whether he/she has a disease. In the context of the Stress Disequilibrium Theory, the environment with reduced control leads to, over time, work related adaptive capacity reductions [2, 35].

These two levels of adaptability have different implications for job strain research. The short-term physiological adaptive responses have implications for the individual's ability to meet short-term demands and maintain homeostasis. It represents what the majority of traditional stress models attempt to measure. Altered adaptive responses are related to increased risk of disease development in these models, with altered response (either called allostatic load, exhaustion or reactivity) leading to this increased risk of disease. However, it is due to changes in the phenotype and phenotype plasticity that these altered responses occur. Changes in phenotype could restrict the short-term adaptive capabilities, either by reducing overall variation, or by reducing adaptive options (directions of the movement within the sphere). Reduced phenotype

plasticity would prevent the system from future long-term change. One might think of aging as a reduction in phenotype plasticity, since the older individual can exercise, but cannot achieve the same improvements in overall fitness as if they were younger. The implications of reduced phenotypic plasticity are that job redesign changes may take longer to change the risk of workers if they have reductions in phenotype plasticity.

Environmental influences on phenotype and phenotype plasticity

Enriched environments in numerous species have been shown to increase an organism's adaptive capacity and improve their phenotypic plasticity [124-127]. It has also been demonstrated that environmental variability is associated with greater degrees of plasticity in plants. Greater plasticity is often referred to as selection for increased fitness homeostasis because selection favors organisms that express fitness under a wide range of environmental conditions. Fitness homeostasis is associated with increases in the plasticity of underlying traits that counter the effects of environmental changes [128]. Organisms demonstrate greater plasticity when they exist in environments that have greater variability. It is our contention that reductions in environmental control placed on individuals within work organization structures lead to phenotypes that have a lower capacity for effective short-term adaptability, and restrict phenotype plasticity. This is consistent with, but is a more macro-level description than Karasek's Stress Disequilibrium Theory which attempts to express these changes in greater detail, through descriptions of successive mechanisms of regulatory failures based on control limits, control system dynamics and changes in equilibrium states.

Physiology and behavior linked neural networks

How do reductions in environmental control exert this adaptation? One mechanism may be through restrictions in behavior that contain neural network linkages to the autonomic nervous system. Blessing proposes that mammals evolved to have coordinated centrally controlled visceral responses to the environment resulting in the development of patterned commands that integrate autonomic and behavioral components of homeostasis [129]. As Blessing puts it: "The walk to the water and drinking are both necessary aspects of fluid balance [129]." The implication of such a hypothesis, that the autonomic nervous system integrates patterns of both behavior and organ responses, is that environmental restrictions to behavior may interrupt the central nervous system's integrated response. Behavioral limitations, through phenotype plasticity and changes in neural networks may lead to work related adaptive capacity that have less system responses options, leading to reduced system variation leading to systems that are less capable of short-term physiological adaptability. Karasek's model would emphasize that environmental restrictions limit physiological coordination, disturbing equilibria, and requiring changes in equilibria that are then equivalent to disease. It is also possible that such environmental restrictions could lead to reductions in phenotypic plasticity itself. Parvizi discusses the role of the brainstem in consciousness and demonstrates that the necessary neural structural connections for such a hypothesis exist. Brainstem activity of the autonomic nervous system nuclei is involved with modulating global activity of the

cerebral cortex [130]. Again, pointing toward a neural network of linked autonomic nervous system activity with central/cerebral brain centers.

Neural networks as evolving dissipative structures. Recordati draws from the work on dissipative structures by Prigogine to describe how the evolution of functional systems of neural networks may be shaped by environmental forces to give rise to organized groups of neurons and their circuitries with specialized common functional goals [131, 132]. According to Recordati neural groupings such as the baroreflex can be taken as separate dissipative structures, or can be aggregated with others to describe more complex (emergent) structures. A dissipative structure is a dynamic state of matter and Prigogine describes them as the result of an irreversible process based on the non-equilibrium free energy concept of the second law of thermodynamics. As dissipative structures, neural networks can evolve demonstrating a certain degree of plasticity in the dynamic functions of the structures. As part of an "open system" the functional organization of dissipative structures is kept in relation to the environmental forces. If the structures have evolved to include centrally integrated patterns to maintain homeostasis with both visceral and behavioral responses, and if they can change, then we can hypothesize that restricting the behavioral responses ultimately restricts the use of the dissipative structure. With time this restriction may change the neural network in subtle ways that affect their ability to coordinate future responses. I am proposing that at a macro-level reduced variation in regulatory systems make it more difficult to coordinate sub-systems thus leading to work related adaptive capacity reductions, making it more difficult to adapt to environmental demands [2].

Implications for job strain research using ECG monitoring

Considerations of environmental control having an effect on work related adaptive capacity of a phenotype or on phenotypic plasticity hold great significance for job strain research due to implications for exposure monitoring and theories regarding disease mechanisms. Exposure monitoring from within the context presented above should identify mechanisms to detect phenotype changes leading to reduced adaptive capacity. Goldberger and Pincus have proposed assessments of heart rate variability based on complexity theory formulations and describe reductions of complexity in HR time series as reduced system capacity. Extending from these theoretical linkages we have conducted a variety of variability assessments of vagal cardiac regulation as a possible indication of work related adaptive capacity reductions. To our knowledge no prior attempts to utilize HRV data in describing the effect of job strain have utilized the perspective of reduced variation as an indication of reduced adaptive capacity. This analysis is performed in addition to coordination assessments based on Berntston's Doctrine of Autonomic Space [39] within the context of the Karasek's Stress Disequilibrium Theory [2]. One aspect of integrating these theoretical formulations is that it is currently unknown, and cannot be tested in this dissertation, is whether (as adaptability theory may propose [116]) reduced variability in the cardiac control systems leads to altered coordination. The full description of these analyses and their discussion is in Chapter Four.

## Summary

Ambulatory electrocardiograph data (ECG) may prove to be a useful monitoring strategy to elucidate cardiac regulatory-based pathological mechanisms in the study of job strain [26, 111]. Despite a growing number of studies few clear conclusions can be drawn because the exact method for either monitoring or analyses of data remain unclear. In our opinion the rationale for using ambulatory electrocardiograph monitoring to measure job strain research exposure is fourfold: first, evidence from cardiovascular epidemiology links alterations that are easily identified by ECG to an increased risk of cardiovascular disease; second, evidence that measures derived from the ECG are related to autonomic nervous system based cardiac regulation; third, the possibility to identify Stress Disequilibrium Theory predicted long term regulatory shifts; and fourth, that the variability of cardiac regulatory mechanisms may reflect work related adaptive capacity reductions. The setting of this dissertation integrates previously disparate research traditions while formulating new theoretical foundations for the practical use of ambulatory electrocardiograph measures and therefore links the subsequent chapters.

## Literature Cited


1. Karasek R A. Heart Rate Variability Project. University of Massachusetts Lowell, 1996.
2. Karasek R.A. The Stress Disequilibrium Theory of Chronic Disease Development. Lowell, 2003.
3. Merriam P.A., Ockene I.S., Hebert J.R., Rosal M.C., Matthews C.E. Seasonal Variations of Blood Cholesterol Levels Study: Study methodology. Journal of Biological Rhythms 1999; 14:330-339.
4. Karsek R.A., Costas K. Job Strain and Heart Rate Variability: Initial Report of Results. International Conference on Behavioral Medicine, 1998.
5. Costas K, Karsek R.A. Recruitment Bias in Occupational Stress Studies with Complex Protocols. University of Massachusetts Lowell, 2003.
6. Karasek R A. The impact of the work environment on life outside the job. Cambridge: Massachusetts Institute of Technology, 1976.
7. Karasek R A. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 1979; 24:285-308.
8. Karasek R A, Theorell T. Healthy Work: Stress Productivity, and the Reconstruction of Working Life. New York: Basic Books, 1990.
9. Sauter S, Brightwell W, Colligan M.J., et al. The Changing Organization of Work and the Safety and Health of Working People. Cincinnati: National Institute of Occupational Safety and Health, 2002.
10. Landsbergis P. The changing organization of work and the safety and health of working people: a commentary. J Occup Environ Med 2003; 45:61-72.
11. Belkic K, Landsbergis P, Schnall P, et al. Psychosocial Factors: Review of the Empirical Data Among Men. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, 2000:7-68.
12. Schnall P, Landsbergis P, Baker D. Job strain and CVD. Annu Rev Public Health 1994; 15:381-411.
13. Johnson J V, Hall E M. Job strain, workplace social support, and cardiovascular disease: a crosssectional study of a random sample of the Swedish working population. American Journal of Public Health 1988; 86:324-331.
14. Schnall P L, Belkic K, Landsbergis P, Baker D. The Workplace and Cardiovascular Disease. Occupational Medicine: State of the Art Reviews. Vol. 15. Philadelphia: Hanley & Belfus, INC., 2000.
15. Kristenson T.S. Job stress and CVD: a theoretic critical review. J Occup Health Psychology 1996; 1:246-260.
16. Kristenson T.S. The demand-control-support model: methodological challenges and future research. Stress Medicine 1995; 11:17-26.
17. Theorell T, Karasek R A. Current Issues Relating to Psychosocial Job Strain and Cardiovascular Disease Research. J Occup Health Psychol 1996; 1:9-26.
18. Grundy SM, Pasternak R, Greenland P, Smith S, Fuster V. AHA/ACC Scientific Statement: Assessment of Cardiovascular Risk by Use of Multiple-Risk-Factor Assessment Equations. A Statement for Healthcare Professionals From the

American Heart Association and the American College of Cardiology. Circulation 1999; 100:1481-1492.
19. 2001 Heart and Stroke Statistical Update. Dallas, Tx: American Heart Association, 2001.
20. Babuty D, Lab M. Mechanoelectrical contributions to sudden cardiac death. Cardiovascular Research 2001; 50:270-279.
21. Gronefeld G, Hohnloser S. What do implantable cardioverter/defibrillators teach us about the mechanisms of sudden cardiac death. Cardiovascular Research 2001; 50:232-241.
22. Lombardi F, Makikallio T.H., Myerburg R, Huikuiri H. Sudden cardiac death: role of heart rate variability to identify patients at risk. Cardiovascular Research 2001; 50:210-217.
23. Myerburg R, Spooner P. Opportunities for sudden cardiac death prevention:directions for new clinical and basic research. Cardiovascular Research 2001; 50:177-185.
24. Mulder G, Mulder L.J.M., Meijman T.F., Veldman J.B.P., van Roon A.M. A Psychophysiological Approach to Working Conditions. In: R.W. Backs WB, ed. Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum Associates, 2000.
25. Lenneman J, Backs R.W. The validity of factor analytically derived cardiac autonomic components for mental workload assessment. In: Backs R.W., Boucsein W, eds. Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum, 2000:161-176.
26. Belkic K. Ambulatory Electrocardiographic Monitoring: Stress-Mediated Clinically Relevant Endpoints. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, Inc., 2000:196-203.
27. Hanson E.K.S., Godaert G.L.R., Maas C.J.M., Meijman T.F. Vagal cardiac control throughout the day: the relative importance of effort-reward imbalance and within-day measurements of mood, demand and satisfaction. Biological Psychology 2001; 56:23-44.
28. van Amelsvoort L.G.P.M., Schouten E.G., Maan A.C., Swene C.A., Kok F.J. Occupational determinants of heart rate variability. Int Arch Occup Environ Health 2000; 73:255-262.
29. Vrijkotte T.G.M., van Dornen L.P.J., de Geus E.J.C. Effects of work stress on ambulatory blood pressure, heart rate, and heart rate variability. Hypertension 2000; 35:880-886.
30. Lovallo WR. Stress and Health: Biological and Psychological Interactions. London: SAGE Publications, 1997.
31. Selye H. Stress and Disease. Science 1955; 122:625-631.
32. Cannon W.B. The emergency function of the adrenal medulla in pain and the major emotions. Am J Physiol 1914; 33:356-372.
33. Selye H. The stress of life. New York: McGraw Hill, 1976.
34. McEwen B.S. Protective and damaging effects of stress mediators. NEJM 1998; 338:171-179.

35. Karasek R A. The Stress Disequilibrium Theory, First NIVA Conference on Biomarkers of Stress in Occupational Health, Copenhagen, Denmark, 2002.
36. Bigger J.T. Jr. Spectral analysis of RR variability to evaluate autonomic physiology and pharmacology and to predict cardiovascular outcomes in humans. Encylopedia of Cardiology 1994:1151-1170.
37. Berntson G.G., Bigger J.T. Jr., Eckberg D, et al. Heart rate variability: Origins, methods, and interpretative caveats. Psychophysiology 1997; 34:623-648.
38. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability standards of measurement, physiological interpretation, and clinical use. European Heart Journal 1996; 17:354-381.
39. Bernston G.G., Cacioppo J.T., Quigley K.S. Autonomic determinism: the modes of autonomic control, the doctrine of autonomic space, and the laws of autonomic constraint. Psychological Review 1991; 98:459-487.
40. Bernston G.G., Cacioppo J.T., Quigley K.S. Cardiac Psychophysiology and Autonomic Space in Humans: Empirical Perspectives and Conceptual Implications. Psychological Bulletin 1993; 114:296-322.
41. Kamath M, Fallen E. Power spectral analysis of heart rate variability: A noninvasive signature of cardiac autonomic function. Critical Reviews in Biomedical Engineering 1993; 21:245-311.
42. Kristal-Boneh E, Raifal M, Froom P, Ribak J. Heart rate variability in health and disease. Scandinavian J of Work Environment and Health 1995; 21:85-95.
43. Bloomfield P. Fourier Analysis of Time Series: An Introduction. New York: Wiley Series in Probability and Mathematical Statistics, 1976.
44. Albrecht P, Cohen R.J. Estimation of heart rate power spectrum bands from real world data: Dealing with ectopic beats and noisy data. Computers in Cardiology 1988; 15:311-314.
45. Bigger J.T. Frequency domain measures of heart period variability and mortality after myocardial infarction. Circulation 1992; 85:164-171.
46. Malliani A, Pagani M, Lombardi F, Cerutti S. Cardiovascular neural regulation explored in the frequency domain. Circulation 1991; 84:482-492.
47. Pagani M, et al. Assessment of the neural control of the circulation during psychological stress. J of Autonomic Nervous System 1991; 35:33-42.
48. Pagani M, et al. Sympathovagal interaction during mental stress: a study using spectral analysis of heart rate variability in healthy control subjects and patients with a prior history of myocardial infarction. Circulation 1991; 83 suppl II:II43-II51.
49. Backs R.W. Going beyond heart rate: autonomic space and cardiovascular assessment of mental workload. The International Journal of Aviation Psychology 1995; 5:25-48.
50. Grossman P, Sveback S. Respiratory sinus arrythmia as an index of parasympathetic cardiac control during active coping. Psychophysiology 1987; 24:228-235.
51. Sinnreich R, Kark J.D., Friedlander Y, Sapoznikov D, Luria M.H. Five minute recordings of heart rate variability for population studies: repeatibilty and age-sex characteristics. Heart 1998; 80:156-162.

52. Maliani A, Pagani M, Furlan R, Guzzetti S, Lucini D, Montano N. Individual recognition by heart rate variability of two different autonomic profiles related to posture. Circulation 1997; 96:4143-4145.
53. Fetsch T, Reinhart L, Wichter T, Borggrefe M, Breithardt G. Heart rate variability and electrical stability. Basic Res in Cardiology 1998; 93:S117-S124.
54. Stein P.K., Kleiger R.E. Insights from the study of heart rate variability. Ann Rev Medicine 1999; 50:249-261.
55. Algra A. Heart rate variability form 24-hour holter ecg and the 2 year risk for sudden death. Circulation 1993; 88:180-85.
56. Dekker J.M. Heart rate variability from short electrocardiographic recordings predicts mortality from all causes in middle aged and elderly men. Am J of Epidemiology 1997; 145:899-908.
57. Tusji H, Larson MG. Impact of reduced HRV on risk for cardiac events: The Framingham Heart Study. Circulation 1996; 94:2850-55.
58. Tusji H, Vendetti FJ. Reduced HRV and mortality risk in an elderly cohort: The Framingham Heart Study. Circulation 1994; 90:878-83.
59. Singh J.P., Larson M.G., Tsuji H., Evans J.C., O'Donnell C.J., Levy D. Reduced heart rate variability and new onset hypertension. Hypertension 1998; 32:293-297.
60. Bigger J.T., La Rovere R.T., Marcus F.I., Mortara A, Schwartz P.J. Baroreflex sensitivity and heart rate variability in prediction of total cardiac mortality after myocardial infarction. Lancet 1998; 351:478-484.
61. Doulalas M.D., Flather D, Pipilis A., Campbell S. Evolutionary pattern and prognositic importance of heart rate variability during the early phase of acute myocardial infarction. Int J of Cardiology 2001; 77:169-179.
62. Kleiger R.E., Miller J.P., Bigger J.T. Jr., Moss A.J. Decreased heart rate variability and its association with increased mortality after acute myocardial infarction. Am J Cardiology 1987; 89:256-262.
63. Lombardi F, Sandrone F, Spinnler M, et al. Heart rate variability in the early hours of an acute myocardial infarction. Am J Cardiology 1996; 77:1037-1044.
64. van Boven A.J., Crijns H, Haaksma J, Zwinderman A.H. Depressed heart rate variability is associated with events in patients with stable coronary artery disease and preserved left ventricular dysfunction. Am Heart J 1998; 135:571-576.
65. Mortara A, LaRovere M.T. Can power spectral analysis of heart rate variability identify a subgroup of high risk patients with CHF with sympathetic activation. Br. Heart J. 1994; 71:422-30.
66. Galinier M, Pathak A, Fourcade J, et al. Depressed low frequency power of heart rate variability as an independent predictor of sudden death in chronic heart failure. European Heart Journal 2000; 21:475-482.
67. Nolan J. Prospective study of heart rate variability and mortality in chronic heart failure. Circulation 1998; 98:1510-1516.
68. Yeragani V.K., Kumar H.V. Heart period and QT variability, hostility, and type-A behavior in normal controls and patients with panic disorder. Journal of Psychosomatic Research 2000; 49:1-7.
69. Sharpley CF. Heart rate reactivity and variability as psychophysiological links between stress, anxiety, depression and cardiovascular disease: Implications for

health psychology interventions. Australian Psychological Society 2002; 37:56-62.
70. Carney RM, Freedland KE, Miller GE, Jaffe AS. Depression as a risk factor for cardiac mortality and morbidity: A review of potential mechanisms. Journal of Psychosomatic Research 2002; 53:897-902.
71. Agelink MW, Boz C, Ullrich H, Andrich J. Relationship between major depression and heart rate variability. Clinical consequences and implications for antidepressive treatment. Psychiatry research 2002; 113:139-149.
72. Grippo A.J., Johnson A.K. Biological mechanisms in the relationship between depression and heart disease. Neuroscience and Biobehavioral Reviews 2002; 26:941-962.
73. Armstrong L.E., van Heest J.L. The Unknown Mechanism of the Overtraining Syndrome: Clues from depression and psychoneuroimmunology. Sports Medicine 2002; 32:185-209.
74. Appels A. Exhausted subjects, exhausted systems. Physiologica Scandinavia 1997; 161:153-154.
75. Appels A. Why do imminent victims of a cardiac event feel so tired? Int J Clin Pract 1997; 51:447-450.
76. Goodkin K, Appels A. Behavioral-neuroendocrine-immunologic interactions in myocardial infarction. Medical Hypotheses 1997; 48:209-214.
77. Wantanabe T, Sugiyama Y, Sumi Y, et al. Effects of vital exhaustion on cardiac autonomic nervous functions assessed by heart rate variability at rest in middle-aged male workers. International Journal of Behavioral Medicine 2002; 9:68-75.
78. Shepard R.J. Chronic Fatigue Syndrome: An update. Sports Medicine 2001; 31:167-194.
79. Uusitalo A.J. Ability of non-invasive and invasive methods autonomic function measurements and stress hormones toindicate endurance training-induced stress. Tampere: University of Tampere, 1998:621.
80. Uusitalo A.L.T., Uusitalo A.J., Rushko H.K. Heart rate and blood pressure variability during heavy training and overtraining in the female athlete. Int J Sports Med 2000; 21:45-53.
81. Theorell T. Neuroendocrine Mechanisms. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, Inc., 2000:139-146.
82. Al'Absi M, Buchanan T, Pncomb G.A., Licinio J, Lovallo W.R. Cardiovascular and neuroendocrine adjustment to public speaking and mental arithmetic stressors. Psychophysiology 1997; 34:266-275.
83. Brody S, Maier C, Montoya P, Rau H. Speaking formthe heart: cardiovasular components of stress rating changes and the relative reactivity of physiological and psychological variables. Eur J Appl Physiol 1994; 69:277-280.
84. Callister R, Suwarno N.O., Seals D.R. Sympathetic activity is influenced by task difficulty and stress perception during mental challenge in humans. J of Physiology 1992; 454:373-387.
85. Hyndman B.W., Gregory J.R. vp. Spectral analysis of sinus arrythmia during mental loading. Ergonomics 1975; 18:255-270.

86. Jiang W. Relation of cardiovascular responses to mental stress and cardiac vagal activity in coronary artery disease. Am J of Cardiology 1993; 72:551-554.
87. Steptoe A, Fieldman G, Evans O, Perry L. Cardiovascular risk and responsivity to mental stress: the influence of age, gender, and risk factors. J Cardiovasc Risk 1996; 3:83-93.
88. Kalsbeek J.W.H. Do you believe in sinus arrhythmia? Ergonomics 1973; 16:99-104.
89. Moriguchi A, Otsuka A, Kohara A, et al. Spectral change in heart rate variability in response to mental arithmetic before and after the beta adrenergic blocker, carteolol. Clin Autonomic Research 1992; 2:267-270.
90. Schwartz P.J., Wolf S. QT Interval prolongation as a predictor of sudden cardiac death in patients with myocardial infarction. Circulation 1978; 57:1074-1077.
91. Berger RD, Kasper EK, Baughman KL, Marban E, Calkins H, Tomaselli GF. Beat-to-beat QT interval variability: Novel evidence for repolarization lability in ischemic and nonischemic dilated cardiomyopathy. Circulation 1997; 96:1557-1565.
92. Guyton A.C., Hall J.E. Textbook of Medical Physiology. Philadelphia: Saunders W B Co, 2000.
93. Negoescu R, Dinca-Panaitescu S. Mental stress enhances the sympathetic fraction of QT variability in an RR independent way. Int Phys Behav Sci 1997; 32:220-226.
94. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Kay J, Igel G. Effect of posture and isoproterenol on beat-to-beat heart rate and QT variability. Neuropsychobiology 2000; 41:113-123.
95. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Ramesh C, Srinivasan K. Increased QT variability in patients with panic disorder and depression. Psychiatry Research 2000; 93:225-235.
96. Pohl R, Yeragani V.K. QT interval variability in panic disorder patients after isoproterenol infusions. International Journal of Neuropsychopharmacology 2001: 4:17-20.
97. Atiga W.L., Calkins H, Lawrence J.H., Tomaselli GF, Smith J.M., Berger R.D. Beat to Beat Repolarization Lability Identifies Patients at Risk for Sudden Cardiac Death. J Cardiovascular Electrophysiology 1998; 9:899-908.
98. Sasaki T, Iwasaki T, Oka I, et al. Effect of working hours on cardiovascular autonomic nervous system functions in engineers in an electronics manufacturing company. Industrial Health 1999; 37:55-61.
99. Kageyama T, Nishikido N, Kobyashi T, Kuokawa Y, Kaneko T, Kabuto M. Long commuting time, extensive overtime, and sympathodominant state assessed in terms of short term heart rate variability among male while collar workers in the Tokyo megalopolis. Industrial Health 1998; 36.
100. Iwasaki T, Sasaki T, Oka I, Hisanaga N. Effect of working hours on biological functions related to cardiovascular system among salesmen in a machinery manufacturing company. Industrial Health 1998; 36.
101. Kobyashi F, Furui H, Akamatsu Y. Changes in psychophysiological functions during night shift in nurses: influences of changing from a full day to half day work shift before night duty. Int Arch Occup Environ Health 1997; 69:83-90.

102. Kobyashi F, Furui H, Akamatsu Y, et al. Autonomic nervous system function in taxi drivers, International Conference of Behavioral Medicine, Copenhagen, Denmark, August, 1998.
103. Matsuzaki I, Mishimura A, Morita N. Autonomic nervous system activity changes due to shift work: an evaluation by spectral components of heart rate variability. J Occup Health 1996; 38:80-81.
104. Riese H. Job strain and risk for cardiovascular disease in female nurses. Amsterdam: Vrije University, 2000.
105. Siegrist J, Junge P.R., Cremer P., Seidel D. Low status control, high effort at work and ischemic heart disease: prospective evidence from blue collar men. Soc Sci Med 1990; 31.
106. Porges S. Cardiac Vagal Tone: A physiological index of stress. Neuroscience and Biobehavioral Reviews 1995; 19:225-233.
107. Pickering T. The effects of occupational stress on blood pressure in men and women. Acta Physiolgica Scandinavica 1997; 640:S125-S128.
108. Cheng Y, Kawachi I, Coakley H, Schwartz J, Colditz G. Association between psychosocial work characteristics and health functioning in American women: prospective study. BMJ 2000; 320:1432-1436.
109. Frankenhaeuser M, Lundberg U, Fredrikson M, Melin F, Tuomisto M, Myrsten A-L. Stress on and off the job as related to sex and occupational status in white-collar workers. J Org Behavior 1989; 10:321-46.
110. Sluiter J.K. How about work demands, recovery and health. Amsterdam: University of Amsterdam, 1999.
111. Collins S M. Emerging methods for the physiological assessment of occupational stress. Work 2001; 17:209-219.
112. Recordati G. A thermodynamic model of the sympathetic and parasympathetic nervous systems. Autonomic Neuroscience: Basic and Clinical 2003; 103:1-12.
113. Schnall P, Pieper C, Schwartz J.E., Karsek R.A., et al. The relationship between "Job Strain," workplace diastolic blood pressure and left ventricular mass index. JAMA 1990; 263:1929-1935.
114. Feder M.E., Bennett A.F., Huey R.B. Evolutionary Physiology. Annu Rev Ecolological Systems 2000; 31:315-341.
115. Gould S.J. The Structure of Evolutionary Theory. Cambridge: The Belknap Press of Harvard University Press, 2002:1433.
116. Conrad M. Adaptability: The Significance of Variability from Molecule to Ecosystem. New York: Plenum Press, 1983.
117. Ashby W.R. Introduction to Cybernetics. London: Chapman & Hall Ltd., 1957:295.
118. Holland J.H. Adaptation in Natural and Artificial Systems: An Introductory Analysis with Applications to Biology, Control, and Artificial Intelligence. Cambridge: MIT Press, 1992.
119. Holland J.H. Hidden Order: How Adaptation Builds Complexity. Cambridge: Perseus Publishing, 1996.
120. Holland J.H. Emergence: From Chaos to Order. Cambridge: Perseus Books, 1998.
121. Morowitz H.I. The Emergence of Everything: How the World Became Complex. Oxford: Oxford University Press, 2002.

122. Schlichting C.D., Pigliucci M. Phenotypic Evolution: A Reaction Norm Perspective. Sunderland: Sinauer Associates, Inc., 1998.
123. Piersma T, Drent J. Phenotypic flexibility and the evolution of organismal design. Trends in Ecology and Evolution 2003; In Press.
124. Renner M.J., Rosenzweig M.R. Enriched and Impoverished Environments: Effects on Brain and Behavior. Recent Research in Psychology. Berlin: Springer-Verlag, 1987.
125. Dufty A.M., Clobert J, Moller A.P. Hormones, developmental plasticity and adaptation. Trends in Ecology and Evolution 2002; 17:190-196.
126. Meyers L.A., Bull J.J. Fighting change with change: adaptive variation in an uncertain world. Trends in Ecology and Evolution 2002; 17:551-557.
127. Rosenzweig M.R., Bennett E.L. Psychobiology of plasticity: effects of training and experience on brain and behavior. Behavioural Brain Research 1996; 78:57-65.
128. Hoffman A.A., Parsons P.A. Evolutionary Genetics and Environmental Stress. Oxford Science Publications. Oxford: Oxford University Press, 1991.
129. Blessing W.W. Inadequate frameworks for understanding bodily homeostasis. Trends in Neuroscience 1997; 20:235-239.
130. Parvizi J, Damasio A. Consciousness and the brainstem. Cognition 2001; 79:135-159.
131. Recordati G. The visceral nervous system and its environment. J Theoretical Biology 2002; 214:293-304.
132. Prigogine I. Time, Structure and Fluctuations. Science 1978; 201:777-785.

Figure 1-1. Integration of Three Research Traditions

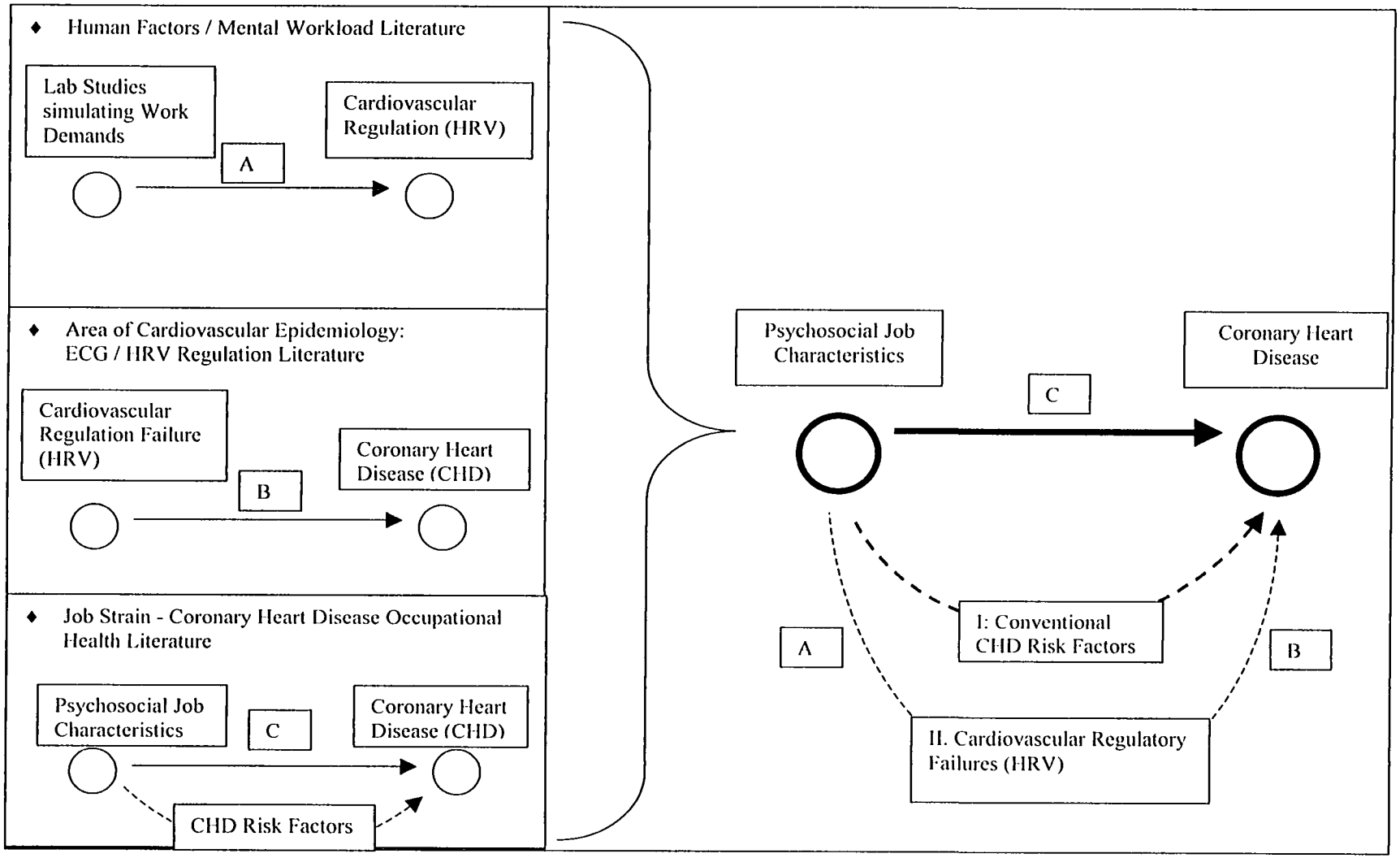


Figure 1-2: Normal Electrocardiogram

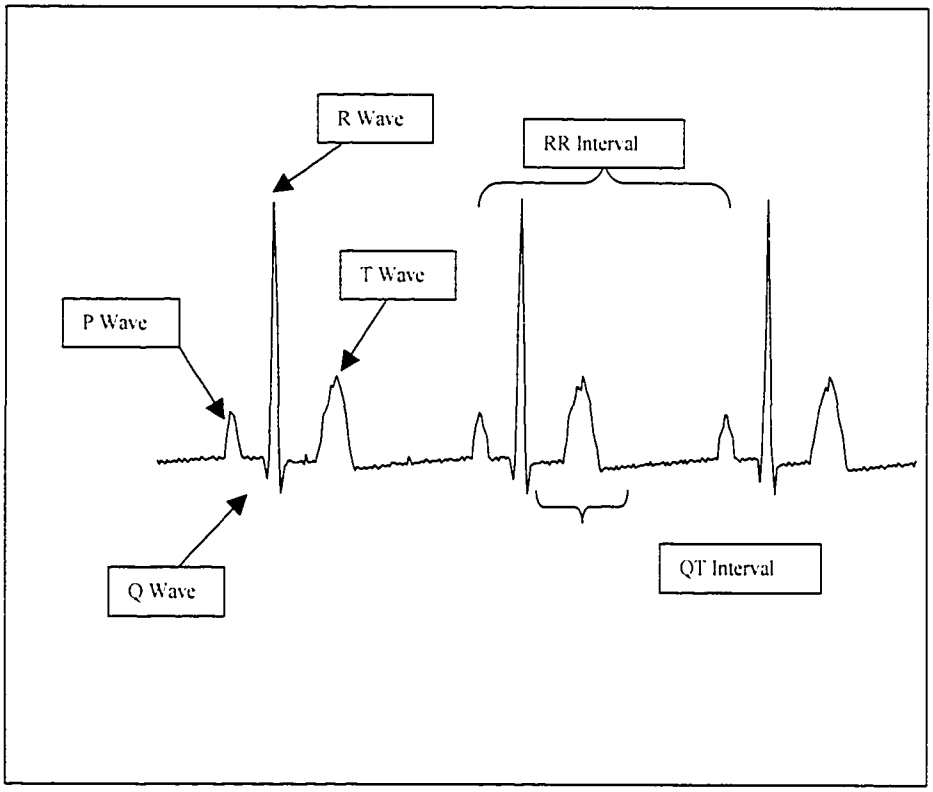


Figure 1-3. Derivation of Electrocardiograph Variables for Testing Hypotheses

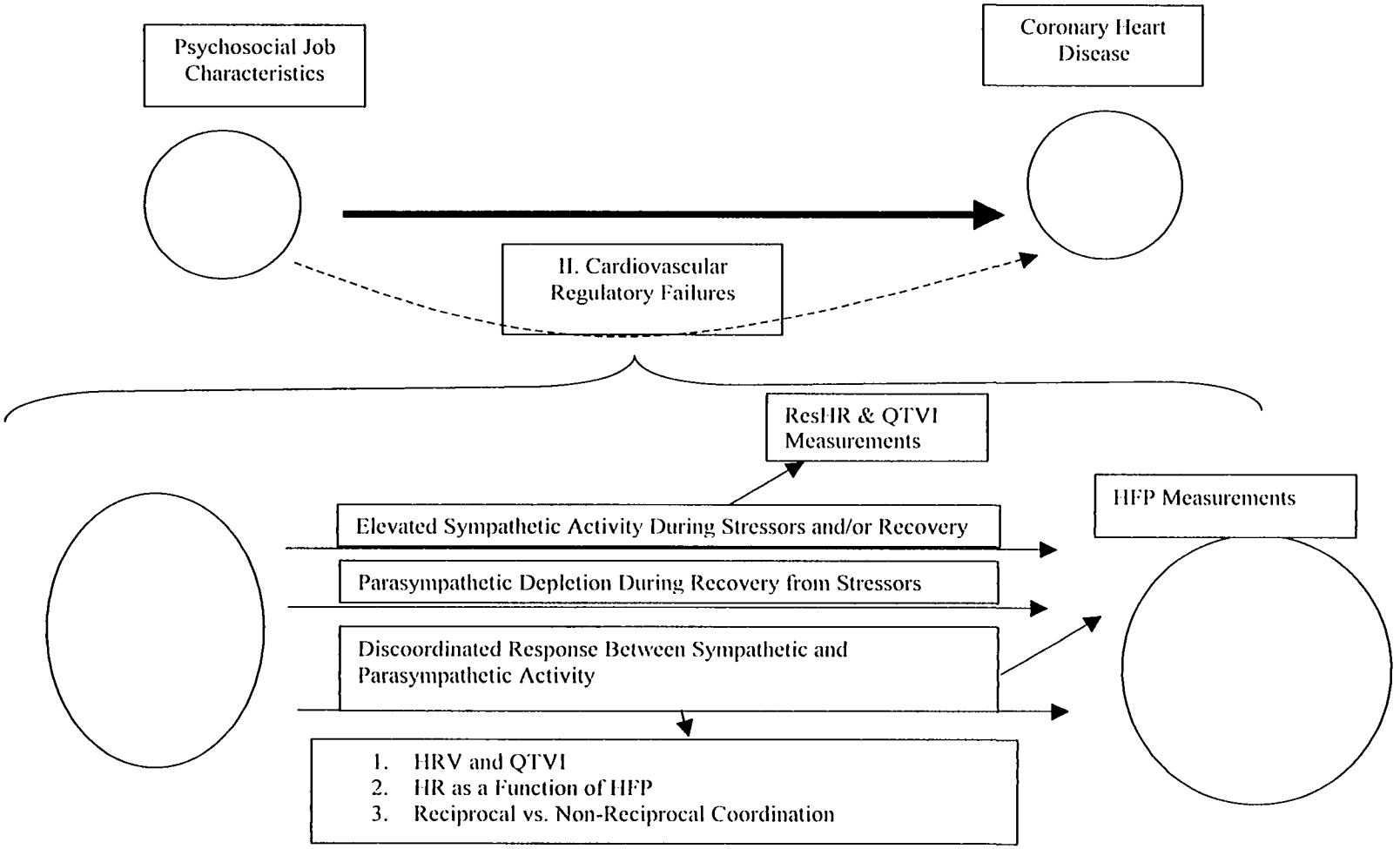
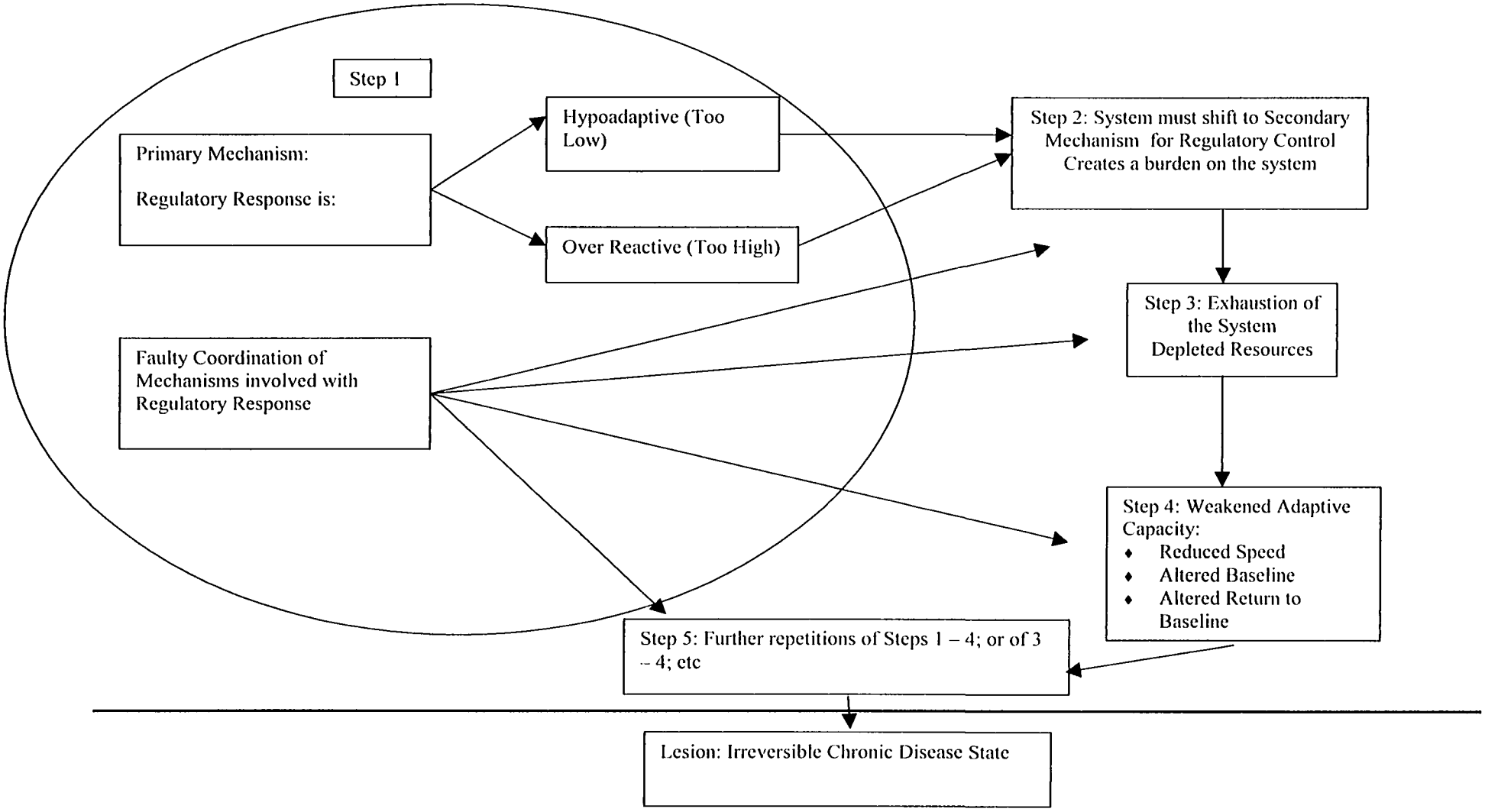


Figure 1-4. Systems Disequilibrium – Hindered Cardiovascular Regulation



# CHAPTER TWO

## IMPACT OF JOB STRAIN ON HEART RATE VARIABILITY AND THE QT INTERVAL VARIABILITY INDEX

### Abstract

This paper tests the hypothesis that heart rate variability and QT interval variability are different in workers in high strain as compared to low strain jobs. High job strain is defined by the demand – control model and based on a weighted score combining assessments with the Job Content Questionnaire, diary self reports during the workday, and imputed job demand and decision latitude scores from each subjects' occupational code. The hypothesis tested was that psychosocial stress from work activity affects cardiac regulation in a pathogenic manner. Thirty-Six males between the ages of 35 and 60 years and free of known heart disease were selected from a larger longitudinal study of healthy subjects and cardiovascular risk based on responses to a subset of questions from the Job Content Questionnaire (JCQ). Subjects were monitored with an ambulatory Holter electrocardiograph. The monitor allowed subjects to continue normal daily activities with minimal distraction while recording cardiac electrical activity. The protocol required monitoring to begin the morning of a workday followed for 48 continuous hours through the following rest day and morning after. Repeated measures

models were constructed to test the hypothesis that subjects with high job strain altered autonomic nervous system activity. High job strain and the main effect of low decision latitude were both associated with significant reductions in cardiac vagal control. These reductions persisted during the entire 48-hour monitoring period. While at work, subjects with high job strain also demonstrated elevations in sympathetic control as measured by three independent ECG based analyses (QTVI, Residual Heart Rate, Lo/Hi Ratio). While job strain does not predict total HRV, low decision latitude is associated with reduced HRV and high demands increase variability (contrary to hypotheses based on HRV and mental load) with a result that these effects appear to cancel the association with job strain. The autonomic nervous system pattern identified in subjects with high job strain may help explain the increased risk of high blood pressure and cardiovascular disease associated with this occupational exposure.

## Introduction

Evidence is accumulating linking psychosocial stress at work to an increasing proportion of occupationally related illness with the majority of the problem stemming from poor work organization [1, 2]. Substantial research exists relating adverse psychosocial work environments with cardiovascular disease. However there is less research underlying the specific causal pathways that may lead to the development of exposure and/or risk assessment [3, 4]. This gap in understanding of the health implications and pathological pathways of poor work organization is highlighted as one of the 21 priority areas addressed in the National Occupational Research Agenda (NORA). Additionally, the need to identify exposure and risk assessment approaches is also listed as a priority in NORA [2].

Perhaps the most commonly utilized hypothesis associating risk of stress-related cardiovascular disease with working conditions is the demand control model (or demand/control/support model) described by Karasek [5, 6] and Karasek and Theorell, [7] and Johnson and Hall [8]. The demand – control model stratifies jobs based on psychological demands (such as work deadline pressure) and decision latitude (control over the work situation). The primary hypothesis involves high psychological demands coupled with low decision latitude (defined as "job strain") increasing the risk of cardiovascular disease. Social support may act as a buffer to high job strain (with high social support decreasing the risk of disease associated with high strain). [9-12] The association between job strain and cardiovascular disease does not seem to be mediated primarily through the routinely associated cardiovascular risk factors. [3, 4]

In light of the rather consistent associations between job strain and cardiovascular disease morbidity and the less convincing picture of job strain and conventional risk factors it is possible that the association between job strain and coronary heart disease is mediated through other physiological pathways besides the conventional cardiovascular risk factors. This possibility increases interest in sudden cardiac causes of coronary disease mortality – which accounts for approximately 50% of total cardiac death. In sudden cardiac death (SCD) defective control of cardiac rhythms is one of the etiological pathways [13-15]. While it must be clearly stated that the common opinion about sudden cardiac death is that it occurs in individuals who often have high levels of the conventional risk factors, it remains true that these factors are not always considered the precipitating cause in the case of sudden cardiac death [16]. These are precisely the pathways to cardiovascular disease that would be implicated by job strain research using heart rate variability to assess pathological cardiovascular stability and control.

Considering the growing body of evidence linking psychosocial stress exposure at work with chronic diseases (in particular heart disease), the implication is of altered autonomic nervous system (ANS) function as a possible etiological pathway for these exposures. Heart Rate Variability (HRV) provides an assessment of the autonomic nervous system as it influences the depolarization of the heart's sino-atrial node. The sino-atrial node is the heart's pacemaker. It has an intrinsic rate, yet is influenced by impulses from the autonomic nervous system through cardiac regulatory reflexes. Variations in the heart rate with a post hoc time series analysis allow identification and quantification of autonomic nervous system components.

HRV has been utilized in clinical studies to predict a variety of cardiovascular outcomes [17-20]. In healthy populations, low HRV has been shown to be a risk factor for all cause and cardiac mortality [21-24], as well as new onset of hypertension [25]. Low high frequency power is also a risk for mortality in patients who have had a myocardial infarction [26-30], have coronary artery disease [31] or congestive heart failure [32-34]. Reduced HRV has also been linked to over-training syndrome in athletes [35]. Over-training syndrome results from training demands that chronically exceed capacity and result in symptoms similar to those related to work stress [36].

In addition to variations in heart rate, variations in the QT interval allow quantification of autonomic nervous system components. The QT interval represents the time for repolarization of the heart following a contraction. QT interval variability is determined from the analysis of the QT interval time series and the QT interval variability index (QTVI) has been proposed as a measure of temporal repolarization lability [37]. The QTVI has been shown to predict risk of sudden cardiac death [37, 38], and has been shown to be a marker of sympathetic nervous system activation [39]. Elevated QTVI has also been linked to certain emotional and mental health disorders such as panic disorder and depression [40].

Well-defined methods exist from the clinical, psychological and human factors literature for analyzing HRV and QTVI as measures of ANS function and cardiovascular disease risk. Utilizing HRV and QTVI in this paper we have integrated the strengths of understanding provided by the above research traditions in an attempt to illuminate how workplace psychosocial stressors contribute to the development of chronic disease – through defined physiological pathways related to cardiac regulation by the autonomic

nervous system.

Studies utilizing HRV measures for investigating the physiological impact of occupational stress in the work place are beginning to appear. A variety of methods have been utilized to categorize occupational stress as well as to measure HRV. A brief review of these studies is available [41]. To our knowledge QTVI has not been utilized to evaluate the physiological impact of occupational stress.

This paper tests the hypothesis that HRV and QTVI are different in high strain as compared to low strain workers. High job strain is defined by the job strain model and based on a weighted score combining assessments with the Job Content Questionnaire, diary self reports during the work day, and imputed job demand and decision latitude scores from each subjects occupational code.

# Methods

## Subjects

A sample of healthy employed men (n = 26) between 35-59 years of age, all members of Fallon Community Health Plan, were recruited from an ongoing investigation of seasonal variance in cholesterol (Seasons) [42]. An additional sample (n=10) of healthy employed men (35 – 59 years of age) was recruited from the University of Massachusetts Worcester Medical Center's stress reduction clinic. Participants of the Seasons study had been screened for job strain using a subset of questions from the Job Content Questionnaire [43]. Investigators reviewed JCQ and current occupational information collected for each subject as part of the intake process and identified a potential sub-population of mixed and high and low strain individuals. Each potential subject was sent a letter explaining the study, how they were selected and informing them that they would be contacted by phone regarding participation.

Subject recruitment occurred in two waves. During the first wave of recruitment the researchers became aware of the difficulties associated with participation in complex protocol studies for participants with high strain jobs. The yield of high strain subjects (8%) was much less than that for low strain subjects (43%). To truly test the intended hypotheses a population-based sample of subjects with high strain jobs was required. Therefore during the second wave of recruitment the research protocol was made more flexible to accommodate the high strain individuals. First, the intake Holter monitor attachment and calibration process occurred at a location and time of the subject's

convenience with respect to the subject's time of arrival at work. Second, monetary incentives were offered for study participation. Subjects were offered up to $250 for completion of the study. Finally, during the recruiting contact, significant attempts were made to be accommodating and flexible in terms of protocol requirements without violating the minimum study requirements. Thirty-six subjects were successfully recruited and monitored according to the entire recruitment period.

## Protocol

The Fallon Community Health Plan and the University of Massachusetts Medical Center Institutional Review Board approved the study for recruitment of Human Subjects. Subjects provided informed consent prior to completing questionnaires and being connected to the monitor. The protocol utilized Holter monitors (Marquette 8500) worn in the workplace and at home to collect 48 hours of continuous ECG recordings beginning in the morning of (prior to) a workday. Subjects were also asked to complete a full version of the Job Content Questionnaire and provide information regarding their occupation at this time. During the first wave of data collection this was a Friday to allow the following day to be a rest day. The second wave of data collection allowed monitoring on any day as long as it was the last day of their regular workweek and the following day was a rest day. While being monitored subjects were asked to complete a diary. Subjects were required to enter the time during the monitoring period on the diary indicating changes in their social situation, such as the start of work, beginning and ending lunch, the end of the workday, eating dinner, etc. These times were utilized to

construct "sociological periods" (Table 1). These periods of the day allow control for potential diurnal rhythm variations (circadian rhythm) that may be related to cardiovascular function. However, they allow for a more precise understanding of the social determinants of daily activity effects, which may be the most salient component [44, 45]. Subjects removed the Holter monitor 48 hours after having had the monitor applied. They were provided with a prepaid envelope to mail the Holter monitor and the diary back to the study team.

Table 2-1. Sociological Periods

| *Day* | *Recording* | *Diary Social Activity and Time of Recording* |
|---|---|---|
| Work Day | 1 | Morning at Work |
| | 2 | Lunch (on Workday) |
| | 3 | Afternoon at Work |
| | 4 | After Work before Dinner |
| | 5 | Dinner until Sleep |
| | 6 | Sleep |
| Rest Day | 7 | Awake on Rest Day |
| | 8 | Time that corresponds to Morning at Work |
| | 9 | Time that corresponds to Lunch at Work |
| | 10 | Time that corresponds to Afternoon at Work |
| | 11 | After Period 10 until Dinner |
| | 12 | Dinner until Sleep |
| | 13 | Sleep |
| | 14 | Awake on Second Rest Day – remove monitor |

## Independent Variables

### *Job Strain*

Kristensen provides a methodological review and rationale for the measurement of job strain based on multiple assessments [46]. In this review Kristensen introduces the concept of applying a triangulation to the assessment of job strain. The basic premise of the triangulation method is to combine the strength of three measures of assessment. The strength of the triangulation method is proposed based on a proposed importance of placing more effort in the repeated and accurate measurement of the proposed risk factors [46, 47]. In his proposal Kristensen refers to a triangulation of stressor measures as the combination of independent, self-rated and average (imputed) measures. The Heart Rate Variability protocol allows the utilization of two of these methods, an imputed occupation average, and two mechanisms of self-report. Kristensen himself reports of the difficulties in determining and implementing valid environmentally based independent measures of stress [46, 48]. Our self-report diary assessment provide a within person, within workday measure where the situation to be assessed is closer in time to the report (almost immediate). This may reduce at least one of the previously discussed concerns in using only self-report measures, being that of accurate recall [46, 48, 49]. For this paper we have utilized a triangulation of stressor assessment based on self-reports to the Job Content Questionnaire and workday diary, as well as imputation of an occupation based average.

Job Content Questionnaire

The Job Content Questionnaire (JCQ) was utilized to compute psychological demands, decision latitude and social support scores and these were compared to a national sample [43, 50]. Psychometric properties of the scales from this sample are presented in Table 2-2.

Table 2-2. Job Strain Scales

| *Scale* | *# of Items* | *Possible Range* | *Valid Range* | *Mean* | *S.D.* | *Cronbach's Apha* |
|---|---|---|---|---|---|---|
| JCQ Psychological Demands | 5 | 12-48 | 21-43 | 31.7 | 5.47 | .73 |
| Decision Latitude | 9 | 24-96 | 36-96 | 72.33 | 11.5 | .87 |
| Diary Demands | 7 | 7-28 | 7-20 | 14.52 | 3.03 | .60 |
| Diary Control | 5 | 5-20 | 6-19 | 13.77 | 2.80 | .55 |

Work Day Diary

During the work day subjects completed a diary with questions relating to job strain concepts. The diary was a modification of one utilized in a study of ambulatory blood pressure monitoring [51]. Diary Scales were constructed for psychological demands and decision latitude using factor analysis with a varimax rotation. The psychometric properties of these scales are presented in Table 2-2.

## Occupational Code Imputation

Subjects provided information regarding their occupation to the researchers before being monitored. Using the subjects' occupational title, their three digit occupational codes were determined as utilized in the 1977 US Quality of Employment Survey [50]. Using the JCQ User's Manual [50] we then determined the average psychological demands and decision latitude associated with each subjects' occupation. These imputed values indicate the mean from a national sample within the designated occupational code.

### *Weighted Job Strain Scores*

To utilize all three of these measures to quantify job demands and decision latitude we needed a method to weight and combine the individual assessment scores. The subsequent sections present the method we have utilized.

## Correlations between Job Strain Assessments

The pattern of correlations for both psychological demands and decision latitude, are consistent with expectations based on their hierarchy and sources of variation within each measure (See Figure 2-1). Occupational code scores represent a national sample of workers with the same occupation. The JCQ represents an individual's self-report of their job (and they are in a certain occupation and therefore should be similar to the occupational code scores). The work diary represents an individual's self report during

one particular day in their occupation. The work diary contains within person, between day and between occupation variations. The JCQ consists of within person and between occupation variation, and the occupational code only includes variation between occupations. Consequently, regarding sources of variance, the JCQ is situated between the diary and the imputed occupational code scores and the correlations reflect this effect.

Correlations between scores for the job strain assessment measures are presented in Figure 2-2 (Psychological Demands) and Figure 2-3 (Decision Latitude). Correlations are greater between the three measures of decision latitude. This may reflect greater stability of the decision latitude scales as compared to the demand scale, also possibility indicating less change in decision latitude in occupations over time (from determination of occupational code scores (1970's) to the this study (1990's)), between subjects within an occupation and their peers, and between individual workdays.

## Determination of Weights

Weights for each measure were determined by computing regression coefficients between the job strain assessment methods and emotion factors determined by scales developed from responses to questions on the JCQ as well as the diary. This is based on the working hypothesis that the demand – control model is an effective predictor of emotion and thus the best weighting formula would be attained by assessing which methods display the strongest association with this "known" emotion in a multiple response relationship [7, 52-54]. Emotion factors were determined based on questions included as part of the job content questionnaire and the diary. Properties of the positive

and negative emotion scales are presented in Table 2-3. Demand and decision latitude scores for each of the three measures were entered into two linear regression models to predict the dependent variables of negative and positive emotion. The standardized coefficients from these regression analyses are presented in Table 2-4. Coefficients for occupational code psychological demands were not significantly associated with mood and therefore were not included to determine weights. There were a total of 10 questions to assess emotion. The standardized regression coefficients for the job strain assessment methods were summed. The percent of the sum accounted for by each particular job strain assessment method was utilized to weight the respective scores. Seventy percent of the total job strain scores were determined by the negative emotion coefficients and 30 percent were determined based on the positive emotion coefficients.

Table 2-3. Emotion Scales

| *Emotion Scale* | *# of Items* | *Possible Range* | *Valid Range* | *Mean* | *S.D.* | *Cronbach's Alpha* |
|---|---|---|---|---|---|---|
| Negative | 7 | 7-28 | 7-16 | 11.12 | 2.83 | .77 |
| Positive | 3 | 3-12 | 5-11 | 9.02 | 1.45 | .66 |

Table 2-4. Standardized Coefficients from Regression Equations

| *Emotion Scale* | *Control Coefficients* | *Demand Coefficients* | *Independent Variable* |
|---|---|---|---|
| Negative | -.339 | .115 | JCQ Scales |
| Negative | -.279 | -.128* | Occupational Coded Imputation |
| Negative | -.479 | .139 | Work Diary |
| Positive | .084 | -.366 | JCQ Scales |
| Positive | .327 | -.063* | Occupational Code Imputation |
| Positive | .248 | -.473 | Work Diary |

* Not included in coefficient based weight

Weighted psychological demands and decision latitude scores were computed using these regression coefficient weights. The values obtained were compared to the scores obtained using the Job Content Questionnaire. Correlations were high for both demands (r = .88), and for decision latitude (r = .86). A descriptive statistical comparison of the job strain scores is provided in Table 2-5. Interestingly, despite the additional steps taken as part of the triangulation process similar results were obtained. The weighted scores were almost identical to their JCQ equivalent, however there are reduced variations in these measures within the sample.

Table 2-5. Comparison of JCQ and Weighted Job Strain Scales

| | *Decision Latitude* | *Weighted Decision Latitude* |
|---|---|---|
| Mean | 72.33 | 73.92 |
| Std. Deviation | 11.57 | 9.03 |
| Minimum | 36.00 | 50.15 |
| Maximum | 96.00 | 87.41 |
| | *Psychological Demands* | *Weighted Psychological Demands* |
| Mean | 31.72 | 31.72 |
| Std. Deviation | 5.47 | 4.84 |
| Minimum | 21.00 | 22.61 |
| Maximum | 43.00 | 40.48 |

*High Strain Classification*

High strain jobs were classified using tertiles from the U.S. national sample with subjects in the three tertiles associated with low weighted decision latitude and high weighted psychological demands categorized as high strain as depicted in Figure 2-4. This method of classification yielded 13 high strain subjects.

*Main Effects of Demand and Control*

To test for the independent effects of psychological demand and decision latitude and the dependent variables we used scores from the Job Content Questionnaire. National US sample means were used as cut-points to split the group into high (n = 15) and low (n = 21) demands as well as high (n=17) and low (n = 19) control groups.

## Covariates

Covariates that have been shown to exert an influence on heart rate variability controlled for in this study include age and gender. The recruitment protocol restricted the pool of potential subjects to men between the ages of 35 and 60 years of age.

## Dependent Variables

### *Data Processing*

Holter data tapes were analyzed for ventricular arrhythmias and RR-interval variability at the Research Holter Laboratory at Columbia University by published methods [55]. In summary, the dependent variables of this study (high frequency power, low frequency / high frequency power ratio, heart rate variability, and heart rate) were computed from 5-minute epochs throughout the 48-hour period. The 48-hour records were digitized using a Marquette 8000 scanner, submitted to Marquette algorithms for QRS identification and editing, and then downloaded to a Sun workstation for further editing using algorithms developed at Columbia University. The power spectrum was computed as previously described [56, 57]. A Fast Fourier algorithm was utilized to compute the power spectrum in the very low frequency (VLF < 0.04 Hz), low frequency (LF, 0.04 – 0.15 Hz), and high frequency (HF, 0.15-0.40 Hz) spectra. Total power was calculated as the sum of very low, low and high frequency power. The low frequency/high frequency power ratio within each subject by total power was then computed for each 5-

minute epoch. From the series of epochs within each subject linear regressions between the natural logarithm of high frequency power and heart rate were computed to determine residual heart rate as initially proposed by Grossman [58] and further utilized and validated by Backs and Lenneman [58-62]. Residual heart rate has been proposed as a relative indicator of sympathetic activation by making use of the expected relationship between vagal activity and heart rate. From knowing these two values for each subject over 48 hours we can estimate the vagal predicted heart rate. Positive residuals indicate that heart rate is higher than estimated by vagal activity, and sympathetic activity is proposed as the major source of this increase. Conversely, when residuals are negative, heart rate is lower than estimated by vagal activity and therefore sympathetic activity may be reduced. Weaknesses of the residual heart rate as an indicator of sympathetic measures have been discussed elsewhere [61]. However, in this study residual heart rate is merely added to demonstrate the similar patterns of estimated sympathetic activation during the workday in high strain subjects as compared to low strain subjects by three estimates of sympathetic activity.

The digitized ECG signals were downloaded to a PC and submitted to algorithms developed at the Johns Hopkins School of Medicine, Department of Cardiology. QTVI was computed by the methods of Berger, as reported in detail elsewhere [37]. In summary, R waves were detected with an automatic peak detection algorithm. The operator then provided a template QT interval by selecting the beginning of the QRS complex and the ending of the T wave for one beat. The algorithm then calculated the QT interval of all following beats by determining how much the ST segment and T wave was stretched or compressed in time to match the template. Due to the long duration of monitoring as part

of our protocol, each subject's QT interval template was updated for every hour of monitoring. An evenly sampled time series from the QT and RR time series was obtained with a sampling frequency of 4 Hz. Long delays in the QT interval time series resulting from noisy signals, artifact or ectopic beats that could not be filtered were eliminated from analysis. If more than 25% of the data in an epoch was eliminated, the entire epoch was eliminated. Eliminated epochs were automatically detected however; their location in the time series was identified and presented to the user for visual inspection.

QTVI was calculated for each epoch as the log ratio of QT interval variance and RR interval variance, both corrected by their respective squared mean:

$$QTVI = \text{Log}\ [(QTV/QTM^2)/(HRV/HRM^2)]$$

The QT interval is known to vary, in part, as a function of the heart rate. The QTVI measures the variability of the QT interval after controlling for the known effect of the heart rate. Therefore the QTVI measures the amount of variance in the QT interval in addition to that of the heart rate. Greater amounts of sympathetic nervous system activation lead to increases in QT interval variation that is independent of the heart rate variability [63].

*Sociological Periods*

Times entered in the diary to indicate daily events were utilized to determine 14 sociological periods (Table 2-1). The ECG data for each subject during each sociological period were aggregated for comparisons between groups within each period. Dependent variables aggregated within each sociological period are described in Table 2-6.

Table 2-6. Dependent Variables Aggregated within Sociological Periods

| *Dependent Variable* | *Aggregation* | *Description* |
|---|---|---|
| Heart Rate (bpm) | Mean Heart Rate | Aggregation provides the Mean HR for each sociological period |
| SDNN (msec) | Mean Standard Deviation of the RR intervals | SDNN reflects all components responsible for variability in the 5-minute epoch. The aggregation provides the mean SDNN for the sociological period |
| HFP $(msec)^2$ | Mean High Frequency Power (.15-.4 Hz) of RR intervals | Quantifies high frequency variability of heart rate. Index of vagal cardiac control. Aggregation provides the mean HFP for the sociological period |
| Lo/Hi Ratio | Mean Ratio of Low to High Frequency Power | Relative index of Sympathetic Activity. Aggregation provides the mean Lo/Hi for the sociological period |
| QTVI | Mean QTVI | Temporal Repolarization Variability, Index of Sympathetic Activity. Aggregation provides the mean for the sociological period |
| Residual HR | Mean Residual HR | Aggregation provides the Mean Residual HR for each sociological period |

## Statistical Analysis

Graphic and descriptive analyses were performed to assess differences between high and low job strain groups during the sociological periods. To test for significant differences between the job strain, control and demand groups we utilized fixed effects, repeated measures models with the SPSS mixed model procedure using an autoregressive covariance matrix. Sociological period was entered as the repeated measure with each subject identified uniquely within each period. Four models were constructed to test hypotheses for each dependent variable. All models controlled for the effect of age. Model I tested the effect of Job Strain. Job strain was entered as a dichotomous variable (0, low strain; 1, high strain). Model II tested the effect of job strain specifically when at work. An interaction variable was entered between location (0, not at work; 1, at work) and strain. Models III and IV tested the main effects of decision latitude and psychological demands entered as dichotomous variables (0 = low demands or high control; 1 = high demands or low control). Due to the number of dependent variables (5) and the number of models constructed for each variable (4) being tested in the study we chose to accept $\alpha \leq .01$ as significant and $\alpha \leq .05$ as a trend.

## Results

Mean, Standard Deviation (S.D.), and ranges of the independent variables for the high and low job strain subjects are presented in Table 2-7. Dependent variables (having been aggregated within each sociological period) are presented in Figures 2-5 through 2-13 to depict the differences between the strain groups within each sociological period. Descriptive statistics for high frequency power stratified by sociological period and job strain group are presented in Table 2-8. Parameter estimates and related statistics from the fixed effects repeated measures models are presented in Tables 2-9 through 2-13 for heart rate variability, high frequency power, QTVI, Lo/Hi ratio, and residual heart rate, respectively.

Table 2-7. Descriptive Comparison between Job Strain Groups

| | | *Low Strain* | *High Strain* |
|---|---|---|---|
| Age | Mean | 48 | 45 |
| | SD | 6.09 | 6.17 |
| | Range | 38-59 | 35-59 |
| Weighted Demands | Mean | 29.06 | 36.61 |
| | SD | 3.63 | 2.59 |
| | Range | 22-36 | 30-40 |
| Weighted Control | Mean | 76.69 | 69.03 |
| | SD | 8.26 | 8.49 |
| | Range | 50-87 | 50 - 79 |

Table 2-8. Strain and Sociological Period Descriptive Statistics for High Frequency Power

| *Sociological Period* | *Strain Classification* | *N* | *Mean* | *SD* |
|---|---|---|---|---|
| 1 | Low | 22 | 211.77 | 205.93 |
| | High | 13 | 149.09 | 95.56 |
| 2 | Low | 23 | 180.51 | 205.14 |
| | High | 12 | 135.32 | 116.20 |
| 3 | Low | 23 | 199.85 | 212.78 |
| | High | 13 | 138.22 | 151.56 |
| 4 | Low | 23 | 201.75 | 245.06 |
| | High | 13 | 149.94 | 106.21 |
| 5 | Low | 23 | 229.46 | 233.73 |
| | High | 13 | 170.38 | 155.15 |
| 6 | Low | 23 | 455.69 | 525.08 |
| | High | 13 | 350.96 | 309.01 |
| 7 | Low | 23 | 257.45 | 259.00 |
| | High | 13 | 190.74 | 155.88 |
| 8 | Low | 23 | 185.49 | 298.88 |
| | High | 13 | 136.97 | 100.49 |
| 9 | Low | 19 | 223.49 | 570.79 |
| | High | 10 | 157.86 | 105.04 |
| 10 | Low | 22 | 181.67 | 309.32 |
| | High | 13 | 147.31 | 98.34 |
| 11 | Low | 22 | 211.65 | 292.31 |
| | High | 13 | 146.63 | 109.76 |
| 12 | Low | 20 | 165.62 | 173.31 |
| | High | 13 | 170.54 | 140.25 |
| 13 | Low | 21 | 406.57 | 394.85 |
| | High | 13 | 354.63 | 332.30 |
| 14 | Low | 10 | 276.50 | 184.10 |
| | High | 9 | 175.35 | 173.93 |

## Heart Rate Variability

As depicted in Table 2-9 there was no significant effect of high job strain on heart rate variability either during the monitoring protocol (Model I) or while at work (Model II). Significant reductions in heart rate variability were found in the low control group (Model III) and paradoxically, elevations were associated with increased demands (Model IV). Increasing age was associated with reductions in HRV in all models.

Table 2-9. Model Parameters for Heart Rate Variability

| | | *Heart Rate Variability* | | |
|---|---|---|---|---|
| *Model I* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain | 3.08 | 1.91 | .108 |
| | Age | -.568 | .151 | .001 |
| *Model II* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain – at Work | -5.34 | 3.44 | .121 |
| | Age | -.57 | .254 | .030 |
| *Model III* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | Low Control | -5.28 | 1.80 | .003 |
| | Age | -.545 | .147 | .001 |
| *Model IV* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Demands | 6.75 | 1.86 | .001 |
| | Age | -.504 | .149 | .001 |

## High Frequency Power – Parasympathetic Activity

Table 2-10 depicts model parameters for high frequency power. Significant differences were identified between the strain groups (Model I) throughout the entire monitoring period with increased age associated with decreased high frequency power. The high strain subjects did not have additional reductions in high frequency power on

the workday as compared to the low strain subjects (Model II). The low control group also demonstrated a significant reduction in high frequency power during the entire monitoring period (Model III). No such finding was identified for psychological demands (Model IV).

Table 2-10. Model Parameters for High Frequency Power

| | | *High Frequency Power* | | |
|---|---|---|---|---|
| *Model I* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain | -93.63 | 45.40 | .040 |
| | Age | -14.28 | 3.59 | .001 |
| *Model II* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain – at Work | -55.20 | 44.94 | .220 |
| | Age | -14.22 | 3.54 | .001 |
| *Model III* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | Low Control | -99.96 | 42.56 | .021 |
| | Age | -11.41 | 3.49 | .002 |
| *Model IV* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Demands | -61.09 | 44.82 | .177 |
| | Age | -13.73 | 3.59 | .001 |

QTVI, Lo/Hi Ratio and Residual Heart Rate – Sympathetic Activity

Tables 2-11 through 2-13 depict model parameters for the sympathetic nervous system measures of QTVI, Lo/Hi Ratio and the residual heart rate. All sympathetic measures displayed the same patter of association with job strain. They were not significantly different between the strain groups when the model considers the entire monitoring

period (Tables 2-11 to 2-13, Model I). However, all sympathetic measures were significantly elevated in the high strain group on the workday (Tables 2-11 to 2-13, Model II).

Table 2-11. Model Parameters for QTVI

| | | *QTVI* | | |
|---|---|---|---|---|
| *Model I* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain | .082 | .105 | .439 |
| | Age | .013 | .008 | .129 |
| *Model II* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain – at Work | .267 | .086 | .003 |
| | Age | .013 | .008 | .132 |
| *Model III* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | Low Control | .005 | .098 | .961 |
| | Age | .015 | .008 | .176 |
| *Model IV* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Demands | -.106 | .102 | .305 |
| | Age | .011 | .008 | .206 |

Table 2-12. Model Parameters for Lo/Hi Ratio

| | | *Lo/Hi Ratio* | | |
|---|---|---|---|---|
| *Model I* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain | .205 | .649 | .753 |
| | Age | .139 | .051 | .008 |
| *Model II* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain – at Work | 1.59 | .599 | .008 |
| | Age | .140 | .052 | .009 |
| *Model III* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | Low Control | -.660 | .617 | .289 |
| | Age | .145 | .057 | .005 |
| *Model IV* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Demands | .563 | .638 | .380 |
| | Age | .145 | .051 | .006 |

Table 2-13. Model Parameters for Residual Heart Rate

| | | *Residual Heart Rate* | | |
|---|---|---|---|---|
| *Model I* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain | 10.66 | 8.06 | .188 |
| | Age | .507 | .639 | .430 |
| *Model II* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Strain – at Work | 68.26 | 11.94 | .001 |
| | Age | .516 | .611 | .400 |
| *Model III* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | Low Control | -2.28 | 7.73 | .768 |
| | Age | .369 | .636 | .562 |
| *Model IV* | *Parameter* | *Estimate* | *SE* | *Sig.* |
| | High Demands | 6.06 | 7.97 | .449 |
| | Age | .438 | .641 | .495 |

Diary data reporting position and exertion were only available at eight discrete times during each day. Only six of the 14 sociological periods have information available regarding position and exertion. In addition, subjects were asked to fill out the diary based on the 20 minutes prior to the time they were completing the diary. Thus, even for the sociological periods where diary information is available it does not truly reflect the exertion or position during the entire period. A within subject analysis has been performed utilizing ECG data during only those times of the day when diary data are available and is presented in Chapter Three [64]. In this analysis, to provide an estimate of whether or not position or exertion may be influencing these findings a fixed effects, repeated measures model was constructed with exertion and position as the dependent variables. There were significant and similar increases in exertion for both groups during the workday as compared to the non-working diary assessments. On the workday posture in the high strain group was more likely to be upright.

## Discussion

Several interesting findings have been identified in this study. The most significant findings include the persistent decrease in vagal cardiac control (HFP) in the high strain group and in the low control group, and the persistent reductions in heart rate variability in the low control group. Slight differences between the high and low strain workers, or high and low control workers, in postures on the workday are an unlikely confounder due to the persistent decrease of high frequency power throughout the entire monitoring protocol. To our knowledge we are the first to attempt to demonstrate differences in the independent effects of job demands and control on heart rate variability measures. Implications for this finding are quite interesting when assessed from the perspective of a new stress physiology model, Karasek's Stress Disequilibrium Theory [65], whereby limitations in external control are hypothesized to result in restrictions in physiological control and adaptability. Further discussion on these findings as they relate to the Stress Disequilibrium Hypothesis is presented in Chapter Four [66].

In the Vrijkotte study[67], utilizing Siegrist's model of effort reward imbalance[68], subjects reporting a high effort-reward imbalance had significantly lower vagal cardiac control (measured with the time domain HRV parameter – RMSSD) and these effects were noted throughout the monitoring period on both the work and rest day. This carryover effect of vagal control measured with high frequency power is similar to our findings. In our data the average within subject correlation between high frequency power and RMSSD was $r = 0.85$, which is very similar to those of previous reports [69].

Our findings for high frequency power (vagal cardiac control) are similar to those for elevated ambulatory blood pressure readings that also demonstrate carry over to rest days in high strain occupations. The studies on ambulatory blood pressure have been linked to increased risk of developing clinical hypertension, as well as development of pathological changes in the myocardium [51]. This association occurs prior to clinic-based blood pressure elevations, indicating the possibility of pre-clinical alterations in physiological control. Extending from this linkage, if sudden cardiac death, of an electrophysiological or arrhythmic basis, is related to alterations in vagal cardiac control and psychosocial characteristics during the day within subjects and macro related occupational traits between subjects, then it is quite possible to hypothesize that pre-clinical alterations in cardiac regulatory activity related to psychosocial strain can increase susceptibility of sudden cardiac death. Additionally, Schnall indicates that the motivation of studying ambulatory blood pressure within the context of the job strain hypothesis is based on the theory that the consistently observed associations between job strain and heart disease is, in part, mediated through elevations in blood pressure [70]. Recent findings by Lucini et al. (2002) of a progressive impairment in autonomic regulation in subjects with blood pressure values below the upper limit for normal [71] extends the possible relevance of our findings to that which includes hypertension as a possible disease end point. It is possible, as Lucini indicates, that cardiac regulatory changes, associated with alterations in high frequency power, represent a spectrum of changes in cardiac autonomic control that may precede clinical hypertension. Then, the increased blood pressure independently results in a cascade of pathological changes resulting in heart disease. However, as the next section of the discussion will present,

cardiac disease related to alterations in regulatory control does not have to be mediated through elevations in blood pressure, but can create risks in and of themselves, more likely in the way of sudden cardiac death or cardiac syndrome X.

The approach utilized in this study is based on the significant body of literature that has investigated and identified a relationship between job strain and cardiovascular disease. However, we have attempted to utilize more sophisticated monitoring strategies and risk assessment methods based on well-defined parameters from the human factors and cardiovascular physiology and epidemiology literature. This process has required a significant effort to recruit a community-based sample with enough variation in job strain measures to provide the necessary contrasts to be assessed. Rather than utilizing a single method of determining job strain, we have attempted to utilize the three assessments available in the data set to select a better high strain group than that which could be identified with only the Job Content Questionnaire.

It is interesting to notice that only vagal cardiac control had residual or carryover effects. The indices of sympathetic control – QTVI, Lo/Hi ratio, and residual heart rate – are only elevated in the high strain group on the workday. As cardiovascular regulatory mechanisms this may help explain, at least in part, elevated BP during the workday (due to increased sympathetic and decreased vagal activity) and then persistent elevations in BP after work (due to persistent drops in vagal control). In addition, this may be partly responsible for the identification of increased cardiac reactivity in high stress subjects [72, 73]. If an initial value of vagal activity is reduced, it is possible that increases in sympathetic activity will have a greater effect on heart rate and blood pressure (hence reactivity). It might also be possible that withdrawal of vagal control is part of a larger

systemic condition of overall fatigue or exhaustion with "brake failure." In brake failure subjects may not be able to utilize the vagal system as a dominant control system for cardiac output regulation, leading to a sympathetic dominant control mechanism. Over time the sympathetic dominant control may influence the heart with chronic changes (in heart failure patients persistent sympathetic stimulation leads to down regulation of sympathetic receptors). This discussion is continued in Chapter Four.

The job strain hypothesis suggests that environmental exposure to job strain can lead to exhaustion (such as vital exhaustion) that has been associated with altered vagal control [74, 75]. Exhaustion may be physiologically related to basic failures in brake mechanisms. In testing the effect of vital exhaustion, Wantanabe (2002) found significant reductions in HFP measured during 5 minutes of supine rest in subjects with a high vital exhaustion score as compared to low and moderate scores [76].

Findings based on sympathetic assessments in this sample are strengthened by the consistency of responses for all three measures (QTVI, Lo/Hi ratio and Residual Heart Rate). However, since significant differences are only identified for the workday, these may reflect, at least in part, the significant difference between the strain groups posture on the workday. Physiologically posture may also explain the reductions in vagal control during the workday for high strain workers, however the persistence in reductions in high frequency power over the entire monitoring period leads us to believe that posture is not accounting for the differences in the groups in vagal control.

In previous investigations, the independent variables shown to influence high frequency power between subjects have been age, sex, and physical fitness [18]. Study selection criteria assured that study subjects were male, between the ages of 35 and 59,

generally healthy and not using medications known to affect HRV. Although level of obesity, cholesterol profiles, and smoking status information was not collected for subjects, the literature does not suggest a relationship between these factors and atypical HRV [77]. There is no evidence in the literature to suggest a relationship between decision latitude and physical fitness. It may be hypothesized that greater decision latitude is conducive to a lifestyle with greater potential for the pursuit of personal fitness. If so, this suggests that high decision latitude individuals might self-select toward better physical fitness. Validation of this possible association should be considered in future investigations.

It is also possible that other work environment exposures influence heart rate variability measures through their influence on autonomic nervous system function. Two possibilities include noise [78, 79] and air particulate exposure [80, 81]. Validation of these possible associations should be considered in future investigations.

As will be discussed in a related work [82], the pool of available subjects had previously committed to participation in the Seasons study, which involved a fairly complicated initial and follow-up protocol [42]. They were then asked to participate in our investigation which required a large time commitment, and a restriction of activities in order to allow the participant to be near home or a place which was conducive to relaxation during weekend evenings. In addition, subjects were required to wear somewhat intrusive hardware, daily diary and telephone contact regimens, and limited participation time scheduling options. Due to these constraints, subjects with high strain jobs may be less likely to participate in such studies. Additionally, the HMO based subject pool from which the majority of our subjects were recruited (n=26) reflects the

population of central Massachusetts, and as such is largely white and middle-class. Our study is restricted to participants living in one relatively restricted geographic area of the country. In our desire to take fullest advantage of our study design we are looking at a considerable number of variables, creating a complex dataset. We believe that we have the expertise and experience to deal with this complexity. On the other hand, it is not possible to evaluate all possible variables obtained from the ECG signal, and it may be that we have overlooked a variable or variables of importance. The Seasons study also excluded individuals who could not speak English because virtually all of their study instruments were developed for use in English-speaking populations and have been calibrated for use in those populations.

## Conclusion

High job strain, and perhaps more interestingly low decision latitude are associated with significant reductions in cardiac vagal control. These reductions persisted during the entire 48-hour monitoring period. While at work, subjects with high job strain also demonstrated elevations in sympathetic control as measured by three independent ECG based analyses. While job strain does not predict total HRV, low decision latitude is associated with reduced HRV. The autonomic nervous system pattern identified in subjects with high job strain may help explain the increased risk of high blood pressure and cardiovascular disease associated with this occupational exposure.

## Literature Cited


1. Landsbergis P. The changing organization of work and the safety and health of working people: a commentary. J Occup Environ Med 2003; 45:61-72.
2. Sauter S, Brightwell W, Colligan M.J., et al. The Changing Organization of Work and the Safety and Health of Working People. Cincinnati: National Institute of Occupational Safety and Health, 2002.
3. Schnall P L, Belkic K, Landsbergis P, Baker D. The Workplace and Cardiovascular Disease. Occupational Medicine: State of the Art Reviews. Vol. 15. Philadelphia: Hanley & Belfus, INC., 2000.
4. Schnall P, Landsbergis P, Baker D. Job strain and CVD. Annu Rev Public Health 1994; 15:381-411.
5. Karasek R A. The impact of the work environment on life outside the job. Cambridge: Massachusetts Institute of Technology, 1976.
6. Karasek R A. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 1979; 24:285-308.
7. Karasek R A, Theorell T. Healthy Work: Stress Productivity, and the Reconstruction of Working Life. New York: Basic Books, 1990.
8. Johnson J V, Hall E M. Job strain, workplace social support, and cardiovascular disease: a crosssectional study of a random sample of the Swedish working population. American Journal of Public Health 1988; 86:324-331.
9. Astrand N E, Hanson B S, Isacsson S O. Job demands, job decision latitude, job support, and social network factors as predictors of mortality in a Swedish pulp and paper company. British Journal of Industrial Medicine 1989; 46:334-340.
10. Falk A, Hanson B.S., Isacsson S.O., Ostergren P.O. Job strain and mortality in elderly men: Social networks, support, and influence as buffers. American Journal of Public Health 1992; 82:1136-1139.
11. Kompier MAJ, di Martino V. Review of bus drivers' occupational stress and stress prevention. Stress Medicine 1995; 11:253-262.
12. Johnson J V, Stewart W, Hall E M, et al. Long-term psychosocial work environement and cardiovascular mortality among swedish men. American Journal of Public Health 1996; 86:324-331.
13. Gronefeld G, Hohnloser S. What do implantable cardioverter/defibrillators teach us about the mechanisms of sudden cardiac death. Cardiovascular Research 2001; 50:232-241.
14. Lombardi F, Makikallio T.H., Myerburg R, Huikuiri H. Sudden cardiac death: role of heart rate variability to identify patients at risk. Cardiovascular Research 2001; 50:210-217.
15. Babuty D, Lab M. Mechanoelectrical contributions to sudden cardiac death. Cardiovascular Research 2001; 50:270-279.
16. Myerburg R, Spooner P. Opportunities for sudden cardiac death prevention:directions for new clinical and basic research. Cardiovascular Research 2001; 50:177-185.
17. Fetsch T, Reinhart L, Wichter T, Borggrefe M, Breithardt G. Heart rate variability and electrical stability. Basic Res in Cardiology 1998; 93:S117-S124.
18. Kristal-Boneh E, Raifal M, Froom P, Ribak J. Heart rate variability in health and

disease. Scandinavian J of Work Environment and Health 1995; 21:85-95.
19. Stein P.K., Kleiger R.E. Insights from the study of heart rate variability. Ann Rev Medicine 1999; 50:249-261.
20. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability standards of measurement, physiological interpretation, and clinical use. European Heart Journal 1996; 17:354-381.
21. Tusji H, Vendetti FJ. Reduced HRV and mortality risk in an elderly cohort: The Framingham Heart Study. Circulation 1994; 90:878-83.
22. Tusji H, Larson MG. Impact of reduced HRV on risk for cardiac events: The Framingham Heart Study. Circulation 1996; 94:2850-55.
23. Dekker J.M. Heart rate variability from short electrocardiographic recordings predicts mortality from all causes in middle aged and elderly men. Am J of Epidemiology 1997; 145:899-908.
24. Algra A. Heart rate variability form 24-hour holter ecg and the 2 year risk for sudden death. Circulation 1993; 88:180-85.
25. Singh J.P., Larson M.G., Tsuji H., Evans J.C., O'Donnell C.J., Levy D. Reduced heart rate variability and new onset hypertension. Hypertension 1998; 32:293-297.
26. Lombardi F, Sandrone F, Spinnler M, et al. Heart rate variability in the early hours of an acute myocardial infarction. Am J Cardiology 1996; 77:1037-1044.
27. Kleiger R.E., Miller J.P., Bigger J.T. Jr., Moss A.J. Decreased heart rate variability and its association with increased mortality after acute myocardial infarction. Am J Cardiology 1987; 89:256-262.
28. Doulalas M.D., Flather D, Pipilis A., Campbell S. Evolutionary pattern and prognositic importance of heart rate variability during the early phase of acute myocardial infarction. Int J of Cardiology 2001; 77:169-179.
29. Bigger J.T. Frequency domain measures of heart period variability and mortality after myocardial infarction. Circulation 1992; 85:164-171.
30. Bigger J.T., La Rovere R.T., Marcus F.I., Mortara A, Schwartz P.J. Baroreflex sensitivity and heart rate variability in prediction of total cardiac mortality after myocardial infarction. Lancet 1998; 351:478-484.
31. van Boven A.J., Crijns H, Haaksma J, Zwinderman A.H. Depressed heart rate variability is associated with events in patients with stable coronary artery disease and preserved left ventricular dysfunction. Am Heart J 1998; 135:571-576.
32. Nolan J. Prospective study of heart rate variability and mortality in chronic heart failure. Circulation 1998; 98:1510-1516.
33. Mortara A, LaRovere M.T. Can power spectral analysis of heart rate variability identify a subgroup of high risk patients with CHF with sympathetic activation. Br. Heart J. 1994; 71:422-30.
34. Galinier M, Pathak A, Fourcade J, et al. Depressed low frequency power of heart rate variability as an independent predictor of sudden death in chronic heart failure. European Heart Journal 2000; 21:475-482.
35. Uusitalo A.L.T., Uusitalo A.J., Rushko H.K. Heart rate and blood pressure variability during heavy training and overtraining in the female athlete. Int J Sports Med 2000; 21:45-53.

36. Sluiter J.K. How about work demands, recovery and health. Amsterdam: University of Amsterdam, 1999.
37. Berger RD, Kasper EK, Baughman KL, Marban E, Calkins H, Tomaselli GF. Beat-to-beat QT interval variability: Novel evidence for repolarization lability in ischemic and nonischemic dilated cardiomyopathy. Circulation 1997; 96:1557-1565.
38. Atiga W.L., Calkins H, Lawrence J.H., Tomaselli GF, Smith J.M., Berger R.D. Beat to Beat Repolarization Lability Identifies Patients at Risk for Sudden Cardiac Death. J Cardiovascular Electrophysiology 1998; 9:899-908.
39. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Kay J, Igel G. Effect of posture and isoproterenol on beat-to-beat heart rate and QT variability. Neuropsychobiology 2000; 41:113-123.
40. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Ramesh C, Srinivasan K. Increased QT variability in patients with panic disorder and depression. Psychiatry Research 2000; 93:225-235.
41. Collins S M. Emerging methods for the physiological assessment of occupational stress. Work 2001; 17:209-219.
42. Merriam P.A., Ockene I.S., Hebert J.R., Rosal M.C., Matthews C.E. Seasonal Variations of Blood Cholesterol Levels Study: Study methodology. Journal of Biological Rhythms 1999; 14:330-339.
43. Karasek R A, Brisson C, Kawakami N, et al. The Job Content Questionnaire: An instrument for internationally comparative assessment of psychosocial job characteristics. J Occup Health Psychol 1998; 3:322-355.
44. Jacob R.G., Thayer J.F., Manuck S.B., et al. Ambulatory blood pressure responses and the circumplex model of mood: a 4-day study. Psychosomatic Medicine 1999; 61:319-333.
45. Schwartz J, Pickering T. The Workplace Blood Pressure Study, American Society of Hypertesion Conference, 1996.
46. Kristenson T.S. Job stress and CVD: a theoretic critical review. J Occup Health Psychology 1996; 1:246-260.
47. Phillips A.N., Smith G.D. The design of prospective epidemiological studies: More subjects or better measurements? Journal of Clinical Epidemiology 1993; 46:1203-1211.
48. Kristenson T.S. The demand-control-support model: methodological challenges and future research. Stress Medicine 1995; 11:17-26.
49. Landsbergis P, Theorell T. Measurement of Psychosocial Workplace Exposure Variables: Self-Report Questionnaires. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Philadelphia: Hanley & Belfus, 2000.
50. Karasek R A, Gordon G, Pietroskovsky C, et al. Job Content Instrument: Questionnaire and User's Guide. Los Angelas, CA./Lowell, MA.: University of Southern California/University of Massachusetts, 1985.
51. Schnall P, Pieper C, Schwartz J.E., Karsek R.A., et al. The relationship between "Job Strain," workplace diastolic blood pressure and left ventricular mass index. JAMA 1990; 263:1929-1935.
52. Van der Doef M, Maes S. The job demand - control (- support) model and

psychological well-being: a review of 20 years of empirical research. Work & Stress 1999; 13:87-114.

53. Bultmann U, Kant I.J., Van den Brandt P.A., Kasl S.V. Psychosocial work characteristics as risk factors for the onset of fatigue and psychological distress: Prospective results from the Maastricht cohort study. Psychological Medicine 2002; 32:333-345.
54. Bultmann U, Kant I.J., Schroer C.A., Kasl S.V. The relationship between psychosocial work characteristics and fatigue and psychological distress. International Archives of Occupational and Environ Health 2002; 75:259-266.
55. Sloan R.P., Shapiro P.A., Bagiella E, et al. Effect of mental stress throughout the day on cardiac autonomic control. Biological Psychology 1994; 37:89-99.
56. Rottman J, Steinman R.C., Albrecht P, Bigger J.T. Jr., Rolnitzky L.M., Fleiss J.L. Efficient estimation of the heart period power spectrum suitable for physiologic or pharmacologic studies. Am J Cardiology 1990; 66:1522-1524.
57. Albrecht P, Cohen R.J. Estimation of heart rate power spectrum bands from real world data: Dealing with ectopic beats and noisy data. Computers in Cardiology 1988; 15:311-314.
58. Grossman P, Sveback S. Respiratory sinus arrythmia as an index of parasympathetic cardiac control during active coping. Psychophysiology 1987; 24:228-235.
59. Backs R.W. Going beyond heart rate: autonomic space and cardiovascular assessment of mental workload. The International Journal of Aviation Psychology 1995; 5:25-48.
60. Backs R.W. A comparison of factor analytic methods of obtaining cardiovascular autonomic components for the assessment of mental workload. Ergonomics 1998; 41:733-745.
61. Backs R.W., Lenneman J, Sicard J.L. The use of autonomic components to improve cardiovascular assessment of mental workload in flight simulation. The International Journal of Aviation Psychology 1999; 9:33-47.
62. Lenneman J, Backs R.W. The validity of factor analytically derived cardiac autonomic components for mental workload assessment. In: Backs R.W., Boucsein W, eds. Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum, 2000:161-176.
63. Negoescu R, Dinca-Panaitescu S. Mental stress enhances the sympathetic fraction of QT variability in an RR independent way. Int Phys Behav Sci 1997; 32:220-226.
64. Collins S M, Karsek R.A., Costas K. Cardiac Vagal Response to Daily Stressors. University of Massachusetts Lowell, 2003.
65. Karasek R.A. The Stress Disequilibrium Theory of Chronic Disease Development. Lowell, 2003.
66. Collins S M. Job Strain, Exhaustion and Cardiac Regulation. University of Massachusetts Lowell, 2003.
67. Vrijkotte T.G.M., van Dornen L.P.J., de Geus E.J.C. Effects of work stress on ambulatory blood pressure, heart rate, and heart rate variability. Hypertension 2000; 35:880-886.
68. Siegrist J, Junge P.R., Cremer P., Seidel D. Low status control, high effort at

work and ischemic heart disease: prospective evidence from blue collar men. Soc Sci Med 1990; 31.

69. Bigger J.T. Jr. Spectral analysis of RR variability to evaluate autonomic physiology and pharmacology and to predict cardiovascular outcomes in humans. Encylopedia of Cardiology 1994:1151-1170.
70. Schnall P, Schwartz J.E., Landsbergis P, Warren K, Pickering T.G. A longitudinal study of job strain and ambulatory blood pressure: results from a three year follow-up. Psychosomatic Medicine 1998; 60:697-706.
71. Lucini D, Mela G.S., Malliani A, Pagani M. Impairment in cardiac autonomic regulation preceding arterial hypertension in humans: Insights from spectral analysis of beat-by-beat cardiovascular variability. Circulation 2002; 106:2673-2679.
72. Steptoe A, Fieldman G, Evans O, Perry L. Cardiovascular risk and responsivity to mental stress: the influence of age, gender, and risk factors. J Cardiovasc Risk 1996; 3:83-93.
73. Steptoe A. Stress, social support and cardiovascular activity over the working day. Int J Psychophysiology 2000; 37:299-308.
74. Appels A, Hoppener P, Mulder P. A questionnaire to assess premonitory symptoms of myocardial infarction. International Journal of Cardiology 1988; 17:15-24.
75. Appels A, Mulder P. Excess fatigue as a precursor of myocardial infarction. European Heart Journal 1988; 9:758-764.
76. Wantanabe T, Sugiyama Y, Sumi Y, et al. Effects of vital exhaustion on cardiac autonomic nervous functions assessed by heart rate variability at rest in middle-aged male workers. International Journal of Behavioral Medicine 2002; 9:68-75.
77. Kuch B, Hense H.W., Sinnreich R, et al. Determinants of short period heart rate variability in the general population. Cardiology 2000; 95:131-138.
78. Tzaneva L, Danev S, Nikolova R. Investigation of noise exposure effect on heart rate variability. Central European Journal of Public Health 2001; 9:130-132.
79. van Amelsvoort L.G.P.M., Schouten E.G., Maan A.C., Swene C.A., Kok F.J. Occupational determinants of heart rate variability. Int Arch Occup Environ Health 2000; 73:255-262.
80. Magari SR, Hauser R, Schwartz J, Williams PL, Smith TJ, Christiani DC. Association of heart rate variability with occupational and environmental exposure to particulate air pollution. Circulation 2001; 14:986-991.
81. Magari SR, Schwartz J, Williams PL, Hauser R, Smith TJ, Christiani DC. The association of particulate air metal concentrations with heart rate variability. Environmental Health Persectives 2002; 110:875-880.
82. Costas K, Karsek R.A. Recruitment Bias in Occupational Stress Studies with Complex Protocols. University of Massachusetts Lowell, 2003.

Figure 2-1. Proposed Hierarchical Relationship between Job Strain Assessment Methods

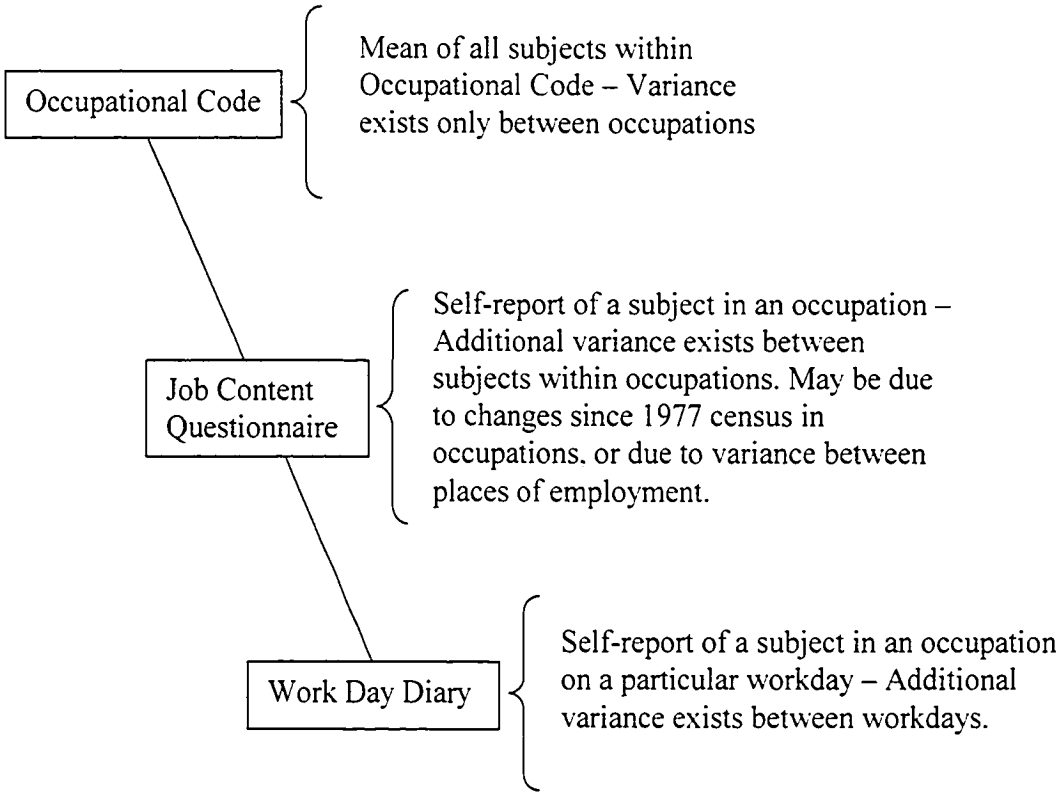


Figure 2-2. Psychological Demands: Correlation Coefficients between Job Strain Assessment Methods

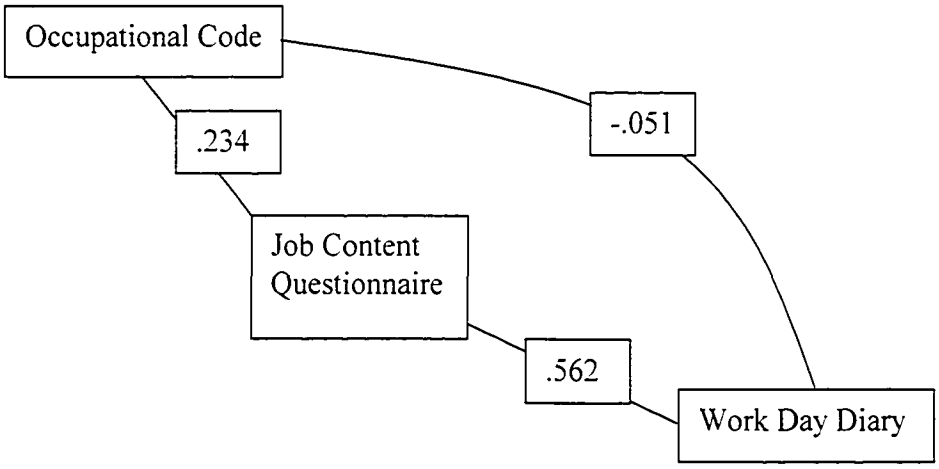


Figure 2-3. Decision Latitude: Correlation Coefficients between Job Strain Assessment Methods

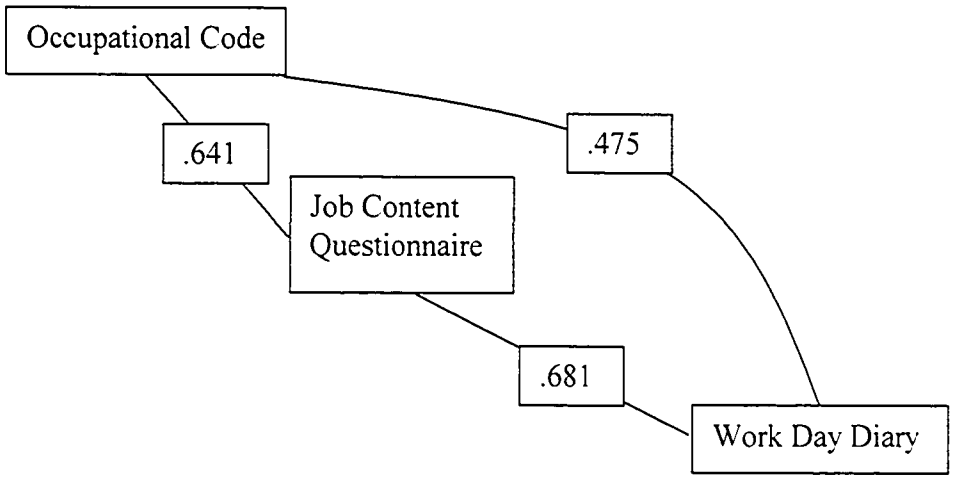


Figure 2-4. High Job Strain Classification using U.S. Quality of Employment Survey Tertiles

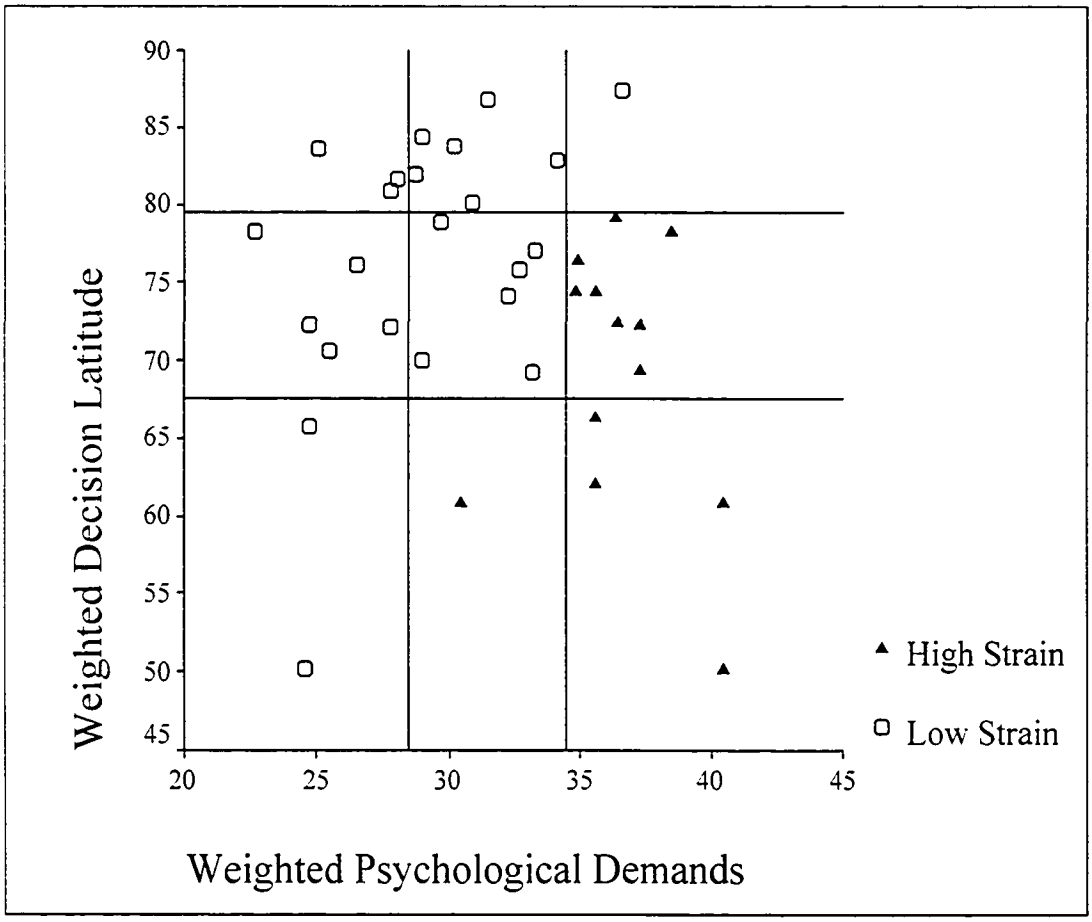
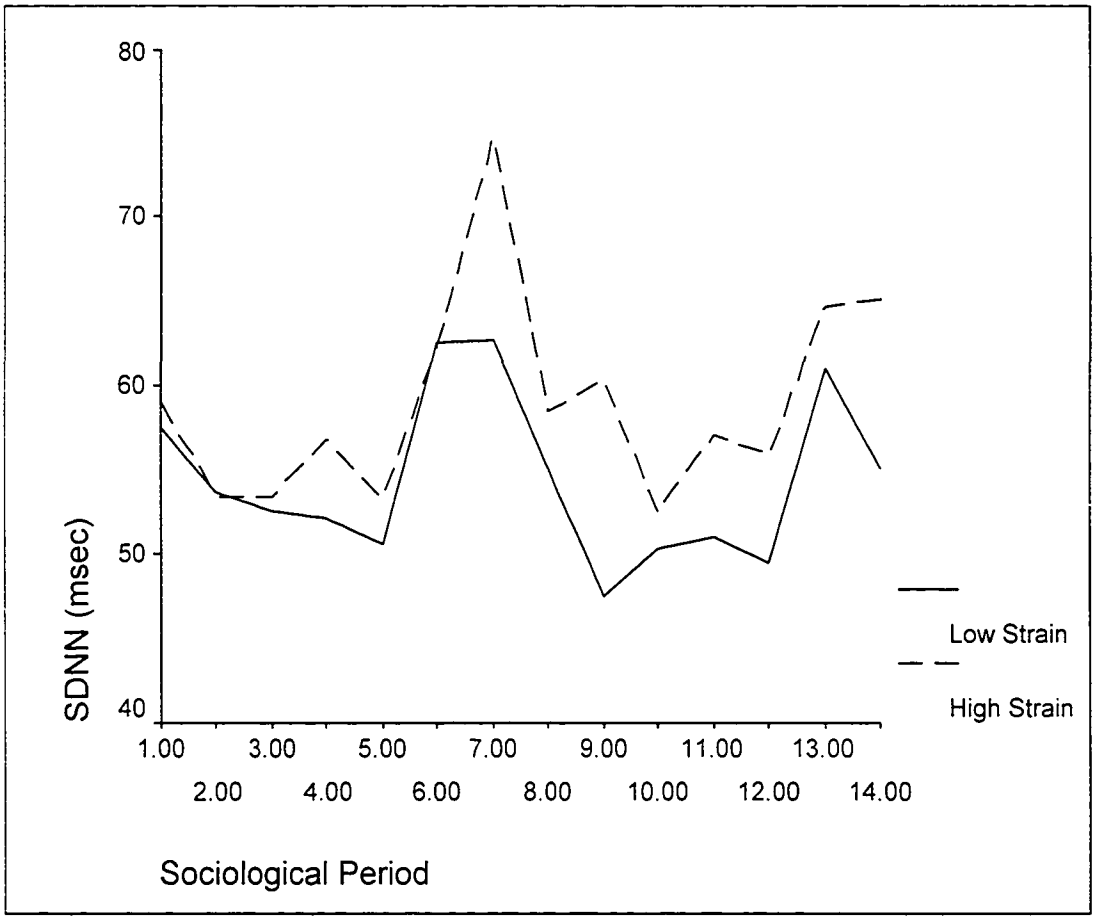
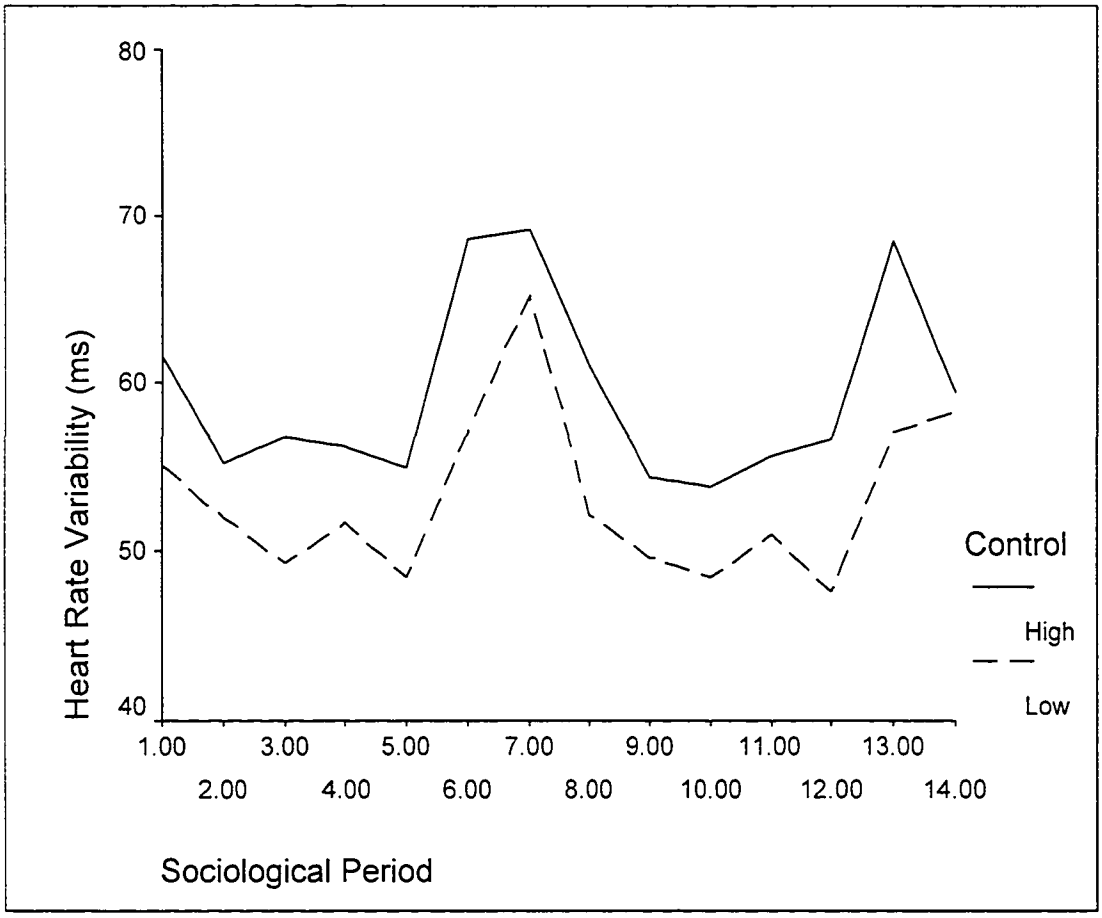


Figure 2-5. Heart Rate Variability by Job Strain



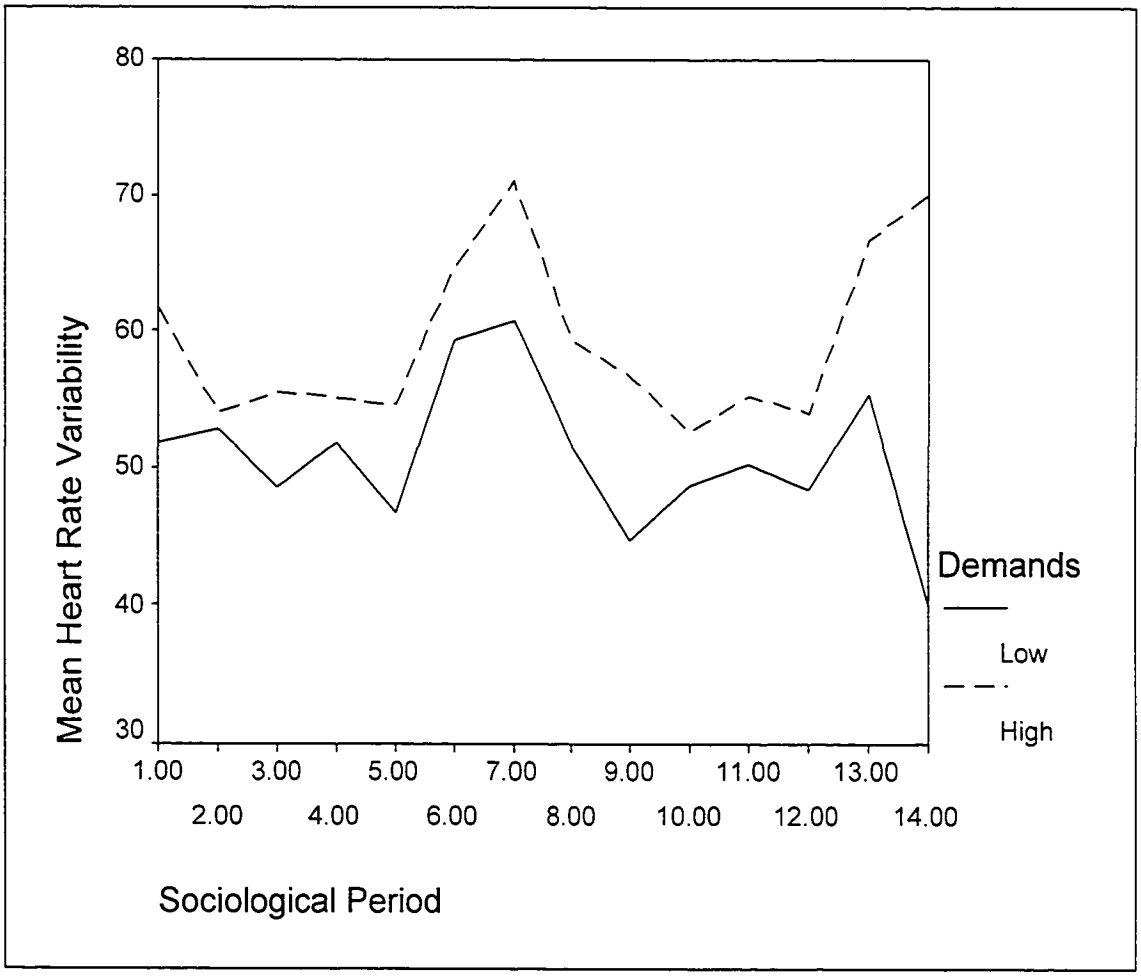

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-6. Heart Rate Variability by Job Control



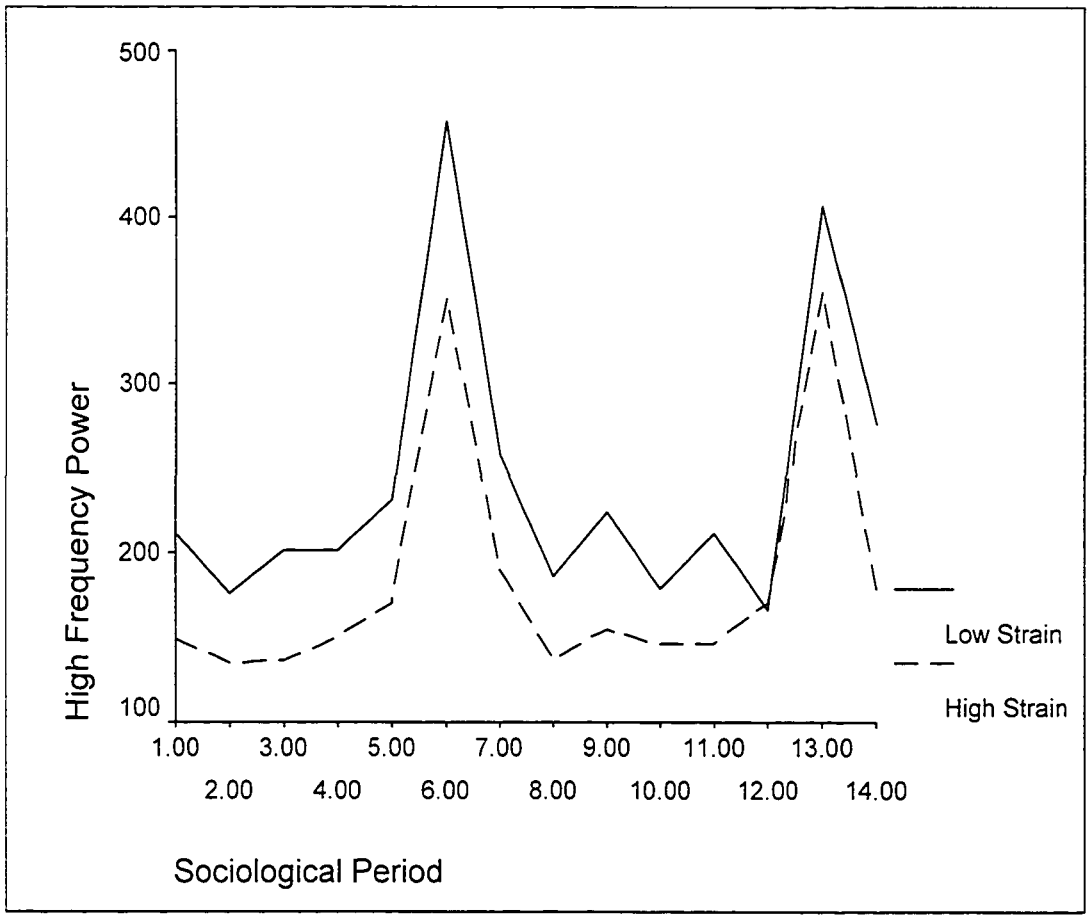

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-7. Heart Rate Variability by Job Demands



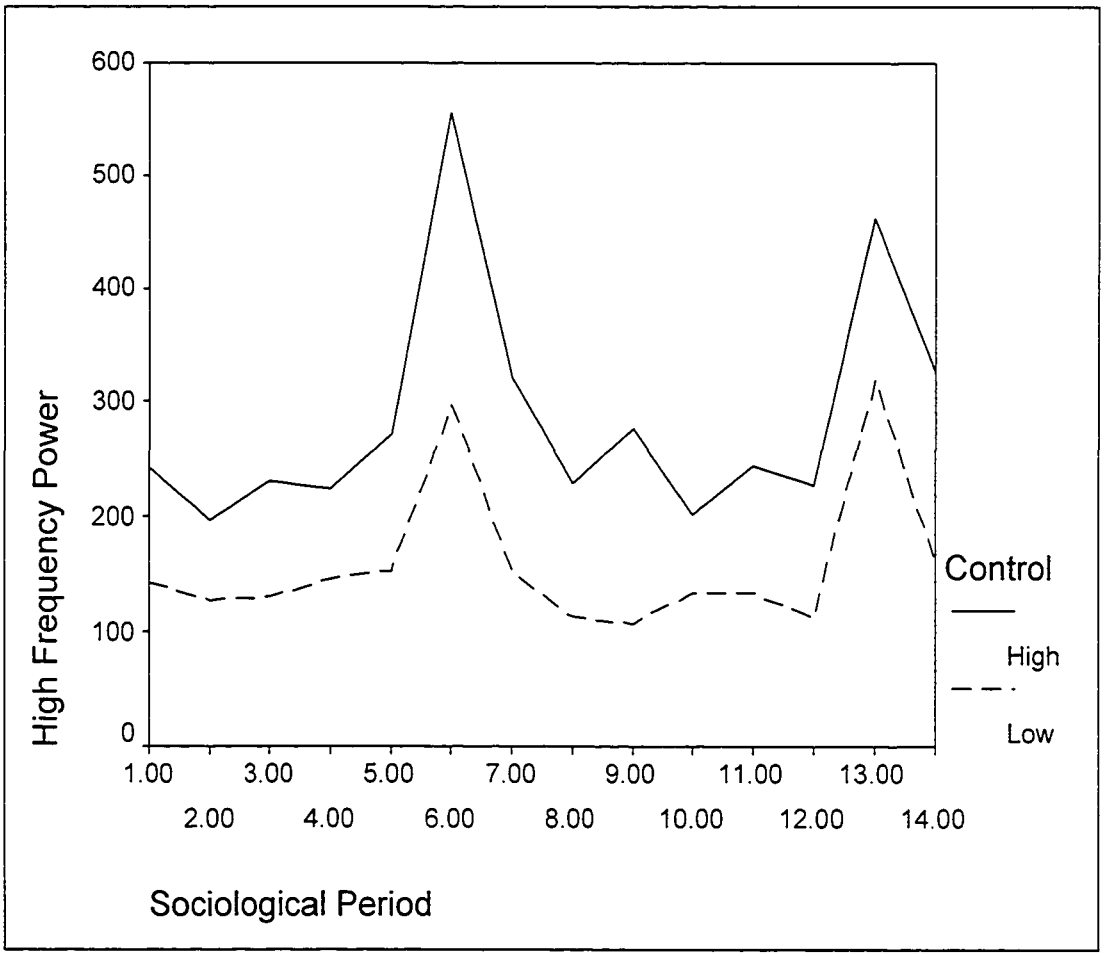

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-8. High Frequency Power by Job Strain



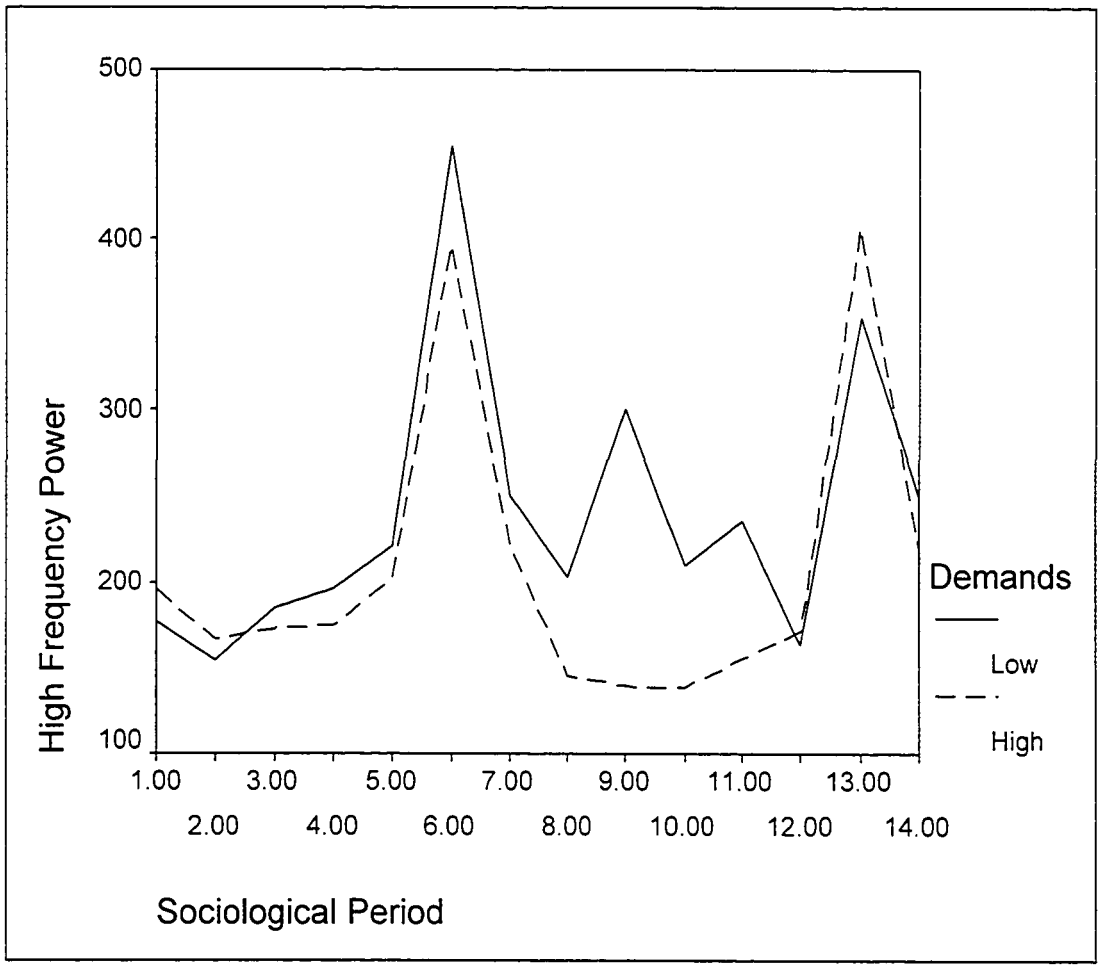

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-9. High Frequency Power by Job Control



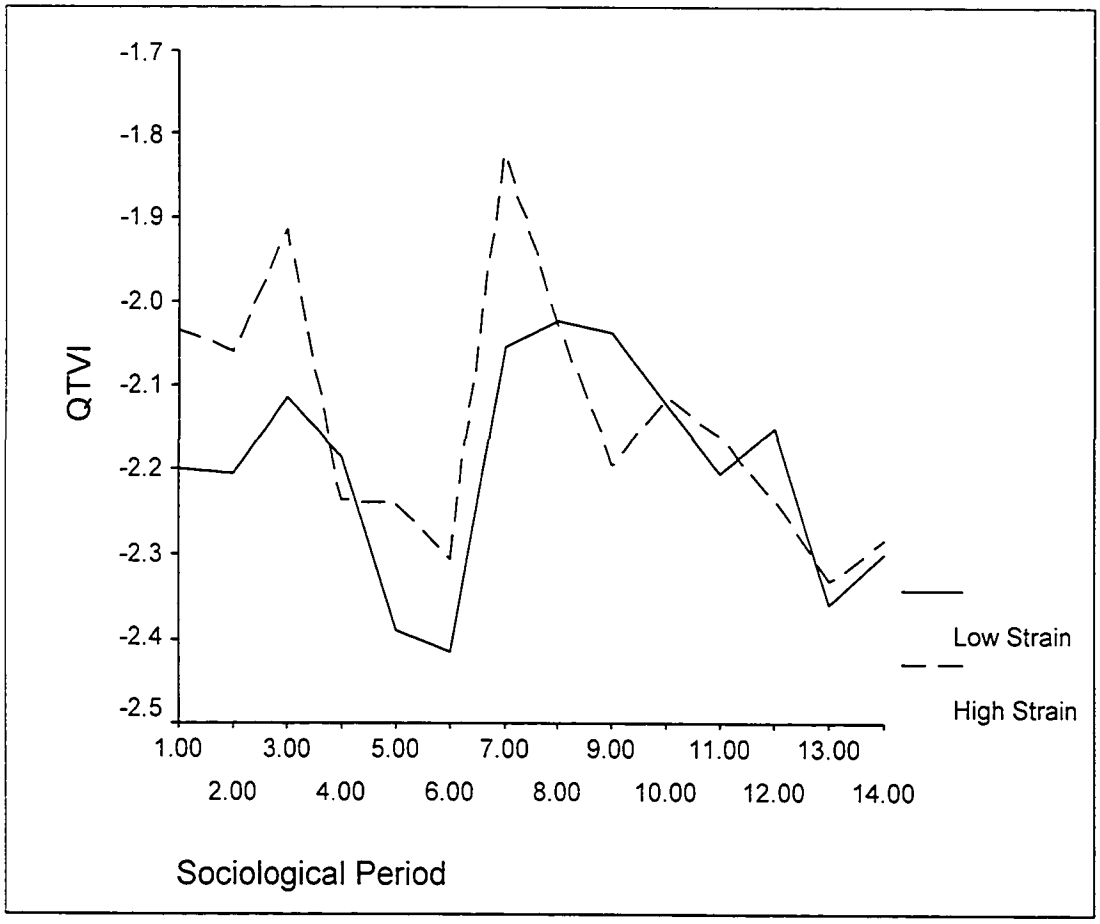

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-10. High Frequency Power by Job Demands



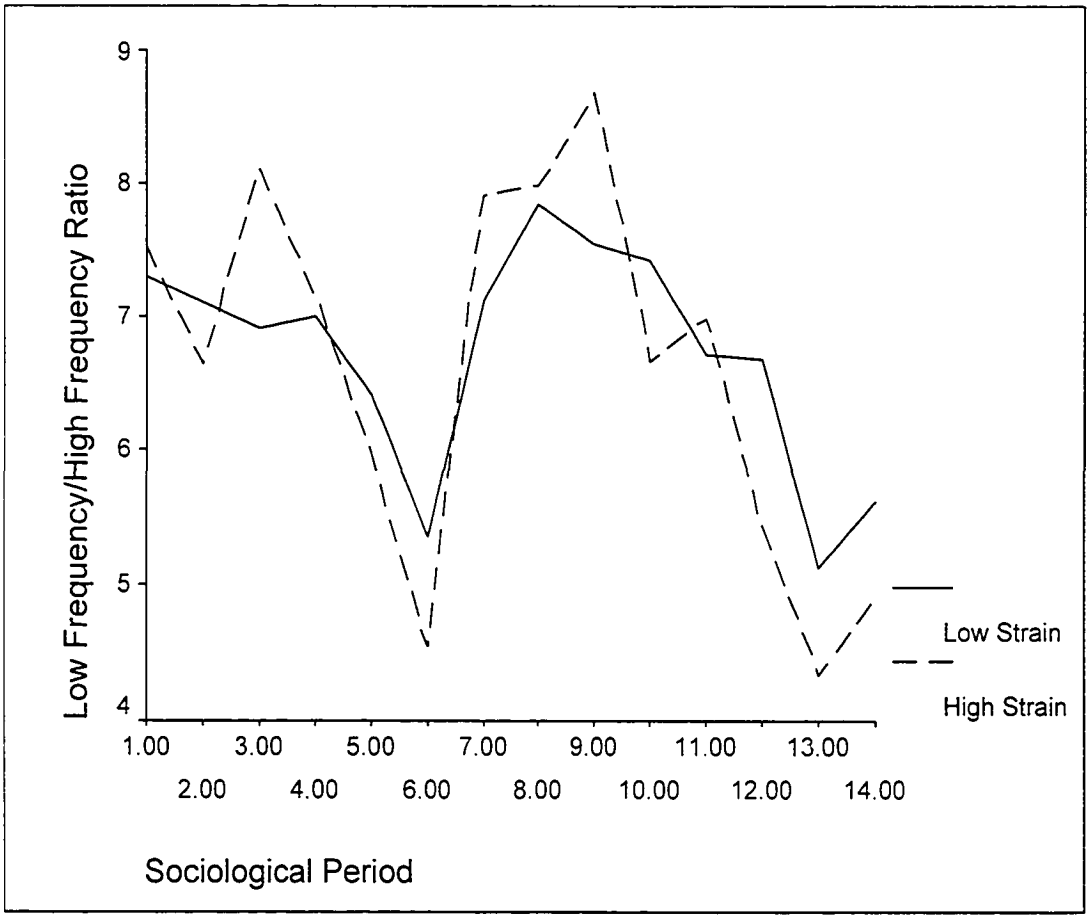

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-11. QTVI by Job Strain



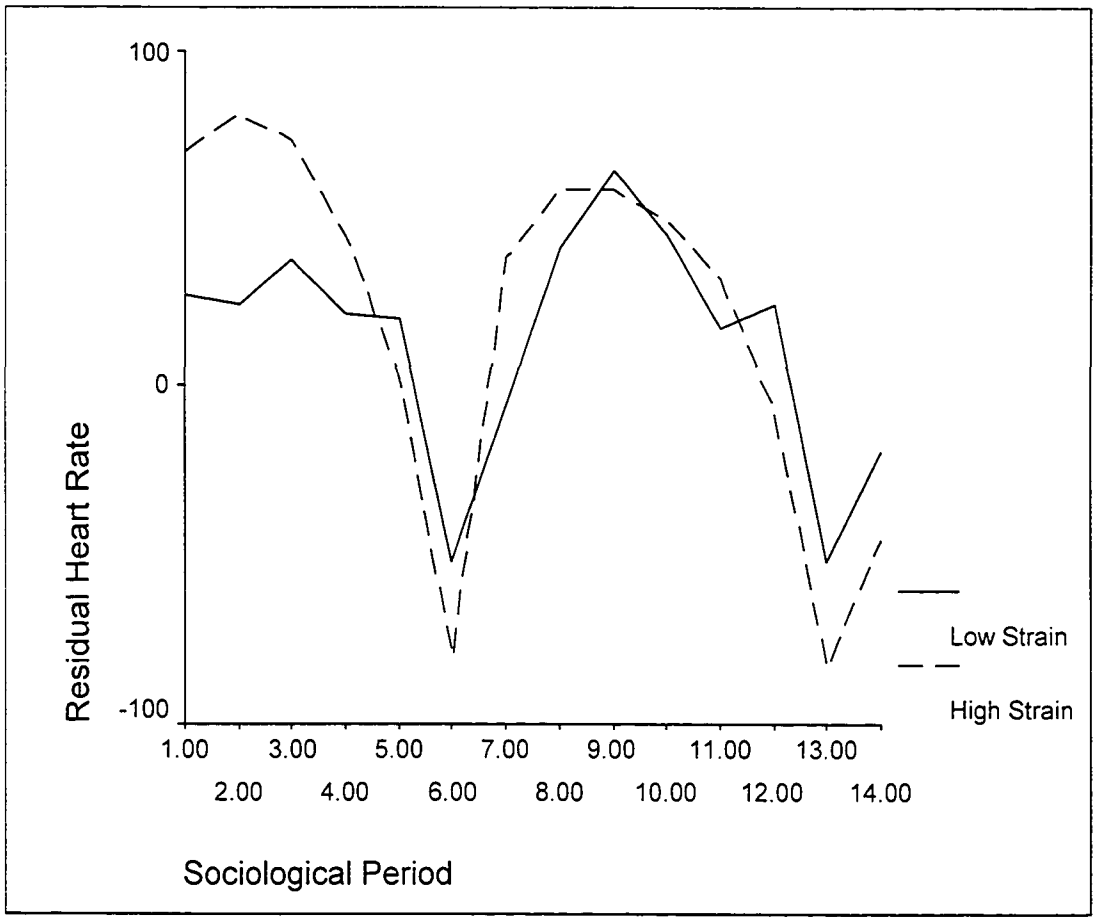

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-12. Low Frequency/High Frequency Ratio by Job Strain




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 2-13. Residual Heart Rate by Job Strain




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

# CHAPTER THREE

# WITHIN SUBJECT CARDIAC VAGAL RESPONSES TO WORK AND REST DAY STRESSORS

## Abstract

This paper tests the hypothesis that cardiac vagal control throughout the day, when controlling for relevant covariates, will be affected by job strain as reported on a diary. In addition, it was hypothesized that cardiac vagal activity will vary between subjects based on macro level job characteristics. Thirty-Six males between the ages of 35 and 60 years and free of known heart disease were selected from a larger longitudinal study of healthy subjects and cardiovascular risk based on responses to a subset of questions from the Job Content Questionnaire (JCQ). Subjects were monitored with an ambulatory Holter electrocardiograph. The monitor allowed subjects to continue on with normal daily activities with minimal distraction while recording cardiac electrical activity. The protocol required monitoring to begin the morning of a workday followed for 48 continuous hours through the following rest day and subsequent morning. Diary questions were based on demand control model assessment of occupational stress.

Psychosocial strain as measured by the diary during two days of monitoring was significantly associated with cardiac vagal control using within and between person based hierarchical modeling analysis. This finding is found after controlling for position, exertion and age. In addition, using the Job Content Questionnaire (JCQ) to assess decision latitude, low workplace control based on a single overall rating of each subjects' permanent job, is significantly associated with reduced between-subject vagal cardiac activity, also controlling for age. The use of multilevel modeling has allowed us to test for the influence of psychosocial conditions within each subject while testing the between subject effects of age and JCQ assessed decision latitude.

## Introduction

Evidence is accumulating linking psychosocial stress at work to an increasing proportion of occupationally related illness with the majority of the problem stemming from poor work organization [1]. Substantial research exists relating adverse psychosocial work environments with cardiovascular disease [2, 3].

Perhaps the most commonly utilized hypothesis associating risk of stress-related cardiovascular disease with working conditions is the Demand Control model (or demand/control/support model) described by Karasek [4, 5] and Karasek and Theorell, [6] and Johnson and Hall [7]. The demand control model stratifies jobs based high and low psychological demands (such as work deadline pressure) and decision latitude (control over the work situation). The primary hypothesis involves high psychological demands coupled with low decision latitude (defined as "job strain") increasing the risk of cardiovascular disease. Social support appears to act as a buffer to high job strain (with low social support increasing the risk) [8-11]. The associations between job strain and cardiovascular diseases do not seem to be completely mediated through the routinely associated cardiovascular risk factors [2, 3].

In light of the rather consistent associations between job strain and cardiovascular disease morbidity and less convincing picture of job strain and the conventional cardiovascular risk factors is that the association between job strain and coronary heart disease is mediated through other physiological pathways besides the conventional cardiovascular risk factors. This possibility increases interest in sudden cardiac causes of coronary disease mortality – which accounts for approximately 50% of total cardiac

mortality [12]. In sudden cardiac death (SCD) defective control of cardiac rhythms is one of the etiological pathways [13-15]. While it must be clearly stated that the common opinion about cardiac death is that it occurs in individuals who often have high levels of the conventional risk factors, it remains true that these factors are not always considered the precipitating cause in the case of sudden cardiac death [16]. SCD that is not related to coronary artery disease risk factors is related to failures of the body to control cardiac output and to support cardiac muscle action.

The most common mechanisms that achieves this cardiac controls is the parasympathetic control mechanism – which coordinates heart activity by slowing it down based on impulses from the central nervous system (the brain stem / autonomic nervous system.) Thus, failures in basic central nervous system control of the heart are implicated in the most substantial cause of cardiovascular disease mortality and yield a conceivable basis for hypotheses related to overall levels of job strain and cardiac risk. These are precisely the pathways to cardiovascular disease that would be implicated by job strain research in the heart rate variability area – which are indeed already documented, albeit in non-epidemiological research, in the human factors literature.

Heart Rate Variability (HRV) refers to an assessment of the variance of the beat-to-beat heart rate intervals with a post hoc time series analysis. HRV represents oscillations within the autonomic nervous system as it influences the depolarization of the heart's sino-atrial node. This analysis of variability allows identification and quantification of autonomic nervous system components. Heart rate variability can be analyzed in the time domain, or spectral (frequency) domain. Time domain analysis uses simple statistical analyses of an RR interval time series [17-19]. However, while time

domain analysis provides information on heart rate variability, which increases with parasympathetic activity and decreases with sympathetic modulation, it does not distinguish the parasympathetic influences nearly as well as frequency domain analysis [18, 20]. Frequency domain analysis (spectral analysis) of HRV decomposes the time series into sinusoidal components, which appear to allow closer identification of physiological contributors [21]. Power (variance) is plotted as a function of frequency and the analysis computes power in defined frequency regions. Thus allowing the quantification of variance within certain frequency ranges. The most commonly used method of performing spectral analysis is the Fast Fourier Transformation (FFT) [18, 20]. The high frequency band of oscillation (.15 to .4 Hz) has known physiological meaning and is related to the activity of the autonomic nervous system's vagal branch in its role controlling heart rate [18, 20, 22]. This range of frequency oscillations thus provides physiologically interpretable information about relaxation, exertion and psychological stress.

Various literatures in the areas of human factors [23-25], and over-training [26] have identified alterations in the autonomic response patterns using HRV that depend on the situation in which the individuals are stressed. These studies are limited by their laboratory based approach and their lack of assessment of job stress.

Considering the growing body of evidence linking psychosocial stress exposure at work and with chronic diseases (in particular heart disease), the implication of altered autonomic nervous system function as a possible etiological pathway for these exposures, the well defined methods from the clinical, psychological and human factors literature for analyzing HRV as a measure of ANS function, and the clinical linkages between altered

HRV and health risks it may be beneficial to combine these research traditions in an attempt to utilize HRV to monitor the physiological impact of occupational stress. Utilizing HRV based measures in this paper we have integrated the strengths of understanding provided by the aforementioned research traditions in an attempt to illuminate how workplace psychosocial stressors contribute to the development of chronic disease – through defined physiological pathways related to cardiac regulation by the autonomic nervous system. A clear benefit of HRV assessment is the capacity to perform ambulatory monitoring during work tasks. This approach has proven quite beneficial in linking work environment characteristics to the risk and development of hypertension and hypertension related structural alterations. In particular it is striking to note that elevated blood pressure during the work-day, in workers with high job strain, has been linked to pathological changes in cardiac structure despite normal resting clinic based measurements. These findings support the contention that alterations in cardiac control produced during the workday that are related to the work environment can be pathological. [27] Similarly, using HRV it may be possible to identify the etiological links and pre-clinical changes in vagal cardiac control as related to the work environment that lead to clinical pathological alterations thus increasing risk for sudden cardiac death [15].

Studies utilizing HRV measures for investigating the physiological impact of occupational stress in the work place are beginning to appear. A variety of methods have been utilized to categorize occupational stress as well as to measure HRV. A brief review of these studies is available. [28]

This paper tests the hypothesis that cardiac vagal activity throughout the day, when controlling for relevant covariates, will be affected by job strain as reported on a

diary. In addition, cardiac vagal activity will vary between subjects based on macro level job characteristics such as decision latitude.

# Methods

## Subjects

A sample of healthy employed men (n = 26) between 35-59 years of age, all members of Fallon Community Health Plan, were recruited from an ongoing investigation of seasonal variance in cholesterol (Seasons) [29]. An additional sample (n=10) of healthy employed men (35 – 59 years of age) was recruited from the University of Massachusetts Worcester Medical Center's stress reduction clinic. Participants of the Seasons study had been screened for job strain using a subset of questions from the Job Content Questionnaire [30]. The Fallon Community Health Plan and the University of Massachusetts Medical Center Institutional Review Board approved the study for recruitment of Human Subjects. Subjects provided informed consent prior to completing questionnaires and being connected to the monitor. Investigators reviewed JCQ and current occupational information collected for each subject as part of the intake process and identified a potential sub-population of mixed and high and low strain individuals. Each potential subject was sent a letter explaining the study, how they were selected and informing them that they would be contacted by phone regarding participation.

Subject recruitment occurred in two waves. During the first wave of recruitment the researchers became aware of the difficulties associated with participation in complex protocol studies for participants with high strain jobs [Costas K, 2003 #209]. The yield of high strain subjects (8%) was much less than that for low strain subjects (43%). To truly test the intended hypotheses a population-based sample of subjects with high strain jobs

was required. Therefore for the second wave of recruitment the research protocol was made more flexible to accommodate the high stress individuals. First, the intake Holter monitor attachment and calibration process occurred at a location and time of the subject's convenience with respect to the subject's time of arrival at work. Second, monetary incentives were offered for study participation. Subjects were offered up to $250 for completion of the study. Finally, in every way possible, during the recruiting contact, attempts were made to be accommodating and flexible in terms of protocol requirements (without violating the minimum study requirements). Thirty-six subjects were successfully recruited and monitored according to the study protocol.

## Protocol

Subjects agreeing to participate in the study provided informed consent prior to being asked to complete questionnaires or being connected to the monitoring apparatus that was approved by the Fallon Community Health Plan and UMASS Medical Institutional Review Board for Human Subjects. The monitoring protocol utilized Holter monitors (Marquette 8500) worn in the workplace and at home to collect 48 hours of continuous ECG recordings beginning in the morning of (prior to) a workday. At this time, subjects were also asked to complete a full version of the Job Content Questionnaire, and provide information regarding their occupation. During the first wave of data collection, this was a Friday, to allow the following day to be a rest day (Saturday). During the second wave of data collection (associated with increasing protocol flexibility), subjects were monitored on any day, as long as it was the last day of

their regular workweek and the following day was a day off (rest day). During the monitoring period subjects were asked to complete an event diary. The subject was required to complete the diary at eight times during each day (Table 3-1). At these times, which were based on events in the subject's day, answers were recorded regarding various psychosocial aspects of their environment (demands, control, socialization), time of day, position, and activity to describe their "state" during the 20 minutes prior to completing the diary. Subjects removed the Holter monitor in the morning 48 hours after having had the monitor applied. Subjects were provided with a prepaid envelope to mail the Holter monitor and the diary to the study team.

Table 3-1. Diary Entry Periods

| Day | Recording | Diary Social Activity and Time of Recording |
|---|---|---|
| Work | 1 | One hour after work starts |
| | 2 | One hour after work starts + 20 minutes |
| | 3 | One hour after lunch |
| | 4 | One hour after lunch + 20 minutes |
| | 5 | 20 minutes prior to leaving work |
| | 6 | Leaving work |
| | 7 | At home after dinner |
| | 8 | Research Call (predetermined time after dinner at relaxation) |
| Rest | 9 | As close to Recording 1 time as possible |
| | 10 | As close to Recording 2 time as possible |
| | 11 | As close to Recording 3 time as possible |
| | 12 | As close to Recording 4 time as possible |
| | 13 | As close to Recording 5 time as possible |
| | 14 | As close to Recording 6 time as possible |
| | 15 | At home after dinner |
| | 16 | Research Call (predetermined time after dinner at relaxation) |

## Measures

### *Job Strain: JCQ Based Measures*

Karasek's Job Content Questionnaire (JCQ) was utilized to compute psychological demands, decision latitude and social support scores and were compared to a national sample. [30, 31] Psychometric properties of the scales from this sample are presented in Table 3-2.

Table 3-2. Questionnaire Scales

| *Scale* | *# of Items* | *Possible Range* | *N Cases* | *Cronbach's Apha* |
|---|---|---|---|---|
| Psychological Demands | 5 | 12-48 | 36 | .73 |
| Decision Latitude | 9 | 24-96 | 36 | .87 |
| Social Support | 8 | 8 – 32 | 36 | .87 |
| Diary Demands | 3 | 3 – 12 | 491 | .75 |
| Diary Control | 3 | 3 – 12 | 491 | .66 |
| Diary Social Support | 2 | 2 – 8 | 491 | .80 |

### *Job Strain: Diary Measures*

The diary was a modification of the one used in the Cornell Ambulatory Blood Pressure studies [27, 32]. Diary questions were utilized to ascertain information regarding the subjects' position, activity, location, time of day, demands, control and support.

Scales were constructed for demands, control and social support using factor analysis with varimax rotation and their psychometric properties are presented in Table 3-2. The demand questions focused on quantitative demands, while the control questions were based primarily on decision authority. These core questions were utilized due to their consistent factor agreement on both the work and home locations. There was inconsistency in factor loadings for cognitive demand questions and skill discretion questions between work and home environments. When at home the control based question regarding organizing people was loaded as a demand, and therefore excluded from the control factor.

Diary strain was assigned based on a priori determined cut points utilizing the pooled samples quartiles and results in 5 levels of strain from lowest strain (1) to highest strain (5). Figure 3-1 shows the cut-points utilized (based on sample derived quartiles) and the numeric strain assignment for the two-dimensional demand control space. Position and location were directly recorded onto the diary while exertion was scored based on recordings for activity.

## Data Processing

Holter data tapes were analyzed for ventricular arrhythmias and RR-interval variability at the Research Holter Laboratory at Columbia University by published methods [33]. In summary, the dependent variable of this study high frequency power, was computed from 5-minute epochs throughout the 48 hour period. The 48 hour records were digitized using a Marquette 8000 scanner, submitted to Marquette algorithms for

QRS identification and editing, and then downloaded to a Sun workstation for further editing using algorithms developed at Columbia University. The power spectrum was computed as previously described [34, 35]. A Fast Fourier spectrum was utilized to compute the power spectrum in the very low frequency (VLF < 0.04 Hz), low frequency (LF, 0.04 – 0.15 Hz), and high frequency (HF, 0.15-0.40 Hz) spectra. Finally the high frequency power was then normalized (HFPnu) to represent the amount of variance in the high frequency range relative to the total power. High frequency power in normalized units (HFPnu) is a more specific index of vagal activity [36].

## Data Linkage

Times entered for the diary were utilized to match the 20-minute period prior to each diary entry to the same 20-minute period for the ECG recording. The ECG data during this 20-minute period was aggregated (average of the four 5 minute epochs) and linked to the appropriate diary data.

It is possible that imprecision in coordination of time intervals between Holter monitor EKG readings and /or diary recordings could lead to diary-time period aggregation of the electrocardiogram data that does not most accurately coincide with diary outcomes. Given that data were aggregated into blocks of 20-minute signal outputs, it is assumed that the majority of the ECG data classified as having occurred in the 20 minutes prior to the diary recording are correct. We utilized this window of data to assess potential worst-case effects of such errors. Utilizing a sliding time boundary window, we tested the models at 20-minute periods surrounding or following the diary

period. Varying the time boundaries created different results. It was possible to alter the boundaries to the point in which statistical measures such as standard errors when applied to these data, would suggest that well documented relationships such as that between postures and high frequency power would no longer be statistically associated (Table 3-3). This sensitivity test result illustrates the importance that future investigations utilize newer technologies to more accurately record and orient the ECG signal as it occurs in real time. Precise associations between ECG events and human activity will afford a stronger opportunity for understanding their relationship.

Table 3-3. Effect of Shifting Linkage Window

| | *20 Minutes Prior* | *Centered* | *20 Minutes Post* |
|---|---|---|---|
| Posture Coefficient | -1.29 | -0.55 | -0.13 |
| Std. Error | .308 | .355 | .412 |
| Significance | .001 | .541 | .753 |

## Statistical Analysis

To test associations between psychosocial conditions and cardiac vagal control, high frequency power normalized by total power (HFPnu) was modeled utilizing multilevel linear regression. This method was chosen for the current analysis due to the multilevel nature of the data considered for the intended hypothesis and the need to account for widely variable between subject baseline values for high frequency power (Figure 3-2). We are interested in the association between the psychosocial conditions reported on the diary with vagal cardiac control at that time of recording within the

individual subjected to the particular condition. In this study, diary entries (measure of environmental conditions) and the associated recording of cardiac vagal control are nested within subjects. However, environmental conditions at the diary entry level (level 1) have a variance structure both within and between subjects while subject level variables (level 2), such as JCQ determined decision latitude and age, have only between subject variance. Table 3-4 shows the variables and their associated level of analysis. The nesting of level 1 variables within an individual has important implications. Variables within a subject are more alike than those chosen from a random sample of the population. The hypothesis being tested is whether a person's psychosocial environmental condition influences that individual's vagal cardiac control. Measures of vagal cardiac control (HFPnu) have a variance structure both within and between subjects, and we are interested in testing the influence of both diary entry level and subject level variables while controlling for the within subject correlation. Figure 3-2 shows the effect of diary strain on HFPnu with a line of best fit for each subject. The multilevel modeling (random intercept) approach utilized in this study accounts for the widely variable intercepts that can be visualized in Figure 3-2, which represent variation in subject baselines. The model thus allows for control for between subject baseline differences while allowing a within and between subject analysis of multi-level predictor variables.

Table 3-4. Variable Levels

| *Variable* | *Diary Entry – Level 1 (Within Subject)* | *Subject – Level 2 (Between Subject)* |
|---|---|---|
| Age | | ● |
| Decision Latitude (JCQ) | | ● |
| Psychological Demands (JCQ) | | ● |
| Job Strain (JCQ) | | ● |
| Location | ● | |
| Diary Strain | ● | |
| Diary Social Support | ● | |
| Position | ● | |
| Exertion | ● | |
| High Frequency Power % | ● | |

The multilevel model (random coefficient) approach also addresses the problems associated with missing cases, differences between subjects' cardiovascular risk factors and baseline differences in high frequency power [37, 38]. Four models will be presented that tested individual demand – control – support hypothesis components (Model A – D), and four models will be presented that tested job strain and social support hypotheses (Model I – IV). Model A is an unconditional means model that was utilized to determine the variance within and between subjects to assess the adequacy of distinguishing two levels of analysis and to provide a baseline against which to compare more complex models.

Model B tests level 2 (between subject) predictors (age, JCQ based decision latitude, demands and social support). Model C adds the level 1 (within subject) predictors diary control, diary demands, diary social support and position. Model D includes all demand – control – support components from both within and between subject levels in addition to the covariates age and position. Model I tests the effects of level 2 predictors (age and JCQ decision latitude) and the hypothesis that age and decision latitude both decrease HFPnu. Model II tests the effects of level 1 predictors and the hypothesis that within each subject position, location, exertion, diary assessed job strain influence HFPnu. Model III tests the effects of all level 1 predictors from Model II with the addition of social support and a social support X diary strain interaction. Finally, Model IV is a two level model with predictors at both levels and tests all the hypotheses in model I and III simultaneously.

Centering was conducted to the group for the level 2 predictor, age, and to the individual for the level 1 predictors diary strain. Implications of centering on interpretation of regression coefficients will be discussed. The SPSS 11.0 mixed model procedure was utilized to perform all statistical analyses.

# Results

## Descriptive Statistics

Mean, Standard Deviation (S.D.), and ranges for the sample are presented in Table 3-5. For repeated measurement variables data are presented as an aggregate both within and between subjects. The mean JCQ based psychological demands (PD) and decision latitude (DL) indicates that this sample provides a fairly good representation of the national sample (sample means PD 30.1 (7.2) and DL 72.6 (15.4). [30] Approximately 61 diary entries were skipped or missed resulting in a compliance rate of 89%. Correlations of main diary based variables are presented in Table 3-6.

Table 3-5. Descriptive Statistics

| | N | | Mean | SD | Observed Range |
|---|---|---|---|---|---|
| | Valid | Missing | | | |
| Age | 36 | 0 | 47 | 6.12 | 35-59 |
| JCQ Decision Latitude | 36 | 0 | 72.33 | 11.57 | 36-96 |
| JCQ Psychological Demand | 36 | 0 | 31.72 | 5.47 | 21-43 |
| Diary Control | 493 | 59 | 9.55 | 2.24 | 3-12 |
| Diary Demands | 491 | 61 | 5.65 | 2.52 | 3-12 |
| Diary Strain Measure | 491 | 61 | 2.84 | 1.49 | 1-5 |
| Diary Social Support | 491 | 61 | 4.02 | 1.97 | 2-8 |
| Diary Exertion | 487 | 65 | 1.35 | .49 | 0-3 |
| Diary Position | 493 | 59 | 1.25 | .52 | 0-2 |
| Diary Strain*Social Support | 491 | 61 | 11.5 | 8.24 | 0-40 |
| High Frequency Power % | 465 | 71 | 6.35 | 4.12 | .27 – 26.25 |

Table 3-6. Diary Variable Correlations

| | *Diary Strain* | *Demands* | *Control* | *Social Support* | *Position* |
|---|---|---|---|---|---|
| *Demands* | .79*** | | | | |
| *Control* | -.72*** | -.31*** | | | |
| *Social Support* | .06 | .11* | -.02 | | |
| *Position* | .19*** | .18*** | -.07 | -.07 | |
| *Exertion* | .186*** | .22*** | -.08 | -.04 | .81*** |

Pearson Correlation
* $\alpha \leq .05$
** $\alpha \leq .01$
*** $\alpha \leq .005$

For comparison purposes on values of HFPnu, in the pooled group sample, HFPnu during sleep, the time period during the day where you expect the highest vagal cardiac control, has a mean of 13. 5% with the group mean while awake of 6.35%.

## Ectopic Beats

As previously mentioned the presence of ectopic beats in the ECG signal represents a physiological disturbance and when present in abundance not only indicates abnormal cardiac function, but also interferes with proper determination of the frequency components of HRV. When aggregating the 5-minute epochs into 20-minute diary periods for analysis the sum of ectopic beats was calculated. For the 16 diary periods in the 36 subjects (n = 576) there were a total of 2168 ectopic beats. One (high strain) subject accounted for 83% of these ectopic beats (1793), with an average within diary

period ectopic beat frequency of 120. The average within diary period ectopic beat frequency for the remainder of the subjects is .59, with a range of 0 – 12. Due to the influence of this one subject, and that the hypotheses being tested are based on a normal healthy population, this subject was not included in the analysis presented in this paper.

## Model A: Unconditional Means Model

Model A is an unconditional means model (Table 3-7) and indicates there is sufficient variation of HFPnu within subjects to allow multilevel modeling. We can estimate the portion of the total variance of HFPnu that occurs between subjects as (8.1/[10.5+8.1] = .43) or 43%. [38] The variation at each level is significantly greater than zero (Within Subjects p = .000; Between Subjects p = .001), thus justifying a 2 level model and suggesting ordinary least squares regression of these data would likely yield misleading results. [37, 38]

Table 3-7. Fixed and Random Effects for the Main Hypotheses[a]

| Fixed Effects | Estimate (95% CI) | | | |
|---|---|---|---|---|
| | Model A | Model B | Model C | Model D |
| Intercept | 6.6 (5.54, 7.57)*** | 6.7 (5.7, 7.7)*** | 9.2 (7.6, 10.7)*** | 9.2 (7.6, 10.7)*** |
| Age | | -.23 (-.42, -.04)* | -.20 (-.37, -.03)* | -.22 (-.40, -.04)* |
| JCQ Decision Latitude | | .08 (-1.2, .18) | | .08 (-1.1, .18) |
| JCQ Demands | | -.05 (-.25, .16) | | -.02 (-.22, .17) |
| JCQ Social Support | | .15 (-.08, .38) | | .16 (-.06, .39) |
| Diary Control | | | -.07 (-.24, .09) | -.07 (-.24, .09) |
| Diary Demands | | | -.16 (-.32, -.05)* | -.16 (-.32, -.05)* |
| Diary Social Support | | | -.29 (-.53, -.07)** | -.30 (-.52, -.08)** |
| Position | | | -1.0 (-1.6, -.38)*** | -1.0 (-1.7, -.38)*** |
| **Random Effects** | **Estimate (95% CI)** | | | |
| Between Subjects | 8.4(4.86-14.38)*** | 6.7 (3.6, 12.7)*** | 7.5 (4.2, 13.6)*** | 6.4 (3.4, 12.1)*** |
| Within Subject | 10.5(9.2-12.1)*** | 10.4 (9.0, 12.5)*** | 9.9 (8.6, 11.5)*** | 9.9 (8.6, 11.5)*** |
| A.I.C.[b] | 2498.84 | 2169.42 | 2153.09 | 2155.62 |

[a] Refer to text for further information regarding model A-D

[b] Akaike's Information Criteria (AIC) are displayed in smaller is better forms.

* $\alpha \leq .05$

** $\alpha \leq .01$

*** $\alpha \leq .005$

## Model B: Random Intercept Model with Level Two Effects

The purpose of Model B was to test the main effect of between subject level predictors (level 2) including age and JCQ measurements of demands, control and social support (Table 3-7). At the JCQ level, the effect of demands, decision latitude and social support are not significant predictors of cardiac vagal activity when entered together into Model B. Since the predictor variables entered into the model are either centered on the group mean, the intercept for model B (6.7) estimates the subjects' mean HFPnu when the remaining predictors are 0. In model B, as in all models constructed to test level 2 predictors, age was estimated to have a statistically significant negative coefficient, and is quite physiologically significant given the age range of subjects in this study. The predicted difference in HFPnu between the youngest subject of 35 years old and the oldest subject of 59 years old is approximately 5 percent (based on a population overall HFPnu of 8%). Despite the insignificant findings of the coefficients for these demand – control – support between subject predictors it is quite interesting that all coefficients are in the hypothesized direction. The interaction terms for JCQ measured demands, decision latitude and social support were tested however no significant interactions were identified. Unlike model A, in the level 2 models the estimates of the random intercepts are now conditional on all level 2 predictors. With the addition of these level 2 predictors the conditional component for the variance within subject has remained essentially unchanged (10.4), however the component for the variation between subjects has decreased (8.4 – 6.7). The addition of these level 2 predictors to the model has reduced the between subjects variance by 20% (8.4-6.7/8.4 = .20) in model B. Meaning, the

addition of age and demand – control – support measures account for 20% of the explainable between subject variance in HFPnu values. Testing age alone there is only an 8% reduction in the explainable variance. The increase in explainable variance to 20% occurs with the addition of decision latitude to the model, which approaches significance ($p = .08$).

## Model C: Random Intercept Model with Level One Effects

Diary control, demands and social support were centered for the subjects' means of their respective values. Changes in diary measures are therefore relative to the subjects' mean values. Centering influences the interpretation of the fixed intercept, which now represents the average of all subjects' mean HFPnu at their mean diary measures (centered at 0), position, social support, demands and control. Interpretation of the coefficients then proceeds as in more traditional linear modeling. The intercept (9.2) represents a subject's mean HFPnu while controlling for all predictors in the model. From this value, each incremental increase of demands will reduce HFPnu by .16%, each increase in social support will decrease HFPnu by .29%, and a change in position from supine to sitting, or from sitting to standing will result in a 1% drop. Diary control does not exert a significant effect in this model. The standard errors are small for these estimates, resulting in large t statistics and low p values. In addition, adding each variable as a random effect does not change the results or result in a significant random estimate. This indicates that the data are best fit with a random intercept model and the estimated slope for the effect of diary strain is consistent between subjects. Interestingly, the

estimate for social support is contrary to theory because it indicates that with increasing social support there is a decrease in HFPnu. This will be discussed later in the paper. Position and exertion were also added to the model to control for their potentially confounding effects. Position was a significant predictor (in the expected direction), while exertion was not important and was subsequently dropped from the model. Position and exertion are highly correlated ($r = .81$, $p = .001$) and this collinearity likely accounts for the lack of evidence of an effect of exertion.

The model C estimates of random intercepts are conditional on all level 1 predictors entered into the model. With the addition of these level 1 predictors the conditional component for the variance within subject has changed (10.4 to 9.9). The addition of these level I predictors to the model reduced the within subject variation by 6% (10.4- 9.9/10.4).

## Model D: Two Level Random Intercept Model

Model D includes both level 1 and level 2 predictors as found in the previous two models (B and C). In this model (Table 3-7) we see that there is essentially no change in the significance or direction of the previous parameter estimates. Cross level interactions of demands and control were tested and found to be unimportant. The random effects indicate that there is a decrease of both the between subject variance (23%), and the within subject variation (6%). When adding the level 1 predictors to the model there was an additional 13% reduction in between subject variation.

## Model I: Random Intercept Model with Level Two Effects

The purpose of Model I was to test the effect of subject level predictors (level 2) age and JCQ decision latitude without social support and demands (Table 3-8). At the JCQ level, the effects of psychological demands or of social support were not significant between subject predictors (Table 3-7. Model B). Since the predictor variables entered into the model are centered on the group mean (age and decision latitude) the intercept for model I (6.80) is an estimate of the subjects' mean HFPnu when the predictors are zero. Age was estimated to have a statistically significant negative coefficient, and as already discussed is physiological significant. The negative coefficient for JCQ based decision latitude (estimate = .1 (95% CI = 0.02, 0.19)) is consistent with other macro level analysis findings between strain groups (See Chapter Two), and is consistent with the job strain hypothesis. Given the range of scores reported for this measure in this sample (36-96), this represents a potential variation of 6% of HFPnu. Recall that the mean HFPnu during sleep (a period of high vagal activity) is 13%. This finding is also consistent with the regular epidemiological evidence for a stronger decision latitude effect on health outcomes [3]

Table 3-8. Fixed and Random Effects for Strain and Support Hypotheses[a]

| Fixed Effects | Estimate (95% CI) | | | |
|---|---|---|---|---|
| | Model I | Model II | Model III | Model IV |
| Intercept | 6.8 (5.8, 7.8)*** | 7.7 (6.23, 9.18)*** | 8.6(7.04, 10.25)*** | 8.8(7.19, 10.43)*** |
| Age | -.25 (-.41, -.08)*** | | | -.25 (-.41, -.09)*** |
| JCQ Decision Latitude | .1 (.02, .19)* | | | .099(.007, .19)* |
| Diary Strain (DS) | | -.24 (-.52, .03) | -.76(-1.38, -.13)* | -.66(-1.3, -.06)* |
| Social Support (SS) | | | -.26(-.47, -.04)* | -.24(-.5, -.04)* |
| DS*SS | | | .14(-.003, .29) | .13(-.03, .28) |
| Position | | -1.74(-2.49, -.39)*** | -1.73(-2.75, -.72)*** | -1.4(-2.5, -.35)** |
| Exertion | | .68 (-.39, 1.74) | .72 (-.43, 1.79) | .59 (-.46, 1.6) |
| Location | | .21(-.46, .89) | .19(-.33, .86) | .29(-.36, .96) |

| Random Effects | Estimate (95% CI) | | | |
|---|---|---|---|---|
| Between Subjects | 6.7 (3.6, 12.2)*** | 8.2 (5.2, 16.2)*** | 8.01(4.6, 13.8)*** | 6.3(3.4, 11.7)*** |
| Within Subject | 10.4 (9.1, 11.9)*** | 10.1 (8.7, 11.7)*** | 9.9 (8.6, 11.4)*** | 9.9(8.6, 11.6)*** |
| A.I.C.[b] | 2286.32 | 2398.36 | 2350.98 | 2150.38 |

[a] Refer to text for further information regarding model I-IV

[b] Akaike's Information Criteria (AIC) are displayed in smaller is better forms.

* $\alpha \leq .05$

** $\alpha \leq .01$

*** $\alpha \leq .005$

Unlike the unconditional means model (Model A in Table 3-7), model I estimates of the random intercepts are now conditional on age and JCQ decision latitude. With the addition of these between-subject (level 2) predictors the conditional component for the variance within-subjects has remained, as would be expected, essentially unchanged (10.5), however the component for the variation between-subjects has decreased from 8.4 to 6.7. The addition of the level 2 predictors has reduced the between-subjects variance by 20% (8.4-6.7/8.4 = .20), meaning, the addition of age and decision latitude accounts for 20% of the explainable between subject variance in HFPnu values. In summary, after dropping JCQ demands and social support from Model B, model I reduced the same amount of the between-subject variation.

## Model II: Random Intercept Model with Level One Effects: Diary Strain

Model II (Table 3-8) combines diary demands and control (as presented in Models C & D, Table 3-7) into diary strain based on the job strain hypothesis as previously discussed. Diary strain was centered at each subjects mean. Changes in diary strain are therefore relative to the subjects mean value so that they are being assessed based on their own level of strain throughout the 48-hour monitoring period. Centering influences the interpretation of the fixed intercept, which now represents the average of all subjects mean HFPnu at their mean diary strain (centered at 0), while controlling for position. Interpretation of the coefficients then proceeds as in more traditional linear modeling. The intercept (7.7) represents a subject's mean HFPnu while controlling for

all predictors in the model. From this value, increasing diary strain will reduce HFPnu by 0.24% for each increment of strain. In addition, reductions in strain from the subject's mean value will result in an increase of HFPnu by 0.24% for each increment of reduction. The standard errors are small for this estimate resulting in large t statistics, and a p value of

0 .08. In addition, adding diary strain as a random effect does not change the results or result in a significant random estimate. This indicates that the data are best fit with a random intercept model and that the estimated slope for the effect of diary strain is consistent between subjects. Since the range of possible strain scores equals 5 increments for all subjects, a total deviation for HFPnu of 1.2% is possible, which represents roughly 15% of the intercept value (7.7). Position, exertion and location (home vs. work) were also added to the model to control for their potentially confounding effects. Position was a significant predictor (in the expected direction), while exertion and location are not significant predictors in the model. Position and exertion are highly correlated ($r = .81$, $p = .001$) and this collinearity likely accounts for the lack of effect of exertion. Significant differences exist between the locations in the pooled sample for diary strain ($t = 6.6$; $p < .001$) with increased strain on while at work (Figure 3-3). Measuring each recording of diary strain most likely explains the differences between work and home locations in this group of subjects.

In testing model II, each fixed effect was tested as a random effect and determined to be insignificant, thus confirming that when considering all level 1 predictors, the coefficients are consistent between subjects and the model is allowing for random intercepts (associated with each subjects baseline HFPnu).

The model II estimates of random intercepts are conditional on all level 1 predictors entered into the model. With the addition of these level 1 predictors the conditional component for the variance within subject has changed (10.5 to 10.1). The addition of these level I predictors to the model reduced the within subject variation by 4% (10.5- 10.1/10.5).

Model III: Random Intercept Model with Level One Effects: Strain and Social Support

Model III (Table 3-8) combines diary strain as in model II with social support to test additional hypotheses. The intercept (8.6) represents a subject's mean HFPnu while controlling for all predictors in the model. In model III the coefficient for Diary Strain increases to .76. Therefore, when controlling for social support and an interaction between strain and support within subjects, increases in diary strain will reduce HFPnu by .76% for each increment of strain. The standard errors are small for this estimate, resulting in large t statistics and low p values. In addition, adding diary strain as a random effect does not change the results or result in a significant random estimate. Since the range of possible strain scores equals 5 increments for all subjects, a total deviation for HFPnu of 4.35% is possible, which is quite significant physiologically given that it represents roughly 50% of the intercept value (8.6%). Interestingly, the estimate for social support (-.26) also has a small standard error, large t statistics and a low p value. However, contrary to theory it indicates that with increasing social support there is a decrease in HFPnu. One possible explanation includes a positive correlation between diary reports of social support and cognitive demands ($r = .245$, $p = .001$). Cognitive

(mental) demands (not included in our assessment of job strain in this model) activate energetic systems and a withdrawal of vagal control. The negative main effect of social support may also indicate an activating personal interaction effect [39]. Social support is thought to buffer against the negative effects of job strain. To test the hypothesis of a buffer effect an interaction term for diary strain and social support was added to the model ($p = 0.06$). This can be interpreted as a blunting of the reduction of vagal activity during times of high strain, when there is also elevated social support. This interaction supports the hypothesis that social support acts as a buffer to the effects of a high strain situation with regard to exposure related cardiac vagal activity. With the interaction, any interpretation of the effect of social support or diary strain requires consideration of the interaction since the effect of each is partially dependent on the other. Therefore, with low diary strain, elevations in social support create some level of demand (perhaps a cognitive energetical / personal interaction based demand) that tends to drive down HFPnu. Using model-based predictions of HFPnu, we can illustrate the effect of this interaction (Figure 3-5). Diary strain is centered, so zero diary strain is interpreted as each subject's mean. HFPnu was then calculated, based on the following equations, for a series of diary strain measures with a range of -3 to 3, in three social support situations: low = 2, No Support = 0, High Support = 8. "No support" illustrates the effect of diary strain in a case of zero social support, however this is not possible based on the scale. It is only included to display the main effect of diary strain.

Eq. 1: HFPnu = 8.6 + (diary strain * -.76) + (social support * - .26) + (Interaction * .14)

If social support is removed from model III and diary strains as well as the social support X diary strain predictors remain, the positive effect of the interaction increases to

(.16; p = .04). Additionally, including the interaction term but removing the main effect of social support the effect of diary strain increases to -.84. Position, exertion and location (home vs. work) were added to the model to control for their potentially confounding effects. Position was a significant predictor (in the expected direction), while exertion and location are not significant predictors in the model for likely reasons already discussed.

In testing model III each fixed effect was tested as a random effect and determined to be insignificant. The model III estimates of random intercepts are conditional on all level 1 predictors entered into the model. With the addition of these level 1 predictors the conditional component for the variance within subject has changed (10.5 to 9.9). Therefore the addition of these level I predictors to the model reduced the within subject variation by 6% (10.5- 9.9/10.5).

## Model IV: Two Level Random Intercept Model

Model IV includes both level 1 and level 2 predictors as found in the previous two models (II and III). In this model (Table 3-8) we see that there is essentially no change in the significance or direction of the parameter estimates of either the level 1 or level 2 predictors. A slight decrease in diary strain is noted, which is likely due to the inclusion of decision latitude. Cross level interactions of diary strain and decision latitude as well as diary strain and psychological demands were tested and found to be insignificant. The random effects indicate that there is a decrease of both the between subject variance

(25%), and the within subject variation (6%). When adding the level 1 predictors to the model there was an additional 15% reduction in between subject variation.

Using the model IV to compute prediction scores for HFPnu allows comparison of the predicted values to the actual values. Table 3-9 compares the actual to predicted HFPnu values.

Table 3-9. Comparison between Model Predictions and Actual Measures of Normalized High Frequency Power

| *Comparison* | *HFPnu* | |
|---|---|---|
| | Actual | Predicted |
| Mean | 6.34 | 6.35 |
| Standard Deviation | 4.12 | 2.69 |
| Correlation | 0.70 | |

## Discussion

There has been little research about associations between questionnaire ratings of job strain during normal daily activities and cardiovascular control responses. The most interesting finding of this study is that psychosocial strain as measured on a diary during two days of monitoring exerts a significant within subject effect on cardiac vagal control. This finding occurs after controlling for position, exertion and age. Diary measures of strain were based on quantitative demands and decision authority. In addition, the Job Content Questionnaire (JCQ) was used to assess decision latitude at work. Thus, a single overall rating of each subject's permanent job, is also significantly associated with between subject vagal cardiac activity, also with concomitant control for age. The use of multilevel modeling has allowed us to test for the influence of psychosocial conditions within each subject while testing the between subject effects of age and JCQ assessed decision latitude. The measures of effect (Table 3-10) which are based on significant predictors in model IV (Table 3-8) demonstrate the influence of job strain indices as compared to well understood and assessed effects such as age and posture. The average HFPnu in our sample is 8.9%. The magnitude of change in high frequency power (normalized by total power) estimated by age is 6% in our sample of males age 35-59. This is determined by multiplying the age range in this sample (24 years) by the coefficient estimated for age (-0.25%). Similarly, calculating the magnitude of effect for JCQ measured decision latitude yields an effect of 6% (Range = 60 * Parameter Estimate = 0.10). Within subjects, the magnitude of effect for diary strain (3.3%) is similar to that of posture (4.2%).

Table 3-10. Estimates[a] for Measure of Effect based on Independent Variables for Normalized High Frequency Power (Sample Mean = 8.9%)

| *Independent Variable* | *Range of Effect* |
|---|---|
| Age | ± 6% |
| JCQ Decision Latitude | ± 6% |
| Diary Strain (DS) | ± 3.3% |
| Diary Social Support (SS) | ± 1.5% |
| Position | ± 4.2% |

[a]Estimates based on Significant Predictors in Model IV of Table 3-8

We are unaware of previous attempts to assess job stress using job strain hypotheses to assess cardiac vagal control (or HRV) with continuous ambulatory ECG monitoring over long time periods using responses to diary assessed strain in a general population based sample. Also, we are unaware that such research has been carried out while assessing both within and between subject job variations. Researchers from Japan have assessed the impact of work characteristics such as working hours, commute time and overtime on HRV measures during a 3-minute [40, 41] or 5 minute [42] single period in the morning of one day. Similar to standard observations of blood pressure in a clinic, this approach makes it impossible to measure the impact of work characteristics on changes in HRV parameters. This makes the measurement of HRV less robust for identifying differences in the contrasted groups and for testing the detailed work hypotheses proposed here. Drawing from experiences in the area of job strain and ambulatory blood pressure, it is understood that a subject's daily measurements of blood pressure often

show elevations before they appear in clinic based measurements. Abnormalities in the clinic which lead to a diagnosis tend to occur only after prolonged exposure and pathological changes associated with the disease in question [43].

There have been studies that have utilized continuous ambulatory monitoring during work shifts [33, 44, 45]. These studies were able to analyze 24-hour patterns of HRV and responses to items such as mood, work, rest and sleep. However, these studies only measure work as a dichotomous variable -- subjects are either at work or not at work. Clearly this lack of measurements pertaining to job characteristics prevents testing of hypotheses relating occupational stress to altered autonomic control and higher risk of heart disease.

Studies in occupational settings that have investigated both job stress and utilized continuous ambulatory monitoring to assess the impact of occupational stress on HRV parameters [46-49] have either not utilized the job strain hypothesis, or have not included measurements of within subject strain during the day, and therefore are difficult to directly compare to the present investigation.

In the Vrijkotte study[67], utilizing Siegrist's model of effort reward imbalance[68], subjects reporting a high effort-reward imbalance had significantly lower vagal cardiac control (measured with the time domain HRV parameter – RMSSD) and these effects were noted throughout the monitoring period on both the work and rest day. This carryover effect of vagal control measured with high frequency power is similar to our findings. In our data the average within subject correlation between high frequency power and RMSSD was $r = 0.85$, which is very similar to those of previous reports [69].

Hanson's findings partly confirm this study's results [Hanson E.K.S., 2001 #41]. Hanson tested the effect of work stress based on the effort reward imbalance theory on parasympathetic control of the heart with the high frequency power of HRV using seventy subjects. Multi-level analysis was performed with level one effects being HRV measures with in a person, and level two representing stable between-subjects differences. After controlling for time of day, "need for control" (a between subjects factor based on hostility and pressured response) had a negative effect on high frequency power of HRV (parasympathetic activity.) However, the ERI "need for control" scale has major similarities to response-based scales such as fatigue and exhaustion.

In testing the effect of vital exhaustion [51], Wantanabe found significant reductions in HFP measured during 5 minutes of supine rest in subjects with a high vital exhaustion score as compared to low and moderate scores [52]. It is difficult to utilize these findings with ours to make conclusions regarding temporal or causal relationships. The job strain hypothesis might suggest that environmental exposure to job strain can lead to exhaustion symptoms (need for control, vital exhaustion) that are associated with altered vagal control since these effects have been observed in demand control research. Hanson did not identify within subject factors as exerting a significant effect on HFP. We were able to identify both within and between subject differences utilizing HFP when normalized for the subjects own total power.

The main effect of social support in our analysis suggests a reduction in vagal cardiac control while the interaction of social support and diary strain supports previous demand – control – support hypotheses of social support acting as a buffer against the physiological reactions associated with stress [11]. As a buffer to strain support is predicted

to reduce the negative health or physiological effects of high job strain. There is confirmation for the engagement of social support to arouse the cardiovascular system as shown with increased ambulatory blood pressure [53, 54]. This is especially noticeable in social engagement that is confrontational [39]. Other studies have shown no cardiovascular effect with social support [55], or decreases in ambulatory blood pressure when family members provide support [56]. The main effect of social support on physiological reactions seems to vary in relation to the familiarity of the supportive individual, gender and other factors [57]. Laboratory support exists for the hypothesis of a buffering effect with numerous studies demonstrating an attenuation of mental stress reactions when support is provided [58]. Additionally, in a field study using ambulatory blood pressure monitoring Steptoe demonstrated that social support buffers the impact of episodic stress on cardiac reactions to daily stressors [59]. Based on our findings from models III and IV in Table 3-8, and the predicted effects of the level 1 diary strain, support and interaction coefficients presented in Figure 3-4 we feel we provide further evidence of a buffer effect of social support with job strain. In ambulatory, real life situations we provide evidence that social support can be engaging and reduce cardiac vagal activity. However, if strain increases, the effect of support prevents or reduces the cardiac vagal reduction effect of strain. In the lowest strain situations depicted in Figure 3-4, no support is associated with greater vagal (relaxation) control. However, during these times of low strain (low demands, and high control) if there is another person that you are speaking with the situation is changed to one of requiring energetical resources and activation. At the other end of the spectrum, with increased strain situations, having support is associated with increased overall vagal

activity. One limitation of Figure 3-5 is the assumption of unchanging social support for descriptive purposes.

Heart Rate Variability Regulatory Failures on the Pathway to Cardiovascular Disease

The approach utilized in this study, based in part on the significant body of literature that has investigated and confirmed a relationship between ambulatory blood pressure and job strain, attempts to assess the job strain hypotheses as related to job strain associated with elevated risk of cardiac disease and sudden cardiac death. Our study can be seen as an extension of cardiovascular output studies illuminating a potential causal linkage in a manner similar to that in studies of ambulatory blood pressure to basic physiological mechanisms governing cardiac function. Elevated ambulatory blood pressure readings, linked with work (and carries over to rest days) in high strain occupations, are also associated with increased risk of the pathological consequences of high blood pressure [27]. In the process of disease development this association occurs prior to elevations in clinic-based blood pressure, indicating the possibility of pre-clinical alterations in physiological control. Extending from this linkage, if sudden cardiac death of an electrophysiological or arrhythmic basis, is related to alterations in vagal cardiac control and psychosocial characteristics during the day within subjects and macro related occupational traits between subjects, then it is quite possible to hypothesize that pre-clinical alterations in cardiac regulatory activity related to psychosocial strain can increase susceptibility for sudden cardiac death. Additionally, Schnall indicates that the motivation for studying ambulatory blood pressure within the context of the job strain

hypothesis is the theory that the consistently observed associations between job strain and heart disease is, in part, mediated through elevations in blood pressure [32]. Lucini has recently reported progressive impairment in autonomic regulation in subjects with blood pressure values below the upper limit for normal [60], extending the possible relevance of our findings to including hypertension as a possible disease end point. Lucini indicates that cardiac regulatory changes associated with alterations in high frequency power, represent a spectrum of changes in cardiac autonomic control that may precede clinical hypertension. Then, the increased blood pressure independently results in a cascade of pathological changes resulting in heart disease. However, as the next section of the discussion will present, cardiac disease related to alterations in regulatory control does not have to be mediated through elevations in blood pressure, but can create risks in and of themselves, more likely in the way of sudden cardiac death or cardiac syndrome X.

When discussing the effects of these risk factors on cardiac disease I think it is important to consider the cardiac disease that is under review, and, perhaps more importantly, the casual pathway that is implicated either from the risk, or to the disease end point. In this discussion, I will consider three major pathways toward cardiac disease that are possibly implicated from our findings. The first is related to the findings associated with elevations in ambulatory blood pressure studies with the primary cardiac disease end point of coronary artery disease (CAD) and possibly myocardial infarction (MI). It is well established that hypertension is a risk for the development of CAD and MI. So, first, we must wonder to what extent alterations in cardiac regulatory control precede the development of hypertension and therefore contribute to the risk of CAD. In addition however, also through the blood pressure mechanisms, it has been shown that

reductions in cardiac vagal activity, which reduce heart rate variability, can concomitantly increase blood pressure variability (BPV)[61]. Extending from this physiological finding, Sloan et al. (1999) hypothesized that with this reduction in BPV, a greater stress in the vascular intimal walls increases the likelihood of atheroma formation in the pathological development of CAD. Such a hypothesis does not contradict these other associations (hypertension as a risk for CAD), but rather proposes additional, perhaps additive, factors to the multiple pathways by which CAD develops.

The second major pathway toward cardiac disease is that which leads to sudden cardiac death. SCD is defined as "...any near instantaneous, electrically-based, cessation of cardiac output in individuals with otherwise uncompromised circulatory function" [62]. SCD accounts for approximately 300,000 deaths in the U.S. annually [63]. Risk of SCD is estimated to have an annual risk of 0.1-0.2% in the general population aged 35 and older. The largest increase in risk occurs between the age of 40-65, and males are at greater risk than females. The age gradient is assumed to be present due to the increased risk of CAD during this time frame. However, individuals with the highest risk for CAD, which develop clinically diagnosed CAD with development of a myocardial infarction or perhaps cardiomyopathy, account for a minority of SCD in the population. Interestingly, the lowest CAD risk group accounts for the larger absolute number of SCD deaths despite a lower relative risk. Decreases in HRV and other HRV markers (such as HFP), when measured over 24 hours and for shorter periods, are consistently associated with elevated risk for SCD post myocardial infarction and in patients with chronic heart failure [15, 64]. Studies conducted on patients during 24-hour Holter recordings consistently demonstrate alterations in HRV parameters prior to arrhythmic or ischemic events [15, 64-67].

While there are consistent alterations, the pattern of the alterations is less consistent [15]. The inconsistencies may be the result of varying underlying cardiac pathological processes in study subjects. Without measuring HRV parameters, Lampert performed a laboratory investigation in 18 patients with implantable cardioverter-defibrillators (ICD) with known inducible but treatment responsive ventricular tachycardia [68]. The researchers evaluated the impact of mental stress (mental arithmetic and anger recall) on the inducibility of ventricular tachycardia (VT) in these patients. Eight patients received a defibrillator shock during the resting state and did not proceed to the mental stress test. For the remaining 10 patients who proceeded to the mental stress, induced VT was faster in 5 patients, and was more difficult to terminate in the other 5 patients (with 4 patients requiring a shock). Lampert concludes that mental stress, via autonomic arousal, may lead to sudden cardiac death through the facilitation of lethal arrhythmias [68].

It is possible that exposures to reduced vagal control (in addition to elevated blood pressure) as a response to job strain exposure can lead to alterations in global (autonomic nervous system) regulation and/or local (cardiac receptor) coupling with regulatory systems. Low total variability may not only reflect alterations in the global state of the autonomic nervous system, but also local changes in receptor morphology. Additionally, alterations in receptor coupling can lead to misinterpretations of HRV determined frequency components, leading to difficulty in the interpretation of results from studies of patients with structural heart disease.

Within the risk pathway of SCD is the possibility of vagal induced alterations leading toward cardiac syndrome X. Cardiac syndrome X (SX) refers to a condition of cardiac ischemia with a normal arteriogram [69]. Previously held assumptions regarding

the pathological processes have focused on the causes of ischemia, altered neural regulation and sensation in patients with SX, and increases in sympathetic tone. However, a recent study by Gulli et al. (2001) provides evidence of reduced cardiac vagal activity in 13 of 20 SX patients using HRV measures [70]. Similar to SX, Prinzmetal's variant angina is a condition caused by a reduction of myocardial blood flow due to coronary artery spasms in the absence of clinically significant coronary artery disease. Lanza et al. (1996) studied 23 patients with variant angina with 24-hour Holter monitoring and demonstrated a significant reduction in HFP in the 2 minute period prior to ST segment alterations indicative of ischemia suggesting that changes in autonomic tone, particularly vagal withdrawal, is a likely trigger of this form of angina [71].

Several overall limitations should be addressed in consideration of the significant findings for this study. In previous investigations, the independent variables shown to influence high frequency power between subjects have been age, sex, and physical fitness [17]. Study selection criteria assured that study subjects were male, between the ages of 35 and 59, generally healthy and not using medications known to affect HRV. Although level of obesity, cholesterol profiles, and smoking status information was not collected for subjects, the literature does not suggest a relationship between these factors and atypical HRV [72]. There is no evidence in the literature to suggest a relationship between decision latitude and physical fitness. It may be hypothesized that greater decision latitude is conducive to a lifestyle with greater potential for the pursuit of personal fitness. If so, this suggests that high decision latitude individuals might self-select toward better physical fitness. Validation of this possible association should be considered in future investigations.

It is also possible that other work environment exposures influence heart rate variability measures through their influence on autonomic nervous system function. Two possibilities include noise [48, 73] and air particulate exposure [74, 75]. Validation of these possible associations should be considered in future investigations.

As will be discussed in a related work [76], the pool of available subjects had previously committed to participation in the Seasons study, which involved a fairly complicated initial and follow-up protocol [29]. They were then asked to participate in our investigation which required a large time commitment, and a restriction of activities in order to allow the participant to be near home or a place which was conducive to relaxation during weekend evenings. In addition, subjects were required to wear somewhat intrusive hardware, daily diary and telephone contact regimens, and limited participation time scheduling options. Due to these constraints, subjects with high strain jobs may be less likely to participate in such studies. Additionally, the HMO based subject pool from which the majority of our subjects were recruited (n=26) reflects the population of central Massachusetts, and as such is largely white and middle-class. Our study is restricted to participants living in one relatively restricted geographic area of the country. In our desire to take fullest advantage of our study design we are looking at a considerable number of variables, creating a complex dataset. We believe that we have the expertise and experience to deal with this complexity. On the other hand, it is not possible to evaluate all possible variables obtained from the ECG signal, and it may be that we have overlooked a variable or variables of importance. The Seasons study also excluded individuals who could not speak English because virtually all of their study

instruments were developed for use in English-speaking populations and have been calibrated for use in those populations.

## Conclusions

Psychosocial strain as measured on a diary during two days of monitoring exerts a significant within subject effect on cardiac vagal control. Adding support to this finding is that it occurs after controlling for position, exertion and age. In addition, using the Job Content Questionnaire (JCQ) to assess decision latitude, thus, a single overall rating of each subjects' permanent job, is significantly associated with between subject vagal cardiac activity also with concomitant control for age with reduced decision latitude associated with reduced vagal tone. The use of multilevel modeling has allowed testing for the influence of psychosocial conditions within each subject while testing the between subject effects of age and JCQ assessed decision latitude.

## Literature Cited


1. Sauter S, Brightwell W, Colligan M.J., et al. The Changing Organization of Work and the Safety and Health of Working People. Cincinnati: National Institute of Occupational Safety and Health, 2002.
2. Schnall P, Landsbergis P, Baker D. Job strain and CVD. Annu Rev Public Health 1994; 15:381-411.
3. Schnall P L, Belkic K, Landsbergis P, Baker D. The Workplace and Cardiovascular Disease. Occupational Medicine: State of the Art Reviews. Vol. 15. Philadelphia: Hanley & Belfus, INC., 2000.
4. Karasek R A. The impact of the work environment on life outside the job. Cambridge: Massachusetts Institute of Technology, 1976.
5. Karasek R A. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 1979; 24:285-308.
6. Karasek R A, Theorell T. Healthy Work: Stress Productivity, and the Reconstruction of Working Life. New York: Basic Books, 1990.
7. Johnson J V, Hall E M. Job strain, workplace social support, and cardiovascular disease: a crosssectional study of a random sample of the Swedish working population. American Journal of Public Health 1988; 86:324-331.
8. Astrand N E, Hanson B S, Isacsson S O. Job demands, job decision latitude, job support, and social network factors as predictors of mortality in a Swedish pulp and paper company. British Journal of Industrial Medicine 1989; 46:334-340.
9. Falk A, Hanson B.S., Isacsson S.O., Ostergren P.O. Job strain and mortality in elderly men: Social networks, support, and influence as buffers. American Journal of Public Health 1992; 82:1136-1139.
10. Kompier MAJ, di Martino V. Review of bus drivers' occupational stress and stress prevention. Stress Medicine 1995; 11:253-262.
11. Johnson J V, Stewart W, Hall E M, et al. Long-term psychosocial work environement and cardiovascular mortality among swedish men. American Journal of Public Health 1996; 86:324-331.
12. 2001 Heart and Stroke Statistical Update. Dallas, Tx: American Heart Association, 2001.
13. Babuty D, Lab M. Mechanoelectrical contributions to sudden cardiac death. Cardiovascular Research 2001; 50:270-279.
14. Gronefeld G, Hohnloser S. What do implantable cardioverter/defibrillators teach us about the mechanisms of sudden cardiac death. Cardiovascular Research 2001; 50:232-241.
15. Lombardi F, Makikallio T.H., Myerburg R, Huikuiri H. Sudden cardiac death: role of heart rate variability to identify patients at risk. Cardiovascular Research 2001; 50:210-217.
16. Myerburg R, Spooner P. Opportunities for sudden cardiac death prevention:directions for new clinical and basic research. Cardiovascular Research 2001; 50:177-185.

17. Kristal-Boneh E, Raifal M, Froom P, Ribak J. Heart rate variability in health and disease. Scandinavian J of Work Environment and Health 1995; 21:85-95.
18. Bigger J.T. Jr. Spectral analysis of RR variability to evaluate autonomic physiology and pharmacology and to predict cardiovascular outcomes in humans. Encylopedia of Cardiology 1994:1151-1170.
19. Kamath M, Fallen E. Power spectral analysis of heart rate variability: A noninvasive signature of cardiac autonomic function. Critical Reviews in Biomedical Engineering 1993; 21:245-311.
20. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability standards of measurement, physiological interpretation, and clinical use. European Heart Journal 1996; 17:354-381.
21. Bloomfield P. Fourier Analysis of Time Series: An Introduction. New York: Wiley Series in Probability and Mathematical Statistics, 1976.
22. Berntson G.G., Bigger J.T. Jr., Eckberg D, et al. Heart rate variability: Origins, methods, and interpretative caveats. Psychophysiology 1997; 34:623-648.
23. Matsuzaki I, Mishimura A, Morita N. Autonomic nervous system activity changes due to shift work: an evaluation by spectral components of heart rate variability. J Occup Health 1996; 38:80-81.
24. Lenneman J, Backs R.W. The validity of factor analytically derived cardiac autonomic components for mental workload assessment. In: Backs R.W., Boucsein W, eds. Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum, 2000:161-176.
25. Meijman T.F., Kompier M. Busy business: how city bus drivers cope with time pressure, passengers, and traffic safety. Journal of Occupational Health Psychology 1998; 3:109-121.
26. Uusitalo A.L.T., Uusitalo A.J., Rushko H.K. Heart rate and blood pressure variability during heavy training and overtraining in the female athlete. Int J Sports Med 2000; 21:45-53.
27. Schnall P, Pieper C, Schwartz J.E., Karsek R.A., et al. The relationship between "Job Strain," workplace diastolic blood pressure and left ventricular mass index. JAMA 1990; 263:1929-1935.
28. Collins S M. Emerging methods for the physiological assessment of occupational stress. Work 2001; 17:209-219.
29. Merriam P.A., Ockene I.S., Hebert J.R., Rosal M.C., Matthews C.E. Seasonal Variations of Blood Cholesterol Levels Study: Study methodology. Journal of Biological Rhythms 1999; 14:330-339.
30. Karasek R A, Brisson C, Kawakami N, et al. The Job Content Questionnaire: An instrument for internationally comparative assessment of psychosocial job characteristics. J Occup Health Psychol 1998; 3:322-355.
31. Karasek R A, Gordon G, Pietroskovsky C, et al. Job Content Instrument: Questionnaire and User's Guide. Los Angelas, CA./Lowell, MA.: University of Southern California/University of Massachusetts, 1985.
32. Schnall P, Schwartz J.E., Landsbergis P, Warren K, Pickering T.G. A longitudinal study of job strain and ambulatory blood pressure: results from a three year follow-up. Psychosomatic Medicine 1998; 60:697-706.

33. Sloan R.P., Shapiro P.A., Bagiella E, et al. Effect of mental stress throughout the day on cardiac autonomic control. Biological Psychology 1994; 37:89-99.
34. Rottman J, Steinman R.C., Albrecht P, Bigger J.T. Jr., Rolnitzky L.M., Fleiss J.L. Efficient estimation of the heart period power spectrum suitable for physiologic or pharmacologic studies. Am J Cardiology 1990; 66:1522-1524.
35. Albrecht P, Cohen R.J. Estimation of heart rate power spectrum bands from real world data: Dealing with ectopic beats and noisy data. Computers in Cardiology 1988; 15:311-314.
36. Malliani A, Pagani M, Lombardi F, Cerutti S. Cardiovascular neural regulation explored in the frequency domain. Circulation 1991; 84:482-492.
37. Goldstein H. Multilevel Statistical Models. London: John Wiley & Sons, 1995.
38. Singer J.D. Using SAS PROC MIXED to fit multilevel models, hierarchical models and individual growth models. Journal of Educational and Behavioral Statistics 1998; 24:323-355.
39. Brondolo E, Karlin W, Alexander K, Bobrow A, Schwartz J.E. Workday communication and ambulatory blood pressure: implications for the reactivity hypothesis. Psychophysiology 1999; 36:86-94.
40. Sasaki T, Iwasaki T, Oka I, et al. Effect of working hours on cardiovascular autonomic nervous system functions in engineers in an electronics manufacturing company. Industrial Health 1999; 37:55-61.
41. Iwasaki T, Sasaki T, Oka I, Hisanaga N. Effect of working hours on biological functions related to cardiovascular system among salesmen in a machinery manufacturing company. Industrial Health 1998; 36.
42. Kageyama T, Nishikido N, Kobyashi T, Kuokawa Y, Kaneko T, Kabuto M. Long commuting time, extensive overtime, and sympathodominant state assessed in terms of short term heart rate variability among male while collar workers in the Tokyo megalopolis. Industrial Health 1998; 36.
43. Landsbergis P, Schnall P, Belkic K, Baker D, Schwartz J.E., Pickering T.G. Work stressors and cardiovascular disease. Work 2001; 17:191-208.
44. Kobyashi F, Furui H, Akamatsu Y. Changes in psychophysiological functions during night shift in nurses: influences of changing from a full day to half day work shift before night duty. Int Arch Occup Environ Health 1997; 69:83-90.
45. Kobyashi F, Furui H, Akamatsu Y, et al. Autonomic nervous system function in taxi drivers, International Conference of Behavioral Medicine, Copenhagen, Denmark, August, 1998.
46. Hanson E.K.S., Godaert G.L.R., Maas C.J.M., Meijman T.F. Vagal cardiac control throughout the day: the relative importance of effort-reward imbalance and within-day measurements of mood, demand and satisfaction. Biological Psychology 2001; 56:23-44.
47. Vrijkotte T.G.M., van Dornen L.P.J., de Geus E.J.C. Effects of work stress on ambulatory blood pressure, heart rate, and heart rate variability. Hypertension 2000; 35:880-886.
48. van Amelsvoort L.G.P.M., Schouten E.G., Maan A.C., Swene C.A., Kok F.J. Occupational determinants of heart rate variability. Int Arch Occup Environ Health 2000; 73:255-262.

49. Riese H. Job strain and risk for cardiovascular disease in female nurses. Amsterdam: Vrije University, 2000.
50. Siegrist J, Junge P.R., Cremer P., Seidel D. Low status control, high effort at work and ischemic heart disease: prospective evidence from blue collar men. Soc Sci Med 1990; 31.
51. Appels A, Mulder P. Excess fatigue as a precursor of myocardial infarction. European Heart Journal 1988; 9:758-764.
52. Wantanabe T, Sugiyama Y, Sumi Y, et al. Effects of vital exhaustion on cardiac autonomic nervous functions assessed by heart rate variability at rest in middle-aged male workers. International Journal of Behavioral Medicine 2002; 9:68-75.
53. Crowther J.H., Stephens M.A., Koss P.G., Bolen K.G. Behavioral predictors of blood pressure variation in hypertensives and normotensives. Health Psychology 1987; 6:569-579.
54. Roy M.P., Steptoe A, Kirschbaum C. Life events and social support as moderators of individual differences in cardiovascular and cortisol reactivity. J Pers Soc Psychol 1998; 75:1273-1281.
55. Kamarack T.W., Shiffman S, Smithline L, et al. Effects of task strain, social conflict, and emotional activation on ambulatory cardiovascular activity: daily life consequences of recurring stress in a multiethnic adult sample. Health Psychology 1998; 17:17-29.
56. Spitzer S.B., Llabre M.M., Ironson G.H., Gellman M.D., Schneiderman N. The influence of social situations on ambulatory blood pressure. Psychosomatic Medicine 1992; 54:79-86.
57. Uchino B.N., Capioppo J.T., Kiecolt-Glaser J.K. The relationship between social support and physiological processes: a review with emphasis on underlying mechanisms and implications for health. Psychological Bulletin 1996; 119:488-531.
58. Lepore S.J. Problems and prospects for the social support-reacitivty hypothesis. Ann Behav Med 1998; 20:257-269.
59. Steptoe A. Stress, social support and cardiovascular activity over the working day. Int J Psychophysiology 2000; 37:299-308.
60. Lucini D, Mela G.S., Malliani A, Pagani M. Impairment in cardiac autonomic regulation preceding arterial hypertension in humans: Insights from spectral analysis of beat-by-beat cardiovascular variability. Circulation 2002; 106:2673-2679.
61. Sloan R.P., Shapiro P.A., Bagiella E, Myers M.M., Gorman J.M. Cardiac autonomic control buffers blood pressure variability responses to challenge: A psychobiological model of coronary artery disease. Psychosomatic Medicine 1999; 61:58-68.
62. Spooner P, Priori S.G., Myerburg R. Spotlight on sudden cardiac death. Cardiovascular Research 2001; 50:173-176.
63. Myerburg R, Kessler K.M., Castellanos A. Sudden cardiac death. Structure, function and time depedence of risk. Circulation 1992; 85 Suppl 1:I2-I10.
64. Huikuiri H, Makikallio H, Airaksinen J, Mitrani R, Castellanos A, Myerburg R. Measurement of heart rate variability: A clinical tool or a research toy? J Am Coll Cardiol 1999; 34:1878-1883.

65. Kop W.J., Verdino R.J., Gottdiener J.S., O'Leary S, Merz C, Krantz D.S. Changes in heart rate and heart rate variability before ambulatory ischemic events. J Am Coll Cardiol 2001; 38:742-749.
66. Shusterman V, Aysin B, Gottipaty V, et al. Autonomic nervous system activity and the spontaneous initiation of ventricular tachycardia. J Am Coll Cardiol 1998; 32:1891-1899.
67. Galinier M, Pathak A, Fourcade J, et al. Depressed low frequency power of heart rate variability as an independent predictor of sudden death in chronic heart failure. European Heart Journal 2000; 21:475-482.
68. Lampert R, Jain D, Burg M.M., Batsford W.P., McPherson C.A. Destabilizing effect of mental stress on ventricular arrhythmias in patients with implantable cardioverter-defibrillators. Circulation 2000; 101:158-164.
69. Rosen S.D. The pathophysiology of cardiac syndrome X - a tale of paradigm shifts. Cardiovascular Research 2001; 52:174-177.
70. Gulli G, Cemin R, Pancera P, Mengegatti G, Vassanelli C, Cevese A. Evidence of parasympathetic impairment in some patients with cardiac syndrome X. Cardiovascular Research 2001; 52:208-216.
71. Lanza G.A., Pedrotti P, Pasceri V, Lucente M, Crea F, Maseri A. Autonomic changes associated with spontaneous coronary spasm in patients with variant angina. J Am Coll Cardiol 1996; 28:1249-1256.
72. Kuch B, Hense H.W., Sinnreich R, et al. Determinants of short period heart rate variability in the general population. Cardiology 2000; 95:131-138.
73. Tzaneva L, Danev S, Nikolova R. Investigation of noise exposure effect on heart rate variability. Central European Journal of Public Health 2001; 9:130-132.
74. Magari SR, Hauser R, Schwartz J, Williams PL, Smith TJ, Christiani DC. Association of heart rate variability with occupational and environmental exposure to particulate air pollution. Circulation 2001; 14:986-991.
75. Magari SR, Schwartz J, Williams PL, Hauser R, Smith TJ, Christiani DC. The association of particulate air metal concentrations with heart rate variability. Environmental Health Persectives 2002; 110:875-880.
76. Costas K, Karsek R.A. Recruitment Bias in Occupational Stress Studies with Complex Protocols. University of Massachusetts Lowell, 2003.

Figure 3-1. Cut Points for Diary Strain Determination

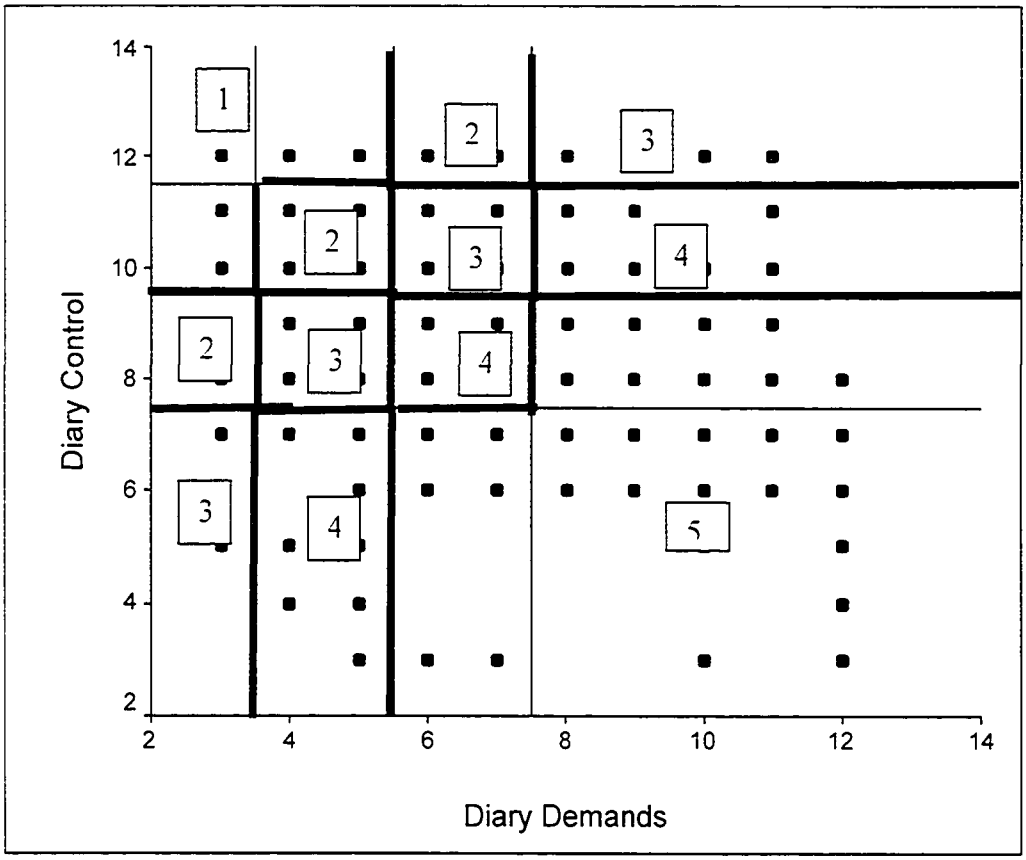


Figure 3-2. Individual Differences in High Frequency Power Response to Diary Strain

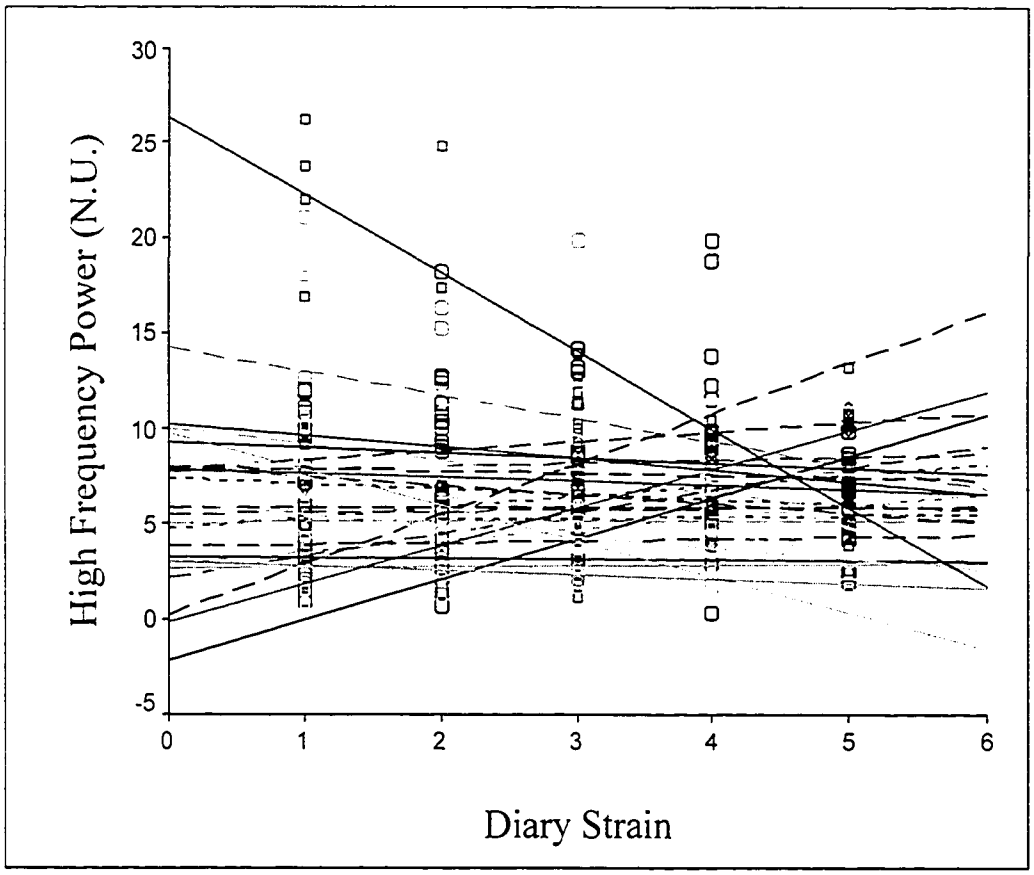


Figure 3-4. Pooled sample differences in Diary Strain while at Work vs. Home

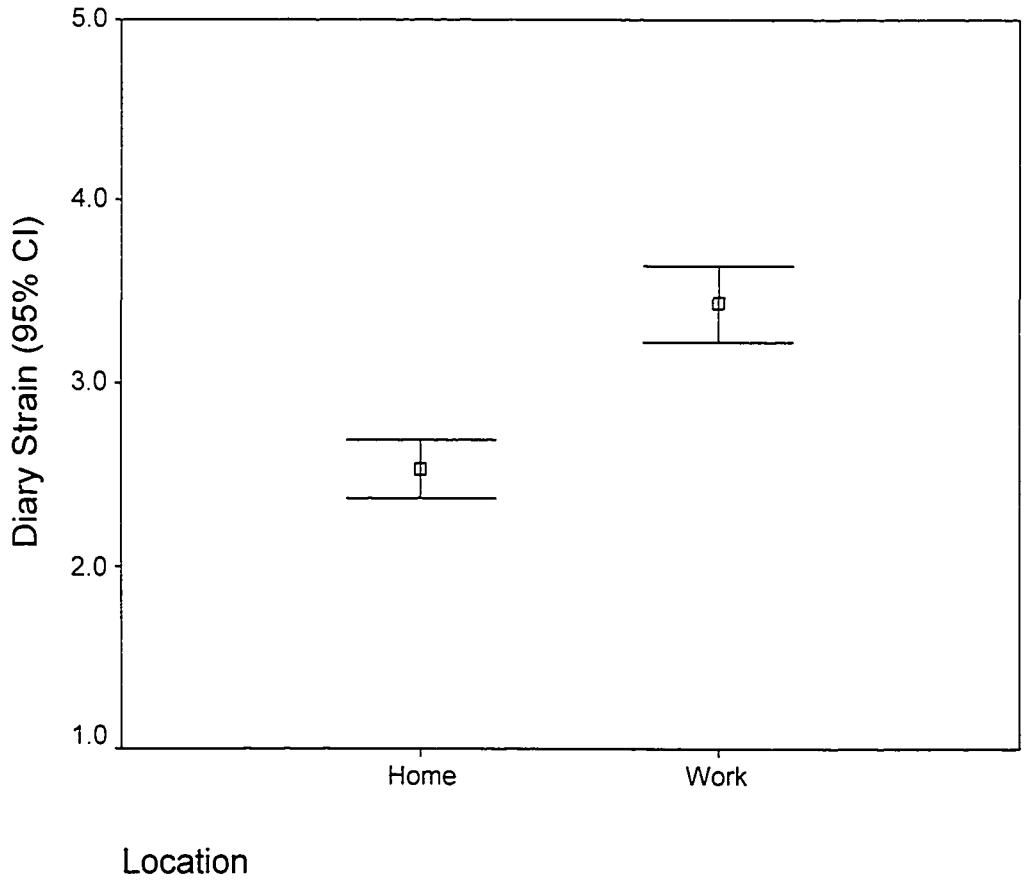
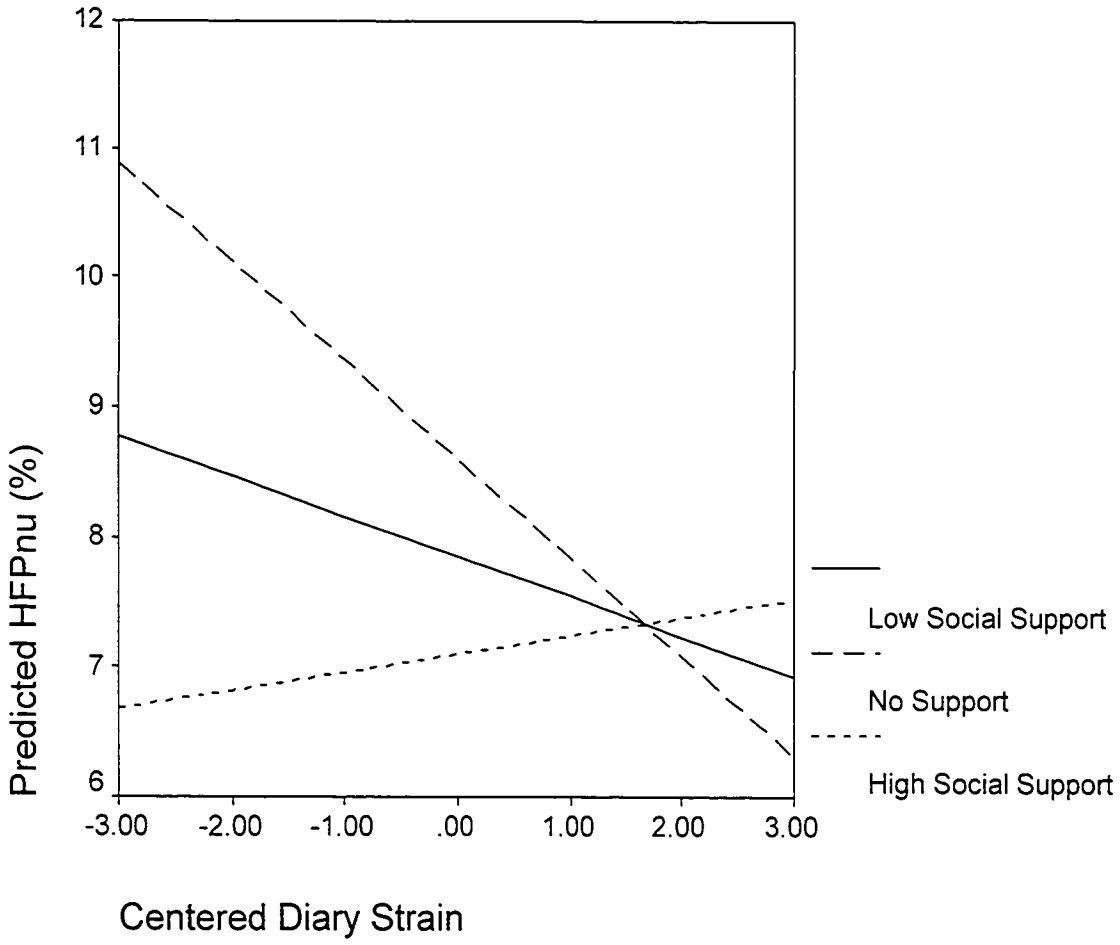


Figure 3-5. Diary Strain X Social Support Interaction: Predicted HFPnu



# CHAPTER FOUR

# JOB STRAIN, EXHAUSTION AND CARDIAC REGULATION

## Abstract

This paper presents methods and proposes hypotheses to test Karasek's Stress Disequilibrium Theory[1, 2] using data obtained through ambulatory electrocardiograph monitoring. The Stress Disequilibrium Theory links job strain with physiological system deregulation with systems principles implicating limitations in environmental control as a cause of restricted physiological control. Cardiac regulation was measured by components of electrocardiograph output: the heart rate variability based measures of high frequency power (HFP) which represents parasympathetic autonomic nervous system (ANS)-based control; the residual heart rate and the QT interval variability index (QTVI) as measures of sympathetic ANS activity. The paper provides a basic theoretical background and is then presented as two parts. Part I provides a validation of ECG indices of sympathetic activity, and Part II proceeds to utilize ECG data to analyze differences between job strain, and exhaustion groups in a population-based sample using three analytical approaches: autonomic coordination in a manner derived from but supplemental to Berntson[3], short term vagal variability and relationships between physiologic variables. Residual heart rate appears to provide a valid assessment of sympathetic activity. QTVI suffers from problems of data drop out during time associated

with elevated sympathetic activity due to movement artifact, but with estimated data imputed to fill these gaps it also appears to provide a valid index of sympathetic activity. The analyses conducted point towards reductions in variance of cardiac vagal activity in high job strain subjects, with further reductions in subjects reporting the highest levels of exhaustion. Additionally, a trend has been identified whereby differences exist in coordination of autonomic nervous system cardiac regulatory branches with differences in high strain subjects, and more severe alterations in those subjects reporting exhaustion. The findings of this paper provide preliminary support for the Stress Disequilibrium Theory, with exhausted subjects demonstrating more severe disequilibrium. Discussion is provided for each analytical method as well as an integration of the methods within the theoretical foundation, ultimately suggesting reductions in adaptive capacity as a function of reduced job control.

## Introduction

This paper presents methods and proposes hypotheses to test Karasek's Stress Disequilibrium Theory[1] using data obtained through ambulatory electrocardiograph monitoring. The Stress Disequilibrium Theory links job strain with physiological system deregulation with systems-theoretic principles, implicating, amongst other hypotheses, limitations in environmental control by the subject as a cause of, his/her restricted internal possibilities for physiological control and physiological deregulation [1]. We hypothesize that deregulation of physiological control increases the risk of exhaustion and disease from stress exposure. The ease of obtaining indicators of cardiac regulation over extended periods is a benefit of ambulatory electrocardiograph monitoring. Cardiac regulation was measured by components of electrocardiograph output: the heart rate variability based measures of high frequency power (HFP) which represents parasympathetic autonomic nervous system (ANS)-based control; the residual heart rate and the QT interval variability index (QTVI) as measures of sympathetic ANS activity. The paper provides a basic theoretical background, a validation of ECG indices of sympathetic activity, and then proceeds to utilize ECG data to analyze differences between job strain, and exhaustion groups in a population-based sample using three analytical approaches: autonomic coordination, short term vagal variability and relationships between physiologic variables.

## Background

Analysis of ambulatory electrocardiograph data (ECG) has been proposed for exposure and/or risk assessment in several recent studies of job stress [4, 5]. Despite a growing number of studies with these and similar research goals, few clear conclusions can be drawn because the exact methods for either monitoring or analysis of data remain unclear [6-16]. Part of the problem may be the lack of a complete theoretical platform providing clear guidance on the use of electrocardiograph data in quantifying job stress exposure. Our rationale for using HRV to measure exposure in job stress research is twofold[17]: first, evidence from cardiovascular epidemiology links reduced HRV to an increased risk of cardiovascular disease [4, 7, 8, 11, 13-16]; and second, evidence that components of HRV are related to autonomic nervous system based cardiac regulation [4, 5]. These are not mutually exclusive since stress based alterations in autonomic nervous system regulation have long been proposed as an etiologic mechanism of chronic disease [18, 19]. Numerous hypothetical mechanisms linking autonomic nervous system activity to a state of exhaustion as a stress-induced precursor to system breakdown in the process of disease development have been proposed [19-23], however, the specific physiological meaning of exhaustion have not often been clearly presented.

The attempt to link occupational exposure to already defined HRV based risks is similar to the approach utilized in psychological and psychiatry literature. Reductions in HRV have been identified in subjects that are depressed, anxious, paranoid, angry, or hostile in an attempt to address pathways of increased cardiovascular disease [24-28]. Exhaustion and similar conditions such as chronic fatigue, and over training are

associated with alterations in both neuroendocrine indices of autonomic nervous system function as well as alterations in autonomic nervous components of heart rate variability [21, 23, 29-35]. Finally, a third well-researched tradition in human factors and ergonomics research on workplace demand, rarely referred to in the cardiology literature clearly associates reduced heart rate variability with mental workload [36-46].

Traditional stress-related etiological models of disease development are "non-specific" in their predictions of relationships between exposure and disease. The best-known theory, Selye's General Adaptation Syndrome [19], was inspired by Selye's frustration as a student with medical school neglect of non-specific symptom information from patients. Selye's subsequent physiological research was based on broadly active hormone systems with cortico-steroids (cortisol) and pitutary-hypothalmic-andreno-cortical response (HPA axis). The 20th century's first stress response model, Cannon's "fight-flight" response (i.e., the rabbit's integrated arousal response when it sees a wolf)[47], implicated the full range of sympathetic autonomic response and adreno-medullary hormone (adrenaline) responses (SAM axis) in the stress process. An implication of Cannon's work is that co-ordination of physiological sub-system response to demands is crucial to organism survival [1]. McEwan's recent "Allostatic Load" extension of Selye's formulation (McEwan, 1998) looks at a summation of chronic stressor loads and response exhaustion. Again, it provides no specificity of exposure response or for health outcome prediction; stress is a generalized risk factor for physiological deregulation and disease.

The use of HRV solely in the manner provided in the context of cardiovascular epidemiology and traditional stress response models may ignore the influence of different

job stress exposure dimensions. Stress on the job would of course be important since it is societies most highly prioritized sphere of life and since it is incorporating an increasingly larger portion of individual's (and even adult family member's time) [48]. The predominant model representing job stress exposure includes two dimensions of the environment, demands and control [2, 49, 50]. Epidemiological literature of job stress has consistently demonstrated stronger associations between low control than between job demands and disease outcomes. Such findings lead to questions of how the concept of control fits with current stress physiology models of chronic disease.

Psychophysiology and human factors applications of HRV have been more attentive in identifying specific responses to arise from specific exposures. Studies in these areas have utilized the "Doctrine of Autonomic Space" initially proposed by Berntson (1991) [3] to specify autonomic nervous system response patterns. Berntson and his colleagues tested and demonstrated modes of autonomic nervous system activation and withdrawal that demonstrated the ability of the sympathetic and parasympathetic branches to act independently [3, 51]. Using this framework studies have attempted to identify ANS response patterns that are specific to the type of mental demand [37, 38, 52-54]. At this time this approach has only been applied in areas focused on mental and cognitive demand exposures and to our knowledge this framework has not been applied in an attempt to identify response patterns in job stress research. Demand control research using physiological and biological measures has attempted to distinguish patterns of responses to exposures. There is some evidence using neuroendocrine measures that exposure to high demands and low control promotes sympathoadrenomedullary and

pituitary adrenocortical axes, while an enjoyable self paced task promotes only sympathoadrenomedullary activation [23].

At this time it is important to point out again, that all of the aforementioned approaches are based on traditional stress physiology models proposing that prolonged exposure to sympathetic arousal (stressors or environmental demands) which result in overworked, or perhaps "exhausted" physiological capabilities which, when linked to cardiovascular epidemiology, may be related to increased risk of disease. As such, these approaches are based on exposure to demands. It is quite possible that using HRV methodologies based only on this theoretical model does not identify all possible effects of exposures related to job strain. Working within the traditional stress response framework studies thus far employ a restricted number of methods to assess – what might actually be a dynamic – exposure response. Without a theory of dynamic physiological changes related to stress response, time aggregation of data may potentially collapse useful information obtained from the HRV data thus limiting the magnitude of the complex data without knowing the impact this might have on identifying an exposure response. Also, from the tradition dating back to Cannon [47], there has not been a model from which we can hypothesize the physiological impact of exposure to environments with low control that is unique from the hypothesized reactions to demands. To develop new approaches based on physiological alterations to low control requires a new theoretical model that allows hypothesis generation and testing.

To address this issue Karasek has proposed the Stress Disequilibrium theory to further account for the fact that "lack of control," as a dimension of stress, appears to be the most consistent contributor to stress-related chronic disease. In the Stress

Disequilibrium Theory the pathogenic process is hypothesized to begin with stress-induced acute deregulation of physiologic control [1]. External demands (as stressors) are proposed to require the ordering of chaos to maintain a stable, homeostatic environment as the organism meets the energy requirements associated with the external demands [1]. Internal homeostasis is critical for higher mammals and represents the platform for complex functioning and interaction of physiological systems. Based on this requirement, external demands call for internal coordination loads (allostatic). External lack of control is proposed to increase coordination difficulty of internal physiological coordination by disrupting parameters. Control over sub-systems creates and organizes the high-level organism response to turn the sub-systems disorganized random activity into ordered work. This is the physiological work at the systems level and it is the essence of the adaptive action in the internal environment that allows the organism to maintain stability. Over the long-term, the deregulatory effects of chronic stress exposures must be compensated for by long-term regulatory shifts. These shifts can reduce adaptive capacity, and thereby increase the risk of morbid events [1].

The action of physiological control systems is cyclical. Cycles of organization and ordered work deplete the organism's control capacity, and must be followed by periods of rest, cut off and protected from environmental challenges during which equilibrium (of disorder with maximum entropy under the 2$^{nd}$ law of thermodynamics) is restricted. This equilibrium must be achieved before the next cycle of ordered work is possible. It is the ordered energy / disordered energy cycle (a "thermodynamic model," consistent with such a proposition by Recordati [55]) that Karasek claims supports understanding of system level control capacity reduction that accompany low external control as restricting

degrees of freedom of the physiological system's adjustment in long term context's leading to chronic disease.

This theory represents a systems-based understanding of disease development and prevention [1]. It implies understanding of the human physiological system's natural abilities to function as an integrated system, of understanding system limitations under complex societal demands and of the attendant physiological warning signals. This approach differs from the currently popular medical research thrust toward molecular-level, reductionist understandings of the disease process.

One approach to the assessment of autonomic nervous system coordination has the "Doctrine of Autonomic Space" [3]. In a series of studies it has been demonstrated that a variety of modes of control exist for the autonomic nervous system. For example, in the most commonly discussed pattern of cardiac response, to elevate heart rate, sympathetic activity increases and parasympathetic activity decreases. Similarly, to reduce heart rate sympathetic activity decreases and parasympathetic activity increases. The most basic understanding of autonomic nervous system control is as a reciprocally coupled system with activation and inhibition of the two branches (sympathetic and parasympathetic) being coupled in their attempt to control visceral responses to the environment. This reciprocal mode of control represents one of many possible coordination patterns of the autonomic nervous system. Berntson et al. predict that different people, under various circumstances, or conditions may have a variety of mechanisms that the branches of the autonomic nervous system respond to stressors in order to maintain homeostasis [51]. It has been demonstrated that during mental stress due to arithmetic tasks a reciprocal pattern is identified, however with an illusion task subject

responses tended to promote parasympathetic activation with no change in sympathetic activity [52]. To date the use of Bernston's categorization has been limited primarily to lab based psychophysiology and human factors studies in an attempt to improve the specificity of determining workload and distinguishing response patterns between sets of subjects [37, 38, 52, 54, 56].

We present a set of methods to analyze data obtained from an ambulatory electrocardiograph signal to test aspects of the stress disequilibrium theory. These methods make use of conventionally determined HRV indices. The first aim is to propose how physiological deregulation as related to job control and eventually exhaustion might appear and to generate hypotheses to be utilized in studies of the effect of job control on physiological control. An additional aim is to demonstrate defective coordination of physiological systems as determined from HRV indices in high strain subjects, and to a greater degree in those subjects with complaints of exhaustion. After stating the overall methodology, the paper is then presented in two parts. Part I validates sympathetic response arousal measure and Part II investigates ECG data to analyze differences between job strain, and exhaustion groups using three analytical approaches: autonomic coordination in a manner derived from but supplemental to Berntson[3], short term vagal variability and relationships between physiologic variables.

## Methods

### Data Collection

A sample of healthy employed men (n = 26) between 35-59 years of age, all members of a community based health maintenance organization, were recruited from an ongoing investigation of seasonal variance in cholesterol (Seasons) [57]. An additional sample (n=10) of healthy employed men (35 – 59 years of age) was recruited from the University of Massachusetts Worcester Medical Center's stress reduction clinic. Details of the subject population and recruitment is presented elsewhere [58, 59]. Subjects agreeing to participate in the study provided informed consent prior to being asked to complete questionnaires or being connected to the Holter monitor that was approved by the Fallon Community Health Plan and the University of Massachusetts Medical Center Institutional Review Board for Human Subjects. The monitoring protocol utilized Holter monitors (Marquette 8500) worn in the workplace and at home to collect 48 hours of continuous ECG recordings beginning in the morning of (prior to) a workday. Subjects were required to the time during the monitoring period on a diary indicating changes in the social situation, such as start work, start lunch, return to work, end work, eat dinner, etc. These times were utilized to construct sociological periods (Table 4-1). Subjects removed the Holter monitor in the morning 48 hours after having had the monitor applied. Subjects were provided with a prepaid envelope to mail the Holter monitor and the diary to the study team.

## Data Processing

Holter data tapes were analyzed for ventricular arrhythmias and RR-interval variability at the Research Holter Laboratory at Columbia University by published methods [45]. In summary, the dependent variables of this study were computed from 5-minute epochs throughout the 48-hour period. The 48-hour records were digitized using a Marquette 8000 scanner, submitted to Marquette algorithms for QRS identification and editing, and then downloaded to a Sun workstation for further editing using algorithms developed at Columbia University. The power spectrum was computed as previously described [60, 61]. A fast Fourier algorithm was utilized to compute the power spectrum in the very low frequency (VLF < 0.04 Hz), low frequency (LF, 0.04 – 0.15 Hz), and high frequency (HF, 0.15-0.40 Hz) spectra. Total power was calculated as the sum of very low, low and high frequency power. The normalized high frequency power (HFP n.u.) was computed within each subject by as the percentage of total power that was in the high frequency range for each 5-minute epoch. From the series of epochs within each subject linear regressions between the natural logarithm of high frequency power and heart rate were computed to determine residual heart rate as initially proposed by Grossman (1987) and further utilized and validated by Backs (1995, 1998, 1999, 2000) and Lenneman (2000) [37, 38, 53, 62, 63]. The digitized ECG signals were then downloaded to a PC and submitted to algorithms developed at the Johns Hopkins School of Medicine, Department of Cardiology. QTVI was computed by the methods of Berger, as reported in detail elsewhere [64]. In summary, R waves were detected with an automatic peak detection algorithm after which the operator provided a template QT interval by selecting the beginning of the QRS complex and the ending of the T wave for one beat. The algorithm

then calculated the QT interval of all following beats by determining how much the ST segment and T wave was stretched or compressed in time so as to match the template. Due to the long duration of monitoring as part of our protocol, each subject's template QT interval was updated for every hour of monitoring. An evenly sampled time series from the QT and RR time series was obtained with a sampling frequency of 4 Hz. Long delays in the QT interval time series resulting from noisy signals, artifact or ectopic beats that could not be filtered were eliminated from analysis, and if more than 25% of the data was eliminated for an epoch, the entire epoch was eliminated. While eliminated epochs were automatically detected their location in the time series was identified and presented to the user for visual inspection.

QTVI was calculated for each epoch as the log ratio of QT interval variance and RR interval variance, both corrected by their respective squared mean:

$$QTVI = \text{Log}\,[(QTV/QTM^2)/(HRV/HRM^2)]$$

## Dependent Variables

### *Heart Rate*

Heart rate (chronotropy) has probably been used more often as a measure of mental workload than any other measure ([62]). While it is true that heart rate changes with the changing activity of the autonomic nervous system, these changes reflect the global state of the autonomic nervous system and not the specific actions of the autonomic nervous system branches. Based on the "doctrine of autonomic space" we know that the autonomic nervous system does not always follow a reciprocal pattern and therefore the

use of HR alone is a limited measure for determining autonomic nervous system activity [37].

*High Frequency Power*

Frequency domain analysis (spectral analysis) of HRV decomposes the time series into sinusoidal components (Bloomfield 1976, Box 1994). Power (variance) is plotted as a function of frequency and the analysis computes power in defined frequency regions. The spectral analysis reveals the frequency of variations in the time series, and also the amount of power (variance) within certain frequency ranges. The high frequency power (HFP) spectrum (.15 to .4 Hz) is thought to be synchronous with respiratory sinus arrhythmia (RSA) (Berntson, 1993). RSA refers to the increase and decrease of RR intervals in phase relationship with inspiration and expiration (Berntson, 1993). RSA is present when the heart is controlled by the parasympathetic branch of the autonomic nervous system. HFP nu (HFP normalized for total power) increases in proportion to parasympathetic activity (Berntson, 1993; Task Force, 1996).

*Residual Heart Rate*

Residual heart rate has been proposed as a relative index of sympathetic activation by making use of the expected relationship between vagal activity and heart rate. From knowing these two values for each subject over 48 hours we can estimate the vagal predicted heart rate. Positive residuals indicate that heart rate is higher than estimated by vagal activity, and sympathetic activity is proposed as the major source of this increase.

Conversely, when residuals are negative heart rate is lower than estimated by vagal activity and therefore sympathetic activity may be reduced. Weaknesses of the residual heart rate as an indicator of sympathetic measures have been discussed elsewhere [38].

Lenneman and Backs (2000) have provided evidence that a factor analytic derived sympathetic factor (determined by the residual heart rate) is highly correlated within subjects with pre-ejection period [62]. Pre-ejection period (PEP) is determined by impedance cardiography and is considered the gold standard of non-invasive cardiac sympathetic activity based on a series of blockade studies [3]. Pearson product-moment correlations were computed for the residual heart rate and PEP. Correlations were computed within subjects. The within subject correlations were computed across ten tasks and five baselines for all subjects. The residual heart rate was correlated with PEP in 14 of 20 subjects ($.68 < r < .86$, $p < .05$).

*QT Interval Variability Index*

The QT interval represents the time for repolarization of the ventricular myocardium and the QT variability index (QTVI) has been proposed as a measure of temporal repolarization lability [64]. The QTVI has been shown to predict risk of sudden cardiac death [64, 65], and has been shown to be a marker of sympathetic nervous system activation [66]. Elevated QTVI has also been linked to certain emotional and mental health disorders such as panic disorder and depression [67].

The QT interval is known to vary, in part, as a function of the RR interval (or heart rate). The QTVI measures the variability of the QT interval after controlling for the

known effect of the RR interval. in essence measuring the amount of variance in the QT interval that is in addition to that of the RR interval. Greater amounts of sympathetic nervous system activation lead to increases in QT interval variation that is independent of the RR interval [42]. Yeragani and colleagues sought to test the hypothesis that QTVI is mediated by sympathetic activity in normal subjects [66]. In 19 subjects QTVI was significantly elevated after a change from supine to standing postures (sympathetic activation) as well as with pharmacological stimulation of $Beta_1$ and $Beta_2$ (primarily cardiac) sympathetic receptors with isoproterenol [66]. These findings occurred independent of changes in QT duration and therefore thought to represent a direct sympathetic effect on the ventricular myocardium influencing repolarization. Pohl (2001) measured QTVI in a group of 6 panic disorder patients and 11 controls during isoproterenol infusions. All subjects demonstrated significant elevations in QTVI during pharmacological sympathetic activation, with greater increases occurring in the panic disorder patients [68].

## Sociological Periods and Diary Entries

During the monitoring period subjects were asked to complete a diary to allow linkage of ECG data to daily events. Subjects were required to enter the time during the monitoring period on a diary indicating changes in the social situation, such as start work, start lunch, return to work, end work, eat dinner, etc. These times were utilized to construct sociological periods (See Table 4-1). These periods of the day allow control for potential diurnal rhythm variations (circadian rhythm) that may be related to cardiovascular function. However, they allow for a more precise understanding of the

social determinants of daily activity effects, which may be the most salient component [69, 70]. Subjects removed the Holter monitor 48 hours after having had the monitor applied. They were provided with a prepaid envelope to mail the Holter monitor and the diary back to the study team. Subjects removed the Holter monitor in the morning 48 hours after having had the monitor applied.

Table 4-1. Sociological Periods

| *Day* | *Recording* | *Diary Social Activity and Time of Recording* |
|---|---|---|
| Work | 1 | Morning at Work |
| | 2 | Lunch |
| | 3 | Afternoon at Work |
| | 4 | After Work before Dinner |
| | 5 | Dinner until Sleep |
| | 6 | Sleep |
| Rest | 7 | Awake on Rest Day |
| | 8 | Time that corresponds to Morning at Work |
| | 9 | Time that corresponds to Lunch at Work |
| | 10 | Time that corresponds to Afternoon at Work |
| | 11 | After 10 until Dinner |
| | 12 | Dinner until Sleep |
| | 13 | Sleep |
| | 14 | Awake on Second Rest Day – remove monitor |

Subjects were also asked to answer diary questions at 16 times during the monitoring period. The diary was a modification of the one used in the Cornell Ambulatory Blood Pressure studies [71, 72]. Diary questions were utilized to ascertain information regarding the subjects' position, activity, location, demands, control, social support, and time of day. Times entered for the diary were utilized to match the 20-minute period prior to each diary entry to the same 20-minute period for the ECG recording. The ECG data during this 20-minute period was aggregated (average of the

four 5 minute epochs) and linked to the appropriate diary data.

## Job Strain

Assessment of job strain was conducted using three assessments, the Job Content Questionnaire, Diary questions on the Workday, and imputed demand and control data from each subjects' three digit occupational code. The exact method utilized has been described in detail in Chapter Two. High strain jobs were classified using tertiles from the U.S. national sample with subjects in the three tertiles associated with the lowest weighted decision latitude and highest weighted psychological demands categorized as high strain. This method of classification yielded 13 high strain subjects, 23 low strain subjects.

## Exhaustion

To categorize subjects as exhausted two questions from the Job Content Questionnaire were utilized ("At the end of the day, I am completely exhausted mentally and physically" and "I am unusually tired in the morning"). Responses were scored as, 1 = hardly ever, never; 2 = sometimes; 3 = most of the time. To consider the most extreme cases of exhaustion only subjects responding "most of the time" to both questions were categorized as being exhausted (n = 4). Of the four subjects categorized, three are also high strain. The exhausted subjects (mean age (S.D.) = 39(3)) are significantly younger than the non-exhausted subjects (48 (5)).

A variety of analyses have been conducted utilizing the dependent variables obtained with the processing of the ECG signal combined with our analysis of subjects' occupations and activities during the monitoring period. These analyses have been conducted as a first step towards understanding possible methods of utilizing ECG data to test the Stress Disequilibrium Theory, and therefore as a bridge between theory and practice, has focused on graphic and pattern based analytical approaches as opposed to hypothesis testing. The remaining sections of this paper are divided into two parts. Part I presents a validation within this data set for sympathetic measures; and, Part II presents analytical steps in our development of approaches with practical to theoretical linkages and hypothesis developments as appropriate.

## Part I: Validation

To test hypotheses related to cardiac regulatory coordination we must be capable of obtaining data on at least two regulatory systems. These systems need to coordinate in order to provide effective organism level function. We have obtained ECG measurement data that quantifies both the sympathetic and parasympathetic branches of the autonomic nervous system. In this section we present an analysis of this data to internally validate the use of these markers. Variables obtained from the electrocardiograph include; heart rate, high and low frequency power of heart rate variability, and the QT interval variability index. The normalized high frequency power (HFPnu) and residual heart rate were calculated after initial data processing. Normalized high frequency power is being utilized as an indicator of parasympathetic nervous system activity [73, 74]. The QTVI [66] and residual heart rate [62, 63] have been proposed as markers of sympathetic nervous system activation.

### Validation Analysis Methods

Validation was based on three graphical analyses and correlations. First, comparisons were made between sleep and wake periods for each of the sympathetic markers. Known differences exist in cardiac control during these distinct physiological states [75]. Next, we assessed differences in the variables based on posture reported during diary assessments with a known effect of posture on sympathetic activation. And finally, we present differences in the dependent variables using measures of exertion (both

physical and psychological) and we include communication with other individuals all as known sympathetic activators [55, 75]. Posture was entered on the diary as either reclined, sitting or upright. Exertion was calculated based on answers to questions about physical activities (heavy work, light work, walking, gardening). Figure 4-1 provides the algorithm for computing exertion scores and the based on diary responses. Exertion based on physical activity was categorized established with consideration to estimated metabolic equivalents [76]. For the physical demanding activities the estimated number of kilo-calories per minute for a 150 pound male are included in parentheses after the description [76]. While the absolute value of these estimates will change with body size, the values relative to each other will not change. Figure 4- 2 illustrates the number of diary periods at each exertion level. To assess the impact of communication the diary entry "number of people talking with" was utilized to create 5 groups (0 = no communication; 1 = communicating with 1 person; 2 = communicating with 2 people; 3 communicating with 3 people; 4 = communicating with 4 or more people). Figure 4- 3 illustrates the number of diary periods categorized within each personal interaction group. Psychological demands were calculated based on diary questions similar to those used in the Job Content Questionnaire and was utilized to create categories based on quartiles within the sample, with sleep periods coded as 0 (0 = no psychological demands; 1 = low to 4 = high). Figure 4- 4 illustrates the number of diary periods categorized for each psychological demand score. Finally, in an attempt to account for all of these sympathetically activating activities we calculated the sum of exertion, communication and psychological demands as the "Sum of Demands." The distribution for the Sum of

Demands is depicted in Figure 4- 5. Aside from sleep (Sum of Demands = 0; there is a fairly even distribution.

## Validation Results

### *Loss of Data: QTVI*

Before continuing with the validation analysis we addressed the presence of selective drop out of QTVI data during active periods of the monitoring protocol. The measurement of QTVI is more sensitive to noisy ECG signals during times of higher physical movement. The QTVI requires determination of the end of the T wave so a small amount of noise, relative to the amount of noise tolerated for determining the R wave (for RR interval determination), leads to an inability to determine the QT interval and thus the QTVI. Figure 4- 6 displays a very noticeable drop in the percentage of QTVI data during active sociological periods. Sociological periods 1 through 4 and 8 through 11 are the active times of the work and rest day, respectively. Whereas, period 5 and 12 are late evening, and 6 and 13 are sleep periods. This selective drop out during periods that are more likely to be associated with elevated sympathetic activation can create underestimations of QTVI assessments of sympathetic activation if physical movement accompanies this activation. Figures 4- 7 and 4- 8 displays the noticeable drop in available QTVI data due to position and exertion, respectively. To address this problem, in addition to utilizing QTVI as previously discussed, we created a variable, QTVI (Imputed). We imputed a single value into missing QTVI data points. The

determination of which value to impute was based on each individuals QTVI range and the assumption that data was missing due to some degree of physical movement, and therefore is potentially activating to the sympathetic nervous system. Since sympathetic activation is hypothesized to increase QTVI (See Page 169) we imputed each subjects $97^{th}$ percentile QTVI value to replace their missing values. The validation analysis then preceded with both the original QTVI and the QTVI (Imputed) variable.

*Results*

Figures 4- 9 to 4-21 graphically display differences in the QTVI, QTVI (Imputed) and residual heart rate for the entire sample between sleep and wake, different postures, and the various demands. Table 4-2 provides the correlation matrix between the QTVI and residual heart rate with our indicators of environmental demands that require sympathetic activation using the Spearman's rank order correlation procedure.

Table 4-2. Validation Correlation Matrix

**Correlations**

| | | | QTVI | QTVI (Imputed) | Residual Heart Rate | Position | Exertion | Personal Interaction | Psychological Demands | Sum of Demands |
|---|---|---|---|---|---|---|---|---|---|---|
| Spearman's rho | QTVI | Correlation Coefficient | 1.000 | .888** | .222** | .168** | .126* | .121 | .188** | .194* |
| | | Sig. (2-tailed) | . | .000 | .000 | .003 | .025 | .152 | .001 | .021 |
| | | N | 325 | 325 | 325 | 320 | 316 | 142 | 325 | 142 |
| | QTVI (Imputed) | Correlation Coefficient | .888** | 1.000 | .342** | .267** | .222** | .166* | .195** | .282** |
| | | Sig. (2-tailed) | .000 | . | .000 | .000 | .000 | .011 | .000 | .000 |
| | | N | 325 | 501 | 493 | 494 | 490 | 236 | 501 | 235 |
| | Residual Heart Rate | Correlation Coefficient | .222** | .342** | 1.000 | .398** | .323** | .436** | .333** | .566** |
| | | Sig. (2-tailed) | .000 | .000 | . | .000 | .000 | .000 | .000 | .000 |
| | | N | 325 | 493 | 493 | 486 | 482 | 231 | 493 | 230 |
| | Position | Correlation Coefficient | .168** | .267** | .398** | 1.000 | .629** | .549** | .477** | .795** |
| | | Sig. (2-tailed) | .003 | .000 | .000 | . | .000 | .000 | .000 | .000 |
| | | N | 320 | 494 | 486 | 494 | 488 | 235 | 494 | 234 |
| | Exertion | Correlation Coefficient | .126* | .222** | .323** | .629** | 1.000 | .459** | .551** | .851** |
| | | Sig. (2-tailed) | .025 | .000 | .000 | .000 | . | .000 | .000 | .000 |
| | | N | 316 | 490 | 482 | 488 | 490 | 235 | 490 | 235 |
| | Personal Interaction | Correlation Coefficient | .121 | .166* | .436** | .549** | .459** | 1.000 | .492** | .760** |
| | | Sig. (2-tailed) | .152 | .011 | .000 | .000 | .000 | . | .000 | .000 |
| | | N | 142 | 236 | 231 | 235 | 235 | 236 | 236 | 235 |
| | Psychological Demands | Correlation Coefficient | .188** | .195** | .333** | .477** | .551** | .492** | 1.000 | .882** |
| | | Sig. (2-tailed) | .001 | .000 | .000 | .000 | .000 | .000 | . | .000 |
| | | N | 325 | 501 | 493 | 494 | 490 | 236 | 501 | 235 |
| | Sum of Demands | Correlation Coefficient | .194* | .282** | .566** | .795** | .851** | .760** | .882** | 1.000 |
| | | Sig. (2-tailed) | .021 | .000 | .000 | .000 | .000 | .000 | .000 | . |
| | | N | 142 | 235 | 230 | 234 | 235 | 235 | 235 | 235 |

**. Correlation is significant at the .01 level (2-tailed).

*. Correlation is significant at the .05 level (2-tailed).

*Validation Discussion*

This analysis demonstrated that both measures provide, at least in a basic sense, an indication of sympathetic nervous system activity. The impact of QTVI drop out is most apparent with the observation of Figure 4- 9, where there is a noticeable drop in the QTVI at higher exertions despite a trend in the expected direction prior to the higher exertion scores. As evident from Figures 4- 7 and 4- 8, there is greater loss of QTVI data in greater sympathetic activities – and significant correlations between data loss and change of position from reclined to standing, and with increases in exertion. While it would be expected that personal interaction and psychological demands would be less impacted by this issue, there are significant positive correlations between both of these indices and exertion, again leading to the possibility of QTVI drop out due to noisy signals and an inability to determine the end of the T wave in ambulatory monitoring. Figures 4- 12 to 4- 14 depict QTVI (Imputed) during wake sleep, with exertion and with posture and further validate the need to consider the potential effect of losing our measure of sympathetic activation during those periods that are sympathetically activating. This is most striking with the complete pattern reversal for posture with QTVI (Imputed) displaying the expected direction of response to changes in posture (Figures 4- 10 & 4- 15). The Imputed QTVI has higher correlation coefficients with residual heart rate as well as our measures that are associated with sympathetic arousal (Table 4-2). A problem with the imputed data is a loss of variation at the high end of the QTVI scale, which reduces its validity. Additionally since there is such high loss of data during time periods with the highest sympathetic activity, it is possible that in some subjects, the 97$^{th}$

percentile is too low as a QTVI value for the amount of sympathetic activity during some of the imputed epochs.

Residual heart rate provides more consistent associations or greater magnitude in the expected direction to the various sympathetic stimulating demands (Table 4-2). Residual heart rate only requires that high frequency power and heart rate measures to be calculated. Therefore, there is no drop out of the residual heart rate during high physical demand periods. A weakness of the residual heart rate is its dependence entirely on the validity of high frequency power as a vagal measure. However, that high frequency power is indicative reflects vagal activity is one area that the literature is extremely clear. It also depends on the assumption that any change in heart rate not related to high frequency power is of sympathetic origin, which is likely to be an oversimplification. However, we are using residual heart rate as it was initially proposed and in the manner in which it was intended as a non-invasive index of sympathetic activity [37]. This information was utilized in helping us to determine which variables we would utilize for further analyses in this paper.

*Validation Conclusions*

The two indices of the sympathetic nervous system we have assessed each provide an indication of sympathetic activity. They are not perfect, laboratory based, gold standard indices of sympathetic activation. However, we believe they are adequate to detect at the very least, relative changes and differences between epochs. The most obvious benefit to their use is the relative ease associated with obtaining an index of

sympathetic activity with ambulatory ECG as compared to A major limitation with the use of QTVI, as a measure in occupational ambulatory monitoring studies, is that of the loss of data during high sympathetic activity time periods, which reduces its predictive power for use as an indicator of the sympathetic state of epochs. For aggregation of larger time periods this presents as a limitation, and a likely underestimation of the QTVI for the aggregated time period, however does not prohibit its use (See Chapter Two). However, for performing analyses of the coordination state of an epoch, or correlations between dependent variables loss of data is a larger problem since in the case of our coordination analysis epochs cannot be categorized, or in the case of correlation analyses epochs with the highest sympathetic activity are not available. The imputed QTVI data provide a slightly better indictor as compared to the residual heart rate, however, its correlations with sympathetic activities was not as strong as the residual heart rate. Even though the imputation prevents the loss of data, it assumes for each individual a set QTVI value for all data loss periods. This loss of variation at the high end of its scale reduces its correlation with residual heart rate and certainly its validity.

The analyses presented in Part II of this chapter utilize high frequency power, heart rate, residual heart rate and QTVI. Data for the QTVI is presented both with and without the imputation.

## Part II: Analysis Methods and Results

### Coordination of Autonomic Control

*Methods*

To analyze cardiac coordination patterns within the framework of the "Doctrine of Autonomic Space" as proposed by Berntson et al. [3] we first categorized each epoch for each subject based on its autonomic space. We developed a basic coordination classification algorithm, based on within subject-normalized values for the primary autonomic control variables. The first step was to determine whether the epoch should be classified as sympathetic or parasympathetic dominant based on heart rate. Epochs greater than the subjects overall mean heart rate were classified as sympathetic and below as parasympathetic. As previous discussed, heart rate is a global measure of sympathetic and parasympathetic activity. While heart rate does not by itself allow determination of the exact mix of ANS activity, it does provide an indication of the systems state – that being either sympathetic or parasympathetic dominant. Then, based on the HFPnu and residual heart rate (ResHR) we categorized the coordination pattern of the autonomic components. We limited the possibilities for categorization to reciprocal control, unclassified, or non-reciprocal (uncoupled, co-inhibition, co-activation). Reciprocal control was determined when there was an increase in HFPnu of greater or less than a threshold value (set at 0.1 standard deviations), with an opposite deflection of the residual

heart rate, again beyond a set threshold value (Figure 4-22). The non-reciprocal patterns either required both HFPnu and ResHR to increase (co-activation) or decrease (co-inhibition) greater than the threshold value, or required variable to stay within the upper and lower bounds of the threshold values (> .1 SD and < -.1 SD), while the other variable required a change to a higher threshold value (.25 SD), in either direction. All those that did not fit into a previously described situation determined the unclassified epochs. Unclassified epochs only accounted for 4% of the data. The algorithm used to classify the epochs is presented in Figure 4- 22. Using two dimensions, sympathetic vs. parasympathetic dominance and coordination we then classified each epoch in each subject as being in one of six states as depicted in Figure 4- 22. With this classification system each epoch is only classified in one of the six possible states. We are considering this a general approach to applying Berntson's doctrine of autonomic space[3] for the analysis of ambulatory monitoring data. For further analysis we calculated the percentage of data in each sociological period for each of the six possible states within a subject. Comparisons of the percentage of data in each coordination state were made between the high and low strain subjects, as well as in exhausted subjects.

*Results and Discussion*

A consistent finding is a greater proportion of reciprocal sympathetic coordination during the workday sociological periods in high strain subjects (Figure 4- 23). This seems to indicate that high strain work requires greater functional coordination of the ANS (Karasek et al, 2003). Additionally, the lower control during this period in the high strain

group may limit the ability of subjects to allow other coordination patterns. While Berntson and colleagues do not attempt to identify what a normal ANS coordination pattern is given an environmental situation, they do comment on the greater effectiveness of the reciprocal control modes for end organ response (i.e. achieving a desired heart rate). It may be true that while reciprocal activation is effective for end organ response, it may require greater energy for coordination (as more states are specified) [1]. It requires activation of two branches of the ANS and therefore a greater number of neural networks. During relaxation other control modes can allow end organ response to meet cardiovascular output needs, with reduced central nervous system energy requirements and are therefore this less centrally demanding. Increased energy may be required for highly coordinated – tightly regulated – reciprocal patterns and with fatigue or exhaustion the mechanism that allows end organ response to follow as needed is based on these higher system requirements. Again, the ability to internally switch modes of control due to control over the external environment is a critical aspect of this analysis in light of the Stress Disequilibrium Theory. Figure 4- 24 compares the non-reciprocal parasympathetic coordination modes in the high and low strain groups. The high strain group has significantly lower non-reciprocal parasympathetic activity, with smaller differences in the amount of reciprocal parasympathetic activity (Figure 4- 25). The finding of greater sympathetic activity in the high strain group could represent the demands of the job, while the differences in control mode might reflect the lower environmental control. In neuromuscular physiology activated motor units are cycled when activation of muscles is required for long periods of time to prevent not only muscular fatigue, but also central fatigue of the neural networks responsible for muscle activation [43, 77-80]. During times of

high demands it might be better for reciprocal control to mobilize the internal sub-system resources to produce effective end organ response, whereas at other times – when the system has the chance to relax, it might be better to allow the sub-systems to not be so tightly coupled leading to equal dominance of reciprocal and non-reciprocal control patterns during parasympathetic dominant activities.

When exhausted subjects are assessed separately there is an even greater reduction in non-reciprocal parasympathetic control and this is most obvious on the workday (Figure 4- 26). There are similar elevations in reciprocal sympathetic activation in the exhaustion and high strain groups as compared to the low strain group (Figure 4-28); however, there is further elevation in non-reciprocal sympathetic control in the exhaustion group (Figure 4- 29). This finding fits nicely with a variety of related hypotheses regarding fatigue and exhaustion. These results imply a greater percentage of data that is sympathetic dominant in the exhausted subjects. Human factors studies consistently demonstrate that with fatigue greater energy costs are induced by subjects to maintain functional output [81, 82]. Additionally, when exhausted subjects must coordinate for the most effective output (high energetic demands and therefore sympathetic activity) they have a greater percentage of probably less precise (in terms of system responses) non-reciprocal coordination [1]. The elevation in non-reciprocal sympathetic coordination in the exhausted subjects persists after working hours into the evening following work (Sociological Period 5). It does not decrease to levels similar to the non-exhausted subjects until sleep. However, by the second day of rest and sleep this patter has actually reversed. This is interesting considering Weiner's theory that psychosomatic diseases are the result of altered feedback control resulting in disturbances in the return to the rest

state [83]. Sudacani et al. reported that an inability to relax after work was associated with increased risk of cardiac disease [84].

## Correlation Analysis

### *Methods*

To investigate system coordination and generate hypotheses we investigated correlation structures between dependent variables that are thought to represent a control sub-system. Correlations were calculated across individuals and across a variety of time scales for both high and low strain subjects (n = 576 per person). Time contrasts considered included the workday ( n = 95 per person), all times after the workday except the active period on the rest day (rest) (n = 384 per person), and sleep n = 1440 per person). Contrasts were observed between the strain groups and between the tasks and times. All physiological variables were normalized within each subject to correct for differences in subject baselines. Heart rate (HR), heart rate variability (HRV), ResHR and QTVI were normalized based on a Z-score transform within each subject. The high frequency (HFPnu) and low frequency power (VLFPnu) were normalized based on total power within each 5-minute epoch. Heart rate and heart rate variability were utilized as general measures of state and variability in the system. ResHR and QTVI are considered indices of sympathetic activity, and HFPnu an index of vagal activity.

*Correlation Based Hypotheses*

High Frequency Power and Heart Rate Variability

Parasympathetic activity is associated with reduced HRV and this occurs most clearly in high strain workers on the workday (Table 4-3). This occurs despite expectations of increased HRV with increased vagal activity. However, since HFP in this analysis is normalized by total power, and since total power equals total variance, as HFP becomes a larger portion of the variance the remaining types of variance are reduced. Therefore the non-vagal variance decreases somewhat as the vagal variance increases. A question can be raised about why this occurs to a greater degree in the high strain workers while at work. One possibility is that in the high strain workers are simply "working harder" and there is tighter overall regulation – a similar finding to results from assessing coordination patterns. If with tighter overall regulation vagal activity increases it is more likely to be associated with less overall variation since these subjects do not have an adequate overall variance capacity and overall vagal power is relatively low (See Chapter Two). This may be one indication of work related reductions in adaptive capacity. High strain subjects may not be capable of letting their cardiac activation system fully disengage to allow increased variation. The low strain workers', with greater adaptive capacity, allow for greater variance and overall freedom so when they increase the percentage of the their vagal variation they also let the system relax thus increasing the total variability. Despite the relative increase in total variability a negative correlation

(although at a lower magnitude) persists because as the normalized HFP increases it is a greater proportion of the total variation.

Table 4-3. Heart Rate, Sympathetic and Parasympathetic Control on the Work Day

| | High Strain | | | | Low Strain | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | HRV | HFPnu | ResHR | HR | HRV | HFPnu | ResHR |
| HRV | .05 | | | | -.19* | | | |
| HFPnu | -.40* | -.43* | | | -.20* | -.28* | | |
| ResHR | .41* | .29* | -.01 | | .65* | .09* | .08* | |
| QTVI | .49* | -.01 | -.20* | .14* | .48* | -.06 | -.06 | .14* |
| QTVI (Imputed) | .53* | .004 | -.22* | .23* | .39* | -.07 | -.16* | .23* |

* $p < .001$

Overall Summary. These results suggest that with increases in vagal activity are associated with less overall variation. This may provide an indication of the workers in high strain jobs "working harder" thus leading to tighter overall regulation.

Detailed Comment. This finding persists, and shows a mild gradient, when stratifying based on strain and exhaustion with the lowest correlations between vagal activity and HRV in the low strain group ($r = -.29$) and the highest correlations in the exhausted group ($r = -.49$). This further demonstrates the relationship between this finding and the findings of the coordination pattern assessment. The trend for the correlations between HFPnu and HRV suggest greater reductions in adaptive capacity in the high strain group and even further in the exhausted group. This is the same trend as identified when classifying epochs based on coordination states during the workday. There is less non-reciprocal parasympathetic control for the exhausted group than for any other group. Each of these

findings supports the concept of work related adaptive capacity despite different analytical approaches.

## Vagal Control of Heart Rate

HFPnu has a relatively greater association with HR in the high strain subjects on the workday (Table 4-3). This is interesting since it occurs despite less overall HFP in the high strain group. Secondary to the tighter overall regulation and the predominant presence of reciprocal coordination, even though there is less vagal control, small changes in the vagal control exert a stronger effect over heart rate. Heart rate is more sensitive to changes in the vagal system activity. This finding is apparent in comparing the aggregated slope coefficients from within subject linear regressions of HR as predicted by HFP. Figure 4- 30 shows these coefficients stratified by job strain computed from the workday only. This finding supports the finding of increased reciprocal control for the high strain subjects. If vagal changes are associated with sympathetic changes in the opposite direction, then HR will demonstrate a larger magnitude of change for any change in vagal activity. This again supports the hypothesis that high strain work is associated with restrictions in control options. High strain work requires continued activation for meet demands without freedom to release activation. The activation is met with reciprocally coupled coordination. The tight regulation of HFP over HR is a consequence of this tight control. We see this overall with reduced absolute cardiac vagal activity in the high strain subjects throughout the entire monitoring period (See Chapter Two), and with reduced variation of vagal activity in both high strain and exhausted

subjects (See Section on Variation Assessment). It is also apparent in the continued elevations in the correlation coefficient and slope coefficient of HR as a function of HFP in the high strain subjects. The differences that exist during the workday persist during the contrasted rest periods, however at a smaller magnitude (Tables 4-4 and 4-5).

Table 4-4. Heart Rate, Sympathetic and Parasympathetic Control during Other Rest

| | High Strain | | | | Low Strain | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | HRV | HFPnu | ResHR | HR | HRV | HFPnu | ResHR |
| HRV | -.06* | | | | -.25* | | | |
| HFPnu | -.36* | -.43* | | | -.24* | -.25* | | |
| ResHR | .54* | -24* | -.19* | | .57* | .07* | -.10* | |
| QTVI | .39* | -.17* | -.02 | .04* | .37* | -.14* | -.09* | .09* |
| QTVI (Imputed) | .35* | -.07* | -.11* | .19* | .33* | -.12* | -.22* | .31* |

- $p < .001$

Table 4-5. Heart Rate, Sympathetic and Parasympathetic Control during Sleep

| | High Strain | | | | Low Strain | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | HRV | HFPnu | ResHR | HR | HRV | HFPnu | ResHR |
| HRV | -.05 | | | | -.20* | | | |
| HFPnu | -.19* | -.52* | | | -.12* | -.35* | | |
| ResHR | .32* | .41* | -.04 | | .52* | .16* | -.02 | |
| QTVI | .16* | -.24* | .13* | -.20* | .19* | -.15* | -.02 | -.07* |
| QTVI (Imputed) | .04 | -.06* | -.05 | -.07* | .14* | -.06* | -.09* | -.04 |

* $p < .001$

Overall Summary. Tighter regulation by the vagal system over heart rate supports the finding of greater reciprocal control during the workday. This may be an additional physiological representation of reductions in external control reducing internal control adjustment capacities.

Detailed Comment. When stratifying based on job strain as well as exhaustion the differences between the strain groups become larger (low strain $r = -.19$, $p < .001$; high strain $r = -.44$, $p < .001$). The coefficient for the exhausted group ($r = .06$) is not significant and demonstrates a lack of vagal control over heart rate in this group of subjects. Vagal activity in exhausted subjects is reduced overall as compared to the other subjects, they have similar elevations in reciprocal coordination, and they have significantly greater non-reciprocal sympathetic activation. A pattern of reciprocal coordination with sympathetic dominance (sympathetic increases, vagal decrease) is more difficult due to the lack of variation in vagal activity and overall reductions in vagal activity. Drawing from the human factors literature that demonstrates elevated compensatory energy activation requirements in fatigued and exhausted subjects [81], combined with our findings of similar increases in reciprocal activation in the exhausted and high strain subjects, and the findings of elevated non-reciprocal sympathetic activation in exhausted subjects, leads to a preliminary conclusion that additional sympathetic activation demand needed by the exhausted subjects is met by non-reciprocal sympathetic activity. In the exhausted subjects greater reductions in vagal control lead to reductions of vagal regulation of heart rate and this leads to greater amounts of non-reciprocal sympathetic coordination. For providing the necessary end organ response for

effectively meeting energetical demands an overall elevation in sympathetic activity is necessary. These findings persist throughout the monitoring time period, however, at a reduced magnitude. Figure 4- 31 demonstrates the slope coefficient of HR as a function of HFP in the strain and exhaustion groups on the workday, and its pattern is similar to that of Figure 4- 32, which shows the coefficients during all rest periods.

Sympathetic Control of HR

Sympathetic control of the HR is higher on the workday in the low strain subjects despite less absolute workday sympathetic activation than both the high strain and exhausted subjects (Table 4-3). This is the opposite pattern to that observed for high strain subjects. Combined with greater vagal activity and increased variation in vagal activity, when they respond to demands on their system effectively with reciprocal activation. This may lead to higher sympathetic – HR correlations. This finding is identified using ResHR and HR correlations, and only on the workday and as such this finding may be tautological. Since ResHR is the residual variance from the association between within-subject high frequency power and heart rate, its association with heart rate is partially dependent the high frequency power – heart rate relationship.

Specifically considering QTVI correlations with HR there are similar correlations between high and low strain subjects during all periods they are reduced during sleep. During sleep there is far less dominance of the sympathetic nervous system, and HR – while more variable – does not change drastically since metabolic/cardiac output demands do not change drastically. This leads to less variation in the dependent variable

(HR) and may make it more difficult to demonstrate associations with potential predictors.

## Vagal System Adaptive Capacity

### *Methods*

The HRV measures that form the majority of the dependent variables in this dissertation are based on the RR interval time series. The RR interval time series represent the most fundamental level of complexity when analyzing cardiac temporal regulation, and as previously described, this analysis of variation yields information related to the autonomic nervous system. RR interval analysis provides information for each of roughly 570 5-minute epochs of data for each of the 36 subjects. The RR series analyzed within each epoch provides data about the state of the subjects physiological system. The statistical analyses performed on aggregated data (usually as a mean) attempt to make use of this information to investigate the differences regarding physiological system status between groups, during work, rest, sleep, and when subjected to demands (position, exertion, strain). However to restrict these analyses to the mean values of the 5-minute epochs ignores the next level of within subject variation that occurs in the subjects physiological state. The analysis of RR interval variation yields information that represents the physiological state. The variation in physiological states of an individual may provide valuable information regarding a subject's adaptive capacity to situational environmental demands. Vagal activity (measured by HFP) is routinely demonstrated to

possess a log normal distribution (Figure 4- 33). Such a skewed distribution pattern is common in systems bound at one end, zero binds vagal activity; you cannot have less than zero vagal activity. Decreases in overall vagal activity can decrease the standard deviation by bringing the mean closer to the bound value of zero. However, decreases in the standard deviation, combined with reductions in extreme values, both the absolute value as well as the amount of data in the tail demonstrate limitations in the range of the vagal control system. Three methods were utilized to focus on the variation in vagal activity in an attempt to describe differences in the adaptive capacity. The first method utilized was determination of the standard deviation for high frequency power within each subject during each sociological period. The second method calculated the percentage of data in the upper tail (> 1 Std. Deviation) of the high frequency power distribution for each subject within each sociological period. The final method presented in this section utilized Poincaré plots for each subject to calculate the short-term variations in high frequency power during each of four approximately spaced periods.

Poincaré (or recurrence) plots provide a two-dimensional analysis of one variable as a function of itself from some lagged earlier value [85]. They allow quantification of short-term variability in the time series. The first step to creating the Poincaré plots for this analysis involved creating a lagged time series of high frequency power data for each subject. To test epoch-to-epoch variations we lagged the data by one 5-minute epoch. Then for each subject a linear regression was performed between the lagged and the original time series. The residuals from this linear regression represents the variation of the lagged data point and the data point to follow (from the original time series) and provided a measure of the short-term, or epoch-to-epoch variations. Poincaré plots were

created for each subject for each of six time periods (Work Day, Evening, Sleep, Rest Day, Evening, Sleep) allowing roughly 100 data points in each time series. The residuals were saved for each epoch of each subject within each period. The standard deviations of the epochs for each subject were calculated during each period that a plot was constructed.

*Results and Discussion*

Standard Deviation

The first analysis aggregated subject data within sociological periods based on standard deviation of vagal activity. Comparisons between high and low job strain subjects, as well as exhausted subjects depict remarkable decreases in the standard deviation as would be predicted by the hypothesis of reduced adaptive capacity. Figure 4-34 depicts the differences in standard deviation of vagal cardiac control between strain and exhaustion groups. It is of course likely that the reductions in standard deviation occur due to the lower absolute vagal activity that then approaches the lower bound value of zero. Evidence for this is the significant correlation ($r = .89$) between mean vagal activity and the standard deviation of vagal activity.

Percentage of Data in Upper Tail

To avoid the possibility that the reduction of variation of the vagal system is due to approaching the lower boundary, we tested the percentage of data above one standard deviation. This value reflects the amount of data in the upper tail of the distribution. The data was initially aggregated within each subject as the percentage of each individual's data greater than their own standard deviation. There is a much lower correlation, albeit still substantial, between mean vagal activity and the percentage of data in the upper tail than that observed with the standard deviation ($r = .41$). Figure 4- 35 displays the percentage of data greater than one standard deviation during each sociological period. The high and low strain subjects have similar percentages of data in the upper tail, however the exhausted group has no data in the upper tail of their distribution during the workday until sleep. Their vagal system is not only reduced in absolute magnitude, but also displays less variation in the upper tail of their own distribution. Physiologically this finding may indicate that they have less adaptive capacity during the workday, and in the hours following the workday. They do not display an increase in variation in the upper tail until sleep. Even with sleep, it takes an additional 24 hours of rest before values reach levels that are comparable to the non-exhausted subjects during sleep of the rest day.

Poincaré Plot Analysis

Poincaré plots of vagal activity as a function of its previous value from the vagal activity time series (Figure 4- 36) allowed determination of the residual variance. The residuals vary perpendicular to the line of identity and reflect short-term variations [85]. The residuals display a Gaussian distribution with a mean of zero (Figure 4- 37). Sample plots for a low strain, high strain and exhausted subject are presented on the same scale in Figures 4- 38 to 4- 40, respectively. The scale is from 0 – 9 for the natural log of high frequency power. The $R^2$ from the linear regression is provided. It is evident in comparing these three subjects that there are reductions in the mean high frequency power, in addition to reductions in short-term variation with a tighter fit to the line in the exhausted subject. To compare groups of subjects quantitatively, residuals were saved and the standard deviation of the residuals was determined for each subject in each time period. A graphic display demonstrates the predicted reductions in short-term variation (Figure 4- 41) in the high strain group, and more significantly in the exhausted group. Very strong decreases in the exhausted subjects are found during the workday, with slight increases for this group on the rest day. There is a slight reduction in the high strain group on the workday, with greater decreases in this group on the rest day. The theory does not provide a rationale for explaining this greater decrease in the high strain group on the rest day. The exhausted subjects demonstrated a greater percentage of non-reciprocal sympathetic control than the non-exhausted subjects. This might be related to the evolution of reduced adaptive capacity in these subjects that has less short-term variation

in vagal activity. If the vagal system is less variable and cannot adapt reciprocal coordination for sympathetic activation may be compromised.

## Discussion

The goal of this paper has been to present a set of analytical methods for testing new theoretical extensions of traditional stress models. The methods have all been based on data obtained from ambulatory electrocardiograph recordings. While several previous job stress studies have employed ambulatory ECG monitoring they have tended to utilize traditional statistical approaches. As Karasek points out, the rationality of stress theories pose a problem to science with a different form of cause and effect, one that includes complex multiple interacting subsystems [2]. "Stress is a systemic concept referring to a disequilibrium of the system as a whole, in particular a system's control capabilities [2]." Analyzing the physiological effect therefore must extend beyond simple approaches of determining whether homeostasis has been altered. As such, static measures may not be the first indication of increased risk. Rather, altered variance or sub-optimal coordination patterns may represent a change from normal that is the beginning of alteration – these may represent the pathological mechanisms between strain and disease.

Existing stress theories usually include some linkage between stress and exhaustion states [1, 18, 20, 23, 31, 83, 86-88]. The General Adaptation Syndrome has the potential to culminate in limitations of the organism leading to adaptation breakdown referred to by Selye as exhaustion [18, 19]. Additional concepts that overlap include burn-out, vital exhaustion and chronic fatigue syndrome [23]. Vital exhaustion represents the second of a two-phase process leading to disease. The first phase includes exposure to long lasting stressful conditions and over exertion leading to a breakdown in adaptation to stressors ultimately leading to vital exhaustion. At this point changes are thought to be part of the

disease process [31]. Clearly, significant theory has been developed and backed by empirical evidence pointing to a role for exhaustion in the development of stress related disease. Exhaustion is vaguely defined as a break down in the system leading to a loss of adaptive capacity [18, 21]. Despite the need to consider complex systems dynamics in the determination of what exhaustion might represent, most attempts have focused on determination of the existence of homeostasis. This is likely due to restrictions imposed in the collection of real time physiological data using blood, urine or even saliva samples of hormone-based systems. Ambulatory ECG monitoring provides a clear benefit in allowing the collection of continuous real time data. Despite this benefit, job stress studies utilizing ambulatory ECG monitoring have adhered to traditional statistical analysis methods as dictated by the human factors and cardiovascular epidemiology literature.

This paper has attempted to remedy two problems we see with the above approach. First, according to Karasek's Stress Disequilibrium Theory [1] and our modification of that in Chapter One, we are considering the step that precedes exhaustion as more than an onslaught of excessive demands with consequential breakdown and loss of adaptive capacity. We are considering the step preceding exhaustion as long-term adaptations of an individual to an environment that has reduced control. Environments with a lot of control are enriched, and as such lead to highly adaptive individuals, while environments that limit personal control and flexibility are restricted, in many ways impoverished, leading to limited adaptive capacity [89-91]. In the concept of biological adaptability, reduced adaptive capacity is associated with reductions in variation. Reductions in variation can exist on multiple scales, such as reductions in HRV at one

level, and reductions in variations of cardiac vagal activity. The reductions in variation may either be the result of or precede altered coordination patterns.

The second problem we have attempted to remedy is the limited use of ambulatory ECG data in studies investigating job stress. While this dissertation is not attempting to claim that there is no value in the traditional (aggregation of the mean HRV and/or HFP) approaches (Chapter Two and Three are based on these approaches), there may be a loss of potential insight when data as rich as that obtained by ECG is not explored. Whereas the analyses of correlations between physiological systems, coordination patterns and an analysis of system level variability as provided in this paper make use of more of the intricate data provided with HRV and ECG analysis.

The theoretical model, the demand – control model and its physiological regulatory explanation in terms of the Stress Disequilibrium Theory [1, 2], is based on the concept that stress is most relevant when viewed using systems principles. As such stress is viewed as a systemic problem where disruptions in one part of the system can affect other parts of the system. This paper has utilized three analytic strategies. A significant strength of this approach has been support for the theory based on the different methods and linkages between the physiological interpretations. The consistent trends of increasing dysfunction in the predicted direction with the exhausted subjects presenting with what might represent some form of system deregulation and work related reductions in adaptive capacity, and the high strain subjects representing some intermediate point between "normal" and exhausted is a powerful set of findings.

The most significant results of this study are that a relationship exists between the categorized coordination patterns, the reduced variation and the correlations. All of these

findings support the contention that high strain work leads to reduced adaptive capacity, and that being exhausted represents a more severe form of reduced capacity. The reduced adaptive capacity may also be associated with alterations in control (changes in autonomic coordination states) – or limitations in control (reduced vagal control of heart rate in exhausted subjects). The exhausted subjects displayed no cardiac vagal control over heart rate during working hours. This inability is likely due to reduced absolute values and reduced variation of vagal activity. This also leads to an inability to engage the system for sympathetic activation with reciprocal coordination.

The finding of reduced vagal control and therefore reduced capacity for reciprocal coordination may be a reason for the additional sympathetic activation found in these subjects while at work. The exhausted subjects have as much reciprocal sympathetic coordination as the high strain subjects; but they have additional non-reciprocal coordination and they have very little vagal coordination either reciprocal or non-reciprocal. Owing to the lack of vagal activity, perhaps the additional non-reciprocal coordination leads to a greater amount of central fatigue of neural networks leading to the symptoms they report in order to be classified as exhausted. In a possible extension of Karasek's Stress Disequilibrium Theory, the reduction of adaptive capacity may be viewed as a adaptive central nervous system neural networks, as dissipative structures briefly discussed in Chapter One, evolving as part of phenotype plasticity. In this dissertation, the terminology "phenotype" has been introduced to allow a linkage to biological adaptability theory (further discussion was presented in Chapter One). Phenotype refers to the biological and physiological internally interacting sub-systems that emerge to create an organism's traits. The phenotype represents the complex

combinations of physicochemical integration within the system and a phenotype possesses a range of capabilities for response to the environment. The ability of the phenotype to adapt to its environment can be referred to as its adaptive capacity. The neural networks forming the basic control mechanisms of the stress response with coordinated patterns integrating visceral function and behavior [92, 93] may evolve to become separated structurally and/or functionally from networks capable of allowing relaxation. Ultimately leading to the finding of an inability to return to a relaxation state as evidenced by the perpetual reductions in short term vagal variation until the second night of sleep in the exhausted subjects.

High strain subjects, while not as severely deregulated as the exhausted subjects, demonstrate changes compared to the low strain subjects, which may imply a deregulation in the direction of exhaustion. They have significantly less vagal activity. However, their reduction in vagal activity is still utilized effectively to control heart rate: and therefore they are capable of reciprocal sympathetic activity. However, this would appear to come at a cost. The cost is a reduced vagal activity reduces the variability that comes with effective adaptation to external demands. They also demonstrate significantly less parasympathetic activity on the workday – in particular less non-reciprocal parasympathetic coordination. The increased use of the non-reciprocal parasympathetic coordination state might reflect a state of letting go, essentially allowing the system to relax. In all subjects the presence of non-reciprocal parasympathetic coordination is most noticeable during sleep accounting for roughly 50% of the epochs during this time. Of course sleep is known for its vagal controlled restorative functions when the system is disengaged from the environment. Times like these during the day may also be

restorative. The high strain group may have less times of non-reciprocal parasympathetic coordination while at work due to an inability to disengage from environmental activity at work because of low control at the workplace. However, what are the long-term consequences. We hypothesize that this environment, which restricts control, leads to the development of a phenotype that is fixed, rigid, and not adaptable. During the life of an individual variation in phenotype is a function of the environment in which the individual exists [94]. Impoverished, restricted, non-variable environments lead to phenotypes that are less adaptive, and to an organism that is less capable to change is phenotype (See Chapter One) [89-91, 95]. Some phenotypes may evolve to present a lack of response from the vagal system, and an inability so switch to relaxation modes after work, or during a rest day, eventually leading to fatigue or exhaustion due to these inabilities to provide optimal physiological coordination for each and every environmental situation.

Demitrack et al have demonstrated impairments in cortisol function similar to the changes we have identified in chronic fatigue patients (CFS). CFS patients had reduced overall cortisol secretion and reduced variation in cortisol values over a 24 hour period. The authors attributed this to an inability of the hypo thalamic – pituitary – adrenal axis to respond to environmental challenges due to being exhausted [96].

A significant limitation in this study is that we cannot show causation. As a cross-sectional study we have no way of demonstrating that any of the processes hypothesized occur due to adaptation to environments have occurred in our sample. The exhausted subjects and high strain subjects may have always been different from the low strain group in the ways we have described. Additionally, traditional hypothesis testing was not conducted as part of this analysis. Therefore there is the potential that the findings of this

study are purely due to chance associations found only in this sample. The intention was to demonstrate the potential to conduct these analyses, and to suggest how the results may be interpreted within the framework of the Stress Disequilibrium Theory [1].

To our knowledge we are the first to utilize the doctrine of autonomic space [3] to classify coordination states based on Holter ECG recordings in a sample of healthy working subjects. Our method of classifying each epoch is based on the within subject mean and variation in both normalized high frequency power and residual heart rate, and the mean heart rate. This method was determined apriori, and represents our best attempt to apply Berntson's [3] approach to this data set. While it is possible that our algorithm (Figure 4-22) misclassifies physiological coordination states, however, given that the approach was applied to all subjects in the same manner, there is no reason to believe that this misclassification would be systematic. It is unlikely that any misclassification would be anything other than random thus with biasing the observations toward the null. Additionally, we feel that with the thresholds we set for triggering the classification of a change in either of the variables (HFPnu and ResHR) the possibility of misclassification is small relative to the volume of data in each sociological period (roughly 48 epochs per period per subject).

In previous investigations, the independent variables shown to influence high frequency power between subjects have been age, sex, and physical fitness [97]. Study selection criteria assured that study subjects were male, between the ages of 35 and 59, generally healthy and not using medications known to affect HRV. Although level of obesity, cholesterol profiles, and smoking status information was not collected for subjects, the literature does not suggest a relationship between these factors and atypical

HRV [98]. There is no evidence in the literature to suggest a relationship between decision latitude and physical fitness. It may be hypothesized that greater decision latitude is conducive to a lifestyle with greater potential for the pursuit of personal fitness. If so, this suggests that high decision latitude individuals might self-select toward better physical fitness. Validation of this possible association should be considered in future investigations.

It is also possible that other work environment exposures influence heart rate variability measures through their influence on autonomic nervous system function. Two possibilities include noise [15, 99] and air particulate exposure [100, 101]. Future studies should control for the effects of these additional environmental exposures. The possibility that autonomic nervous system alterations occur due to a multitude of environmental demands adds to the importance for methodologies for the assessment of ECG data in an attempt to determine system-based adaptations that lead to disease. Additionally, with an implication of reduced adaptive capacity due to reduced environmental control there could be an exacerbation of the physiological response to these physical and/or chemical exposures.

As will be discussed in a related work [102], the pool of available subjects had previously committed to participation in the Seasons study, which involved a fairly complicated initial and follow-up protocol [57]. They were then asked to participate in our investigation which required a large time commitment, and a restriction of activities in order to allow the participant to be near home or a place which was conducive to relaxation during weekend evenings. In addition, subjects were required to wear somewhat intrusive hardware, daily diary and telephone contact regimens, and limited

participation time scheduling options. Due to these constraints, subjects with high strain jobs may be less likely to participate in such studies. Additionally, the HMO based subject pool from which the majority of our subjects were recruited (n=26) reflects the population of central Massachusetts, and as such is largely white and middle-class. Our study is restricted to participants living in one relatively restricted geographic area of the country. In our desire to take fullest advantage of our study design we are looking at a considerable number of variables, creating a complex dataset. We believe that we have the expertise and experience to deal with this complexity. On the other hand, it is not possible to evaluate all possible variables obtained from the ECG signal, and it may be that we have overlooked a variable or variables of importance. The Seasons study also excluded individuals who could not speak English because virtually all of their study instruments were developed for use in English-speaking populations and have been calibrated for use in those populations.

## Conclusion

The Stress Disequilibrium theory provides a new approach to the analysis of job stress based primarily on the concept of environmental based control affecting internal physiological control. This study has successfully demonstrated methods utilizing data collected by ambulatory electrocardiograph monitoring to test aspects of this theory. Preliminary results support the use of ECG data for such analyses in job stress research and longitudinal studies on larger populations in a wide variety of occupations are needed to fully explore the methods and theories presented.

## Literature Cited


1. Karasek R.A. The Stress Disequilibrium Theory of Chronic Disease Development. Lowell, 2003.
2. Karasek R A, Theorell T. Healthy Work: Stress Productivity, and the Reconstruction of Working Life. New York: Basic Books, 1990.
3. Bernston G.G., Cacioppo J.T., Quigley K.S. Autonomic determinism: the modes of autonomic control, the doctrine of autonomic space, and the laws of autonomic constraint. Psychological Review 1991; 98:459-487.
4. Belkic K. Ambulatory Electrocardiographic Monitoring: Stress-Mediated Clinically Relevant Endpoints. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, Inc., 2000:196-203.
5. Collins S M. Emerging methods for the physiological assessment of occupational stress. Work 2001; 17:209-219.
6. Belkic K, Landsbergis P, Schnall P, et al. Psychosocial Factors: Review of the Empirical Data Among Men. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, 2000:7-68.
7. Hanson E.K.S., Godaert G.L.R., Maas C.J.M., Meijman T.F. Vagal cardiac control throughout the day: the relative importance of effort-reward imbalance and within-day measurements of mood, demand and satisfaction. Biological Psychology 2001; 56:23-44.
8. Iwasaki T, Sasaki T, Oka I, Hisanaga N. Effect of working hours on biological functions related to cardiovascular system among salesmen in a machinery manufacturing company. Industrial Health 1998; 36.
9. Kageyama T, Nishikido N, Kobyashi T, Kuokawa Y, Kaneko T, Kabuto M. Long commuting time, extensive overtime, and sympathodominant state assessed in terms of short term heart rate variability among male while collar workers in the Tokyo megalopolis. Industrial Health 1998; 36.
10. Kobyashi F, Furui H, Akamatsu Y. Changes in psychophysiological functions during night shift in nurses: influences of changing from a full day to half day work shift before night duty. Int Arch Occup Environ Health 1997; 69:83-90.
11. Kobyashi F, Furui H, Akamatsu Y, et al. Autonomic nervous system function in taxi drivers, International Conference of Behavioral Medicine, Copenhagen, Denmark, August, 1998.
12. Matsuzaki I, Mishimura A, Morita N. Autonomic nervous system activity changes due to shift work: an evaluation by spectral components of heart rate variability. J Occup Health 1996; 38:80-81.
13. Sasaki T, Iwasaki T, Oka I, et al. Effect of working hours on cardiovascular autonomic nervous system functions in engineers in an electronics manufacturing company. Industrial Health 1999; 37:55-61.
14. Sluiter J.K. How about work demands, recovery and health. Amsterdam: University of Amsterdam, 1999.

15. van Amelsvoort L.G.P.M., Schouten E.G., Maan A.C., Swene C.A., Kok F.J. Occupational determinants of heart rate variability. Int Arch Occup Environ Health 2000; 73:255-262.
16. Vrijkotte T.G.M., van Dornen L.P.J., de Geus E.J.C. Effects of work stress on ambulatory blood pressure, heart rate, and heart rate variability. Hypertension 2000; 35:880-886.
17. Collins S M. Job Strain and Electrocardiographically Assessed Pathophysiological Mechanisms. Department of Work Environment. Lowell: University of Massachusetts Lowell, 2003.
18. Selye H. Stress and Disease. Science 1955; 122:625-631.
19. Selye H. The stress of life. New York: McGraw Hill, 1976.
20. McEwen B.S. Protective and damaging effects of stress mediators. NEJM 1998; 338:171-179.
21. Appels A. Exhausted subjects, exhausted systems. Physiologica Scandinavia 1997; 161:153-154.
22. Porges S. Cardiac Vagal Tone: A physiological index of stress. Neuroscience and Biobehavioral Reviews 1995; 19:225-233.
23. Theorell T. Neuroendocrine Mechanisms. In: Schnall P, Belkic K, Landsbergis P, Baker D, eds. The Workplace and Cardiovascular Disease. Vol. 15. Philadelphia: Hanley & Belfus, Inc., 2000:139-146.
24. Yeragani V.K., Kumar H.V. Heart period and QT variability, hostility, and type-A behavior in normal controls and patients with panic disorder. Journal of Psychosomatic Research 2000; 49:1-7.
25. Sharpley CF. Heart rate reactivity and variability as psychophysiological links between stress, anxiety, depression and cardiovascular disease: Implications for health psychology interventions. Australian Psychological Society 2002; 37:56-62.
26. Carney RM, Freedland KE, Miller GE, Jaffe AS. Depression as a risk factor for cardiac mortality and morbidity: A review of potential mechanisms. Journal of Psychosomatic Research 2002; 53:897-902.
27. Agelink MW, Boz C, Ullrich H, Andrich J. Relationship between major depression and heart rate variability. Clinical consequences and implications for antidepressive treatment. Psychiatry research 2002; 113:139-149.
28. Grippo A.J., Johnson A.K. Biological mechanisms in the relationship between depression and heart disease. Neuroscience and Biobehavioral Reviews 2002; 26:941-962.
29. Armstrong L.E., van Heest J.L. The Unknown Mechanism of the Overtraining Syndrome: Clues from depression and psychoneuroimmunology. Sports Medicine 2002; 32:185-209.
30. Appels A. Why do imminent victims of a cardiac event feel so tired? Int J Clin Pract 1997; 51:447-450.
31. Goodkin K, Appels A. Behavioral-neuroendocrine-immunologic interactions in myocardial infarction. Medical Hypotheses 1997; 48:209-214.
32. Wantanabe T, Sugiyama Y, Sumi Y, et al. Effects of vital exhaustion on cardiac autonomic nervous functions assessed by heart rate variability at rest in middle-aged male workers. International Journal of Behavioral Medicine 2002; 9:68-75.

33. Shepard R.J. Chronic Fatigue Syndrome: An update. Sports Medicine 2001; 31:167-194.
34. Uusitalo A.J. Ability of non-invasive and invasive methods autonomic function measurements and stress hormones toindicate endurance training-induced stress. Tampere: University of Tampere, 1998:621.
35. Uusitalo A.L.T., Uusitalo A.J., Rushko H.K. Heart rate and blood pressure variability during heavy training and overtraining in the female athlete. Int J Sports Med 2000; 21:45-53.
36. Al'Absi M, Buchanan T, Pncomb G.A., Licinio J, Lovallo W.R. Cardiovascular and neuroendocrine adjustment to public speaking and mental arithmetic stressors. Psychophysiology 1997; 34:266-275.
37. Backs R.W. Going beyond heart rate: autonomic space and cardiovascular assessment of mental workload. The International Journal of Aviation Psychology 1995; 5:25-48.
38. Backs R.W., Lenneman J, Sicard J.L. The use of autonomic components to improve cardiovascular assessment of mental workload in flight simulation. The International Journal of Aviation Psychology 1999; 9:33-47.
39. Callister R, Suwarno N.O., Seals D.R. Sympathetic activity is influenced by task difficulty and stress perception during mental challenge in humans. J of Physiology 1992; 454:373-387.
40. Hyndman B.W., Gregory J.R. vp. Spectral analysis of sinus arrythmia during mental loading. Ergonomics 1975; 18:255-270.
41. Moriguchi A, Otsuka A, Kohara A, et al. Spectral change in heart rate variability in response to mental arithmetic before and after the beta adrenergic blocker, carteolol. Clin Autonomic Research 1992; 2:267-270.
42. Negoescu R, Dinca-Panaitescu S. Mental stress enhances the sympathetic fraction of QT variability in an RR independent way. Int Phys Behav Sci 1997; 32:220-226.
43. Newsholme TR, et al. Physical and mental fatigue:metabolic mechanisms and importance of plasma amino acids. Br Med Bull 1992; 48:477-489.
44. Pagani M. et al. Sympathovagal interaction during mental stress: a study using spectral analysis of heart rate variability in healthy control subjects and patients with a prior history of myocardial infarction. Circulation 1991; 83 suppl II:II43-II51.
45. Sloan R.P., Shapiro P.A., Bagiella E, et al. Effect of mental stress throughout the day on cardiac autonomic control. Biological Psychology 1994; 37:89-99.
46. Steptoe A, Fieldman G, Evans O, Perry L. Cardiovascular risk and responsivity to mental stress: the influence of age, gender, and risk factors. J Cardiovasc Risk 1996; 3:83-93.
47. Cannon W.B. The emergency function of the adrenal medulla in pain and the major emotions. Am J Physiol 1914; 33:356-372.
48. Landsbergis P. The changing organization of work and the safety and health of working people: a commentary. J Occup Environ Med 2003; 45:61-72.
49. Karasek R A. The impact of the work environment on life outside the job. Cambridge: Massachusetts Institute of Technology, 1976.

50. Karasek R A. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 1979; 24:285-308.
51. Bernston G.G., Cacioppo J.T., Quigley K.S. Cardiac Psychophysiology and Autonomic Space in Humans: Empirical Perspectives and Conceptual Implications. Psychological Bulletin 1993; 114:296-322.
52. Bernston G.G., Cacioppo J.T., Fieldstone A. Illusions, arithmetic and the bidirectional modulation of vagal control of the heart. Biological Psychology 1996; 44:1-17.
53. Backs R.W. A comparison of factor analytic methods of obtaining cardiovascular autonomic components for the assessment of mental workload. Ergonomics 1998; 41:733-745.
54. Melis C, van Boxtel A. Differences in autonomic physiological responses between good and poor inductive reasoners. Biological Psychology 2001; 58:121-146.
55. Recordati G. A thermodynamic model of the sympathetic and parasympathetic nervous systems. Autonomic Neuroscience: Basic and Clinical 2003; 103:1-12.
56. Friedman B.H., Thayer J.F. Anxiety and autonomic flexibility: a cardiovascular approach. Biological Psychology 1998; 47:243-263.
57. Merriam P.A., Ockene I.S., Hebert J.R., Rosal M.C., Matthews C.E. Seasonal Variations of Blood Cholesterol Levels Study: Study methodology. Journal of Biological Rhythms 1999; 14:330-339.
58. Collins S M, Karasek R A, Costas K. Impact of Job Strain on Heart Rate Variability & QT Interval Variability Index. University of Massachusetts Lowell, 2003.
59. Collins S M, Karsek R.A., Costas K. Cardiac Vagal Response to Daily Stressors. University of Massachusetts Lowell, 2003.
60. Rottman J, Steinman R.C., Albrecht P, Bigger J.T. Jr., Rolnitzky L.M., Fleiss J.L. Efficient estimation of the heart period power spectrum suitable for physiologic or pharmacologic studies. Am J Cardiology 1990; 66:1522-1524.
61. Albrecht P, Cohen R.J. Estimation of heart rate power spectrum bands from real world data: Dealing with ectopic beats and noisy data. Computers in Cardiology 1988; 15:311-314.
62. Lenneman J, Backs R.W. The validity of factor analytically derived cardiac autonomic components for mental workload assessment. In: Backs R.W., Boucsein W, eds. Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum, 2000:161-176.
63. Grossman P, Sveback S. Respiratory sinus arrythmia as an index of parasympathetic cardiac control during active coping. Psychophysiology 1987; 24:228-235.
64. Berger RD, Kasper EK, Baughman KL, Marban E, Calkins H, Tomaselli GF. Beat-to-beat QT interval variability: Novel evidence for repolarization lability in ischemic and nonischemic dilated cardiomyopathy. Circulation 1997; 96:1557-1565.
65. Atiga W.L., Calkins H, Lawrence J.H., Tomaselli GF, Smith J.M., Berger R.D. Beat to Beat Repolarization Lability Identifies Patients at Risk for Sudden Cardiac Death. J Cardiovascular Electrophysiology 1998; 9:899-908.

66. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Kay J, Igel G. Effect of posture and isoproterenol on beat-to-beat heart rate and QT variability. Neuropsychobiology 2000; 41:113-123.
67. Yeragani V.K., Pohl R, Jampala V.C., Balon R, Ramesh C, Srinivasan K. Increased QT variability in patients with panic disorder and depression. Psychiatry Research 2000; 93:225-235.
68. Pohl R, Yeragani V.K. QT interval variability in panic disorder patients after isoproterenol infusions. International Journal of Neuropsychopharmacology 2001; 4:17-20.
69. Jacob R.G., Thayer J.F., Manuck S.B., et al. Ambulatory blood pressure responses and the circumplex model of mood: a 4-day study. Psychosomatic Medicine 1999; 61:319-333.
70. Schwartz J, Pickering T. The Workplace Blood Pressure Study, American Society of Hypertesion Conference, 1996.
71. Schnall P, Pieper C, Schwartz J.E., Karsek R.A., et al. The relationship between "Job Strain," workplace diastolic blood pressure and left ventricular mass index. JAMA 1990; 263:1929-1935.
72. Schnall P, Schwartz J.E., Landsbergis P, Warren K, Pickering T.G. A longitudinal study of job strain and ambulatory blood pressure: results from a three year follow-up. Psychosomatic Medicine 1998; 60:697-706.
73. Berntson G.G., Bigger J.T. Jr., Eckberg D, et al. Heart rate variability: Origins, methods, and interpretative caveats. Psychophysiology 1997; 34:623-648.
74. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability standards of measurement, physiological interpretation, and clinical use. European Heart Journal 1996; 17:354-381.
75. Guyton A.C., Hall J.E. Textbook of Medical Physiology. Philadelphia: Saunders W B Co, 2000.
76. Katch F, Katch V, McCardle W.D. Appendix C: Energy Expenditure in Household, Occupational, Recreational, and Sports Activities. In: Katch F, Katch V, McCardle W.D., eds. Exercise Physiology: Energy, Nutrition and Human Performance. Philadelphia: Lippincott Williams and Wilkins, 2001.
77. McKardle F, Katch F, Katch V. Neural Control of Human Movement. In: McKardle F, Katch F, Katch V, eds. Exercise Physiology: Energy, Nutrition, and Human Performance. Philadelphia: Lippincott Williams & Wilkins, 2002.
78. Febbraio M.A., Dancey J. Skeletal muscle energy metabolism during prolonged fatiguing exercise. J Appl Physiol 1999; 87:2341-2347.
79. Davis J.M., Bailey S.P. Possible mechanisms of central nervous fatigue during exercise. Med Sci Sports Exerc 1997; 29:45-55.
80. Baldwin J, et al. Muscle IMP accumulation during fatiguing submaximal exercise in endurance trained men. Am J Physiol 1999; 277:R295.
81. Hockey R. Compensatory control in the regulation of human performance under stress and high workload: A cognitive energetic framework. Biological Psychology 1997; 45:73-93.
82. Mulder G, Mulder L.J.M., Meijman T.F., Veldman J.B.P., van Roon A.M. A Psychophysiological Approach to Working Conditions. In: R.W. Backs WB, ed.

Engineering Psychophysiology: Issues and Applications. London: Lawrence Erlbaum Associates, 2000.
83. Weiner H. Psychobiology of Human Disease. New York: American Elsevier, 1977.
84. Suadicani P, Hein HO, Gyntelberg F vp. Are social inequalities as associated with the risk of ischaemic heart disease as a result of psychosocial working conditions? Atherosclerosis 1993; 101:165-175.
85. Brennan M, Palaniswami M, Kamen P. Do existing measures of Poincare plot geometry reflect nonlinear features of heart rate variability? IEEE Transactions on Biomedical Engineering 2001; 48:1342-1347.
86. Appels A, Mulder P. Excess fatigue as a precursor of myocardial infarction. European Heart Journal 1988; 9:758-764.
87. Henry J.P., Stephens P.M. Stress, Health, and the Social Environment: A Sociobiologic Approach to Medicine. New York: Springer - Verlag, 1977.
88. Lovallo WR. Stress and Health: Biological and Psychological Interactions. London: SAGE Publications, 1997.
89. Piersma T, Drent J. Phenotypic flexibility and the evolution of organismal design. Trends in Ecology and Evolution 2003; In Press.
90. Renner M.J., Rosenzweig M.R. Enriched and Impoverished Environments: Effects on Brain and Behavior. Recent Research in Psychology. Berlin: Springer-Verlag, 1987.
91. Rosenzweig M.R., Bennett E.L. Psychobiology of plasticity: effects of training and experience on brain and behavior. Behavioural Brain Research 1996; 78:57-65.
92. Blessing W.W. Inadequate frameworks for understanding bodily homeostasis. Trends in Neuroscience 1997; 20:235-239.
93. Blessing W.W. The Lower Brainstem and Bodily Homeostasis. New York: Oxford, 1997.
94. Schlichting C.D., Pigliucci M. Phenotypic Evolution: A Reaction Norm Perspective. Sunderland: Sinauer Associates, Inc., 1998.
95. Piersma T, Lindstrom A. Rapid reversal changes in organ size as a component of adaptive behaviour. Trends in Ecology and Evolution 1997; 12:134-138.
96. Demitrack M.A., Dale J.K., Straus S.E. Evidence of impaired activation of the hypothalamic - pituitary - adrenal axis in patients with chronic fatigue syndrome. J Clin Endroc Metab 1991; 73:1224-1234.
97. Kristal-Boneh E, Raifal M, Froom P, Ribak J. Heart rate variability in health and disease. Scandinavian J of Work Environment and Health 1995; 21:85-95.
98. Kuch B, Hense H.W., Sinnreich R, et al. Determinants of short period heart rate variability in the general population. Cardiology 2000; 95:131-138.
99. Tzaneva L, Danev S, Nikolova R. Investigation of noise exposure effect on heart rate variability. Central European Journal of Public Health 2001; 9:130-132.
100. Magari SR, Hauser R, Schwartz J, Williams PL, Smith TJ, Christiani DC. Association of heart rate variability with occupational and environmental exposure to particulate air pollution. Circulation 2001; 14:986-991.

101. Magari SR, Schwartz J, Williams PL, Hauser R, Smith TJ, Christiani DC. The association of particulate air metal concentrations with heart rate variability. Environmental Health Persectives 2002; 110:875-880.
102. Costas K, Karsek R.A. Recruitment Bias in Occupational Stress Studies with Complex Protocols. University of Massachusetts Lowell, 2003.

Figure 4- 1. Calculation of Exertion

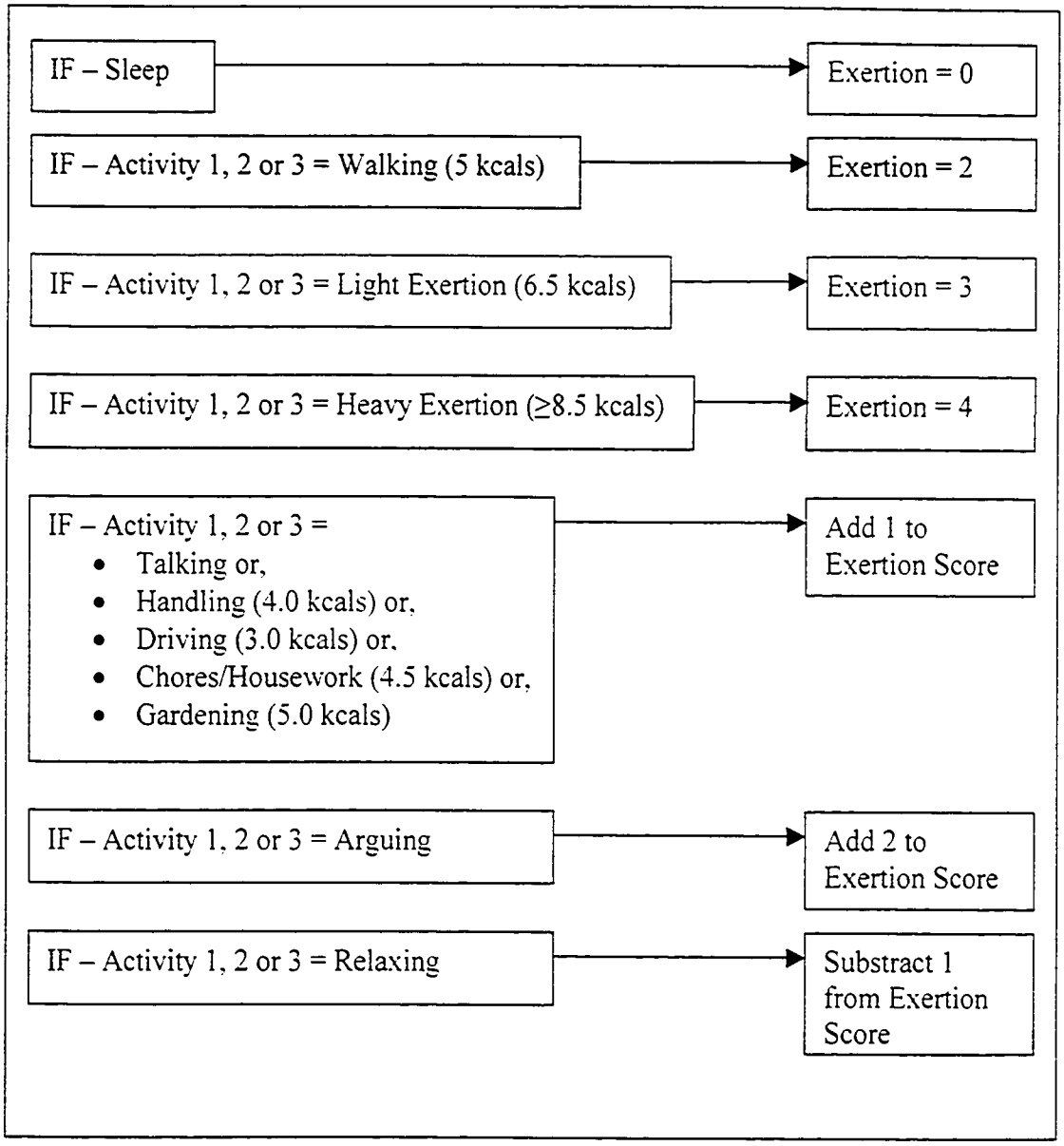


Figure 4- 2. Exertion

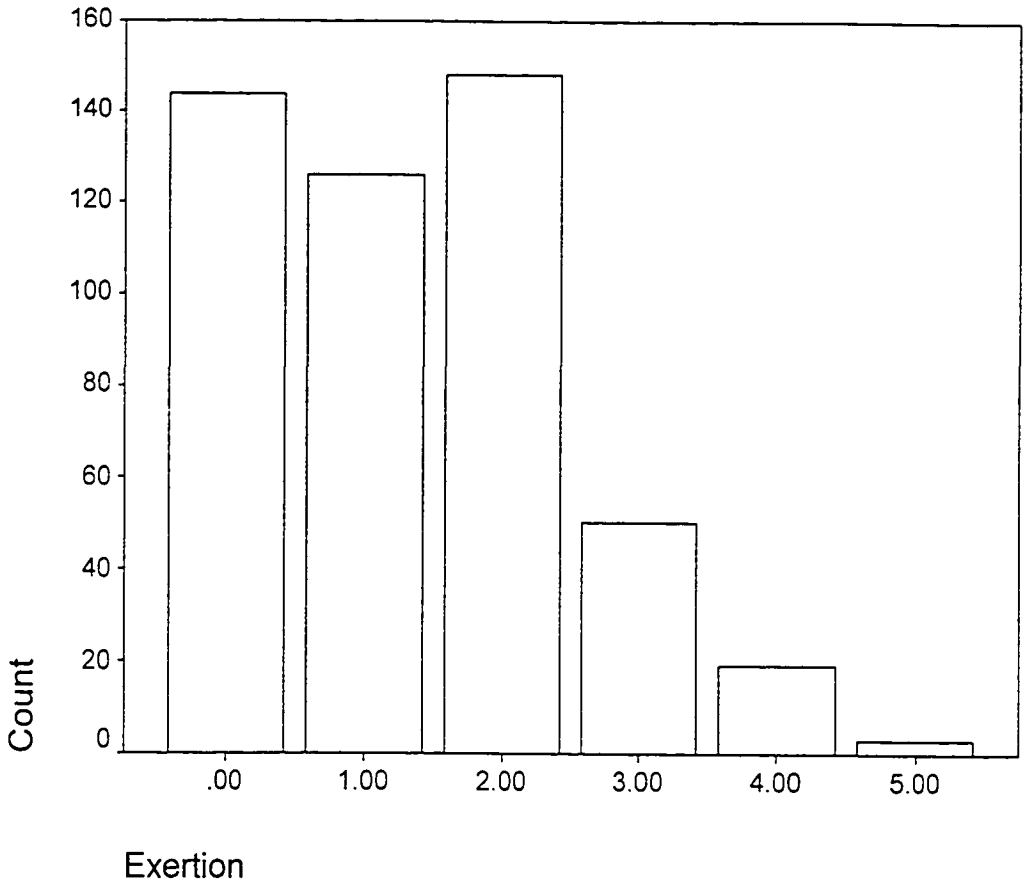


Figure 4- 3. Personal Interaction: Categories

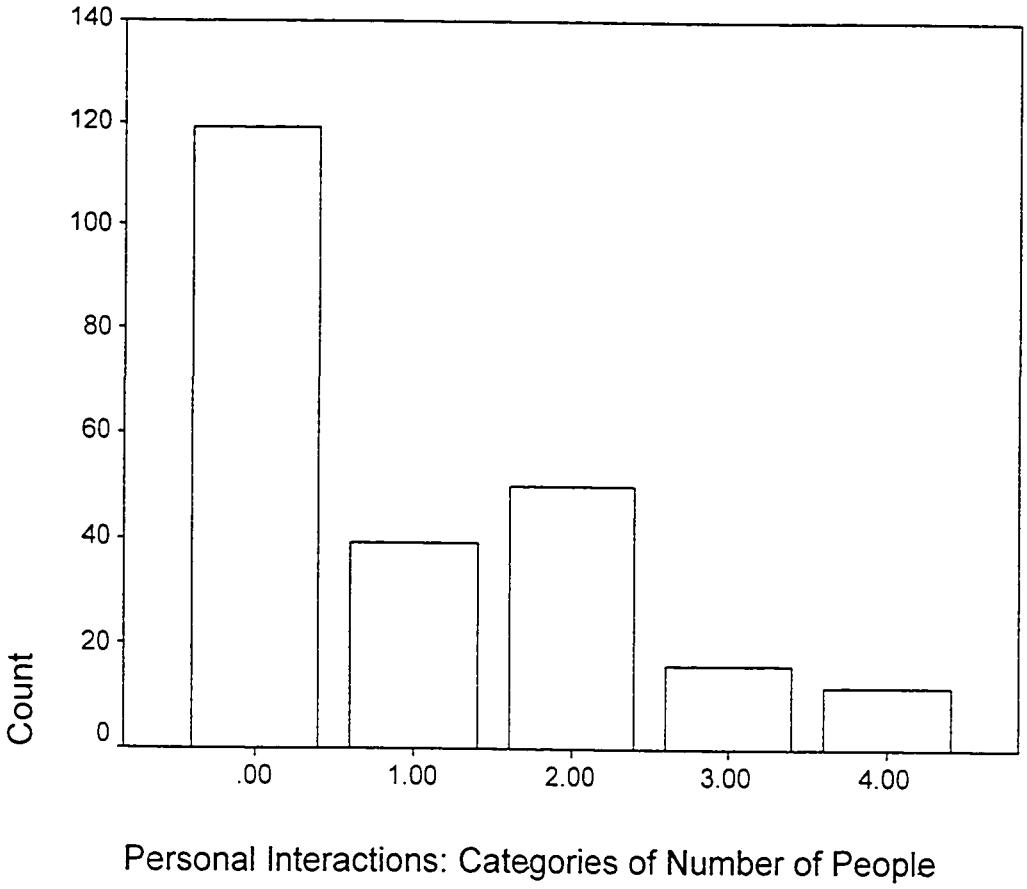


Figure 4- 4. Psychological Demands

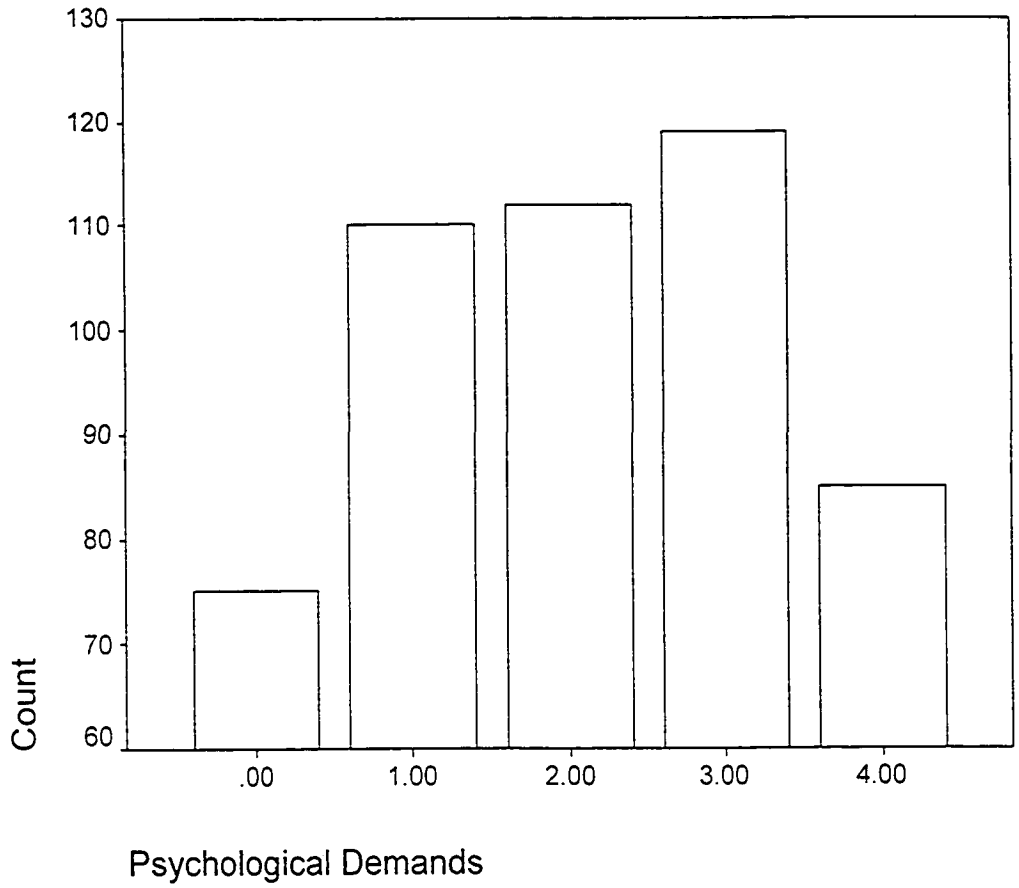


Figure 4- 5. Sum of Demands (Exertion + Psychological + Communication)

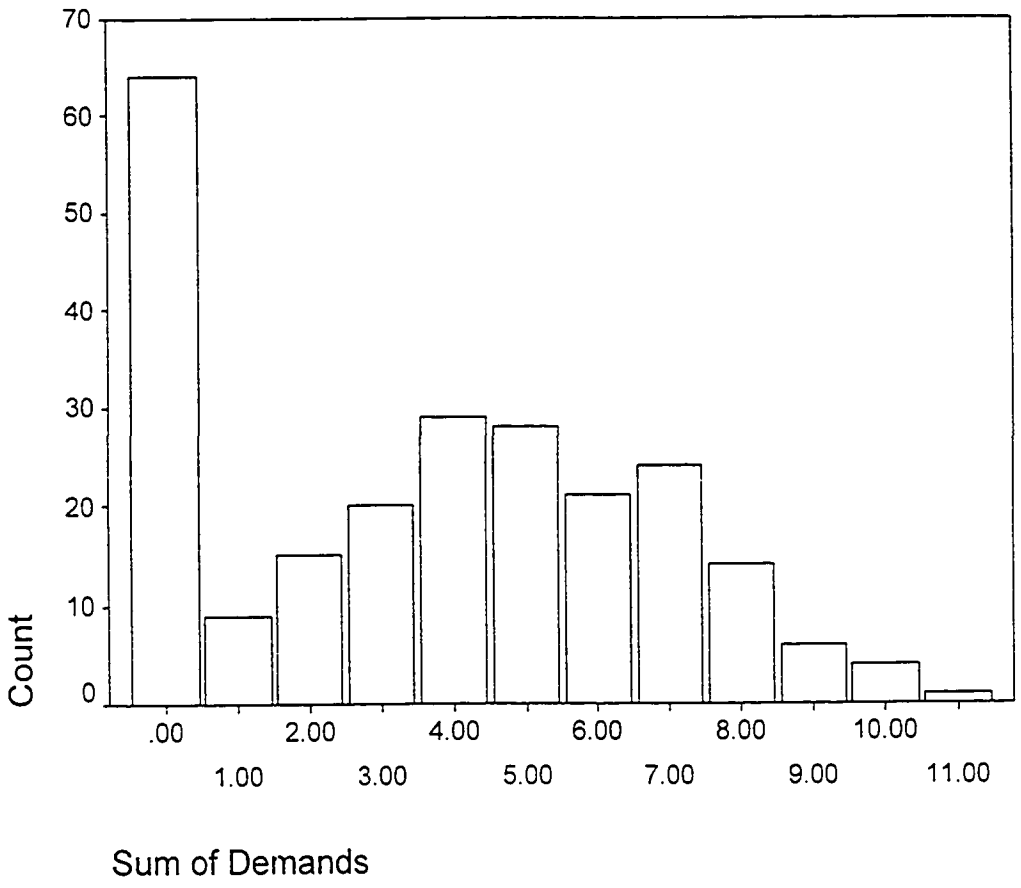
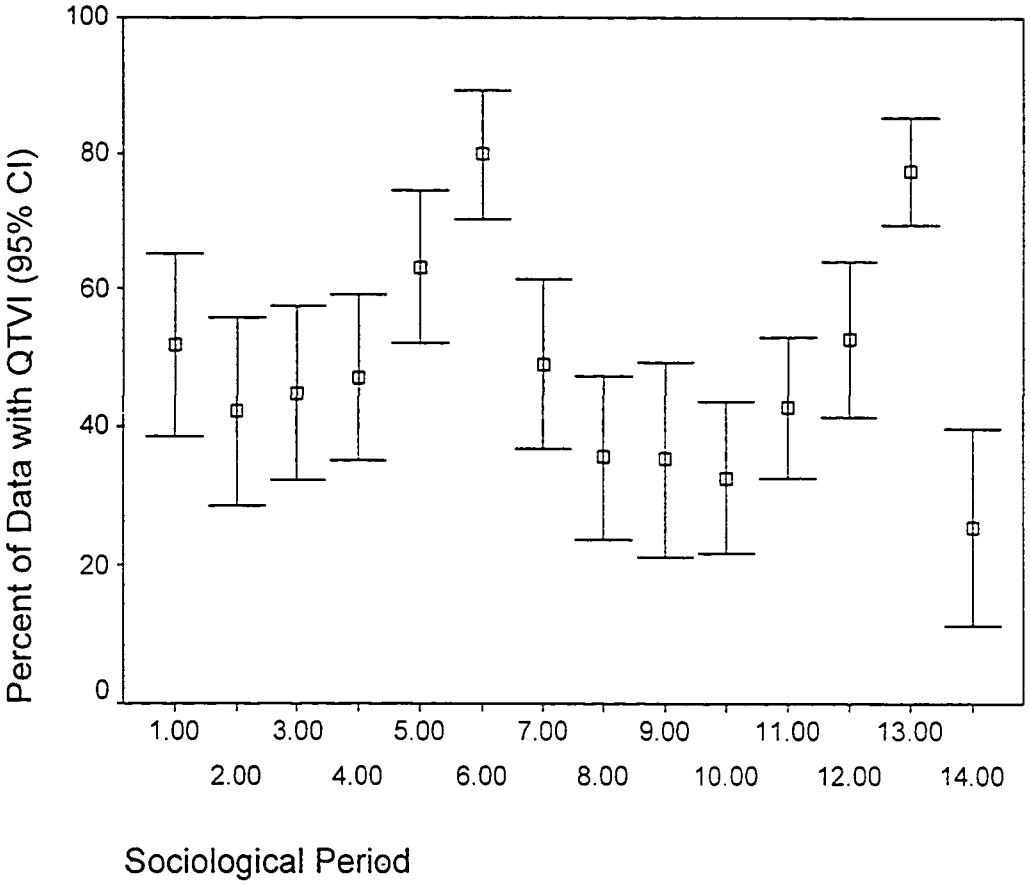


Figure 4- 6. Percent of QTVI Data available during the Monitoring Period




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 7. Percent of QTVI Data available in Various Positions

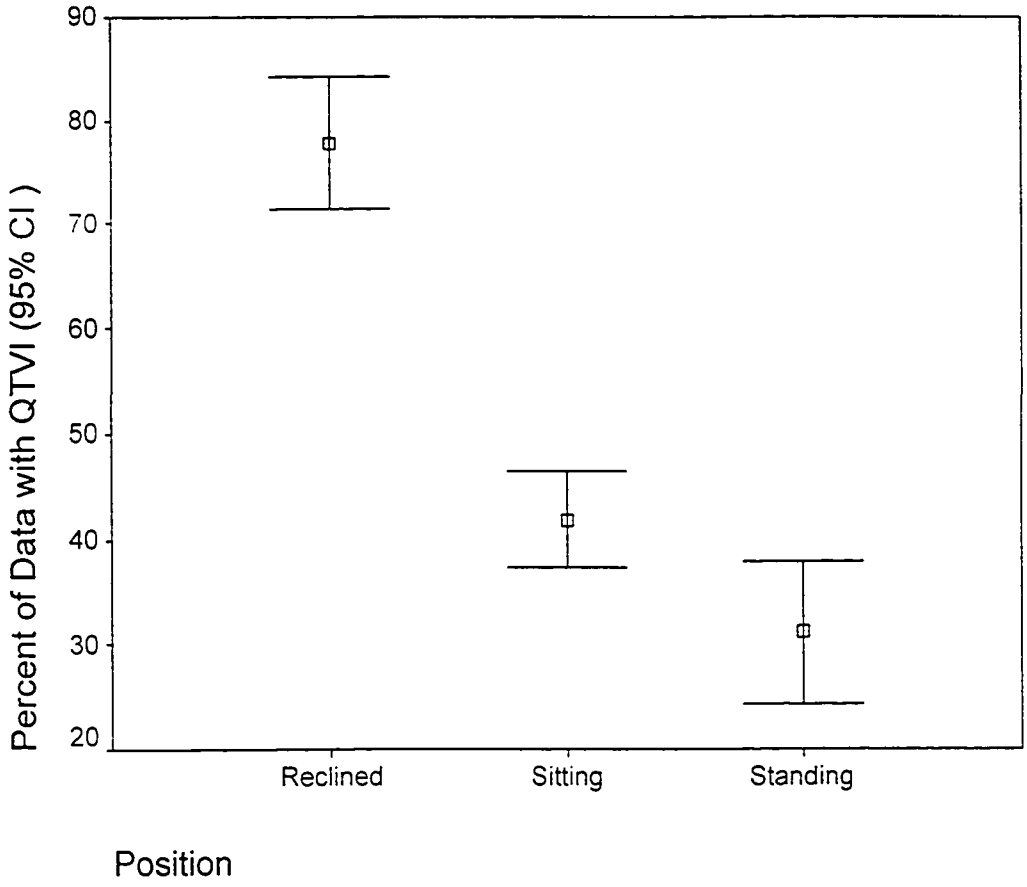


Figure 4- 8. Percent of QTVI Data available in with Increasing Exertion

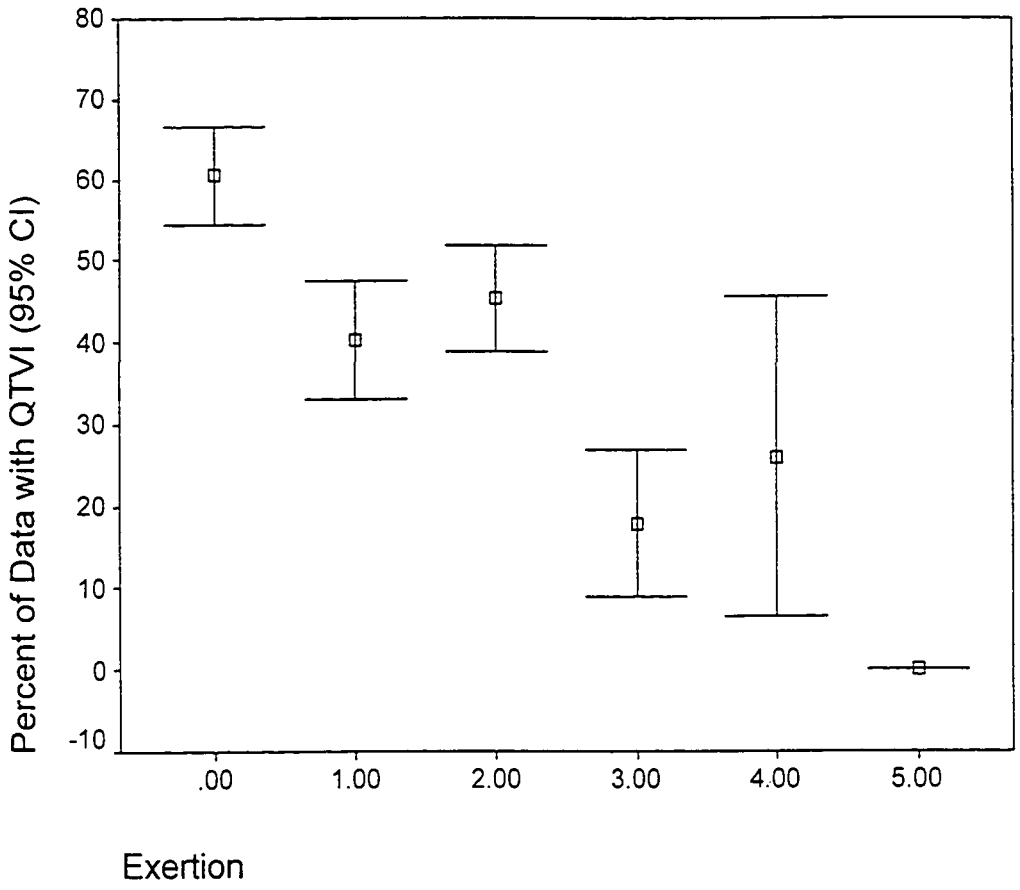


Figure 4- 9. QTVI During Wake and Sleep

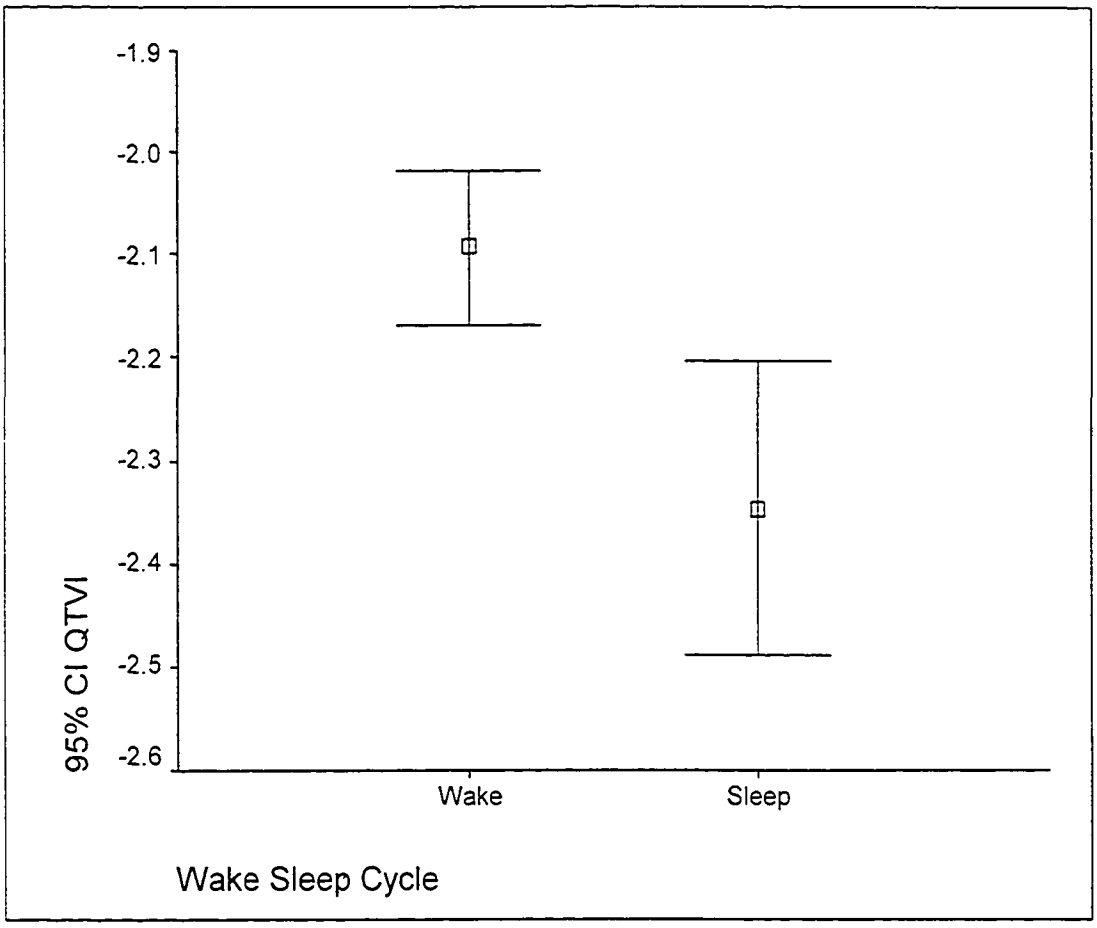


Figure 4- 10. QTVI with Posture

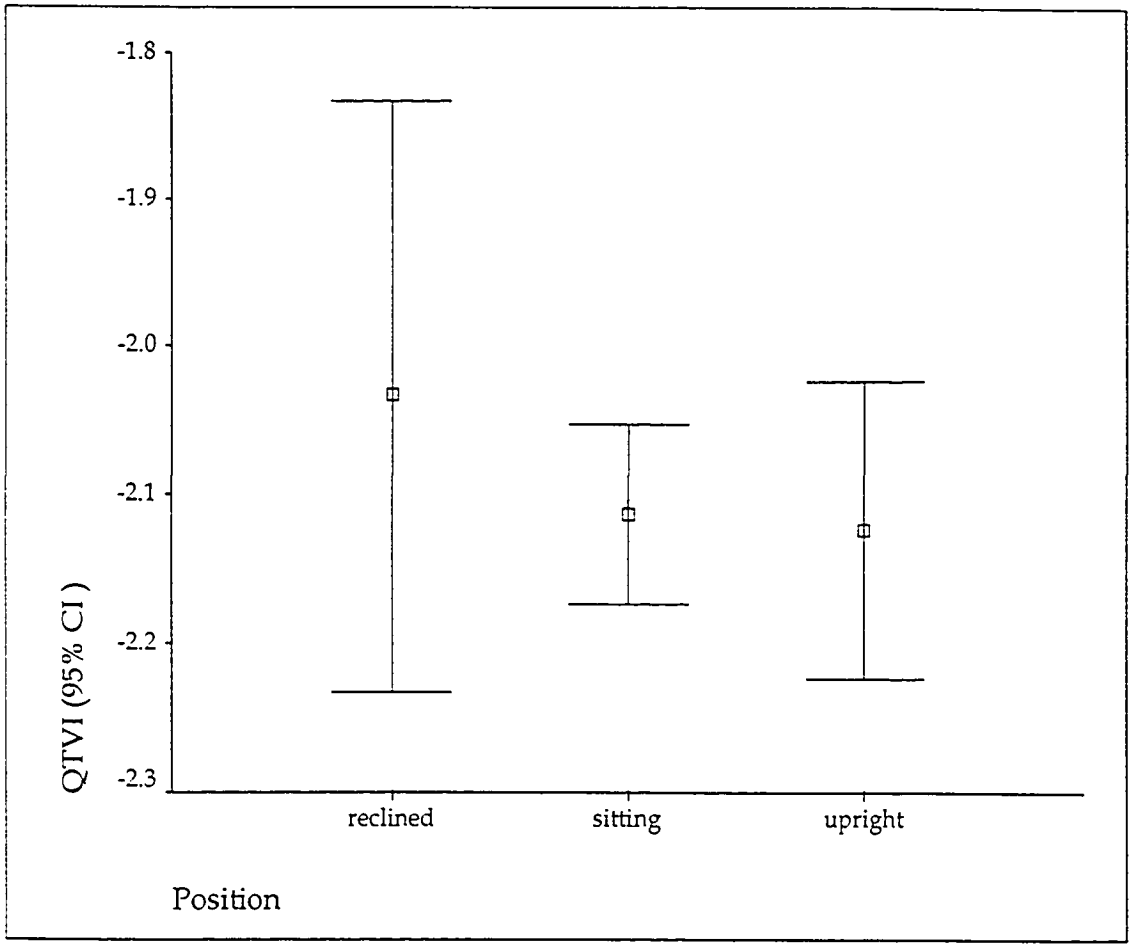


Figure 4- 11. QTVI with Exertion

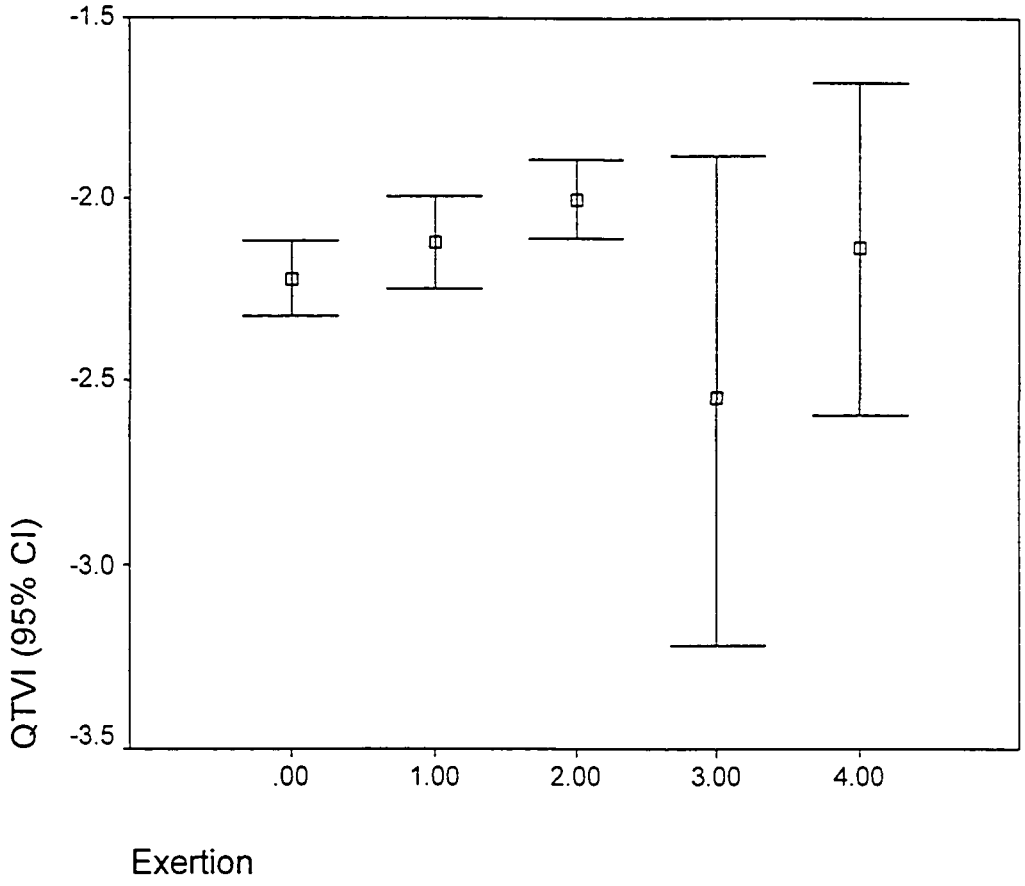


Figure 4- 12. QTVI with Personal Interaction Categories

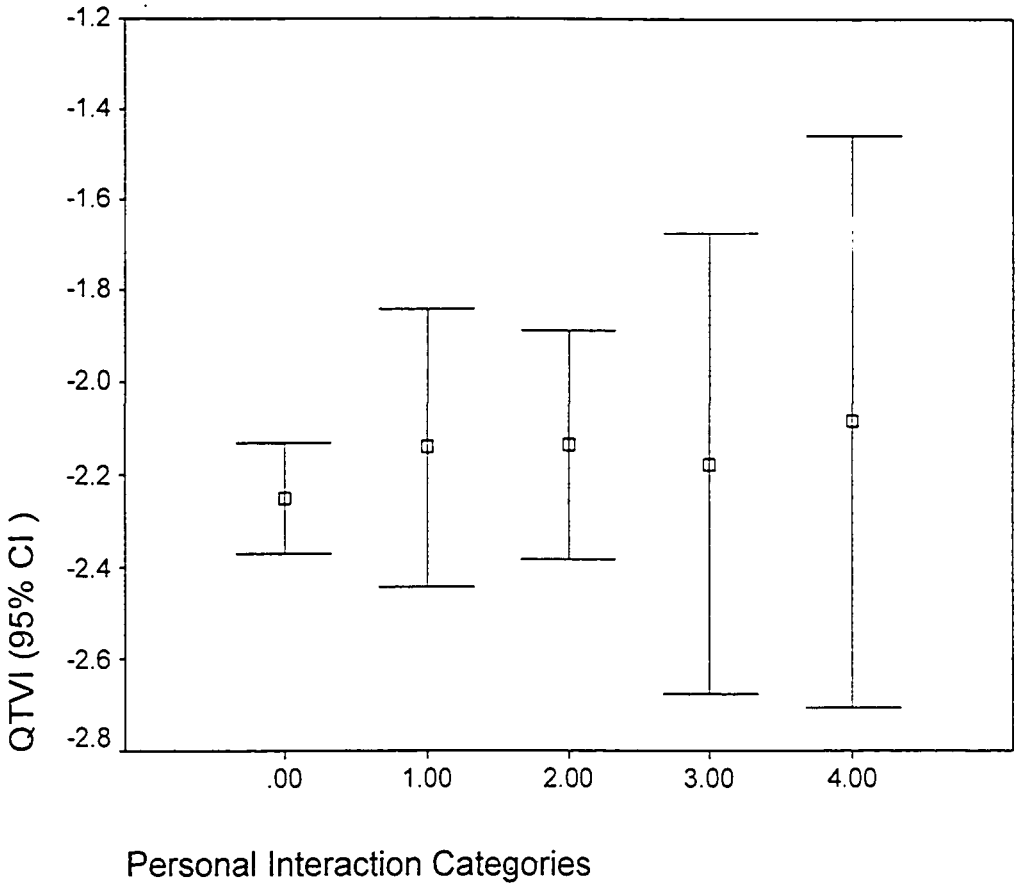


Figure 4- 13. QTVI with Psychological Demands

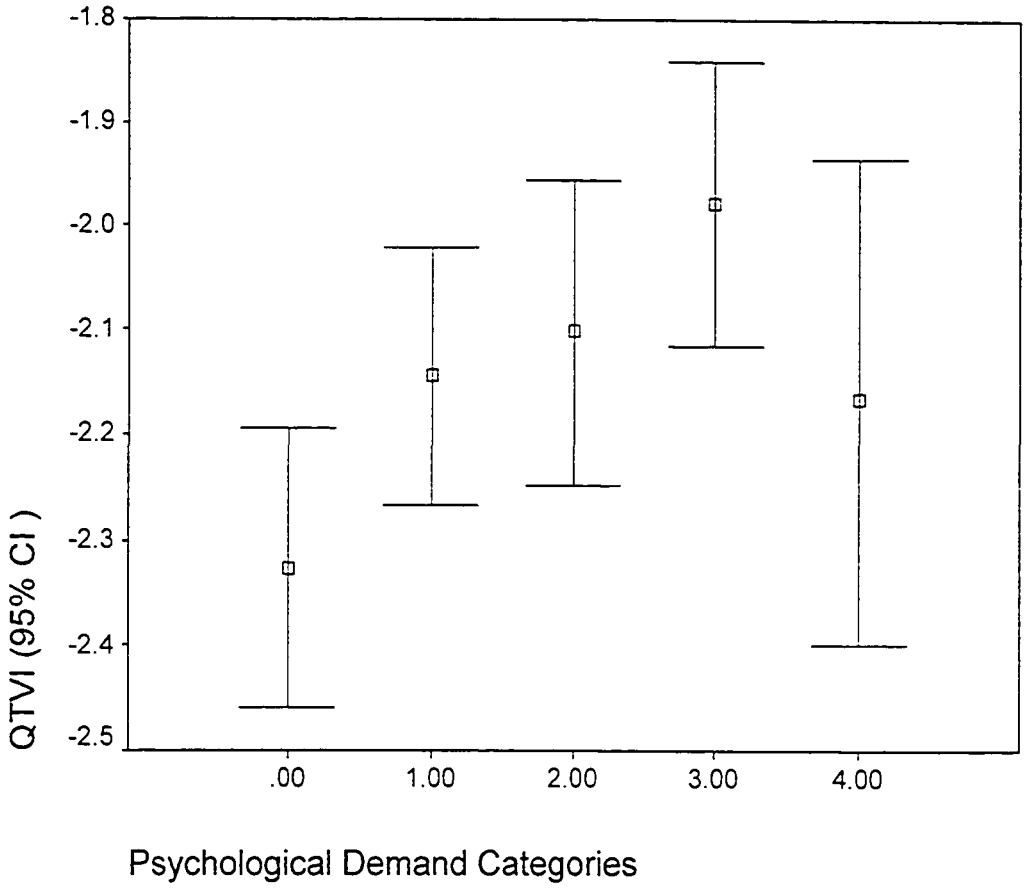


Figure 4- 14. QTVI (with Imputed Data) during Wake Sleep

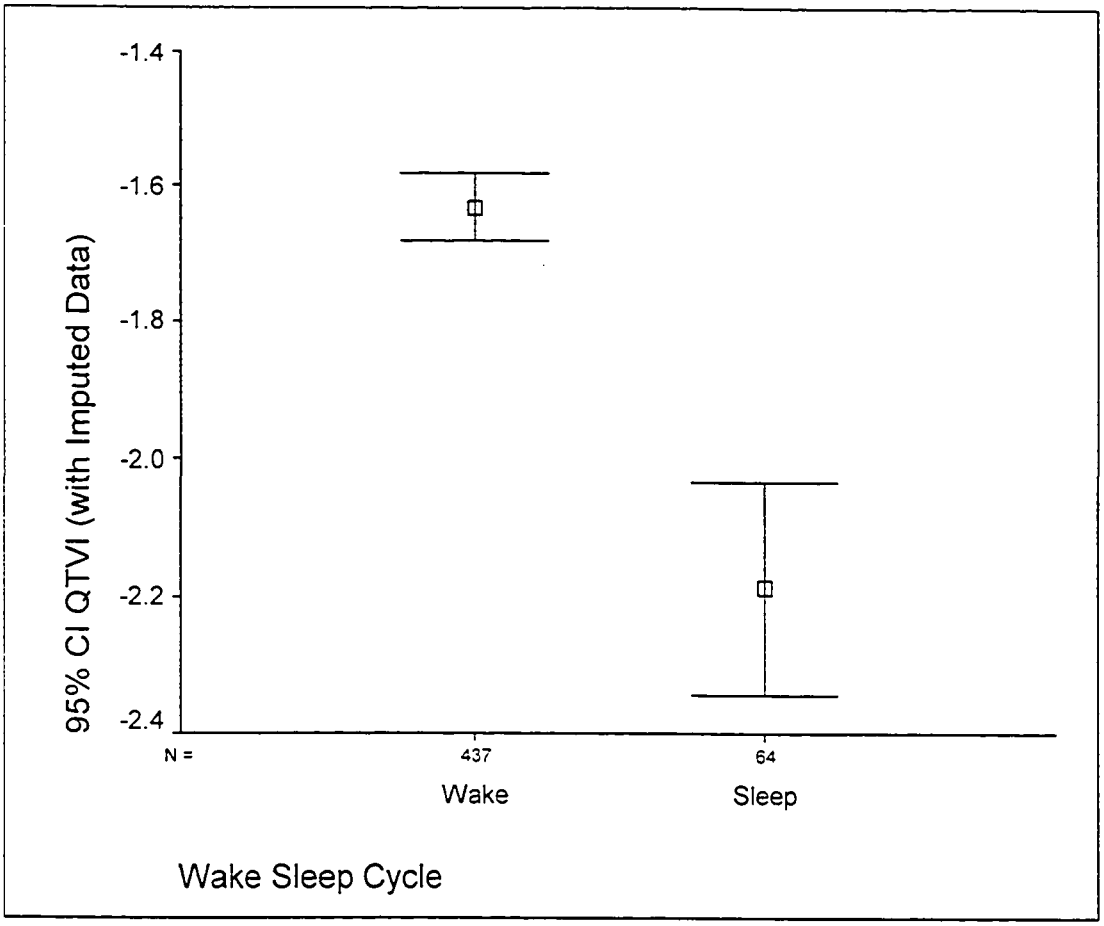


Figure 4- 15. QTVI (Imputed Data) with Position

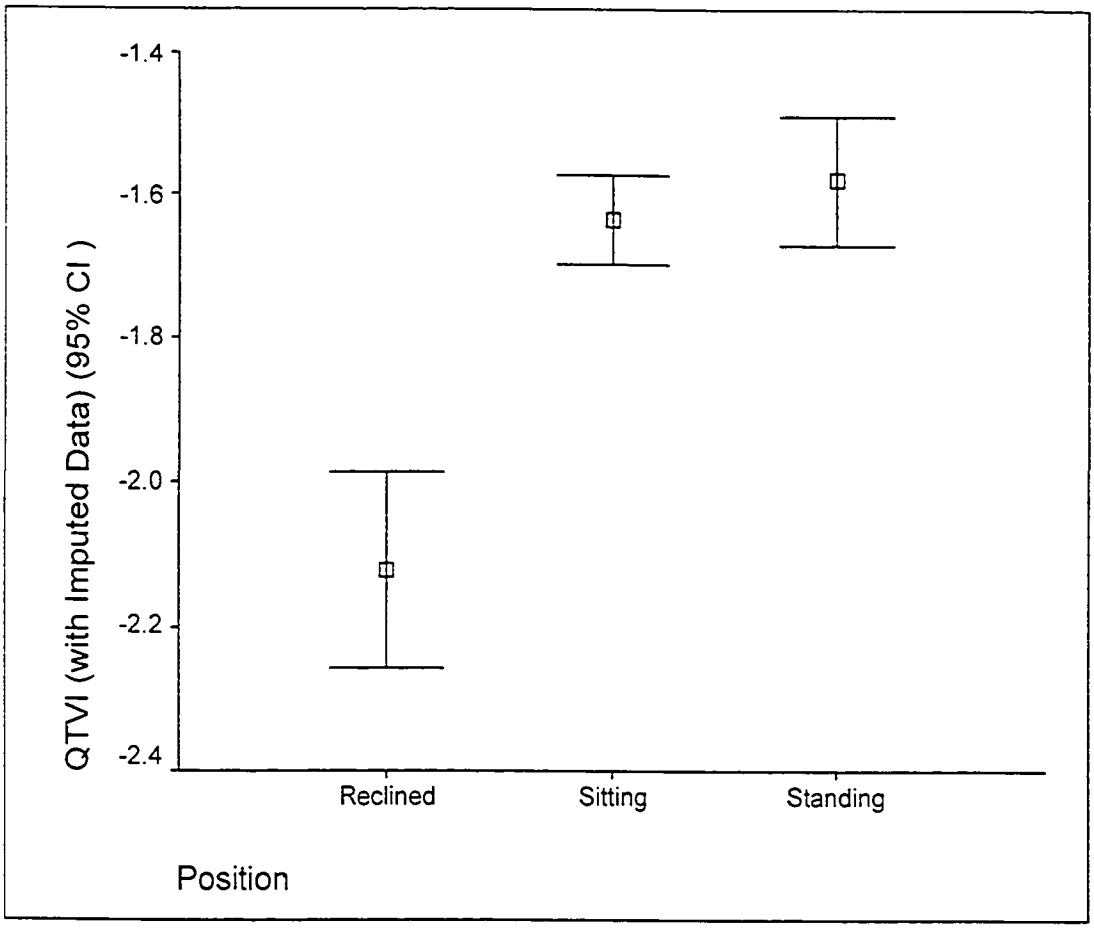


Figure 4- 16. QTVI (with Imputed Data) with Exertion

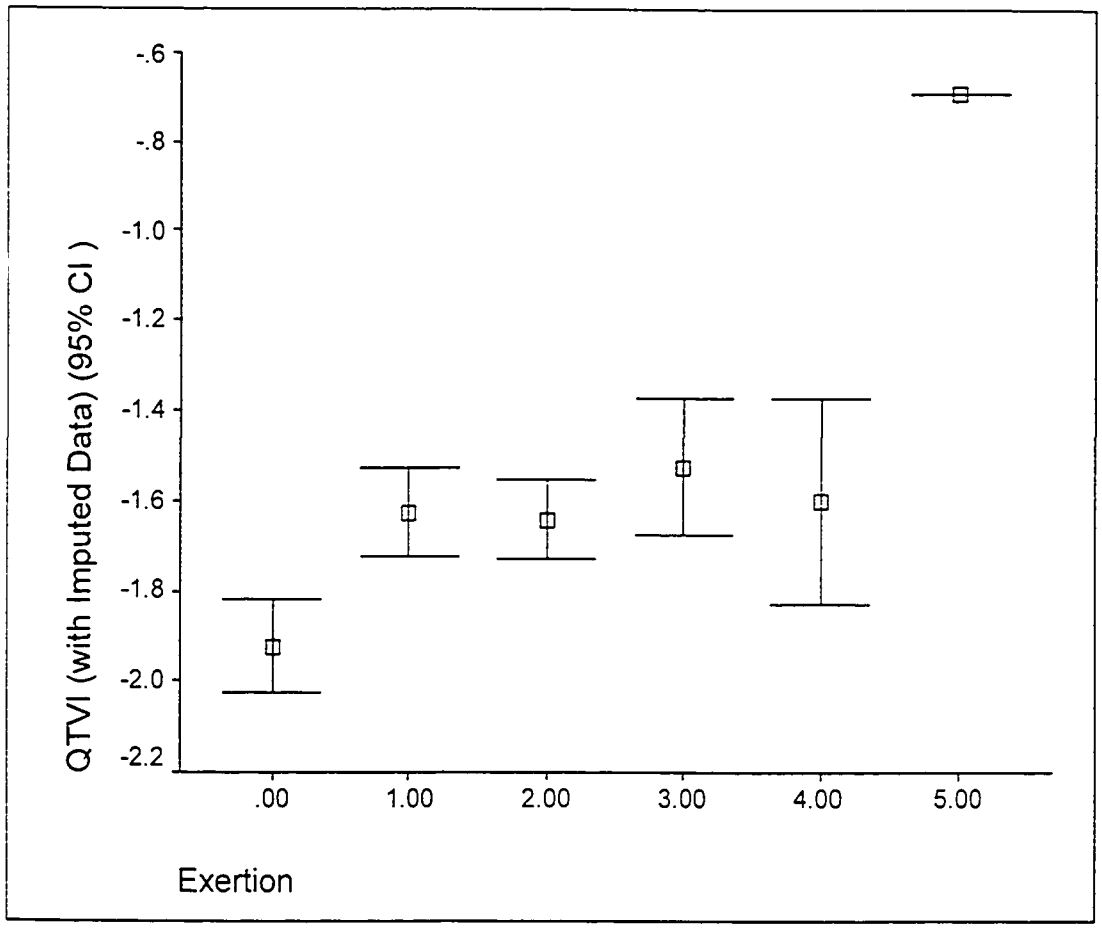


Figure 4-s 17. Residual Heart Rate with Wake and Sleep

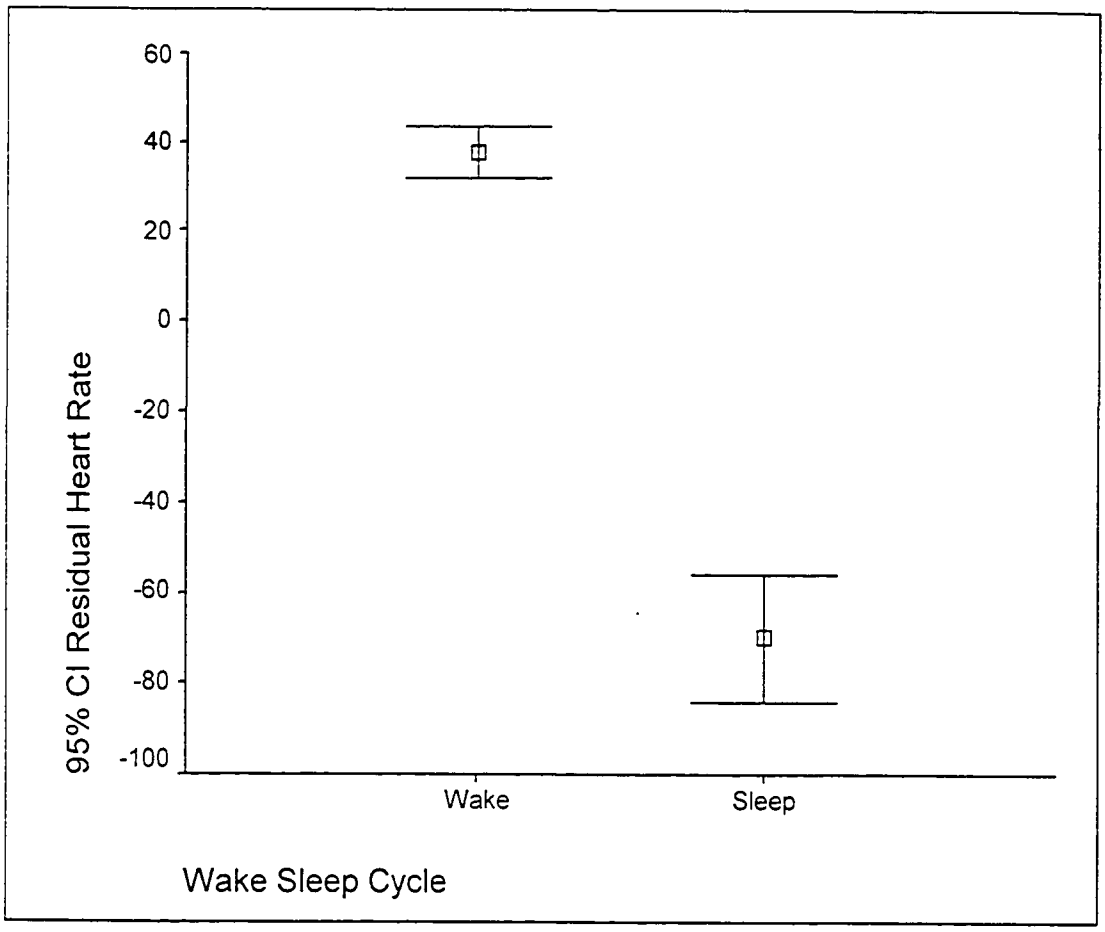


Figure 4- 18. Residual Heart Rate with Position

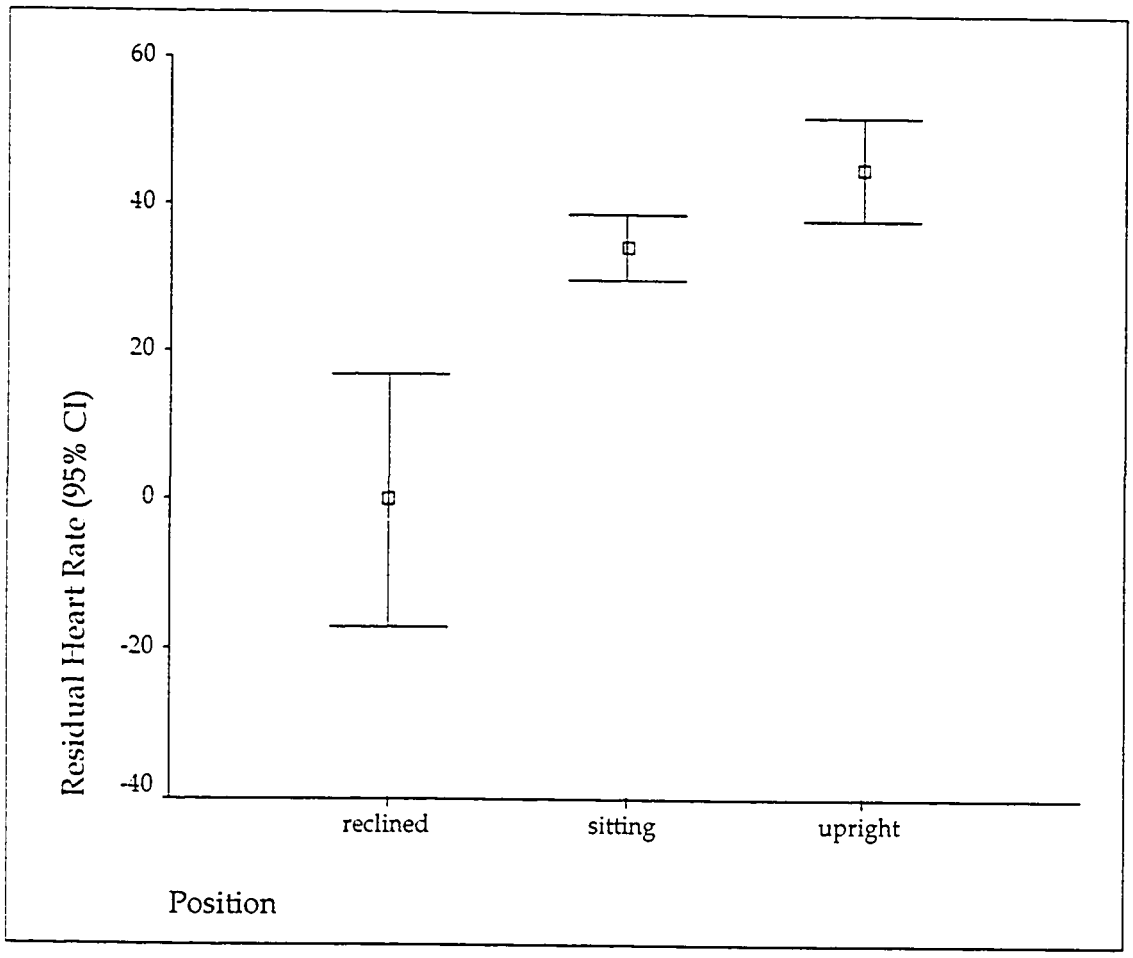


Figure 4- 19. Residual Heart Rate with Exertion

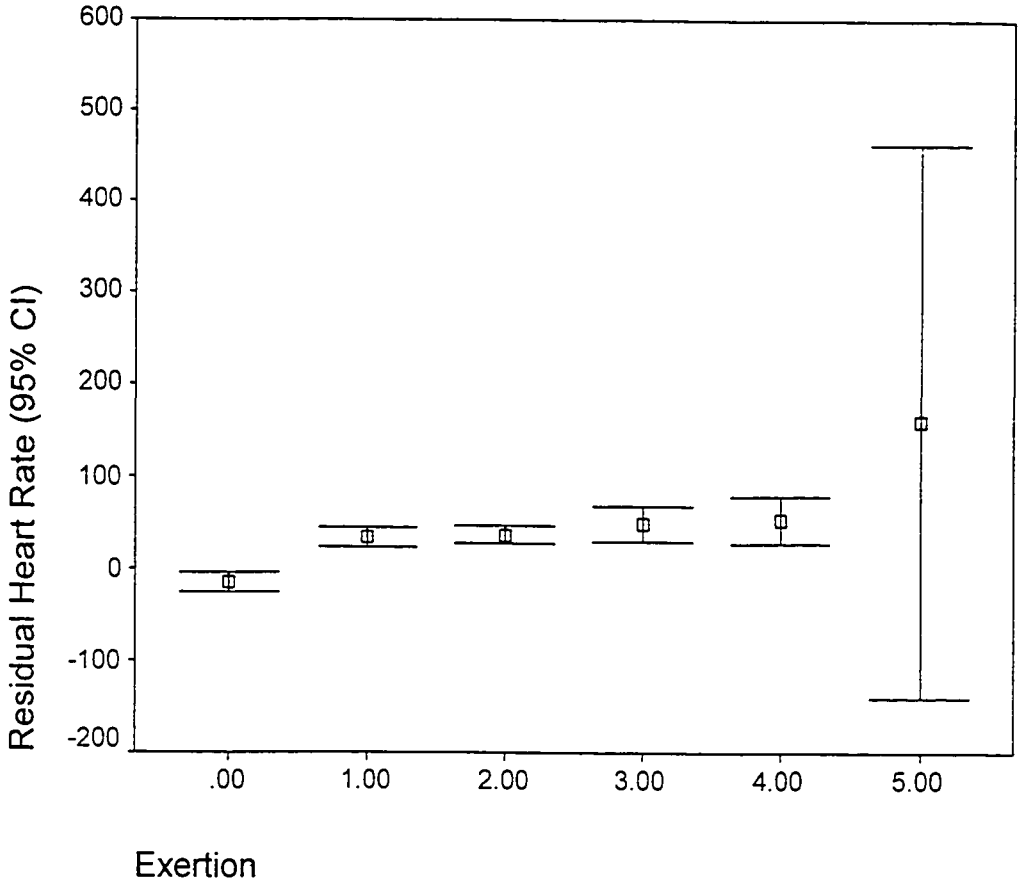


Figure 4- 20. Residual Heart Rate with Personal Interaction

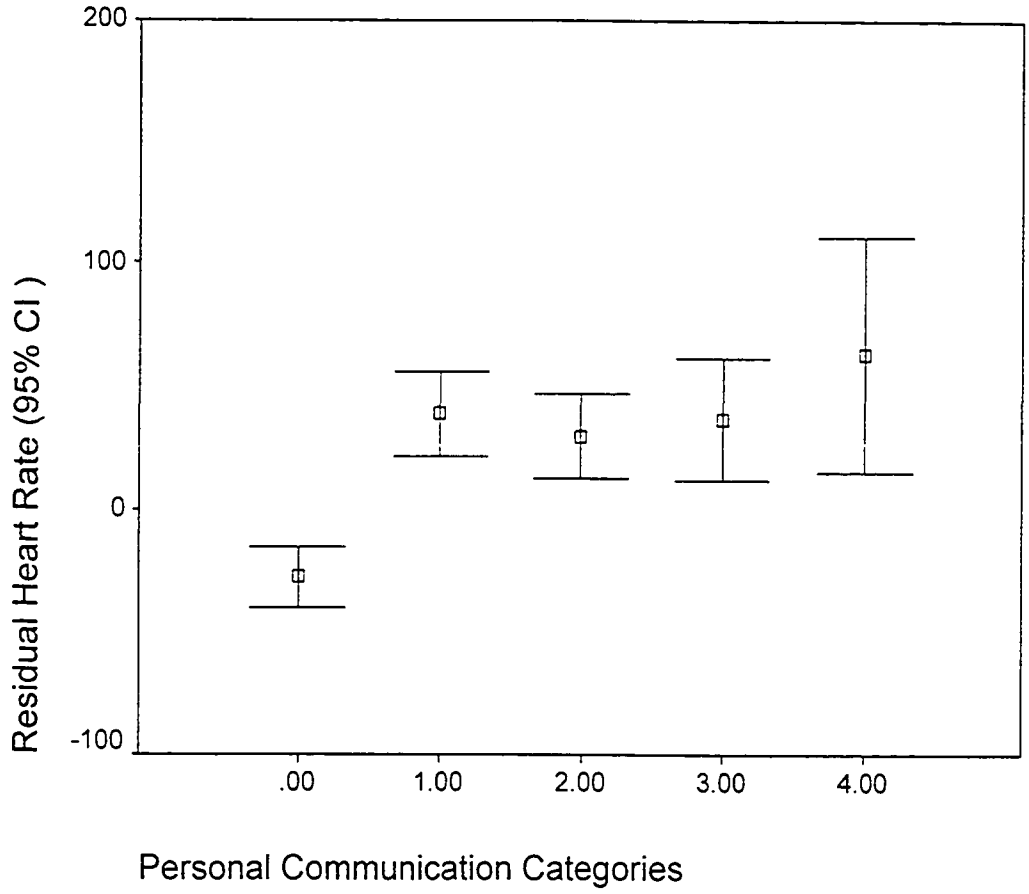


Figure 4- 21. Residual Heart Rate with Psychological Demands

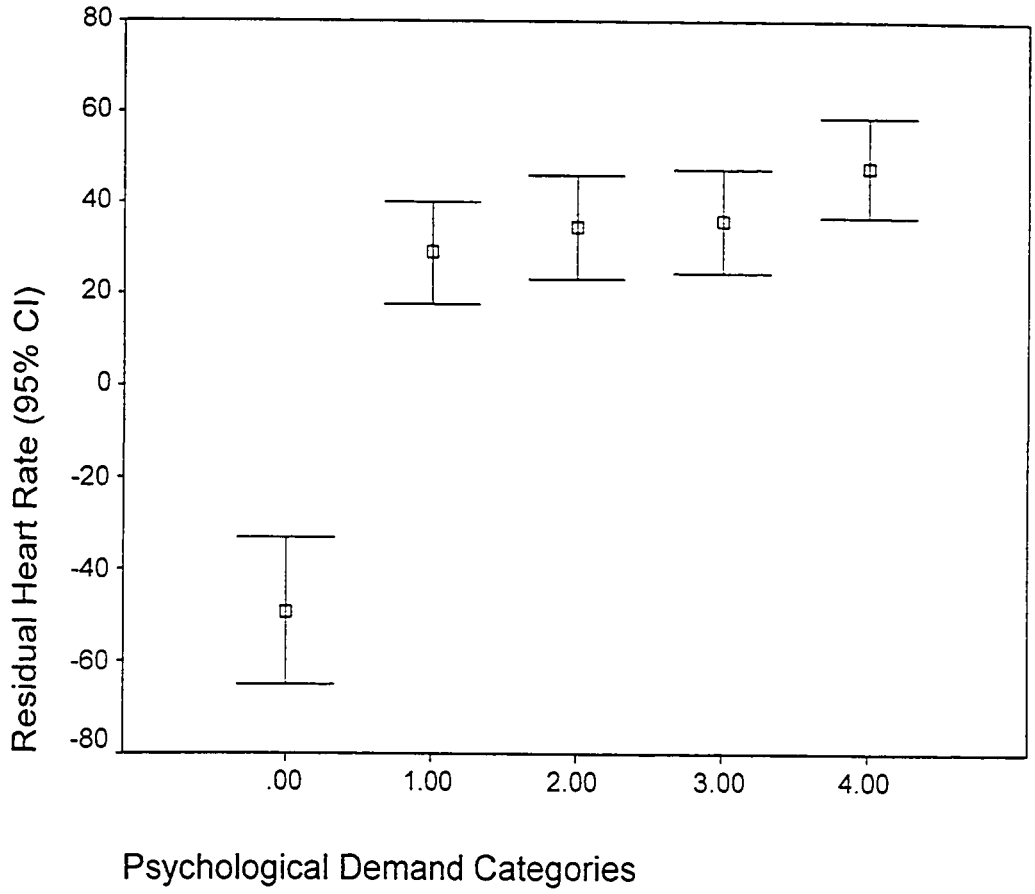


Figure 4- 22. Algorithm for Autonomic Space Categorization based on High Frequency Power and Residual Heart Rate

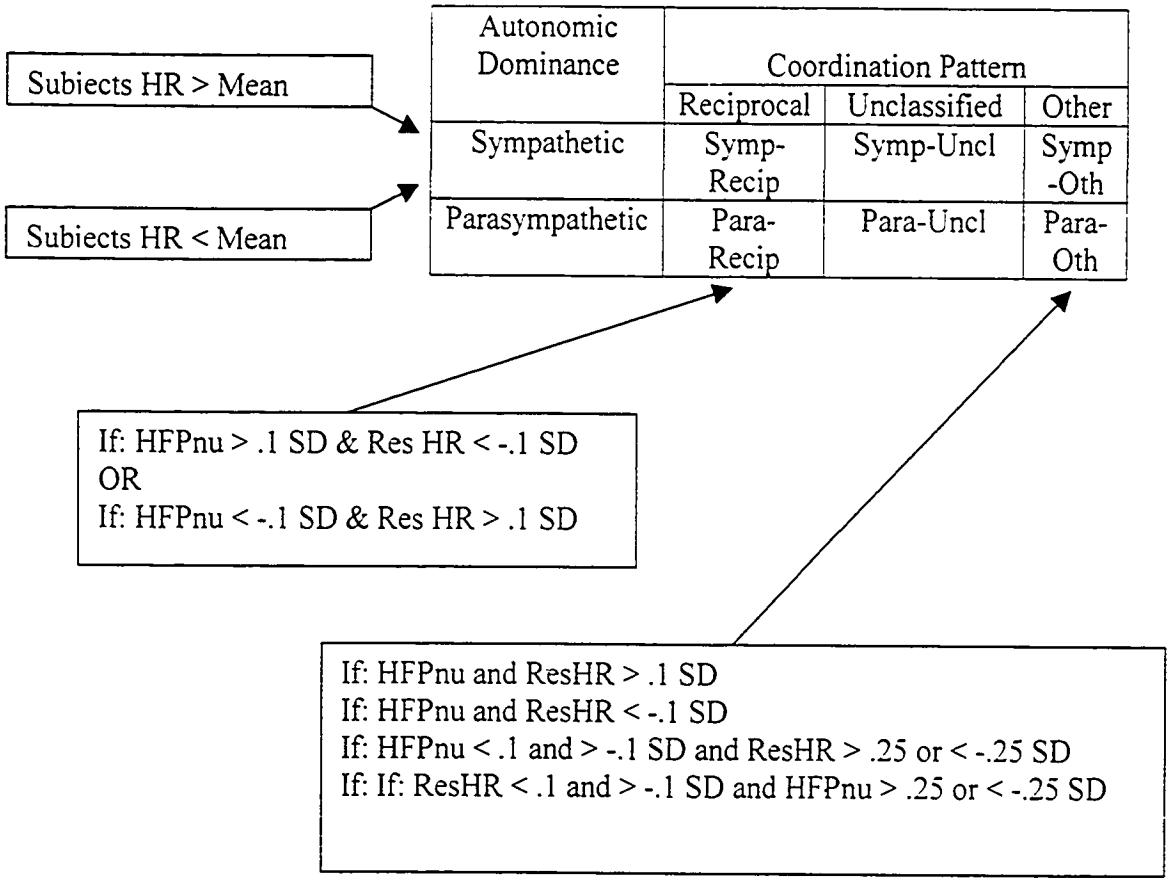
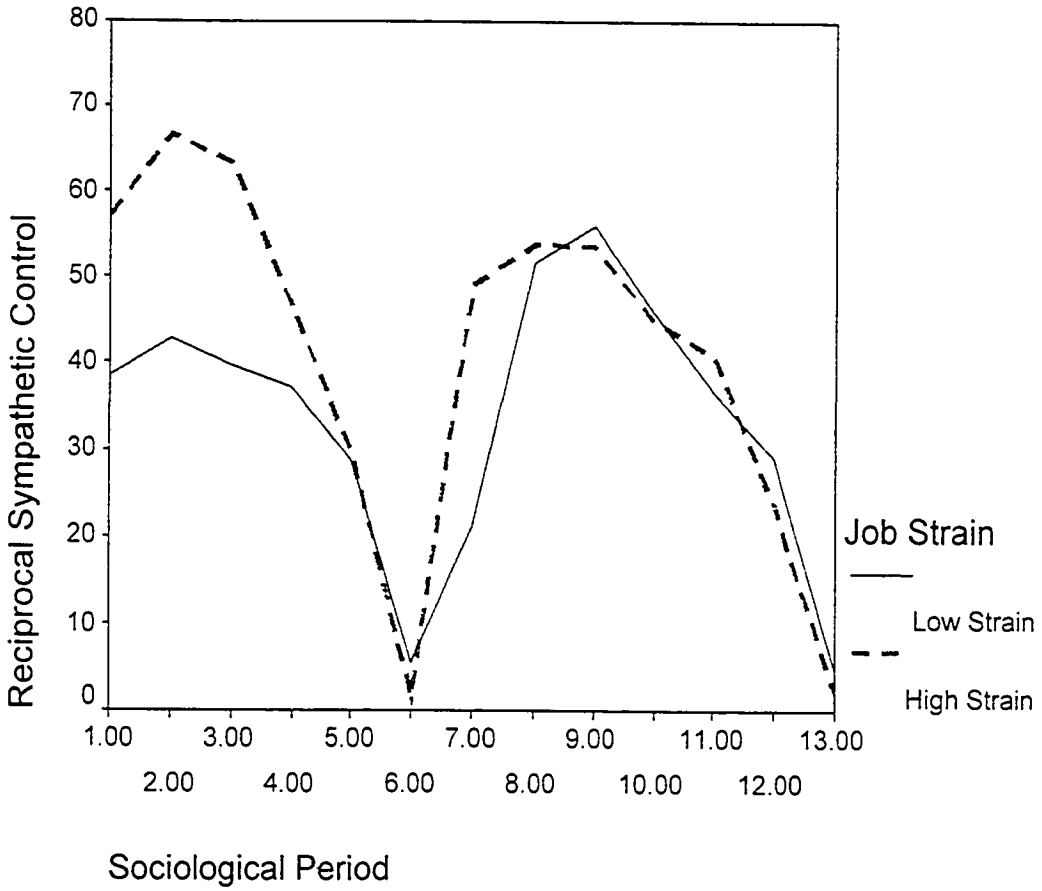
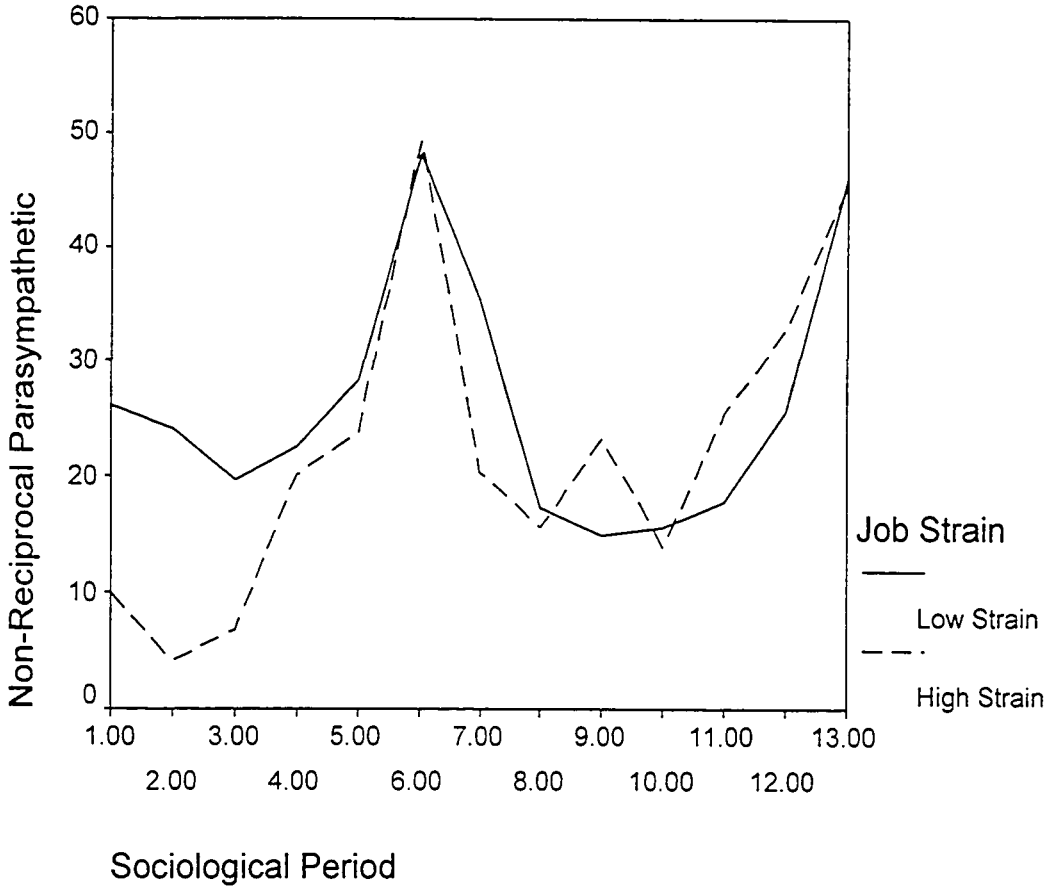


Figure 4- 23. Sympathetic Reciprocal Control During the Monitoring Period



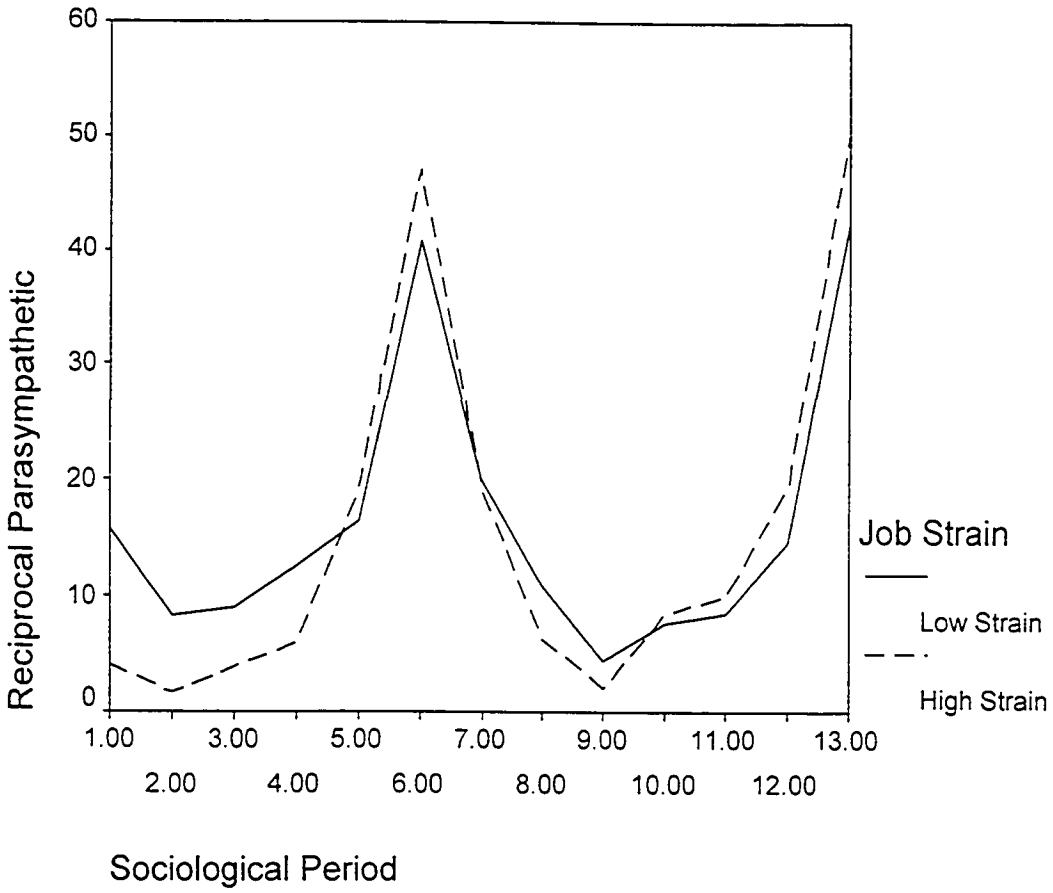

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 24. Non-Reciprocal Parasympathetic Control During the Monitoring Period




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 25. Reciprocal Parasympathetic Control During the Monitoring Period




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
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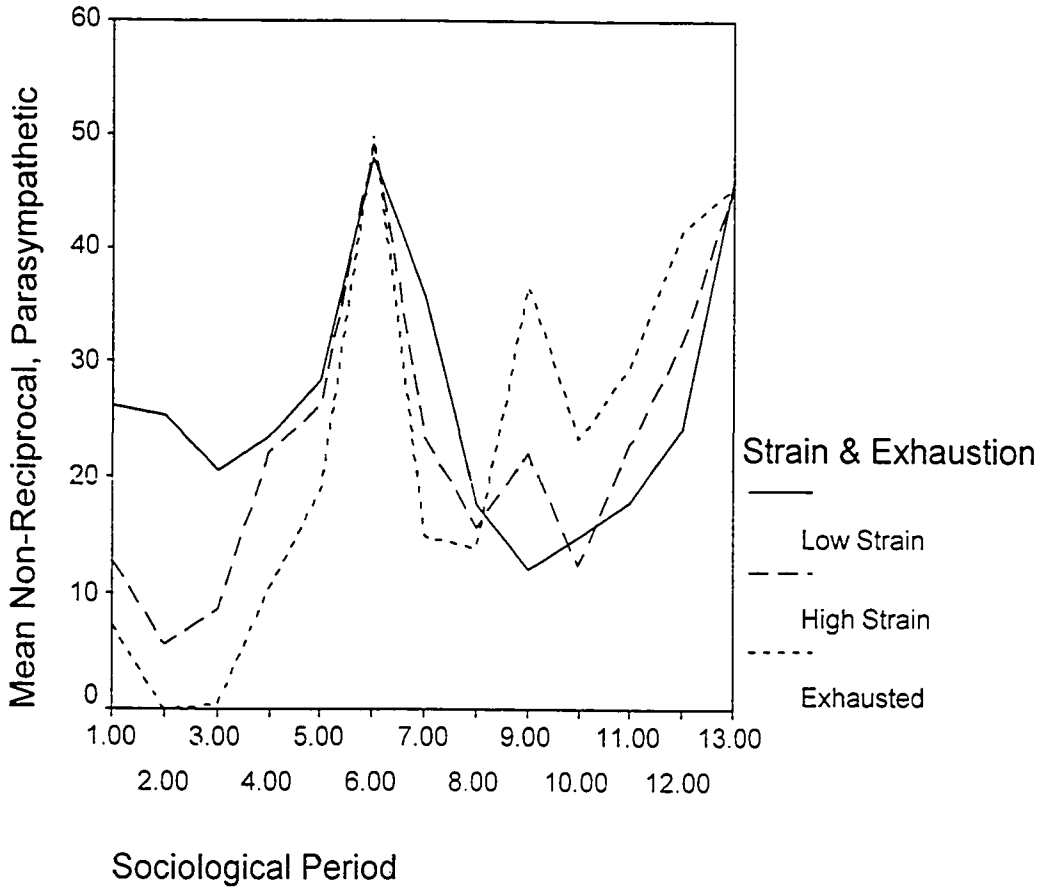
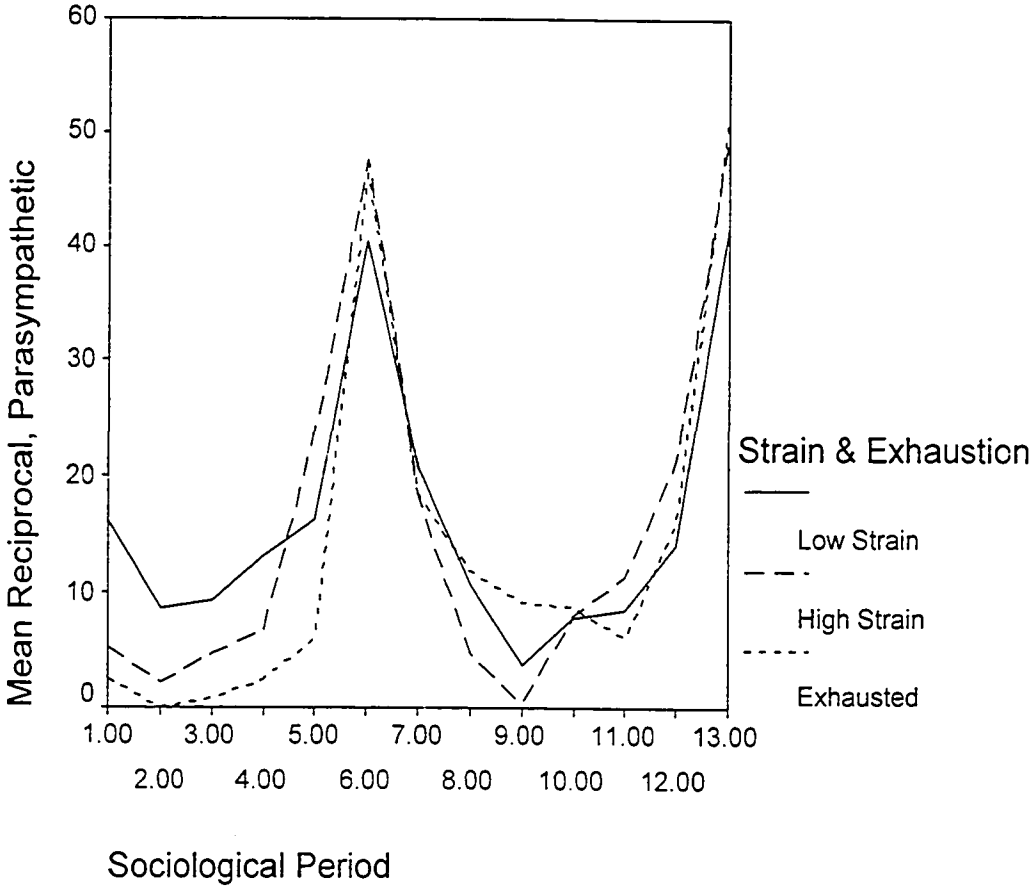
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 26. Parasympathetic Non-Reciprocal Control: The Effect of Exhaustion



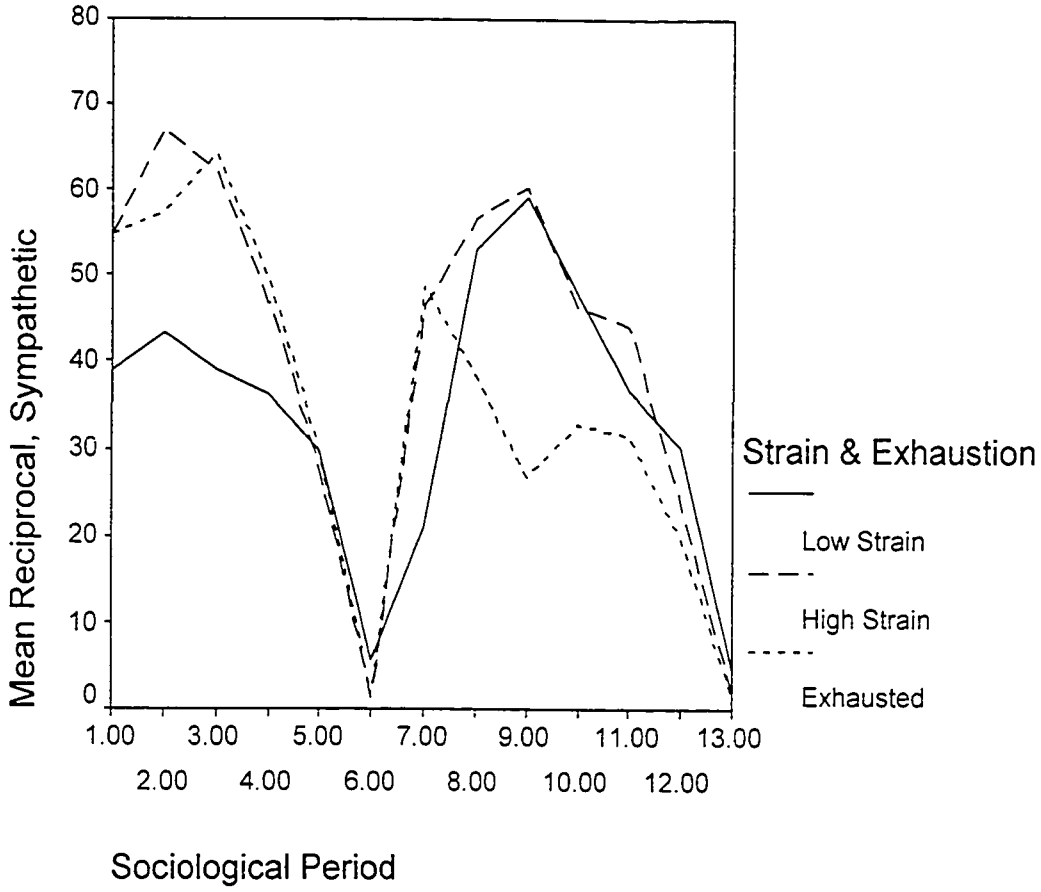

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 27. Parasympathetic Reciprocal Control: The Effect of Exhaustion



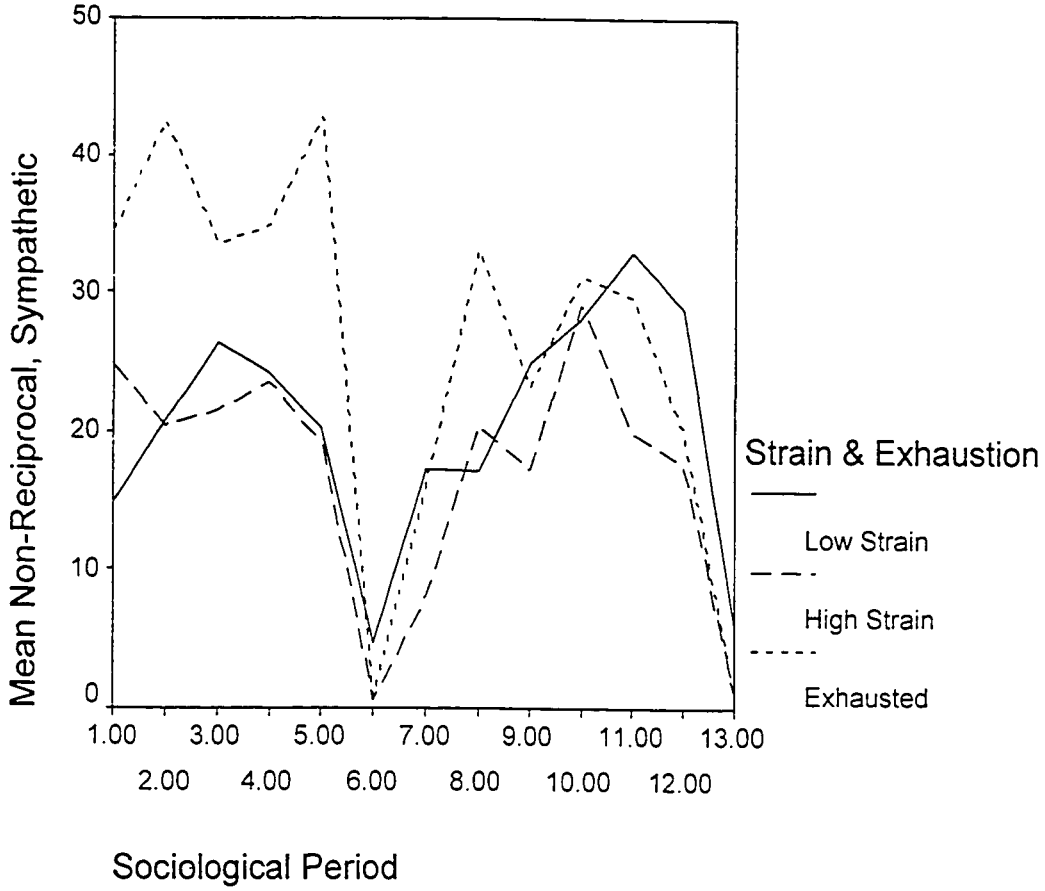

| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 28. Sympathetic Reciprocal Control: The Effect of Exhaustion




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 29. Sympathetic Non-Reciprocal Control: The Effect of Exhaustion




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 30. Workday Slope Coefficients of HR as a Function of HFP

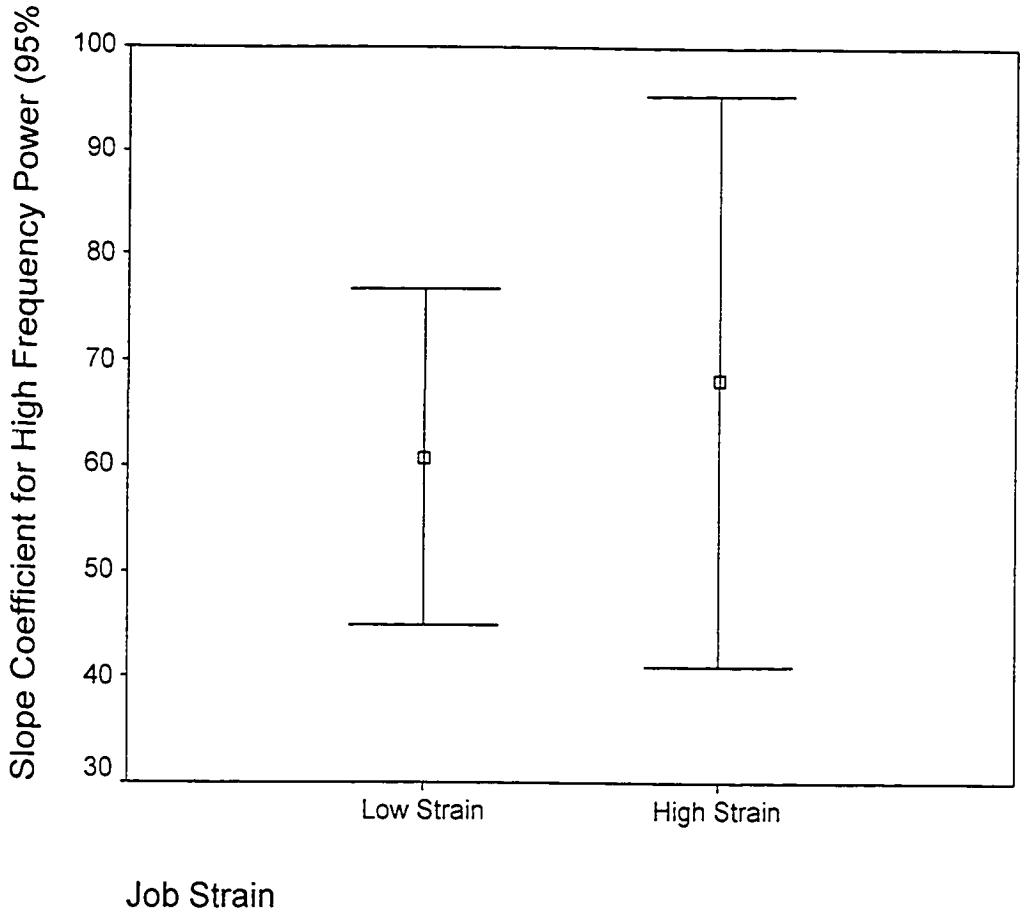


Figure 4- 31. Workday Slope Coefficients of HR as a Function of HFP

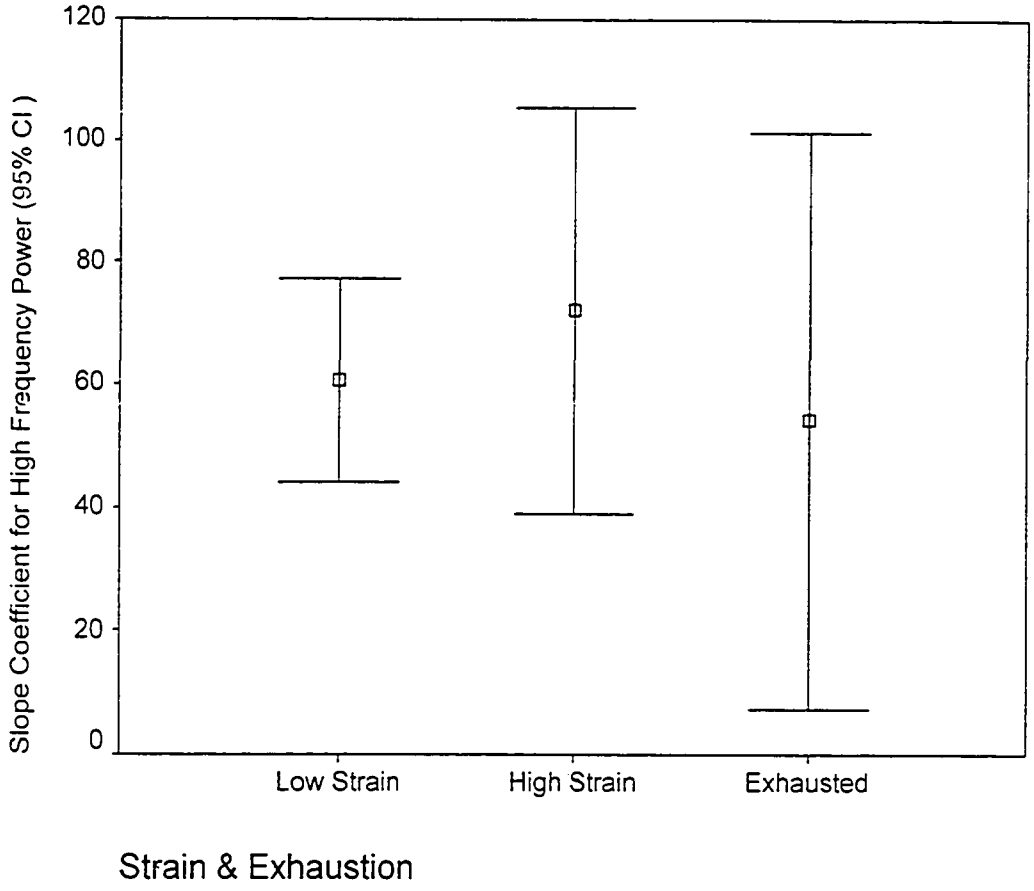


Figure 4- 32. Rest Time Slope Coefficients of HR as a Function of HFP

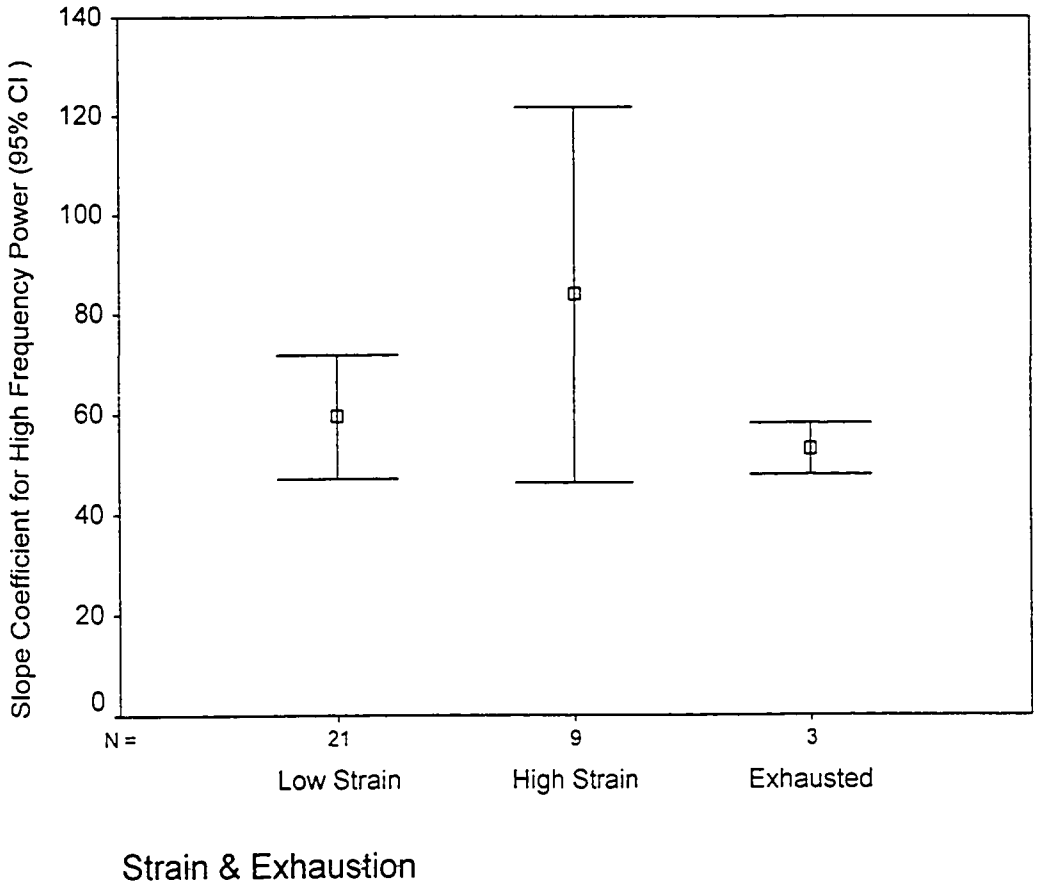


Figure 4- 33. Distribution of Vagal Activity

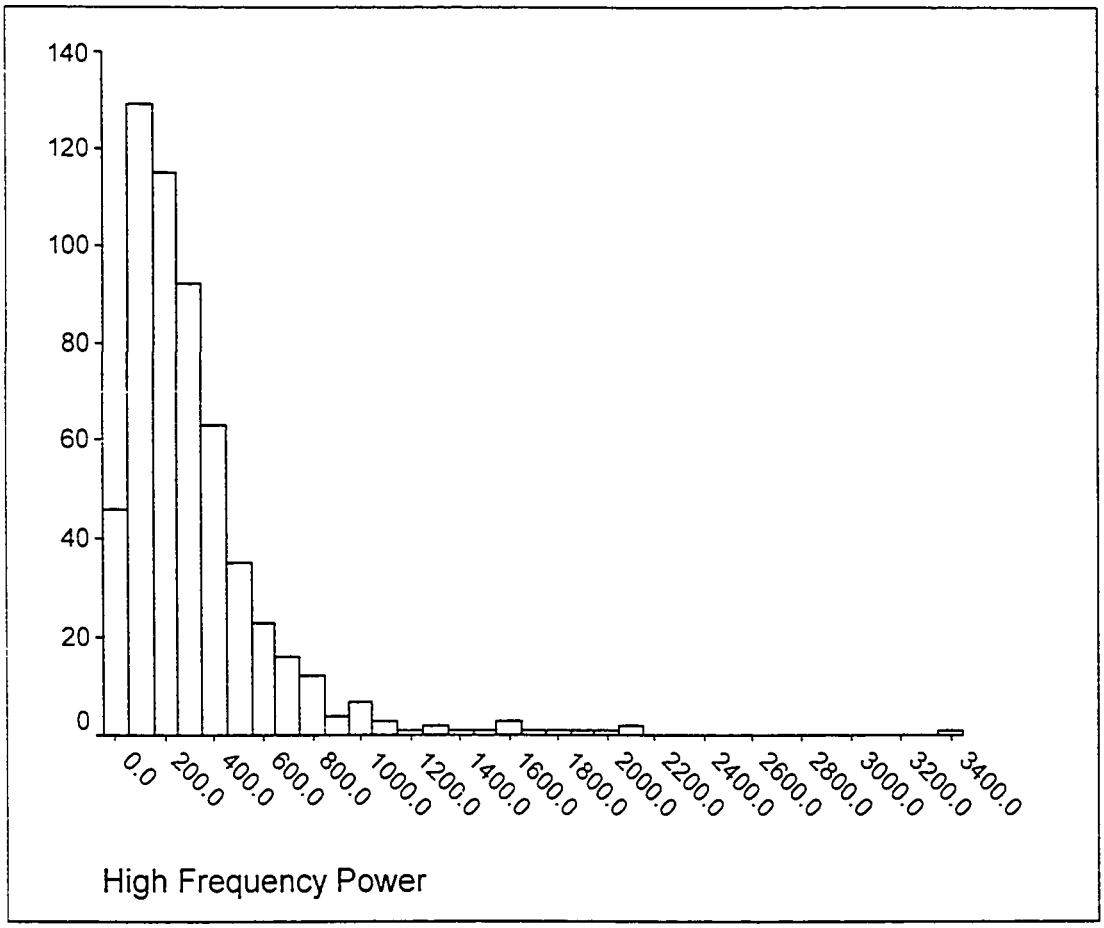
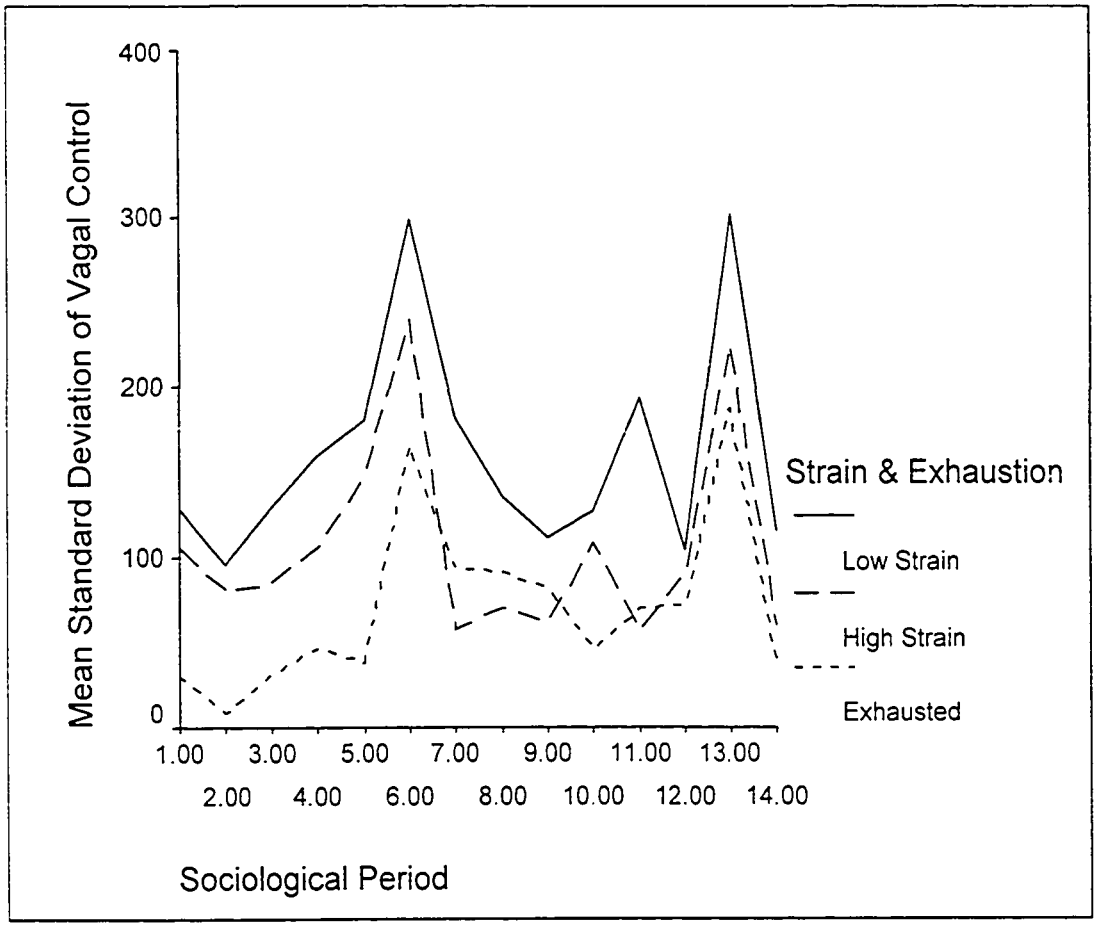
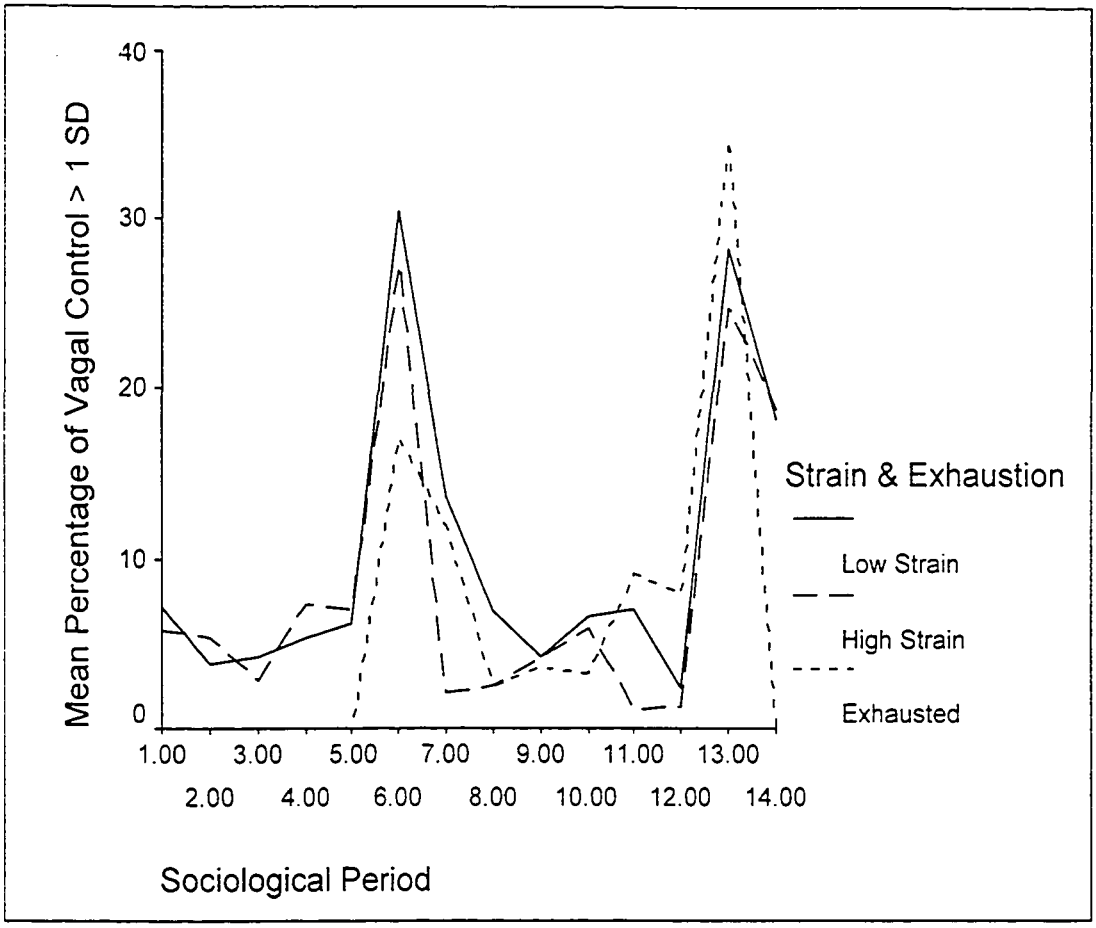


Figure 4- 34. Standard Deviation of Vagal Cardiac Control




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
| 5 | Dinner until Sleep |
| 6 | Sleep |
| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 35. Percentage of Vagal Control Greater Than 1 Standard Deviation




| Sociological Periods | |
|---|---|
| 1 | Morning at Work |
| 2 | Lunch (on Workday) |
| 3 | Afternoon at Work |
| 4 | After Work before Dinner |
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| 7 | Awake on Rest Day |
| 8 | Time that corresponds to Morning at Work |
| 9 | Time that corresponds to Lunch at Work |
| 10 | Time that corresponds to Afternoon at Work |
| 11 | After Period 10 until Dinner |
| 12 | Dinner until Sleep |
| 13 | Sleep |
| 14 | Awake on Second Rest Day – remove monitor |

Figure 4- 36. Sample Poincaré Plot for Entire Monitoring Period (Low Strain Subject)

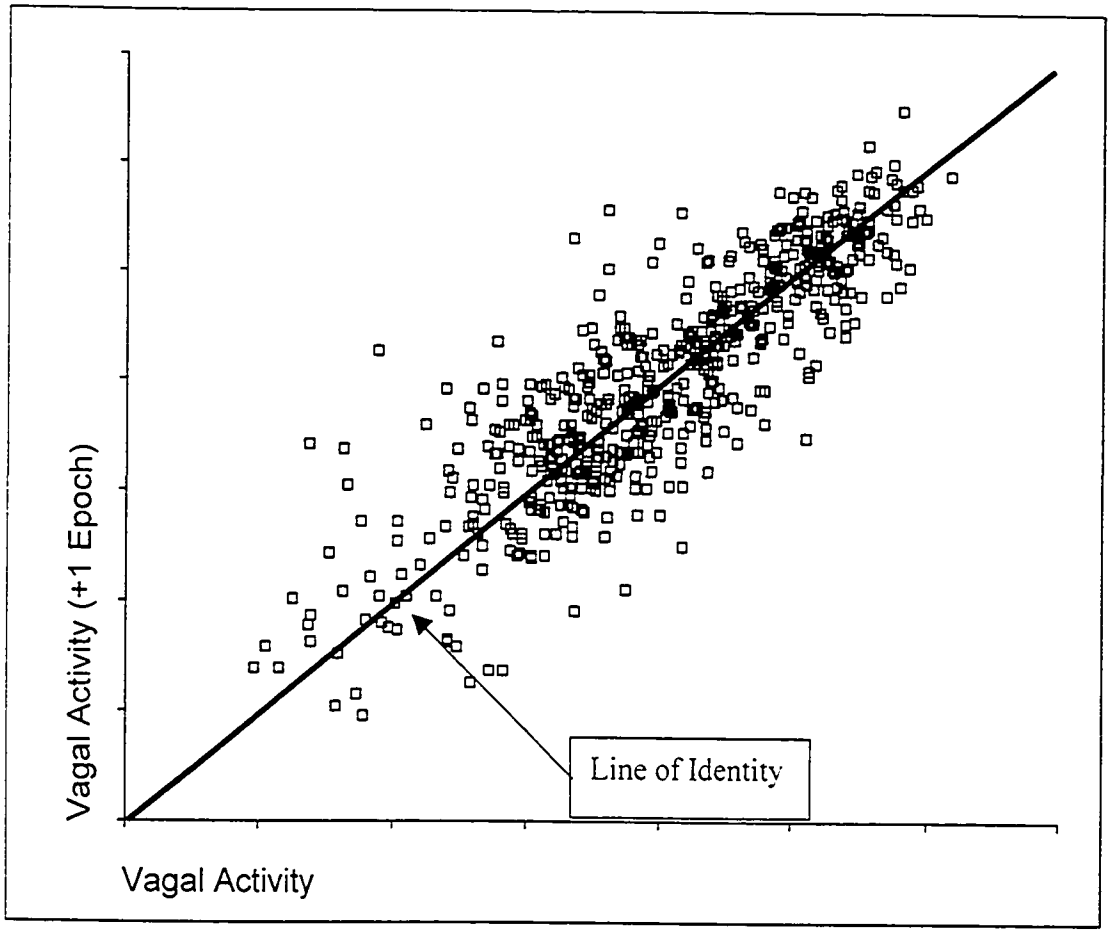


Figure 4- 37. Distributions of Residuals Displaying Short Term Variation in Vagal Activity

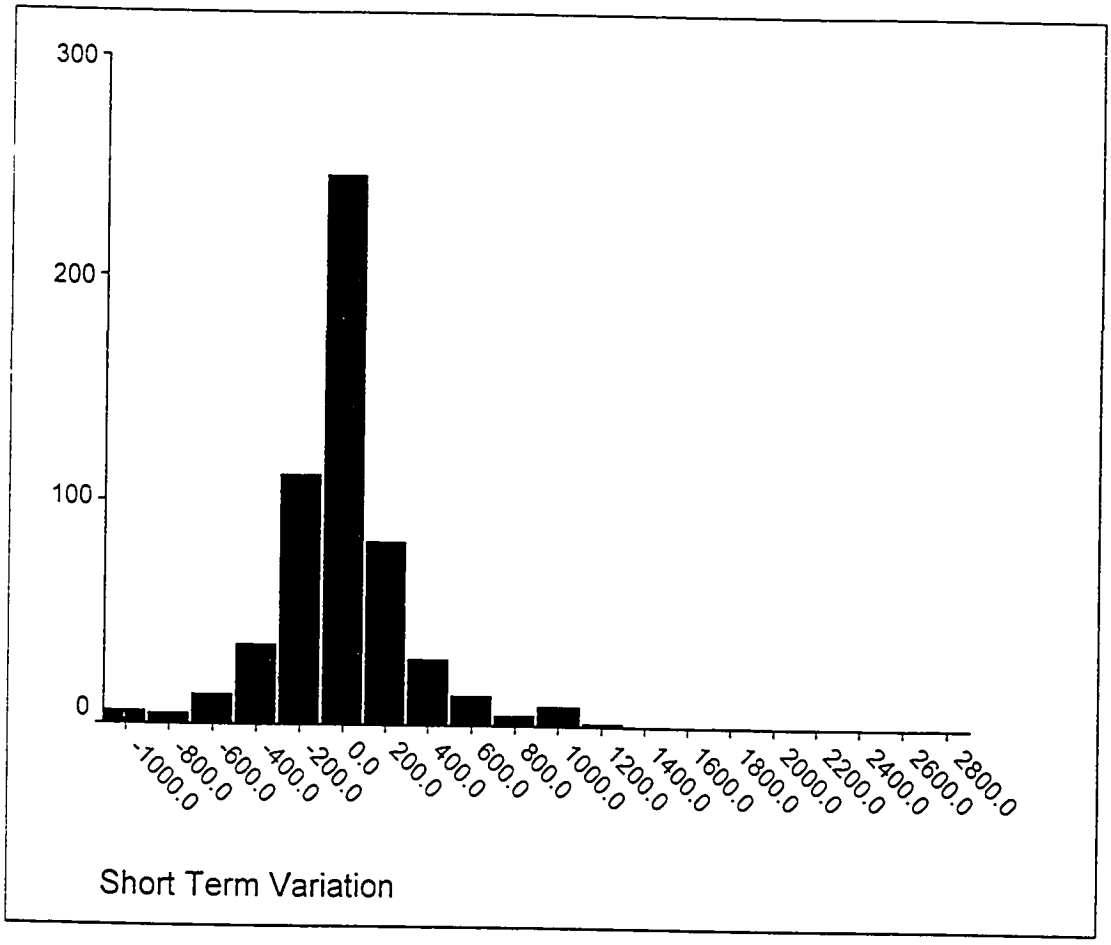


Figure 4- 38. Sample Poincaré Plot for Work Day (Low Strain Subject)

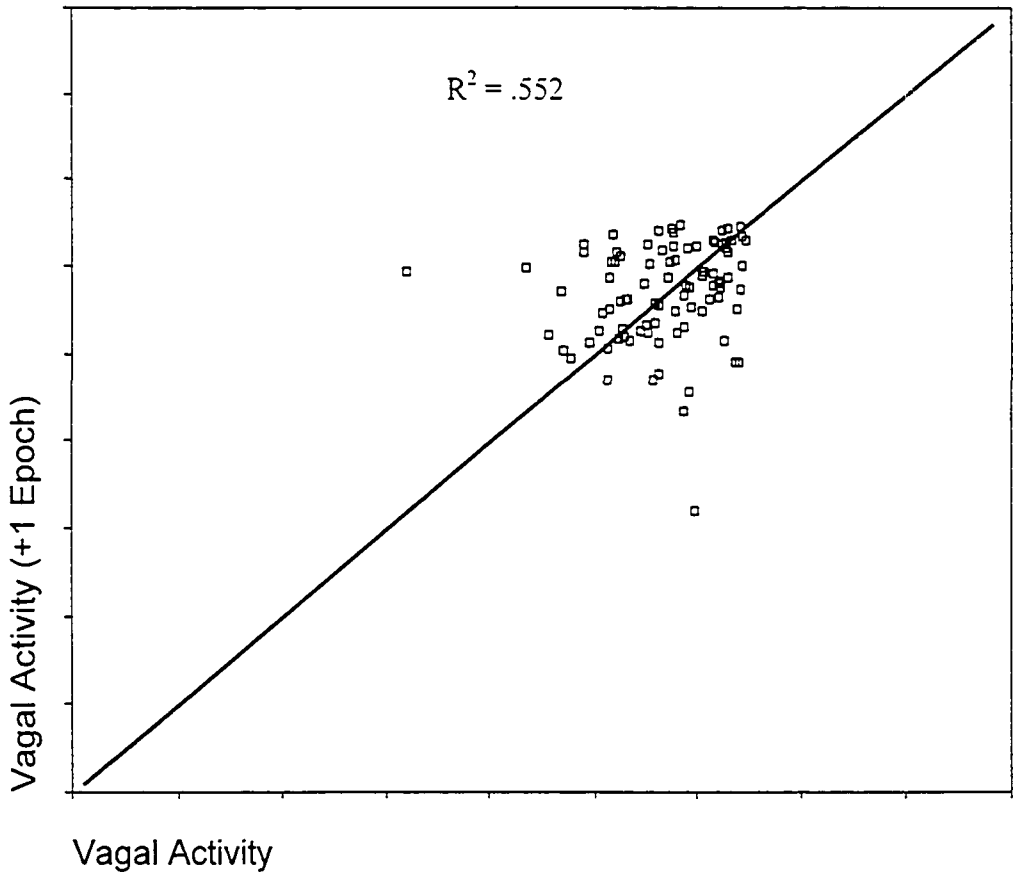


Figure 4- 39. Sample Poincaré Plot for Work Day (High Strain Subject)

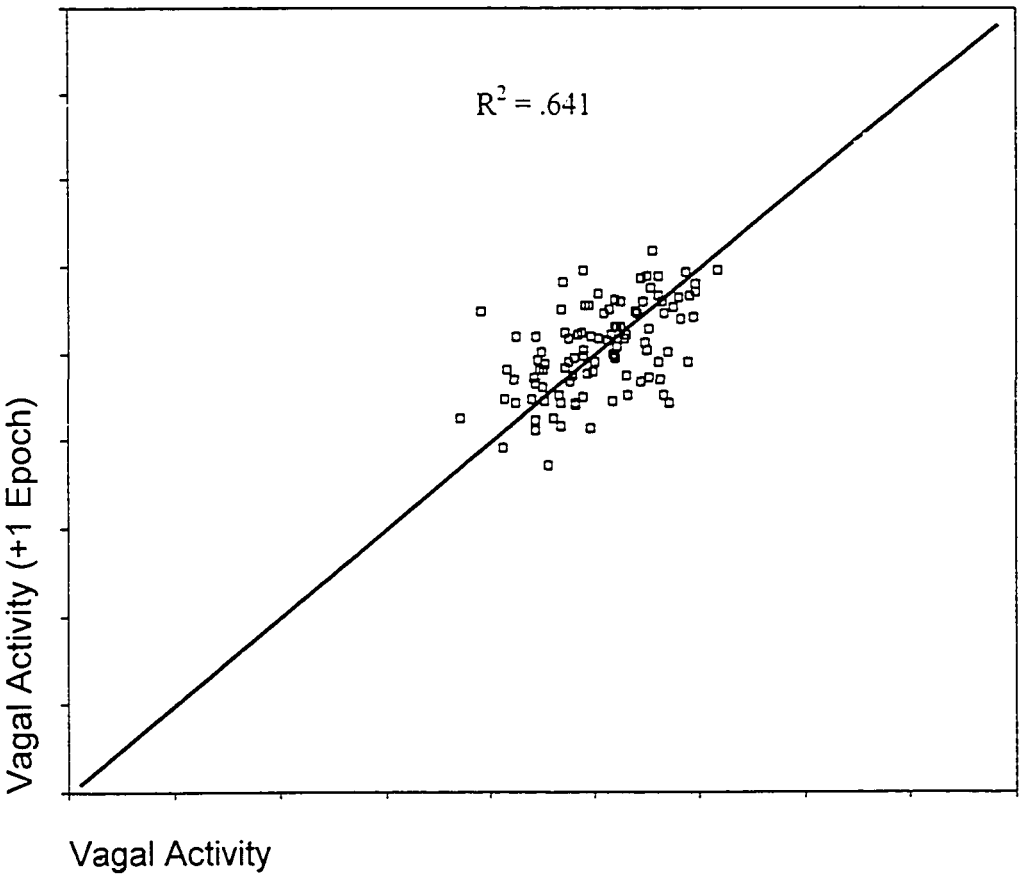


Figure 4- 40. Sample Poincaré Plot for Work Day (Exhausted Subject)

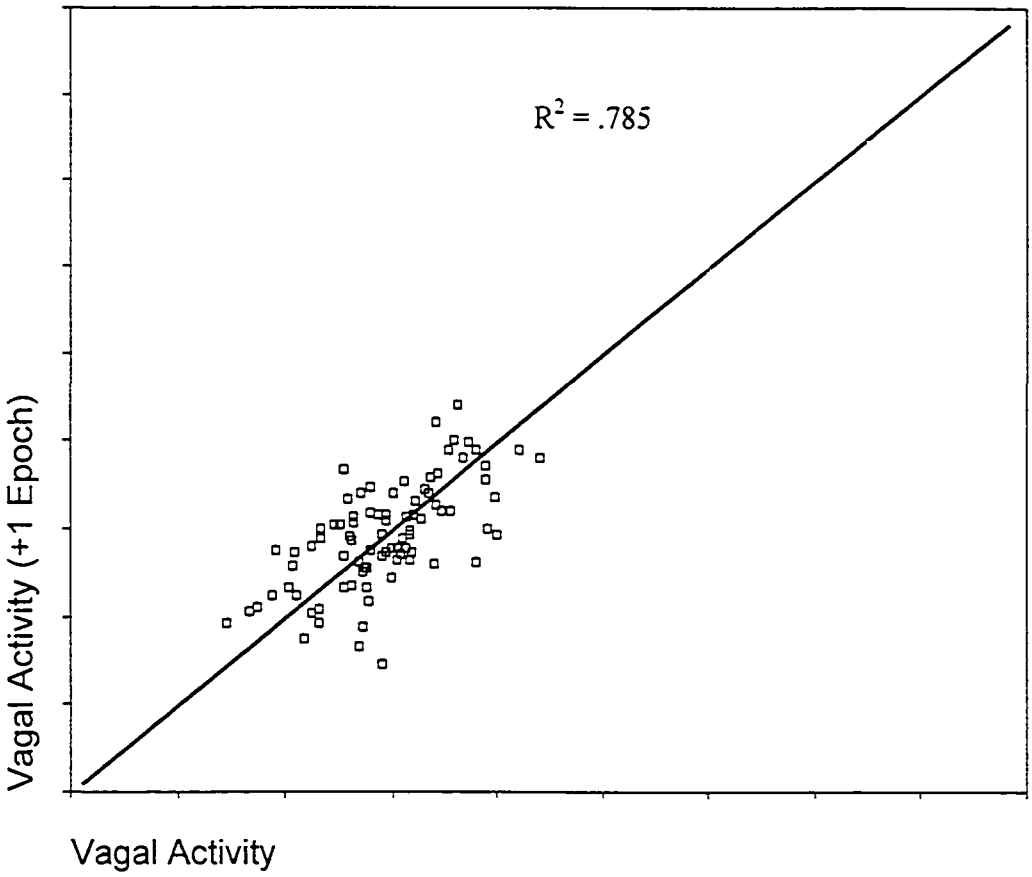
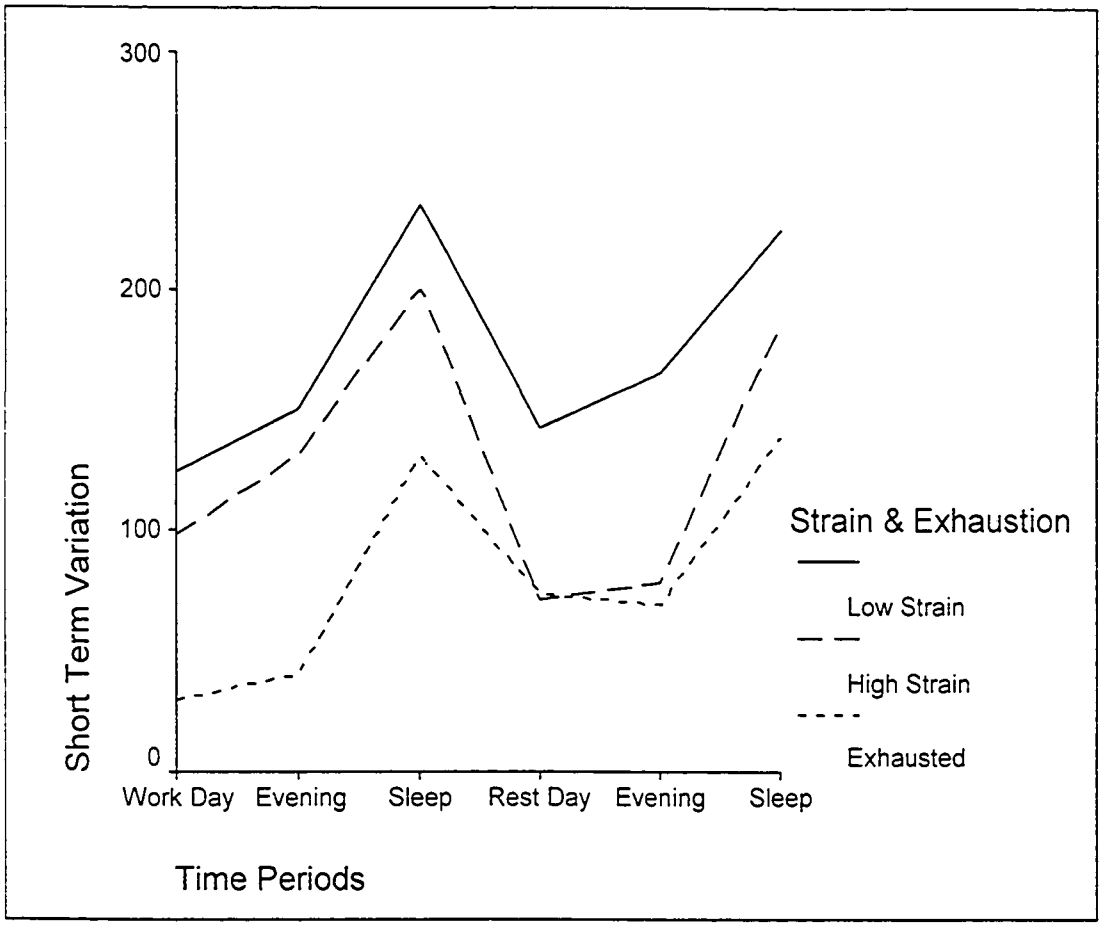


Figure 4- 41. Short Term Variation as Determined by Poincaré Plot Residuals



# CHAPTER FIVE

## OVERALL CONCLUSIONS

The overall goal of the Heart Rate Variability project was to assess the effect of stress in the work environment on cardiovascular health through ambulatory electrocardiograph (ECG) monitoring [1]. The overall hypothesis was that psychosocial stress from work activity effects cardiac regulation in a pathogenic manner. Chapters Two through Four have provided evidence from a variety of analyses that support this overall hypothesis. We utilized well-understood measures (demand – control model [2-4]) to quantify job stress. These chapters have assessed altered autonomic nervous function as measured with heart rate variability and QT interval variability with a variety of methods. Overall these analyses provide support to the hypothesis that the pathogenic process associated with job strain begins with stress-induced acute deregulation of cardiac control. Chapters Two and Three provide evidence using traditional methods to assess HRV data from the small but growing literature in occupational health research, and the large body of literature of human factors, psychophysiology and cardiovascular epidemiology. Chapter Four utilizes a new theoretical model, Karasek's Stress Disequilibrium Theory [5], to provide evidence that over the long-term, deregulatory effects of chronic stress exposures may reduce adaptive capacity. While reductions of HRV are known to be predictors of increased disease and mortality, the implication that

these reductions are a result of reduction adaptive capacity due to reduced environmental control has not been previously made.

## Addressing the Specific Aims

More specifically this dissertation was conducted to address five specific aims as outlined in Chapter One. Each aim will be addressed separately with a short discussion on overall implications.

### Investigate Possible Electrocardiograph Determined Etiologic Factors Linking Work Related Stress with Cardiovascular Disease

The linkage created by the Heart Rate Variability Project [1] between cardiovascular epidemiology, the human factors mental load literature and the demand – control model literature has provided the framework for this dissertation identifying etiological factors. Chapter Two provides evidence that exposure to job strain, and more specifically to low control on the job, reduces vagal cardiac control in what appears to indicate a long term adaptation. This finding does not change with removal of the exposure (work). In addition, there is evidence utilizing three estimates of sympathetic activity that high strain work is associated with elevated sympathetic tone during working hours. These electrocardiograph determined autonomic nervous system based etiological risk factors could help explain the increased risk of job strain and hypertension, cardiac mortality, and coronary artery disease.

Identify the Effect of Daily Fluctuations of Stress on Cardiac Regulatory Control

This dissertation has been able to demonstrate that within subjects there is a statistically significant association between daily fluctuations in vagal cardiac regulation and stress (Chapter Three). These associations were identified while controlling for the effects of age, exertion, position. Extending beyond the demand – control model to the demand – control – support hypotheses we have also demonstrated evidence that when assessing within subject responses to job strain the impact of social support may reduce the negative effect of strain if there is high strain. However, in times of low strain, the presence of support may actually create a demand associated with reduced cardiac vagal activity. The results point to the issue of changes in physiological regulation throughout two days of monitoring that are influenced by self-reports of diary assessed job strain and social support, in addition to the known factors such as age and posture. The present results add to the understanding of mechanisms provided above by which psychosocial factors may in the long-term contribute to cardiovascular disease development.

Identify Patterns of Autonomic Nervous System Response Indicative of Altered Cardiac Regulatory Coordination or Reduced Adaptive Capacity that Might Indicate Exhaustion

Despite the limitations associated with this cross-sectional data set we feel we have been able to extend the use of ambulatory ECG data in a manner capable of detecting altered coordination or reduced adaptive capacity. The approaches suggested in Chapter Four are build on platform created by the Stress Disequilibrium Theory [5] using systems principles to identify not only changes in the absolute level of ANS activity provided by HRV indices (although we feel these are also important markers – Chapter Two and Three) but also to identify patterns of altered coordination or reduced adaptive capacity. Patterns identified, which we have discussed in Chapter Four, include increased negative correlation between vagal cardiac control and heart rate, reduced vagal non-reciprocal control, elevated reciprocal sympathetic control, and reductions in the short-term variation of vagal activity. This is a preliminary finding that supports propositions from the Stress Disequilibrium Theory, however, further work with longitudinal protocols is required to test whether these alterations are first of all reproducible, and secondly whether they precede further deregulation ultimately leading to disease.

## Investigate the Possibility of Utilizing Ambulatory Electrocardiograph Monitoring in Occupational Stress Research

This dissertation provides support for the use of ambulatory ECG monitoring to identify biomarkers for exposure and/or risk assessment of job stress. Future longitudinal research should focus identifying the temporal trends of cardiac deregulation to confirm the associations and patterns we have identified in this dissertation. Another possibility is to conduct intervention studies that would allow determination of changes in ECG indices of ANS coordination and/or control.

## Literature Cited


1. Karasek R A. Heart Rate Variability Project. University of Massachusetts Lowell, 1996.
2. Karasek R A. The impact of the work environment on life outside the job. Cambridge: Massachusetts Institute of Technology, 1976.
3. Karasek R A. Job demands, job decision latitude and mental strain: Implications for job redesign. Adm Sci Q 1979; 24:285-308.
4. Karasek R A, Theorell T. Healthy Work: Stress Productivity, and the Reconstruction of Working Life. New York: Basic Books, 1990.
5. Karasek R.A. The Stress Disequilibrium Theory of Chronic Disease Development. Lowell, 2003.